

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement Standard Performance 2.5V IO, Operated at 1.8V, MOSFET SPICE Model Document

**Version 0.5 Phase 1
2016/09/07**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
v0.2_p1	2011/03/12	YZ Wang (NMOS) Laney Fan (PMOS)	Original release.
V0.3_p1	2011/09/02	YZ Wang	Revised the wmin from 0.16um to 0.35um and target values.
V0.3_p2	2012/01/17	YZ Wang	Revised the section 1.2 general information in the document.
0.4_p1	2012/5/15	Ho Cheng Hu	Add bpw NMOS model card
0.5_p1	2016/09/07	Denny Hung	1. Add jtsswgs parameter which value align with jtsswgd.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced
Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.8V n-ch MOSFET
1.8V bpw n-ch MOSFET
1.8V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of these compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

Note: Three choices of thick oxide IO devices are available, 2.5V, 3.3V and HG 3.3V, which cannot coexist on the same chip. If any design requirement is necessary, please discuss in advance with UMC through CE department.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9SP2
Cadence Spectre version 13.1.0.066
Mentor Graphics ELDO version 2013.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNFP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "/ModelFileName.lib.eldo" CaseName
, where CaseName can be tt, ff, ss, fnsp, or snfp.
- Define device condition by 'Mxx' statement;

Example:

M1 Vdd Vgg Vss Vbb DeviceName W=WDES L=LDES

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
- Add the following statement in the input file to the simulator.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName can be tt, ff, ss, fnsp, or snfp.
- Define device condition by 'Mxx' statement;

Example:

M1 (Vdd Vgg Vss Vbb) DeviceName W=WDES L=LDES

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName can be tt, ff, ss, fnsp, or snfp.
- Define device condition by 'Mxx' statement;

Example:

M1 Vdd Vgg Vss Vbb DeviceName W=WDES L=LDES

ModelFileName for 1.8V MOSFET: l110ae_25ud18_v051

DeviceName for 1.8V n-ch MOSFET: n_25ud18_l110ae

DeviceName for 1.8V bpw n-ch MOSFET: n_bpw_25ud18_l110ae

DeviceName for 1.8V p-ch MOSFET: p_25ud18_l110ae

As stated above, one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range (After 90% shrink):

For 1.8V n-ch and bpw n-ch MOSFET:

$$0.234 \text{ um} \leq L_{\text{DES}} \leq 45 \text{ um}$$

$$0.35 \text{ um} \leq W_{\text{DES}} \leq 90 \text{ um}$$

For 1.8V p-ch MOSFET:

$$0.234 \text{ um} \leq L_{\text{DES}} \leq 45 \text{ um}$$

$$0.35 \text{ um} \leq W_{\text{DES}} \leq 90 \text{ um}$$

Adopt multi-finger structures if $W_{\text{DES}} > 9 \text{ um}$.

$$0.318 \text{ um} \leq S_{\text{A}}, S_{\text{B}} \leq 1.98 \text{ um}$$

Note: $S_{\text{AREF}} = S_{\text{BREF}} = 1.98 \text{ um}$ is the reference dimension of the STI-stress (LOD) effect.

Voltage range:

For 1.8V n-ch and bpw n-ch MOSFET:

$$0\text{V} \leq V_{\text{GS}} \leq 1.8\text{V}(*1.1)$$

$$0\text{V} \leq V_{\text{DS}} \leq 1.8\text{V}(*1.1)$$

$$-1.8\text{V} \leq V_{\text{BS}} \leq 0\text{V}$$

For 1.8V p-ch MOSFET:

$$0\text{V} \geq V_{\text{GS}} \geq -1.8\text{V}(*1.1)$$

$$0\text{V} \geq V_{\text{DS}} \geq -1.8\text{V}(*1.1)$$

$$1.8\text{V} \geq V_{\text{BS}} \geq 0\text{V}$$

Temperature range:

$$-50 \text{ C} \sim +125 \text{ C}$$

2 1.8V N-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0QA	BSW0QA	BSW0QA	BSW0QA
Lot ID	D5FHR.11	D5FHR.11	D5FHR.11	D5FHR.11
Wafer Number	11	11	11	11

2.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
 $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and
 $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS, respectively.
The dimension is after 90% shrink and 12nm sizing channel length back.

All parameters are given for $T=25\text{ C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Table 2-2 Electrical parameters for NMOS

All parameters are given for T=25 °C , V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W/L= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.512	0.506	0.516	0.512
I _{ON}	V _{DS} =V _{GS} = 1.8 V	μA/ μm	13.38	13.26	13.29	13.38

Wide and short channel device (W/L= 10 μ m / 0.246 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.5	0.497	0.498	0.500
V _{TSAT}	V _{DS} = 1.8 V	V	0.432	0.430	0.434	0.431
I _{ON}	V _{DS} =V _{GS} = 1.8 V	μA/μm	351.57	352.30	349.09	351.01
I _{OFF}	V _{DS} = 1.8 V, V _{GS} =0 V	A/μm	--	2.8E-13	5.6E-12	5.9E-12*

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.512	0.508	0.505	0.512
V _{TSAT}	V _{DS} = 1.8 V	V	0.508	0.503	0.502	0.509
I _{ON}	V _{DS} =V _{GS} = 1.8 V	μA/μm	11.48	11.42	11.33	11.48

Narrow and short channel device (W/L= 0.35 μ m / 0.246 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.487	0.478	0.489	0.487
V _{TSAT}	V _{DS} = 1.8 V	V	0.417	0.407	0.413	0.416
I _{ON}	V _{DS} =V _{GS} = 1.8 V	μA/μm	414.28	413.26	402.63	414.34
I _{OFF}	V _{DS} = 1.8 V, V _{GS} =0 V	A/μm	--	8.8E-12	8.6E-12	7.4E-12*

Capacitance

C _J	V _J = 0 V	F/m ²	--	7.812E-04	7.812E-04	7.812E-04
C _{JSW}	V _J = 0 V	F/m	--	1.125E-10	1.125E-10	1.125E-10
C _{JSWG}	V _J = 0 V	F/m	--	2.250E-10	2.155E-10	2.155E-10
C _{OV}	V _{GS} = -0.5 V	F/m	--	2.65E-10	2.55E-10	2.55E-10

*gmindc=1e-15

2.3 MOSFET DC Model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

WL	10	1	0.8	0.6	0.507	0.435	0.417	0.399	0.318	0.3	0.282	0.264	0.246
10	X	X	X	X	X	X	X	X	X	X	X	X	X
1											X	X	X
0.8	X	X		X	X		X		X		X	X	X
0.6				X			X		X				X
0.43	X	X			X		X		X		X	X	X
0.35	X			X	X		X		X				X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	0.1	1.8	1.7	V_{DS} (V)	0	2	0.025
V_{GS} (V)	-0.5	2	0.025	V_{GS} (V)	0.7	1.8	0.275
V_{BS} (V)	0	-1.8	-0.45	V_{BS} (V)	0	-1.8	-1.8

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

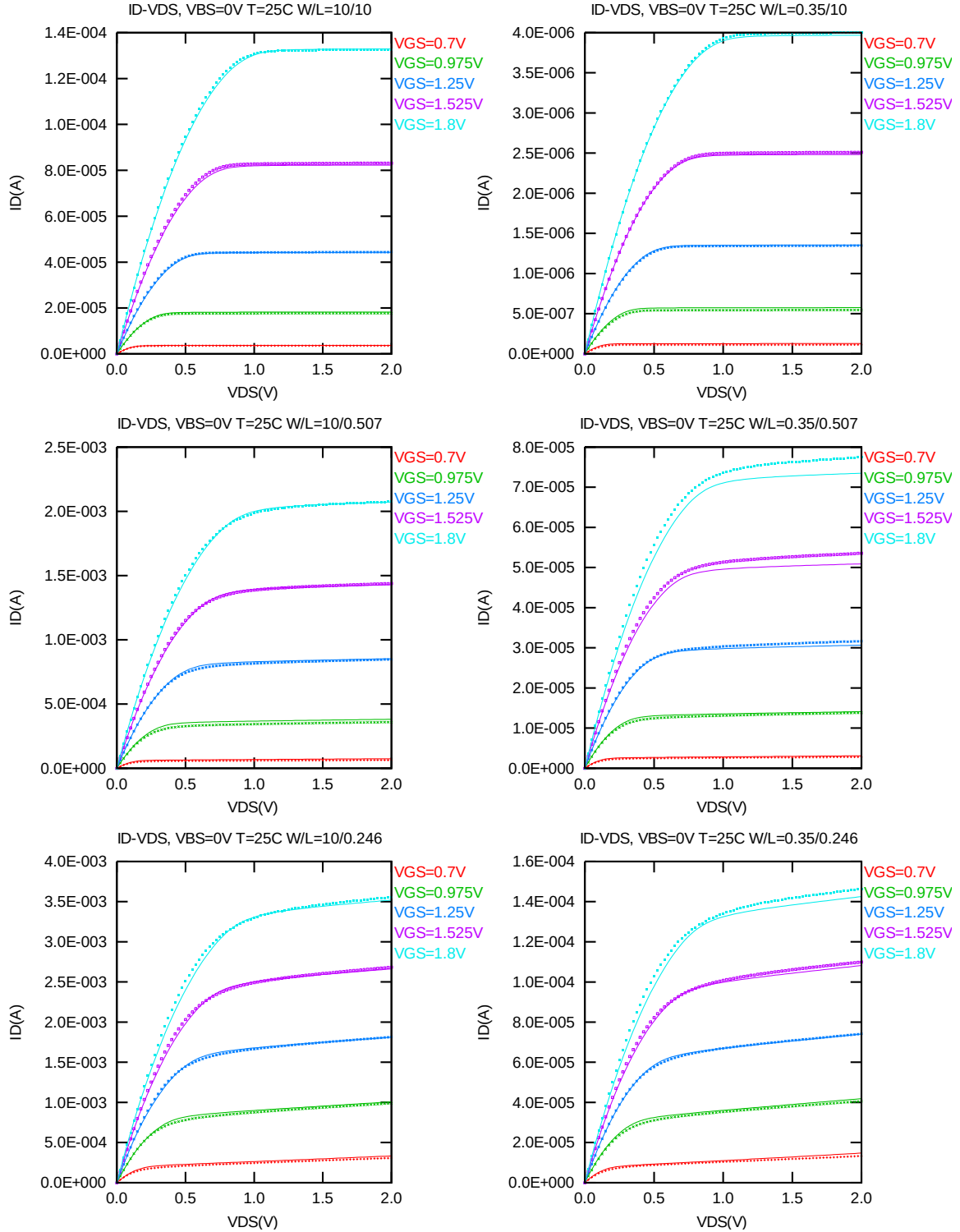


Figure 2-1 I_D vs. V_{DS} at $V_{BS} = 0$ V for NMOS

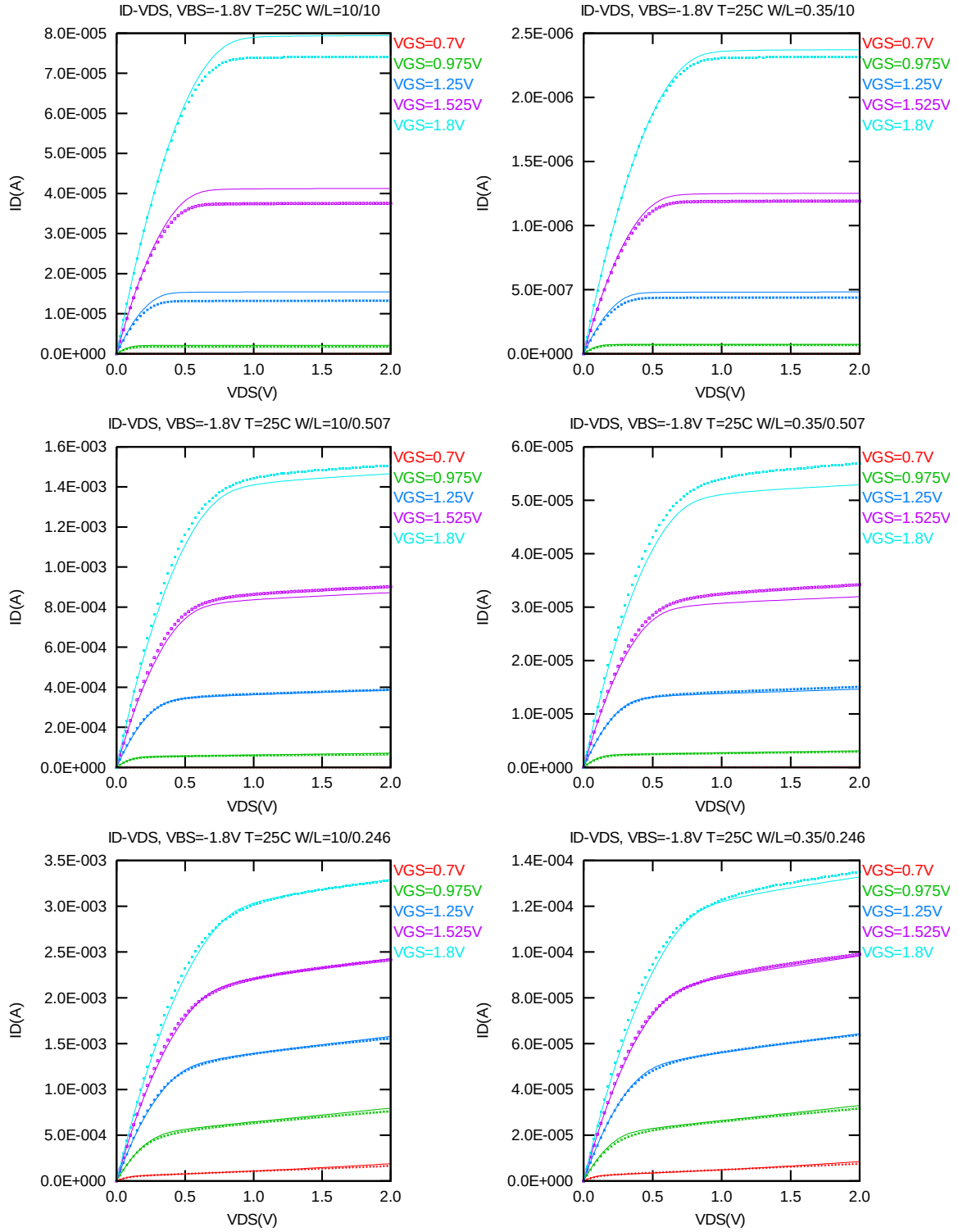


Figure 2-2 I_D vs. V_{DS} at $V_{BS} = -1.8V$ for NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

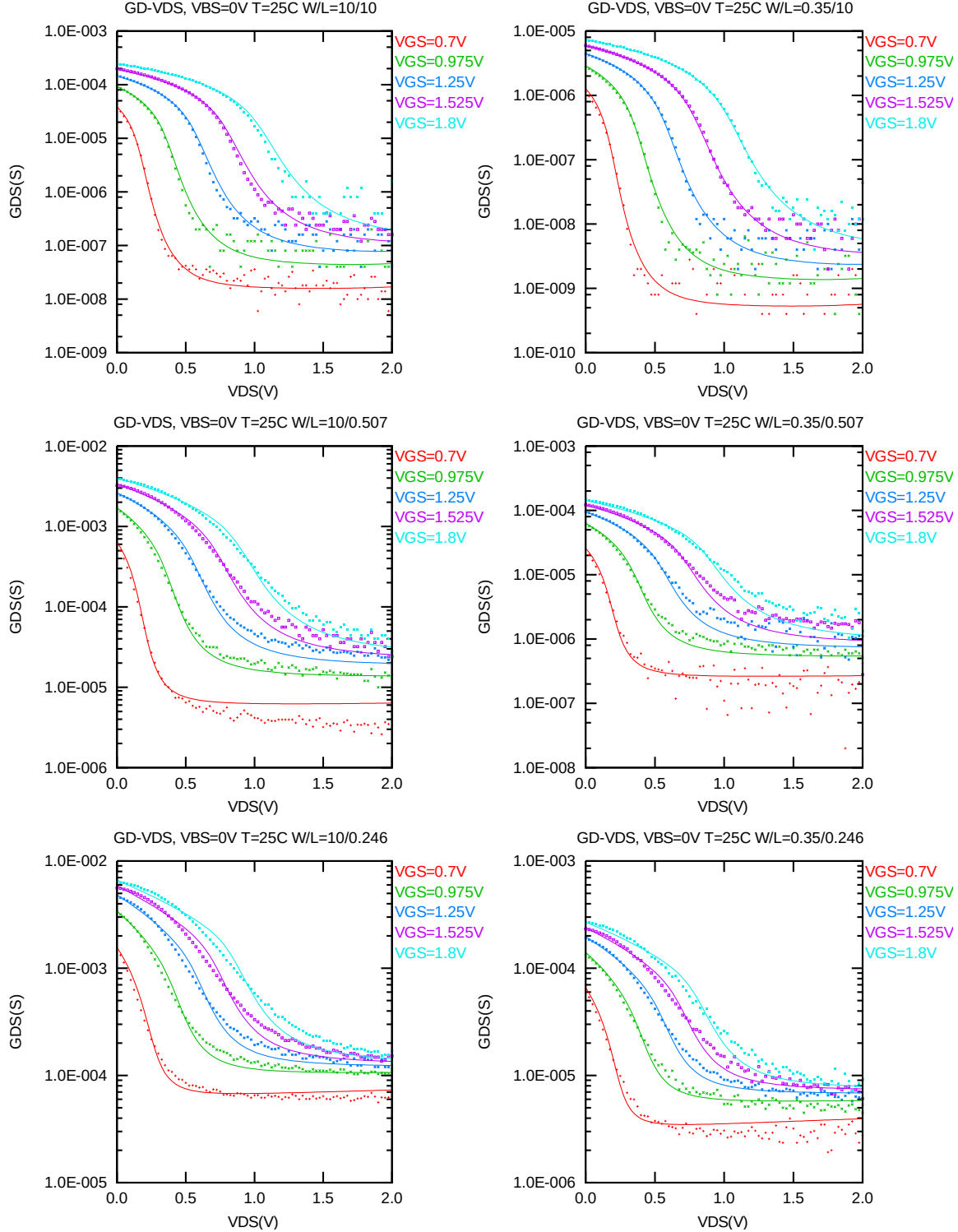


Figure 2-3 G_{DS} vs. V_{DS} at $V_{BS} = 0$ V for NMOS

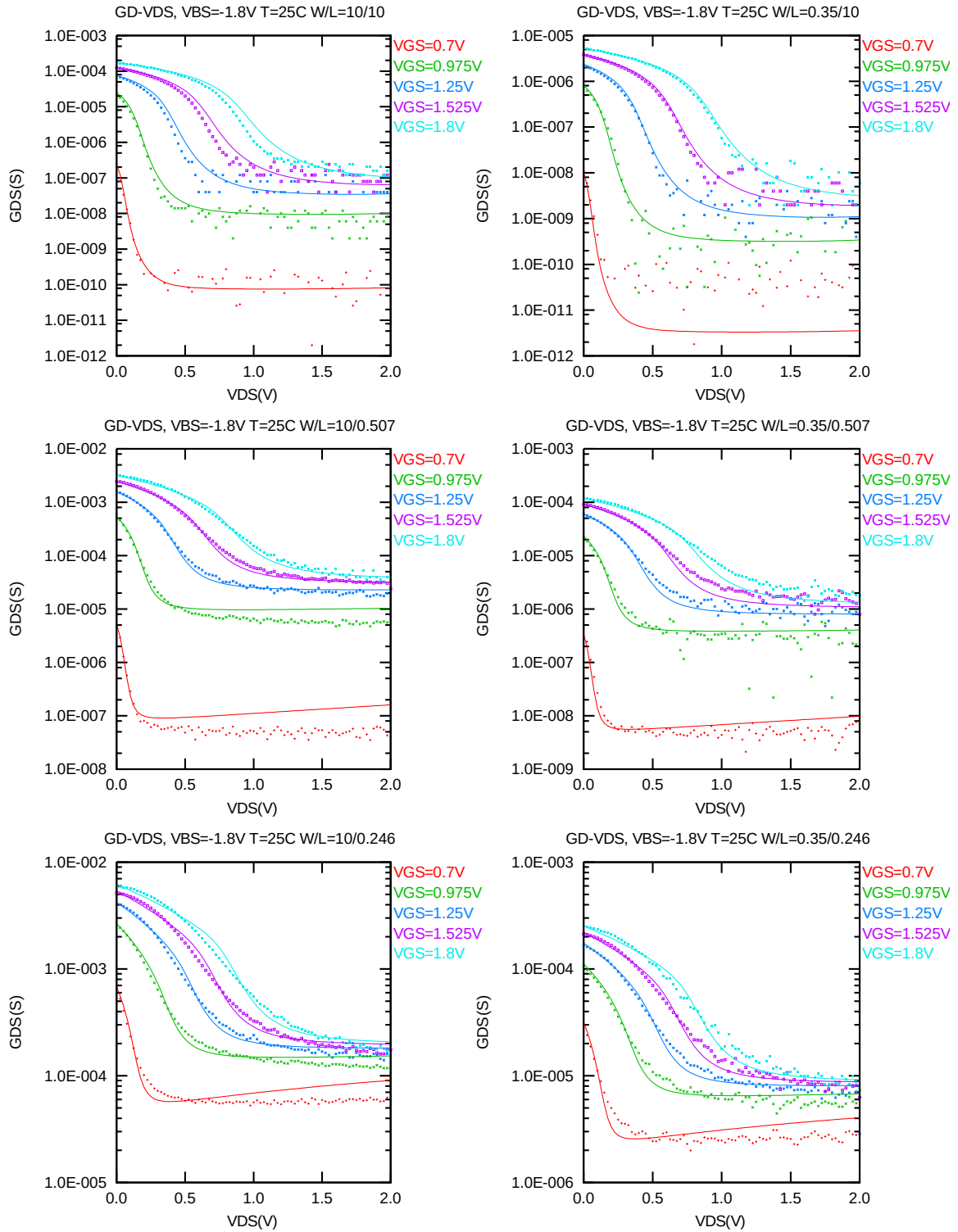


Figure 2-4 G_{DS} vs. V_{DS} at $V_{BS} = -1.8V$ for NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

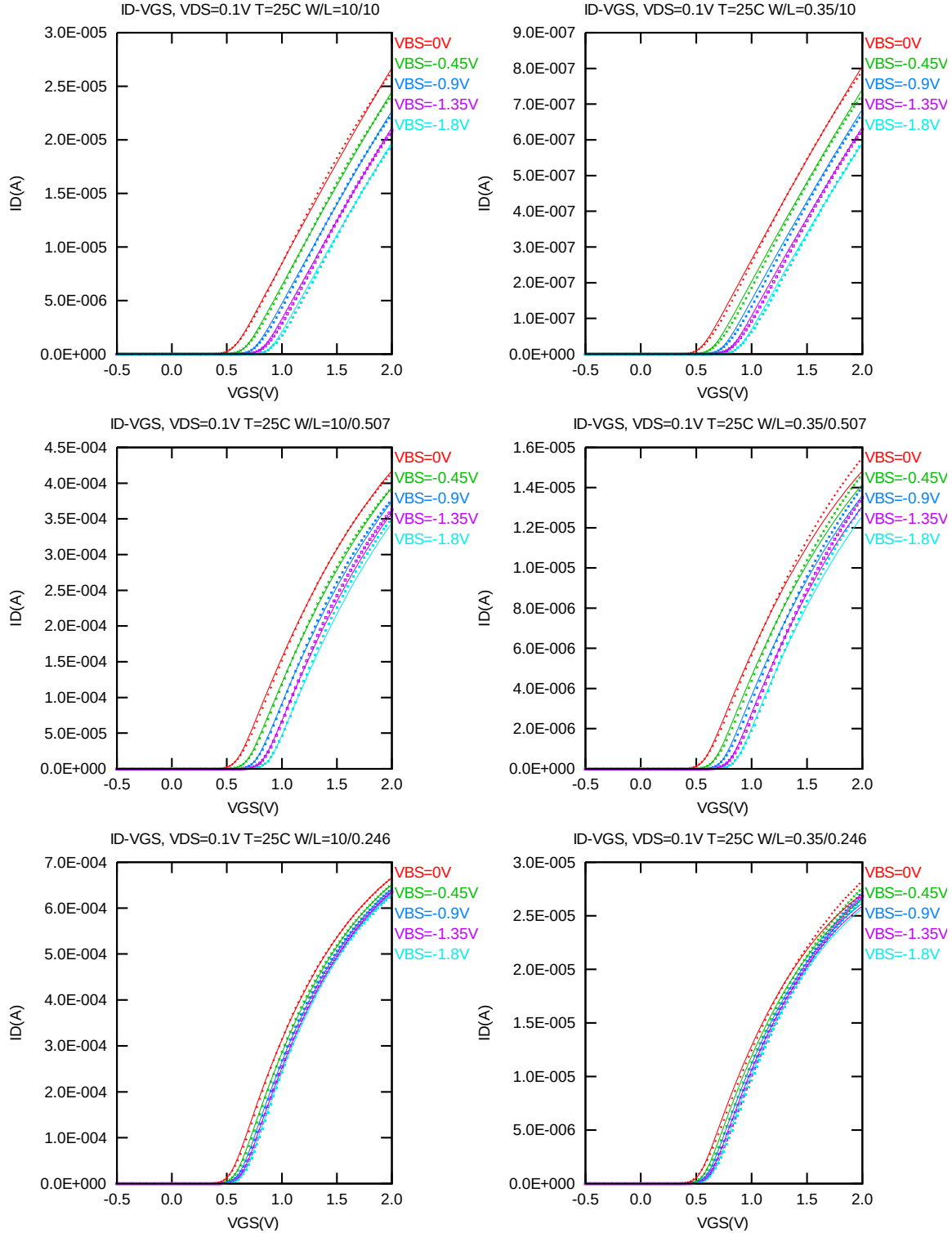


Figure 2-5 I_D vs. V_{GS} at $V_{DS} = 0.1V$ for NMOS

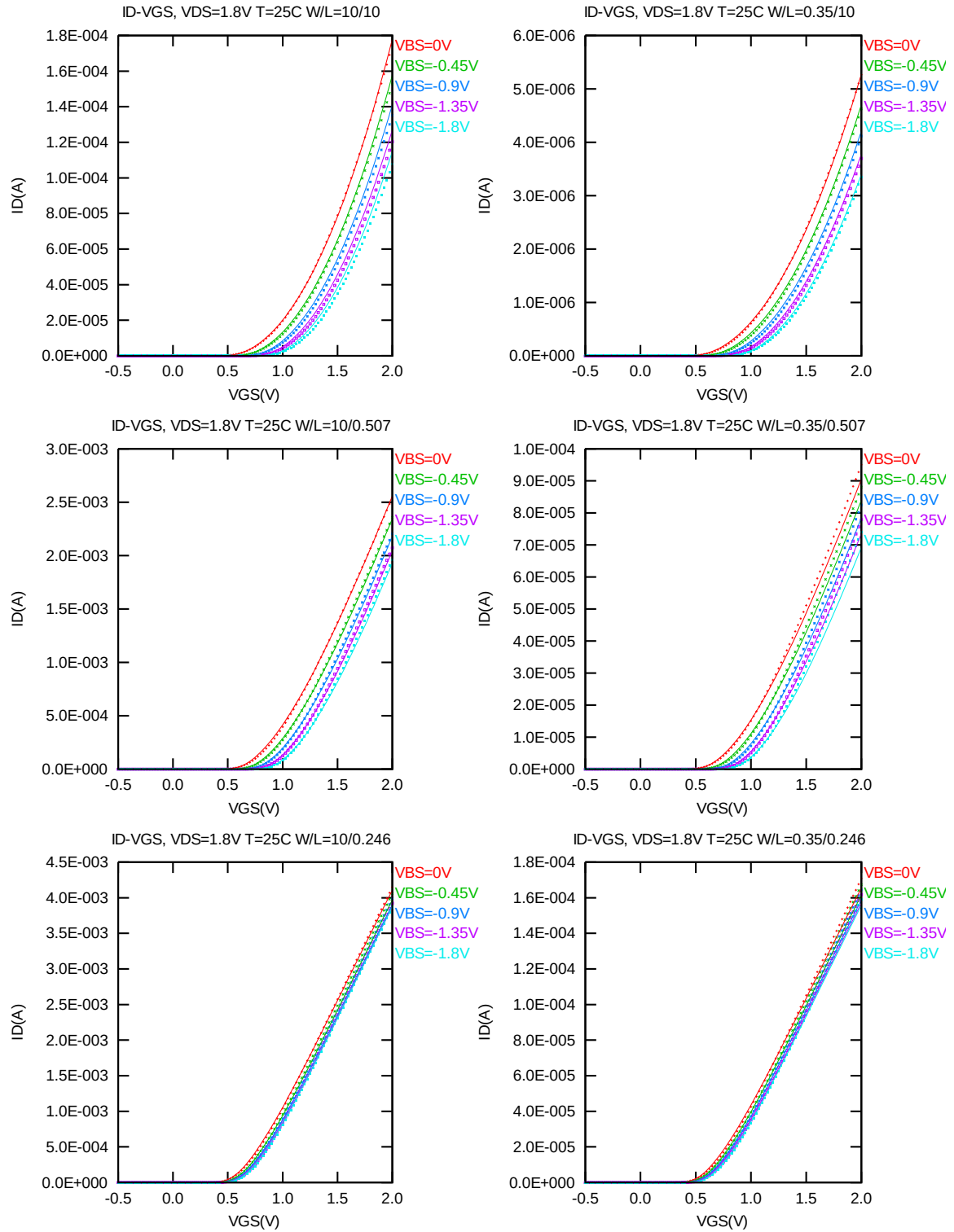


Figure 2-6 I_D vs. V_{GS} at $V_{DS} = 1.8V$ for NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

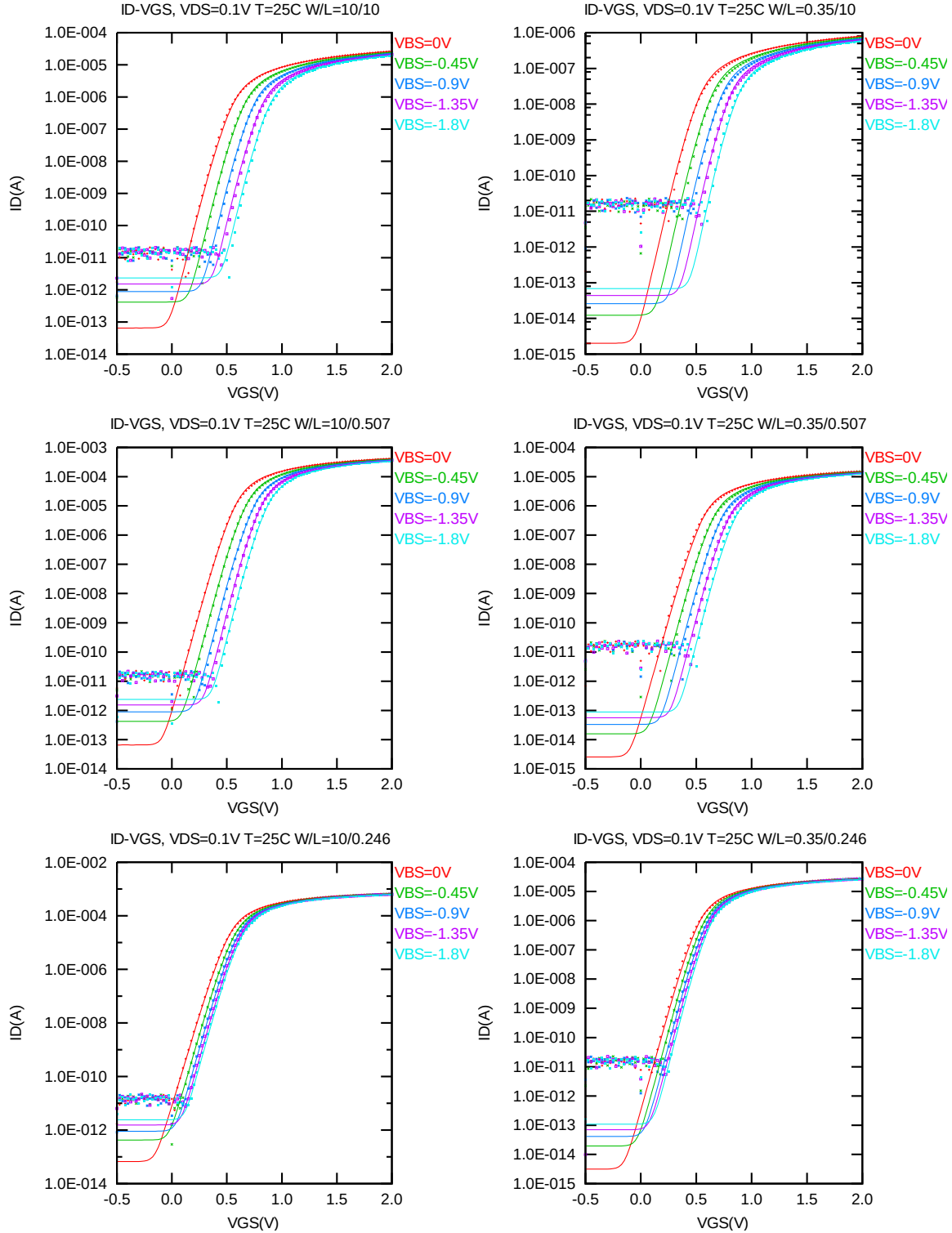


Figure 2-7 Log scaled I_D vs. V_{GS} at $V_{DS} = 0.1$ V for NMOS

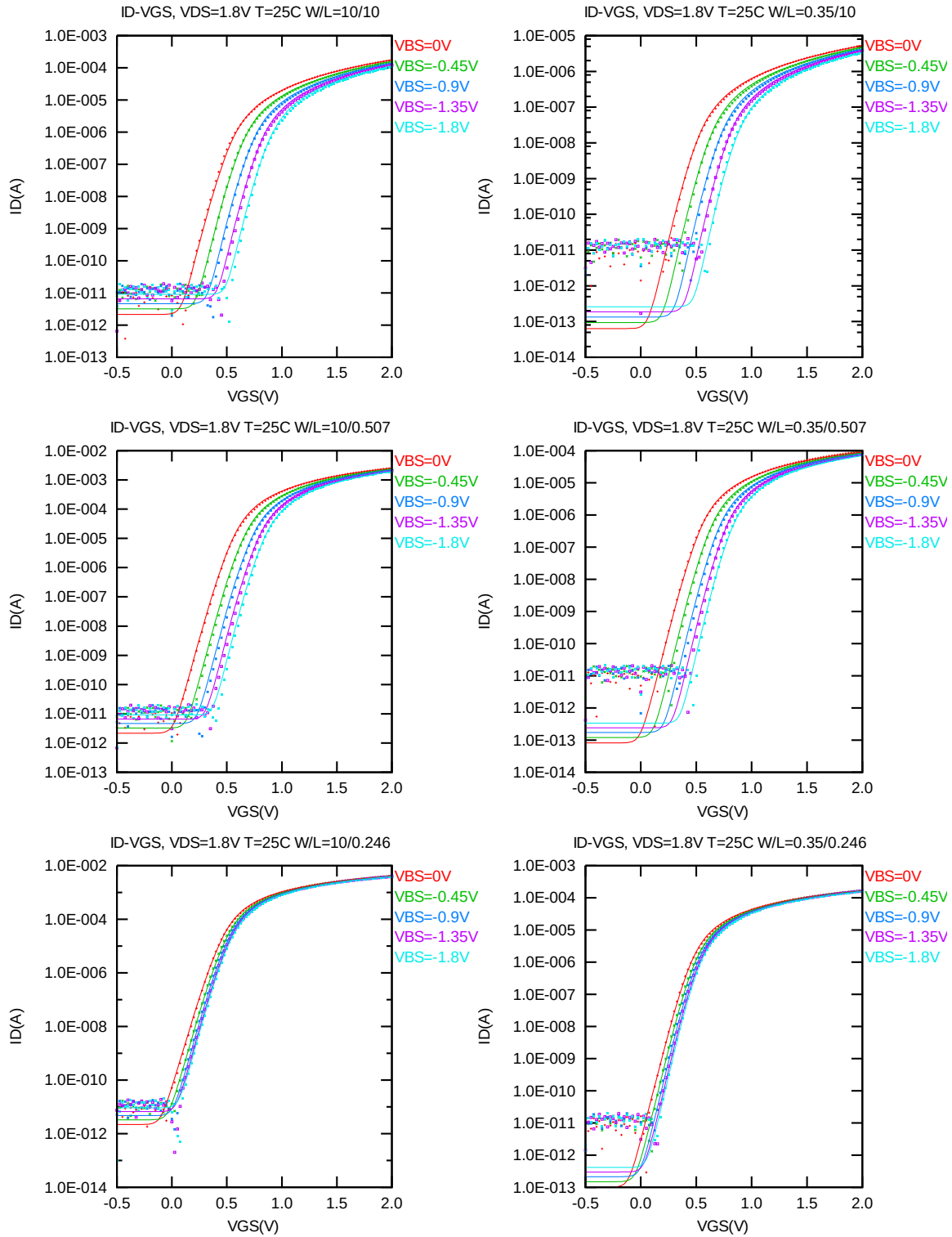


Figure 2-8 Log scaled I_D vs. V_{GS} at $V_{DS} = 1.8V$ for NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

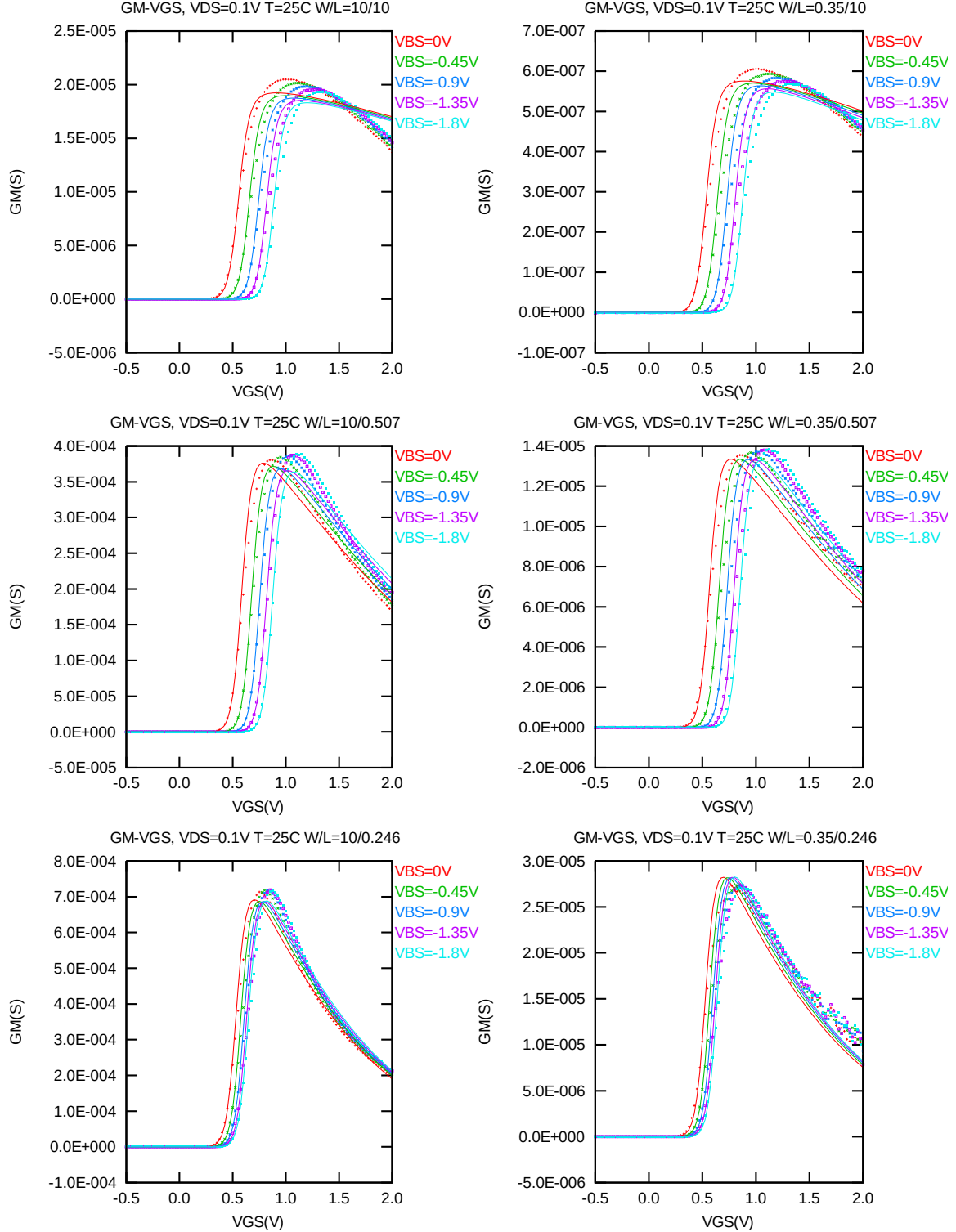


Figure 2-9 G_M vs. V_{GS} at $V_{DS} = 0.1 V$ for NMOS

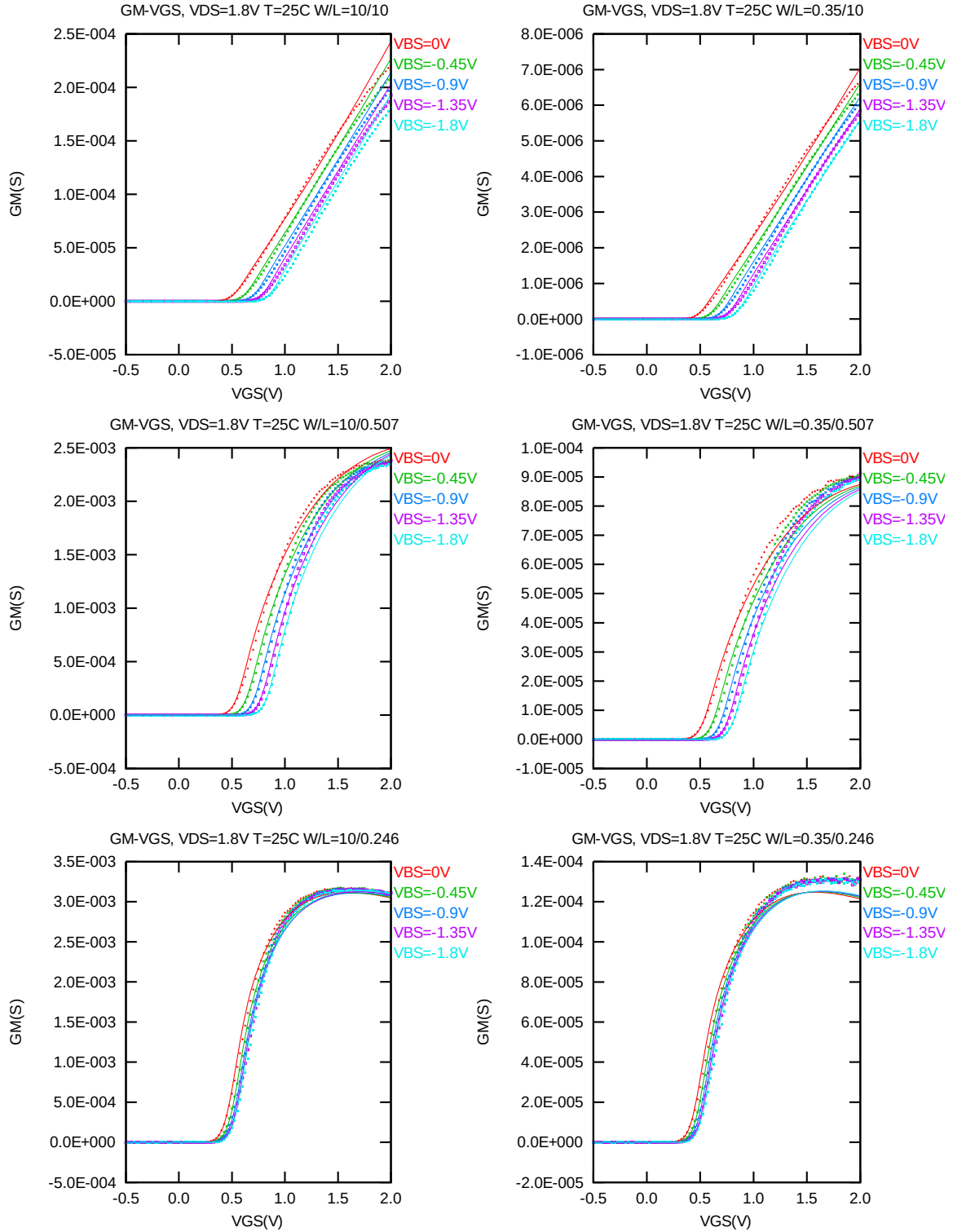


Figure 2-10 G_M vs. V_{GS} at $V_{DS} = 1.8V$ for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu m)$ at $25^\circ C$. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

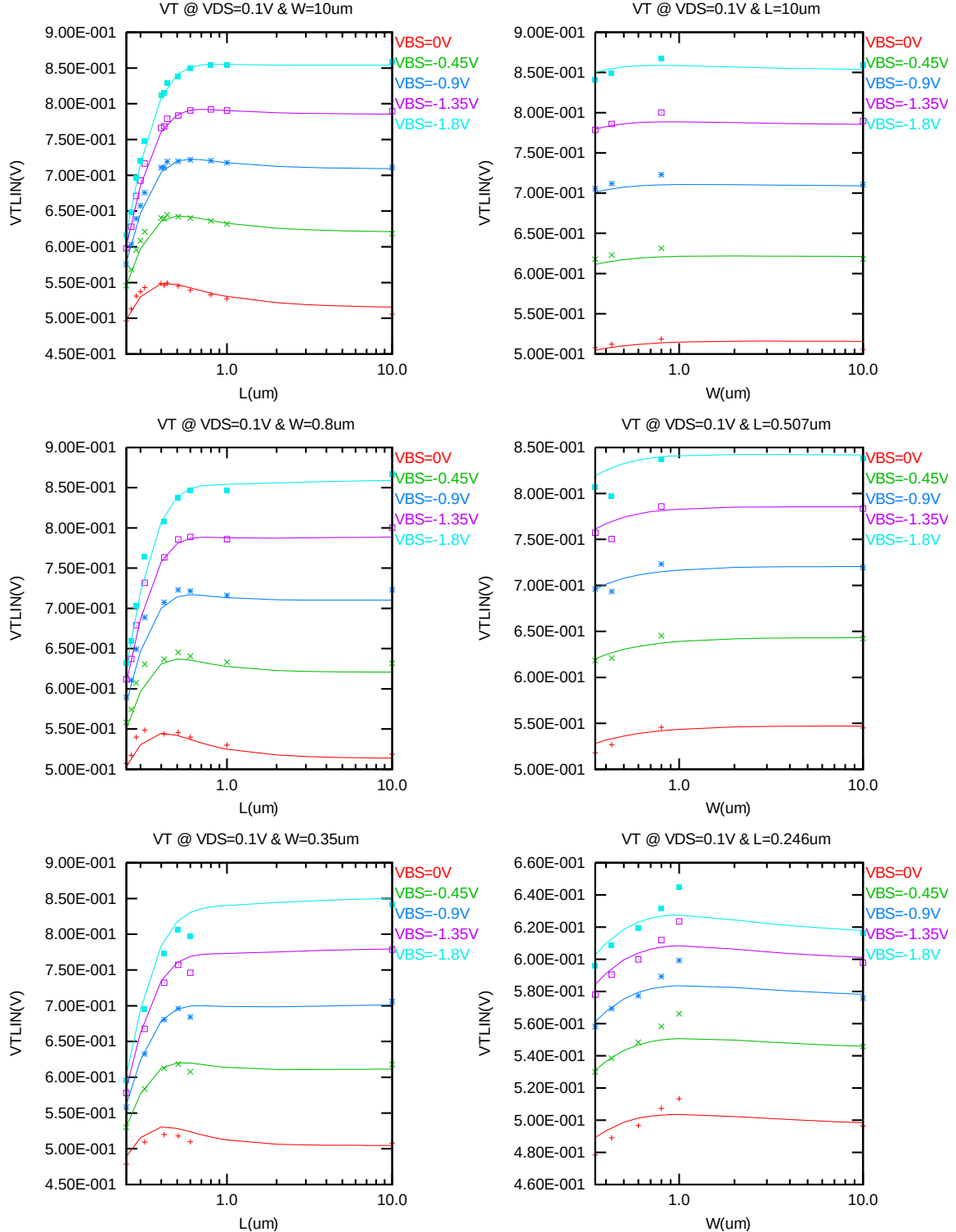


Figure 2-11 V_{TLIN} vs. channel length & channel width for low drain bias ($V_{DS} = 0.1V$) for NMOS

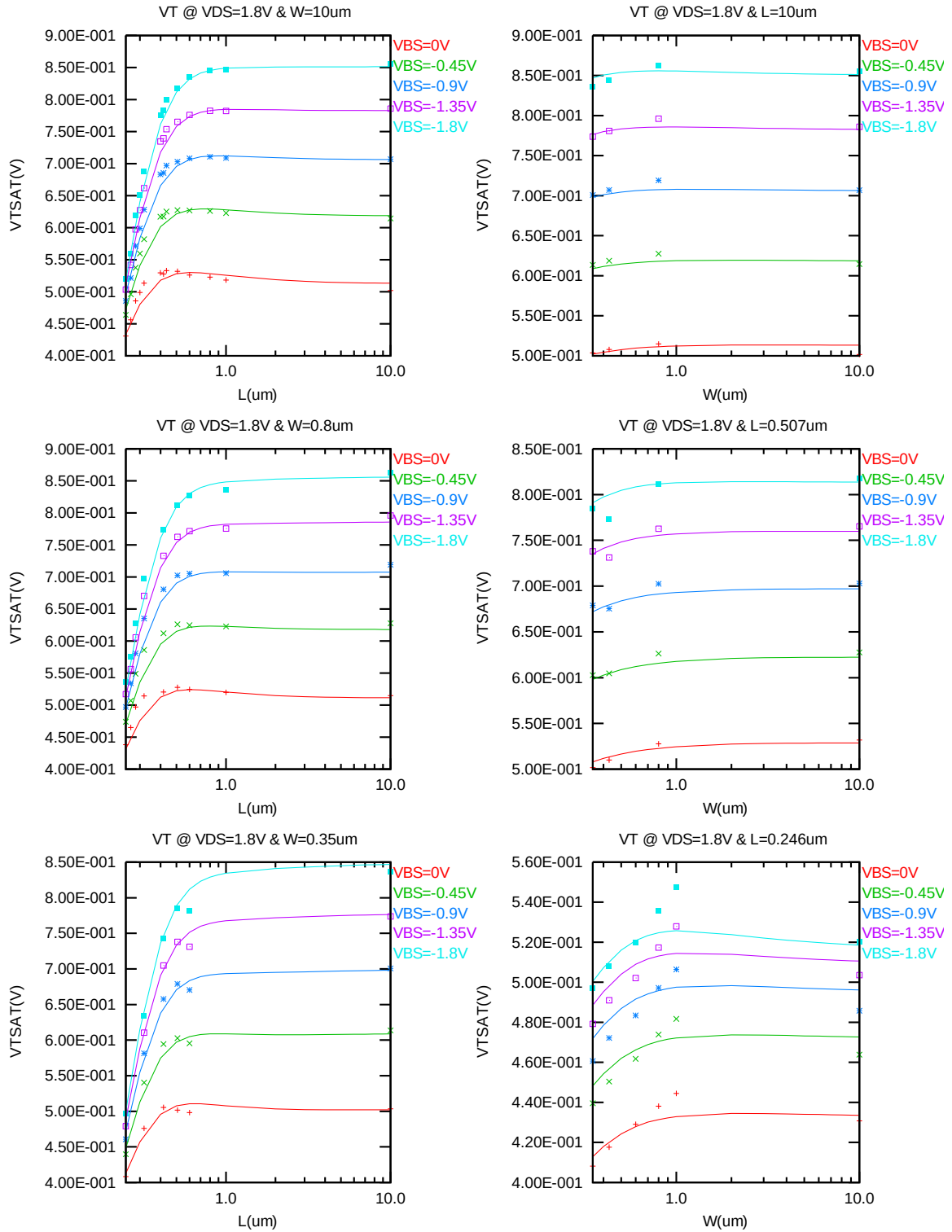


Figure 2-12 V_T vs. channel length & channel width for high drain bias ($V_{DS} = 1.8V$) for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.8V$ for $V_{BS} = 0V$ and $V_{BS} = -1.8V$ at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

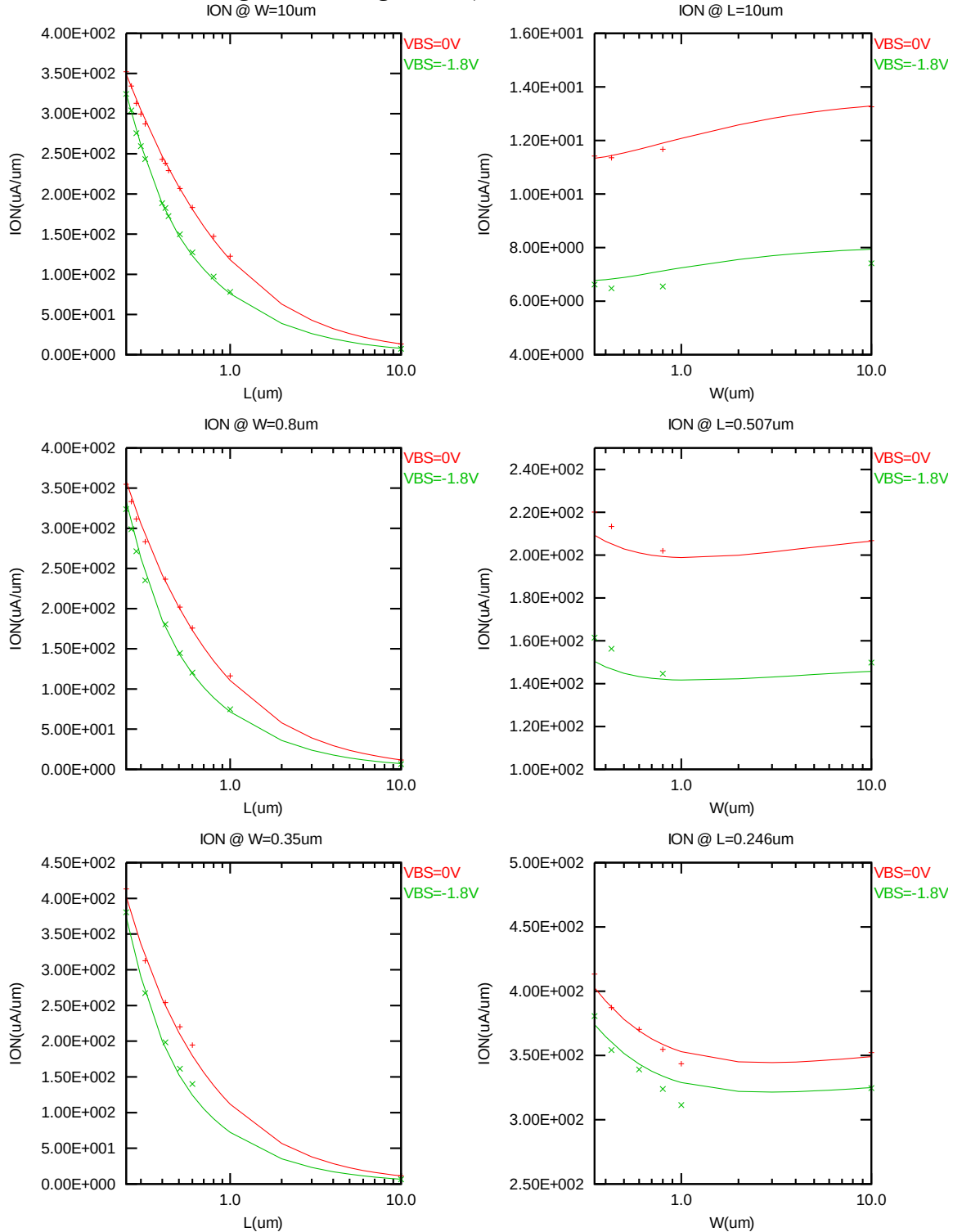


Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0V$ to $-1.8V$ at 25 C. A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

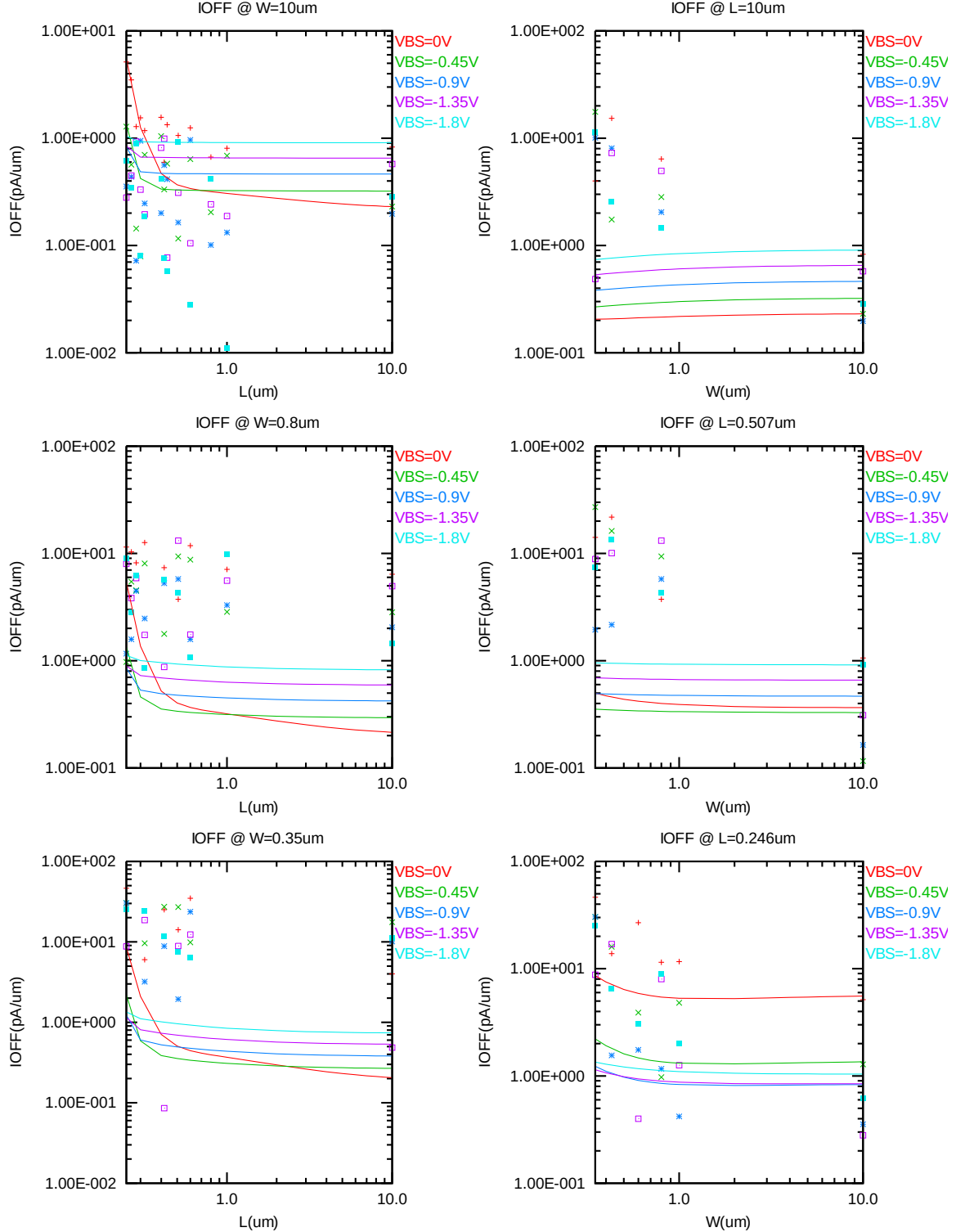


Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu m)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

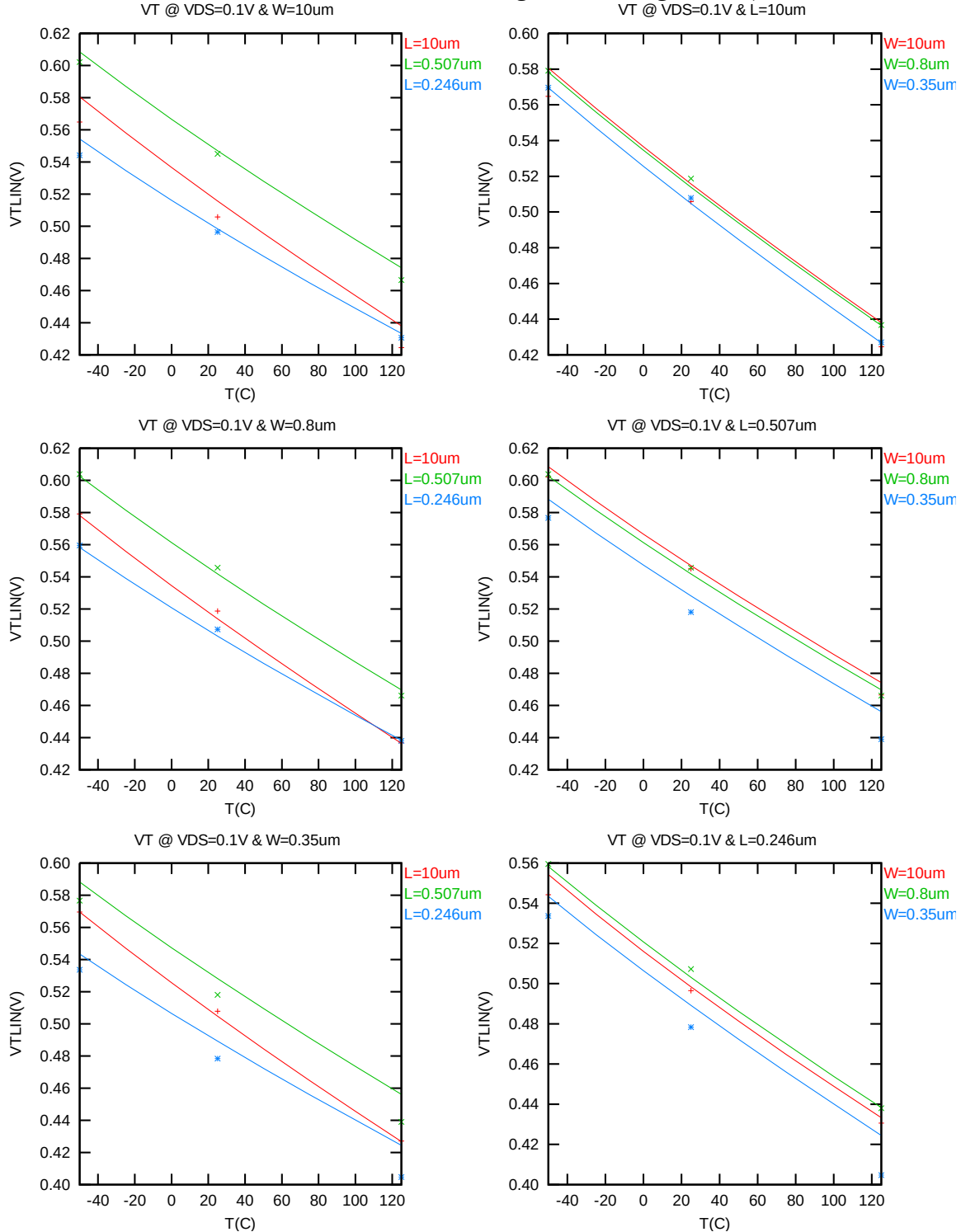


Figure 2-15 V_{TLIN} vs. temperature for NMOS

V_{TSAT} vs. temperature

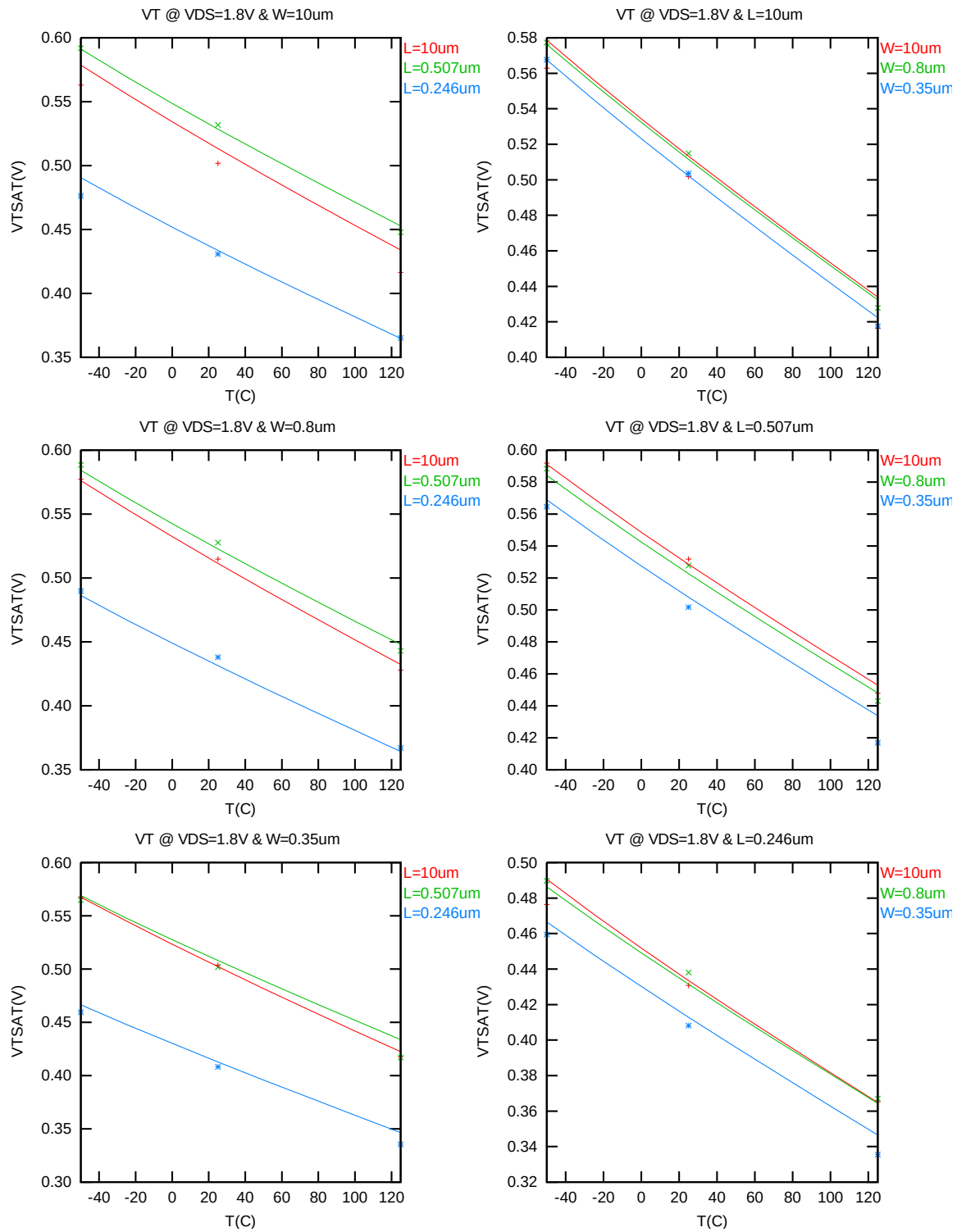


Figure 2-16 V_{TSAT} vs. temperature for NMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.8V$, $V_{BS} = 0V$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

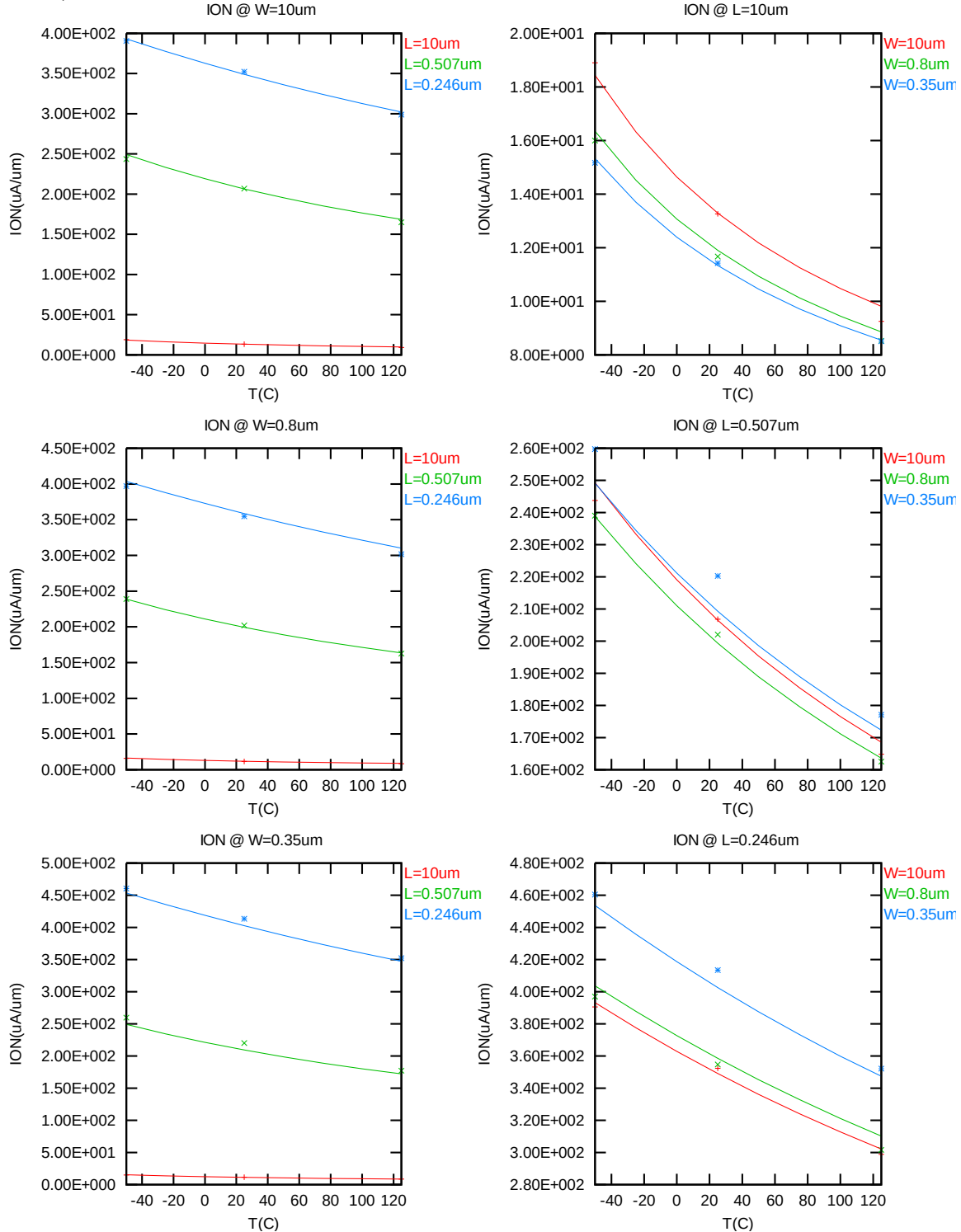


Figure 2-17 Saturation current vs. temperature for NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{DS} = 1.8V$, $V_{GS} = V_{BS} = 0V$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

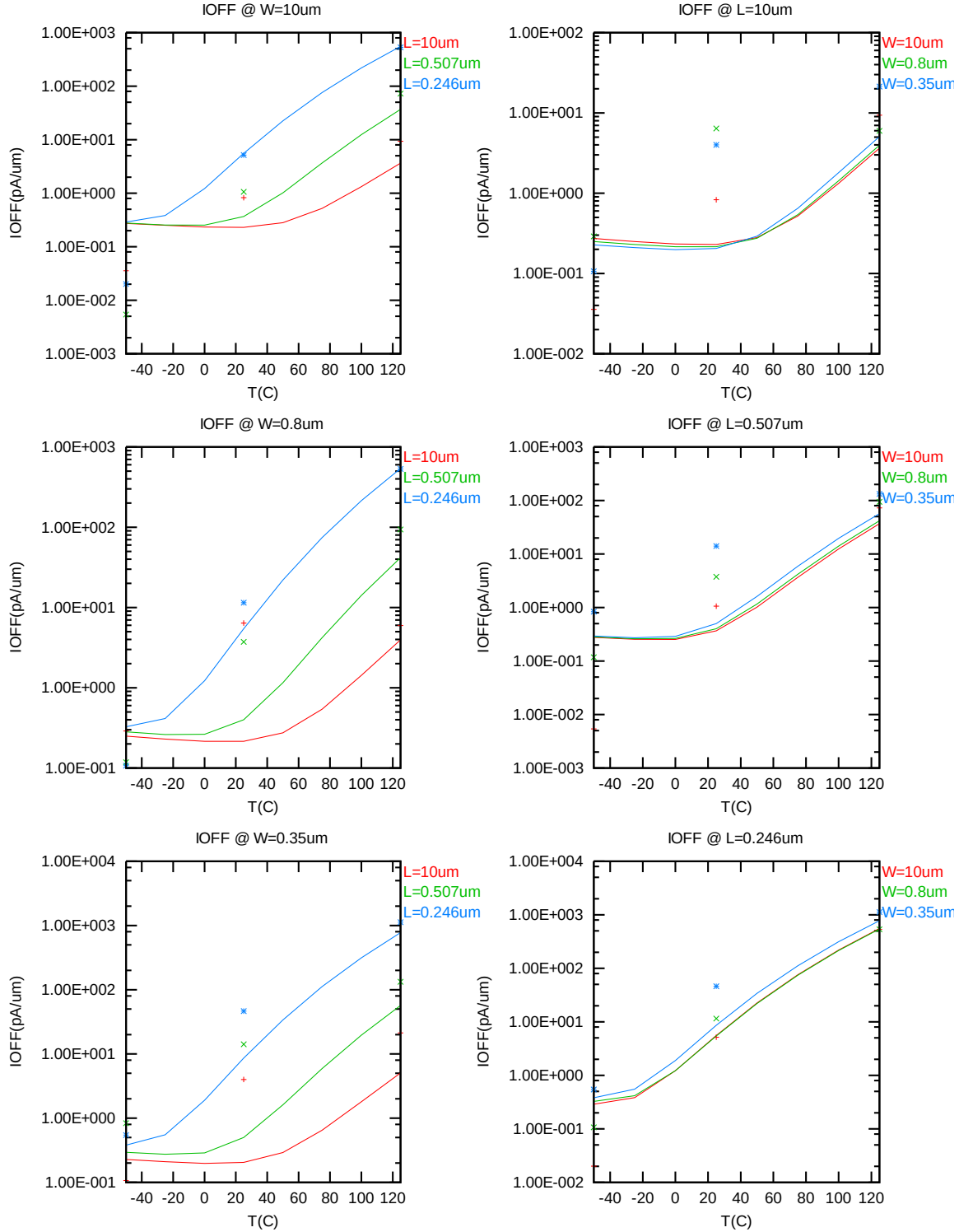


Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- b. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- c. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- e. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 2-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

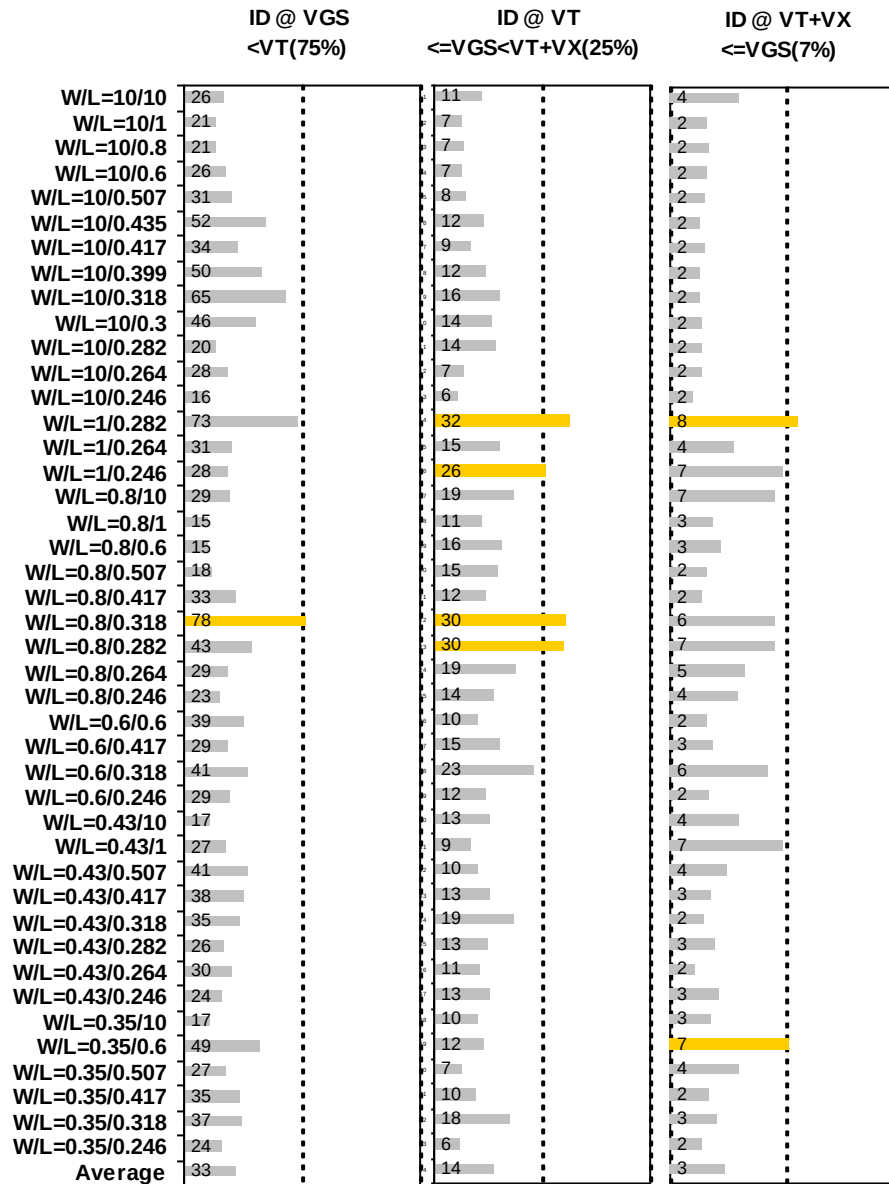


Figure 2-19 I_D accuracy at 25 C for NMOS

G_M and G_{DS} accuracy

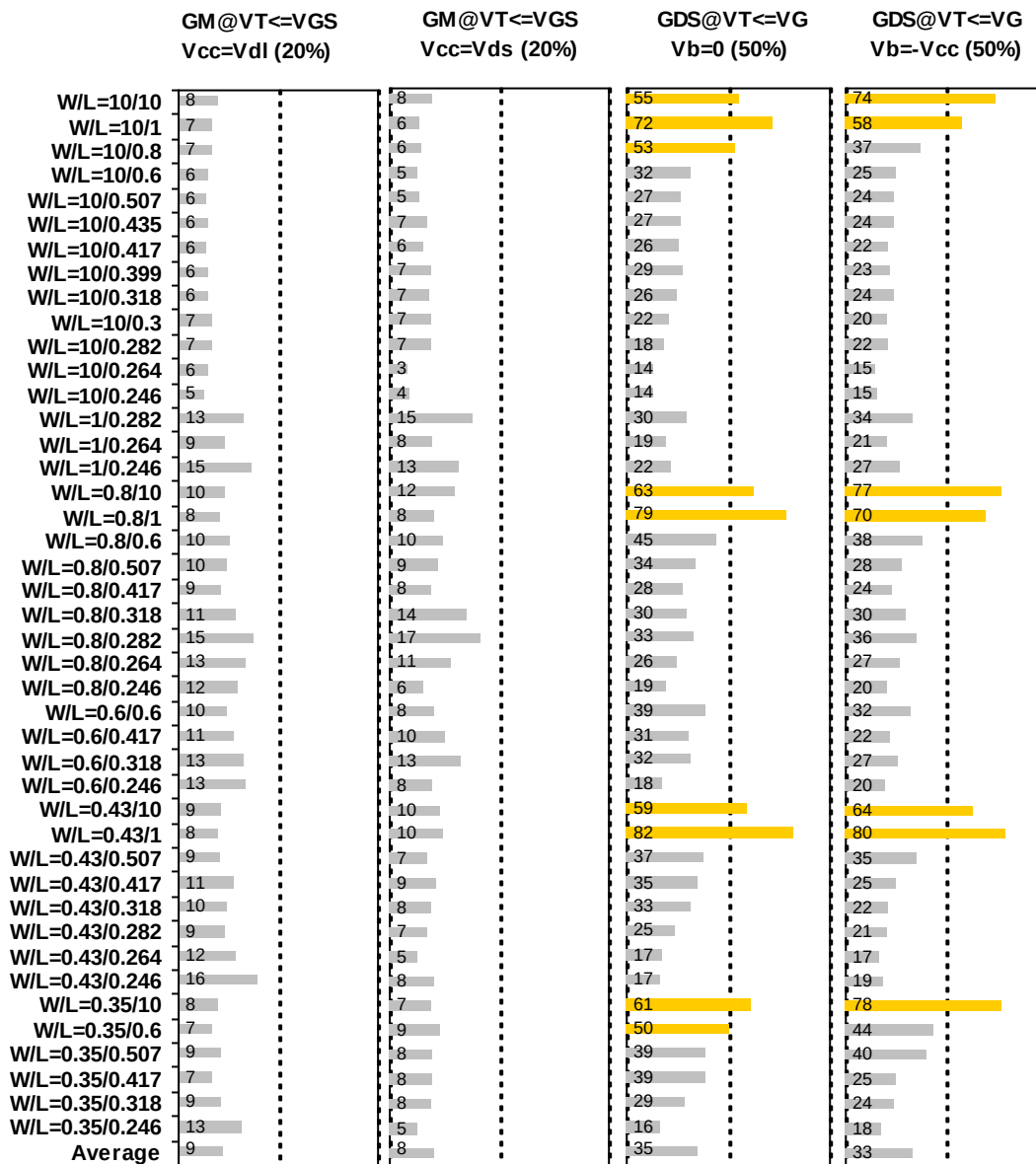


Figure 2-20 G_M and G_{DS} accuracy at 25 C for NMOS

2.4 MOSFET Capacitance Model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS C_{gg} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C _{GDS}	80	80

Table 2-7 Large area measurement structure for NMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	10	10
	1	10
	.5	10
	.246	10
	10	1
	.417	10
	.417	1
	10	.5
	1	1
	.246	.5
	.5	.5
	.417	.25
	10	.16
	10	.25
	.417	.16
	.246	.16
	.5	.16

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 2-8 Measurement conditions for NMOS capacitance model

V_{BIAS} [V]	From -3.63 to 3.63 V in 0.066 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{OX} extraction

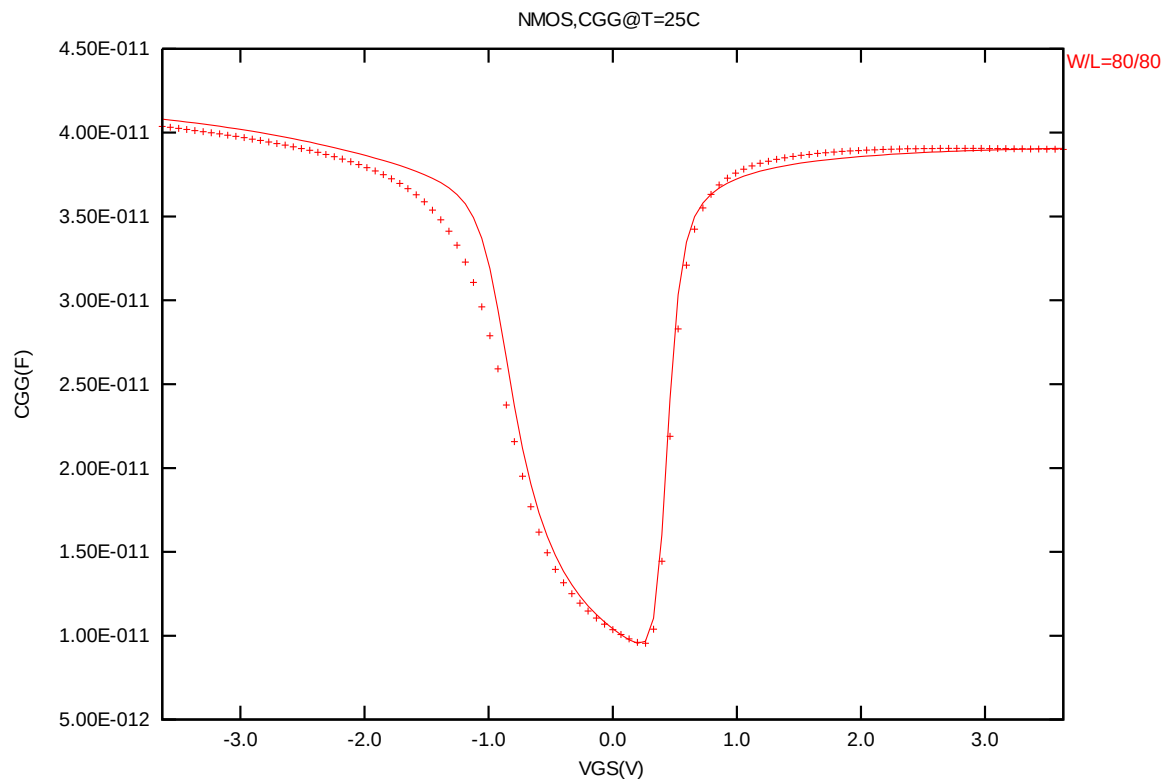


Figure 2-21 Measured and simulated C_{GG} curves

2.4.4 Width and length AC model accuracy

Width and length AC model accuracy

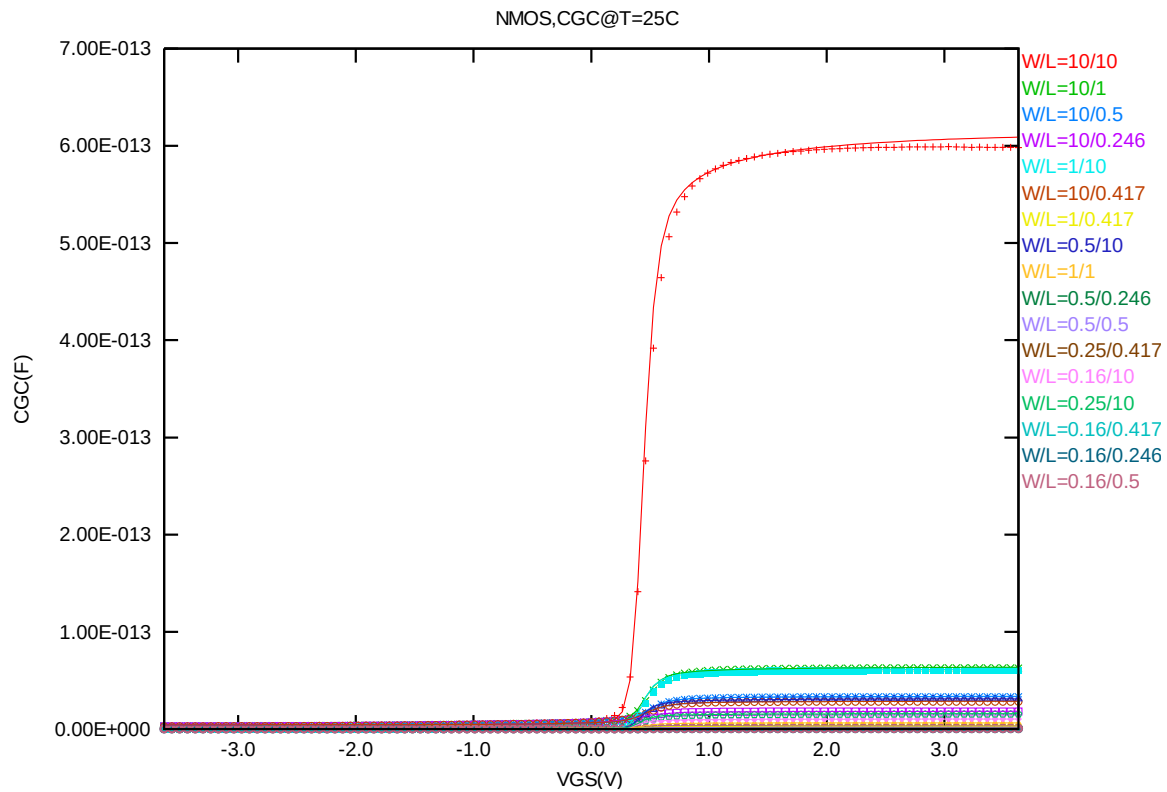


Figure 2-22 Measured and simulated C_{GC} curves for MOSFET measurement structure for NMOS

2.5 Parasitic Source / Drain Junction Capacitance Model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.75E-08	1.80E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V _{BIAS} [V]	VH: pwell & VL: n+ From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,125

2.5.3 Extracted model parameters and verification plots

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

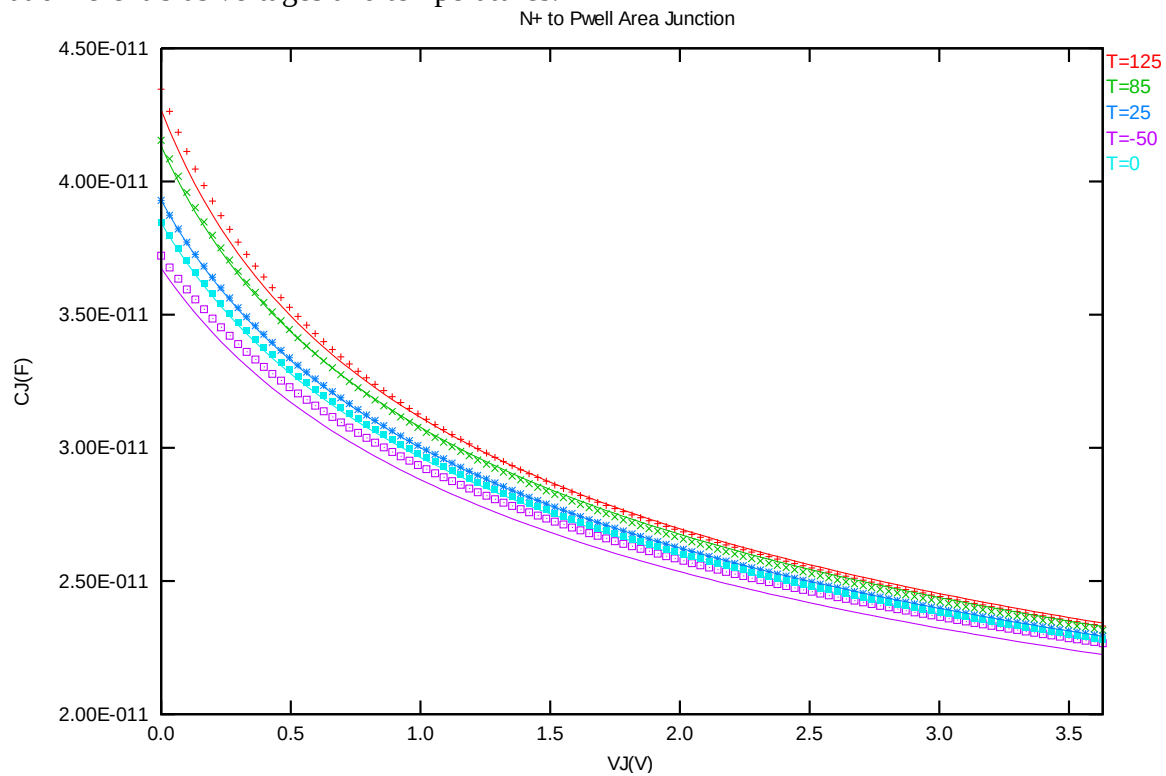


Figure 2-23 Bottom area junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for NMOS

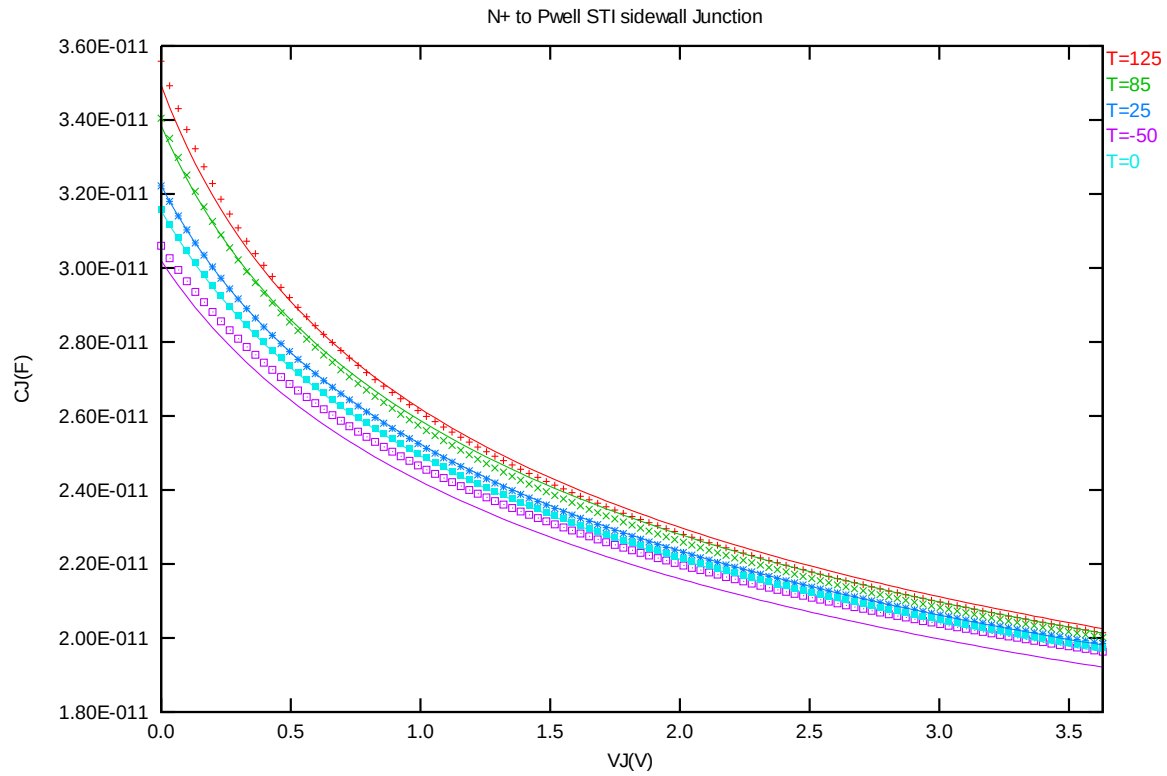


Figure 2-24 STI bounded sidewall junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for NMOS

3 1.8V bpw N-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for bpw NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0QA	BSW0QA	BSW0QA	BSW0QA
Lot ID	D5FHR.11	D5FHR.11	D5FHR.11	D5FHR.11
Wafer Number	11	11	11	11

3.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
 $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS.
 The dimension is after 90% shrink and 12nm sizing channel length back.

Table 3-2 Electrical parameters for bpw NMOS

All parameters are given for $T = 25^\circ\text{C}$, $V_{BS} = 0\text{ V}$ unless specified otherwise.

Target & Centered Value are given for $SA = 2.2\mu\text{m}$ $SB = 2.2\mu\text{m}$

Measured & Extracted Value are given for $SA = 2.2\mu\text{m}$ $SB = 2.2\mu\text{m}$

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device ($W/L = 10\mu\text{m} / 10\mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.1\text{ V}$	V	0.512	0.510	0.511	0.512
I_{ON}	$V_{DS} = V_{GS} = 1.8\text{ V}$	$\mu\text{A} / \mu\text{m}$	12.85	12.84	12.87	12.85

Wide and short channel device ($W/L = 10\mu\text{m} / 0.246\mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.1\text{ V}$	V	0.503	0.501	0.501	0.503
V_{TSAT}	$V_{DS} = 1.8\text{ V}$	V	0.442	0.440	0.440	0.442
I_{ON}	$V_{DS} = V_{GS} = 1.8\text{ V}$	$\mu\text{A} / \mu\text{m}$	343.5	343.4	344.6	343.5
I_{OFF}	$V_{DS} = 1.8\text{ V}$, $V_{GS} = 0\text{ V}$	$\text{A} / \mu\text{m}$	$3.5\text{E-}12$	$3.6\text{E-}12$	$4.3\text{E-}12$	$3.5\text{E-}12^*$

Narrow and long channel device ($W/L = 0.35\mu\text{m} / 10\mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.1\text{ V}$	V	0.510	0.506	0.505	0.510
V_{TSAT}	$V_{DS} = 1.8\text{ V}$	V	0.506	0.502	0.503	0.507
I_{ON}	$V_{DS} = V_{GS} = 1.8\text{ V}$	$\mu\text{A} / \mu\text{m}$	11.32	11.34	11.14	11.32

Narrow and short channel device ($W/L = 0.35\mu\text{m} / 0.246\mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.1\text{ V}$	V	0.493	0.491	0.489	0.493
V_{TSAT}	$V_{DS} = 1.8\text{ V}$	V	0.425	0.431	0.428	0.425
I_{ON}	$V_{DS} = V_{GS} = 1.8\text{ V}$	$\mu\text{A} / \mu\text{m}$	411.5	406.6	403.9	411.5

I_{OFF}	$V_{DS} = 1.8 \text{ V}, V_{GS} = 0 \text{ V}$	$A/\mu\text{m}$	4.6E-12	4.9E-12	7.0E-12	4.6E-12*
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Capacitance

CJ	$V_J = 0 \text{ V}$	F/m^2	--	7.812E-04	7.812E-04	7.812E-04
CJSW	$V_J = 0 \text{ V}$	F/m	--	1.125E-10	1.125E-10	1.125E-10
CJSWG	$V_J = 0 \text{ V}$	F/m	--	2.250E-10	2.155E-10	2.155E-10
C_{OVL}	$V_{GS} = -0.5 \text{ V}$	F/m	--	2.65E-10	2.55E-10	2.55E-10

*gmindc=1e-15

3.3 MOSFET DC Model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 3-3 Geometry of bpw NMOS devices measured for DC parameter extraction

WL	10	1	0.8	0.6	0.507	0.435	0.417	0.399	0.318	0.3	0.282	0.264	0.246
10	X	X	X	X	X	X	X	X	X	X	X	X	X
1											X	X	X
0.8	X	X		X	X		X		X		X	X	X
0.6				X			X		X				X
0.43	X	X			X		X		X		X	X	X
0.35	X			X	X		X		X				X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of bpw NMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	0.1	1.8	1.7	V_{DS} (V)	0	2	0.025
V_{GS} (V)	-0.5	2	0.025	V_{GS} (V)	0.7	1.8	0.275
V_{BS} (V)	0	-1.8	-0.45	V_{BS} (V)	0	-1.8	-1.8

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

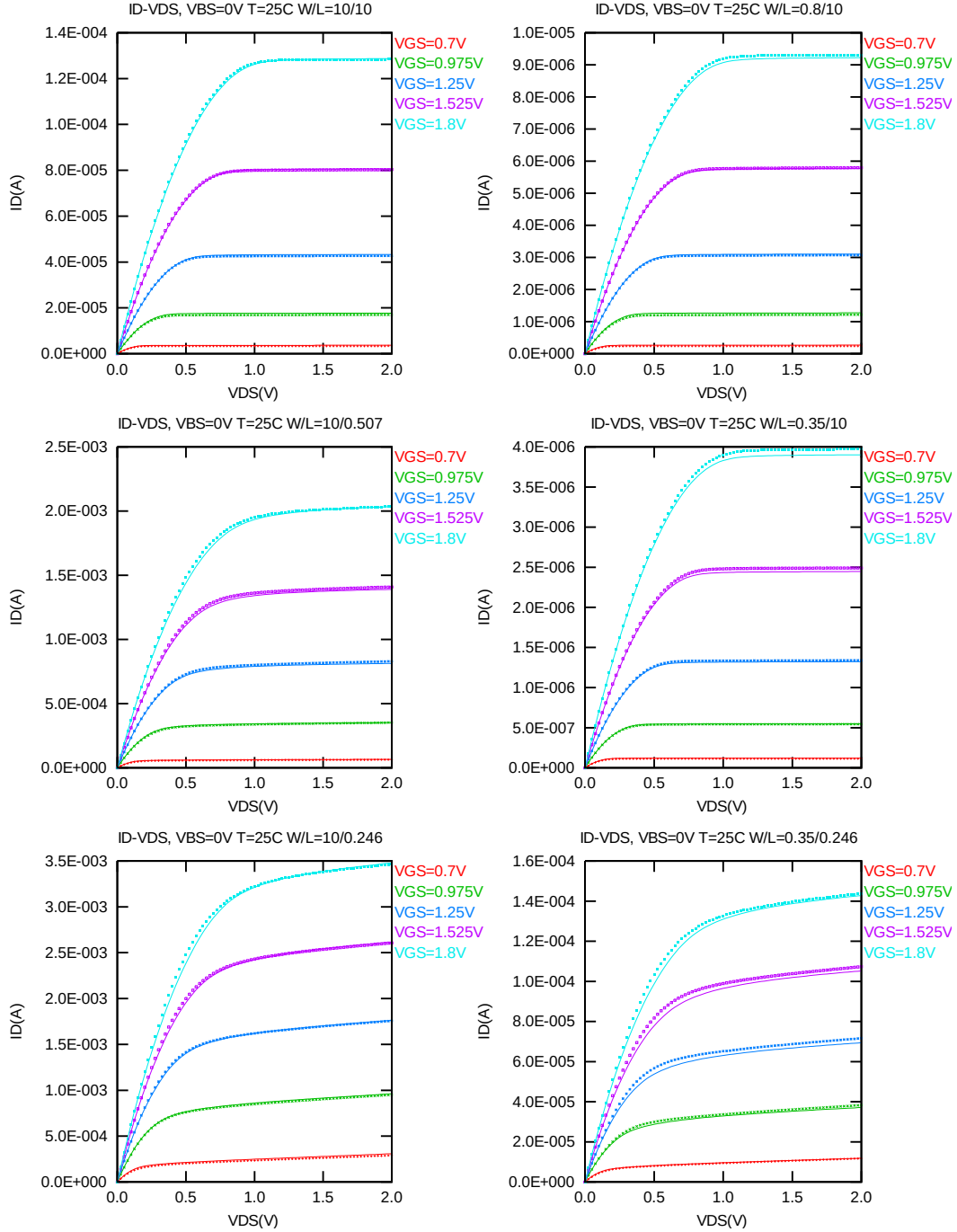


Figure 3-1 I_D vs. V_{DS} at $V_{BS} = 0\text{V}$ for bpw NMOS

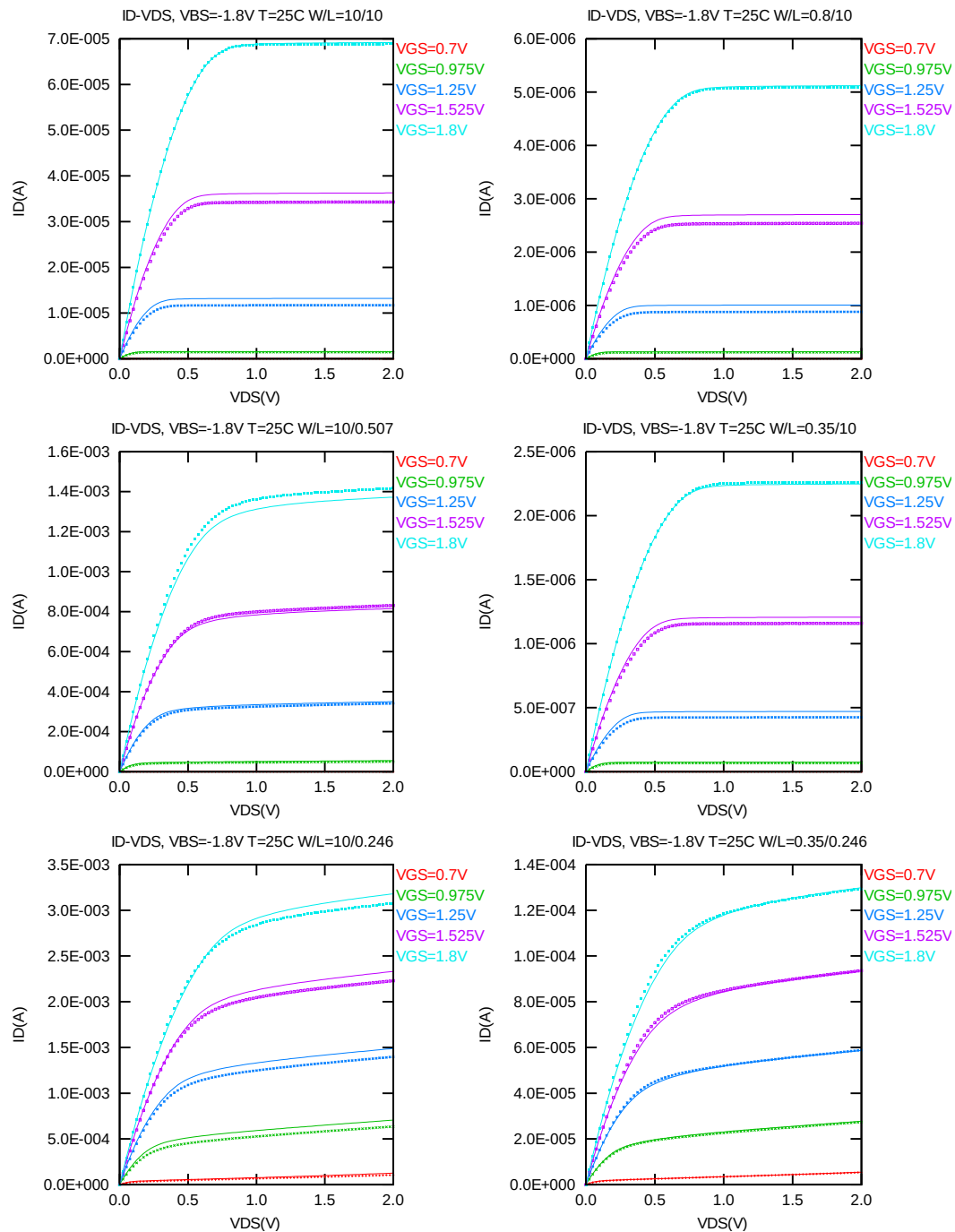


Figure 3-2 I_D vs. V_{DS} at $V_{BS} = -1.8V$ for bpw NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

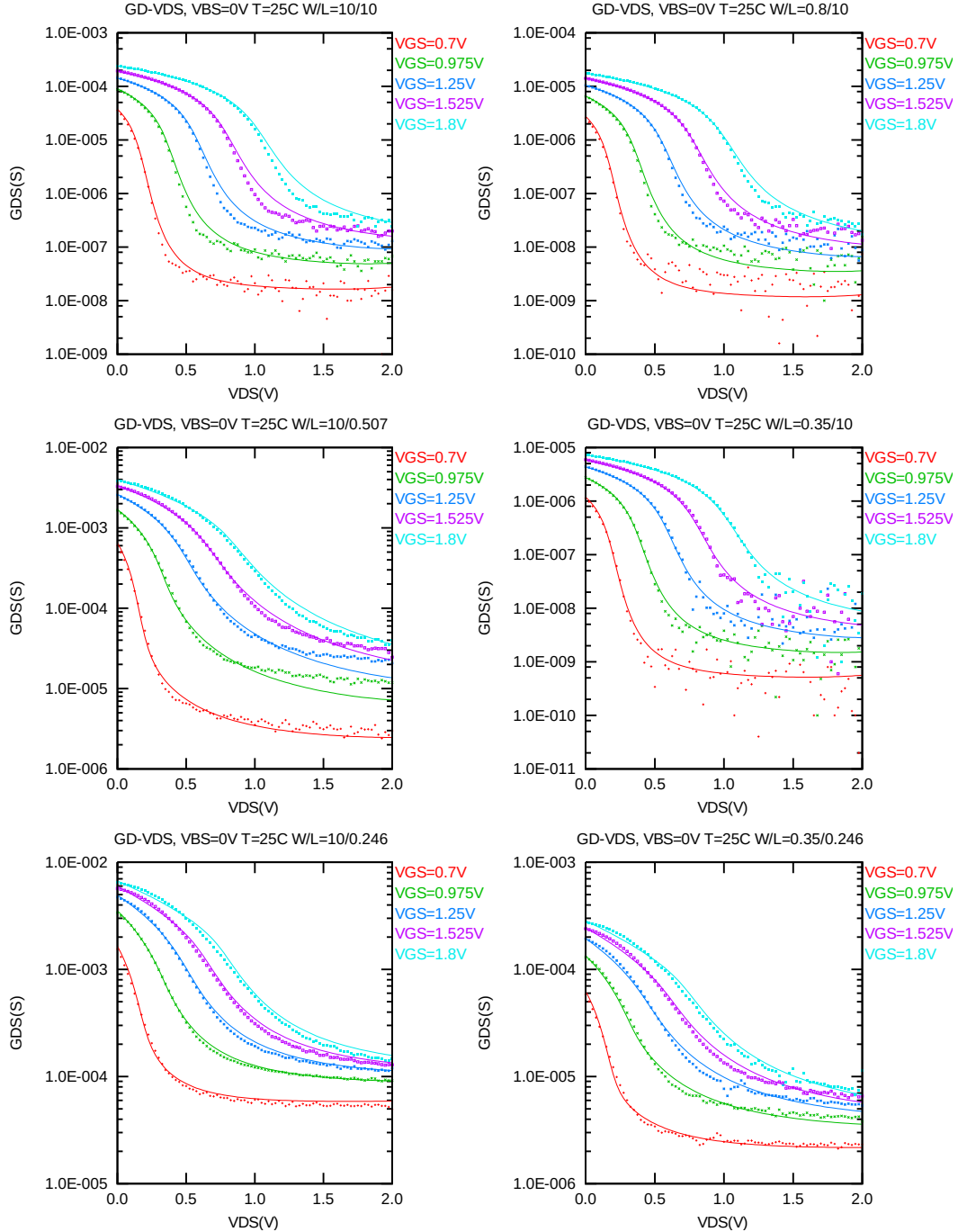


Figure 3-3 G_{DS} vs. V_{DS} at $V_{BS} = 0V$ for bpw NMOS

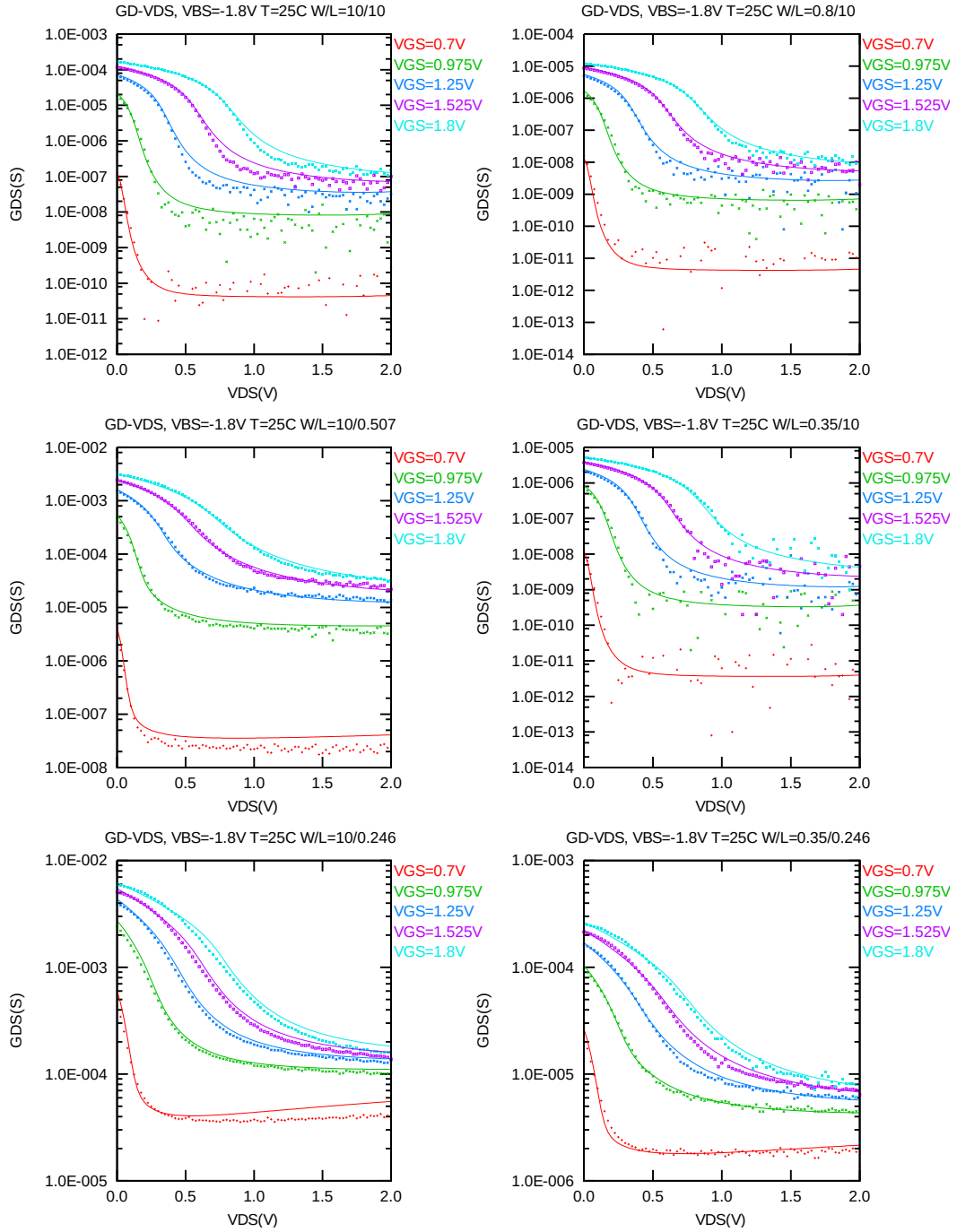


Figure 3-4 G_{DS} vs. V_{DS} at $V_{BS} = -1.8V$ for bpw NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

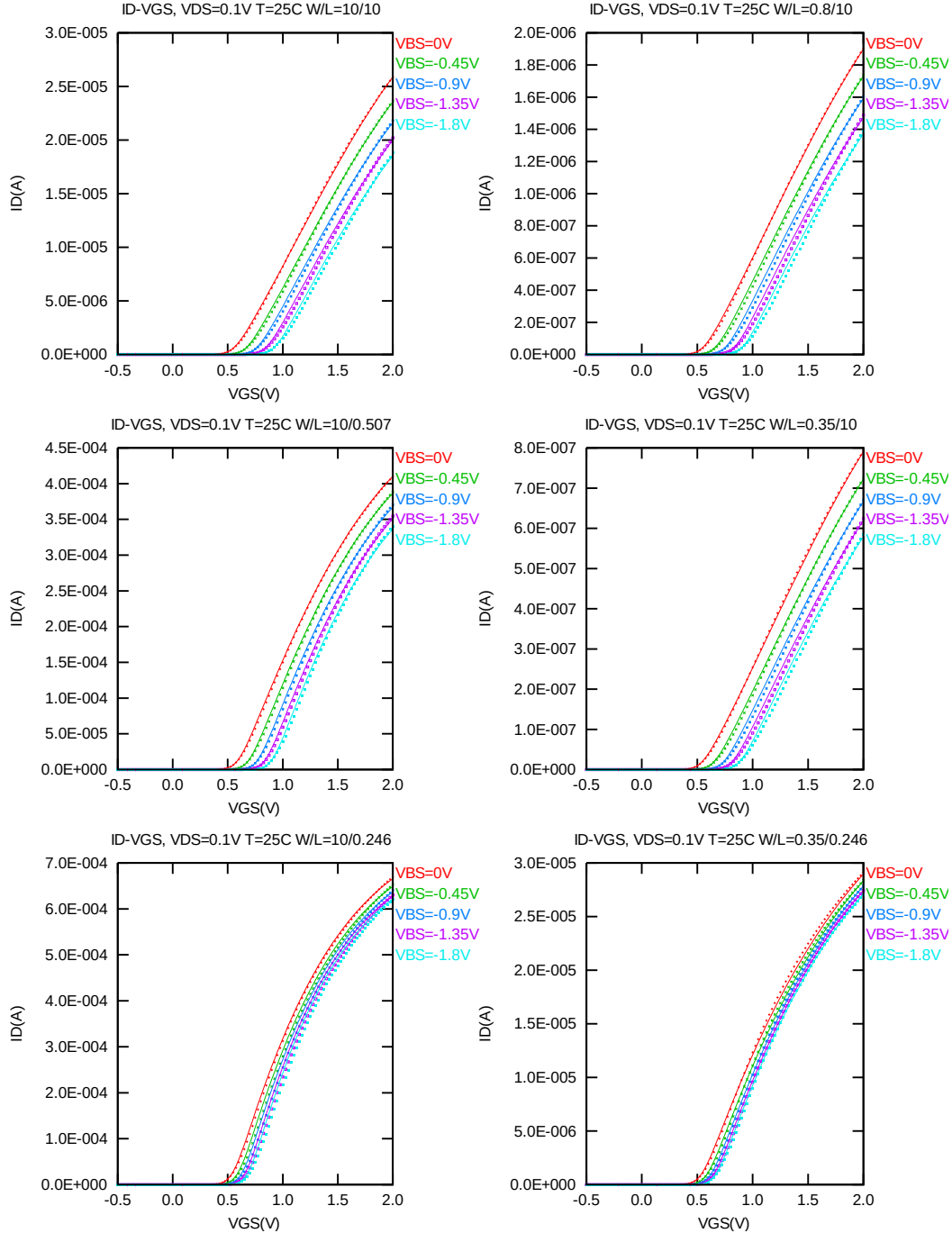


Figure 3-5 I_D vs. V_{GS} at $V_{DS} = 0.1$ V for bpw NMOS

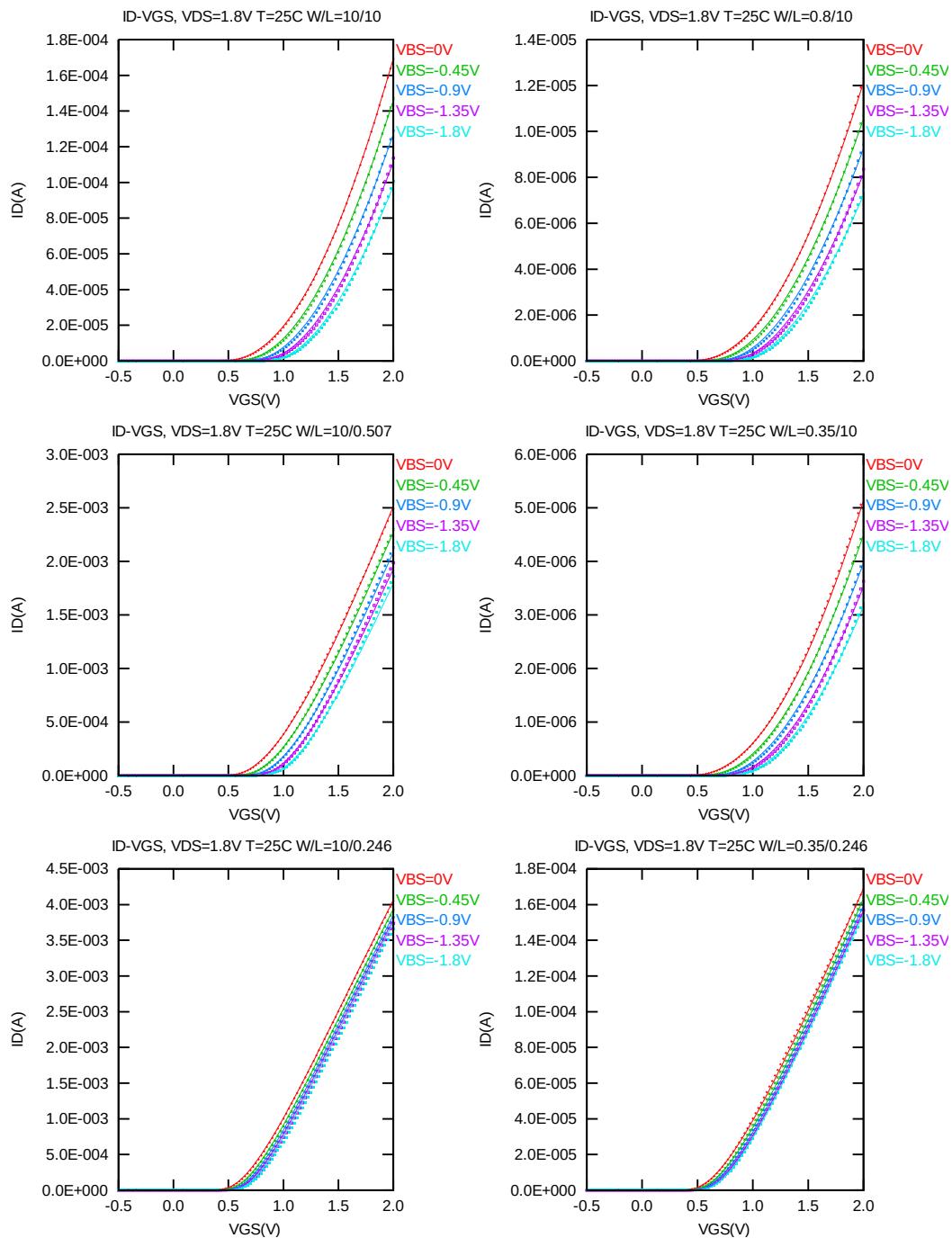


Figure 3-6 I_D vs. V_{GS} at $V_{DS} = 1.8V$ for bpw NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

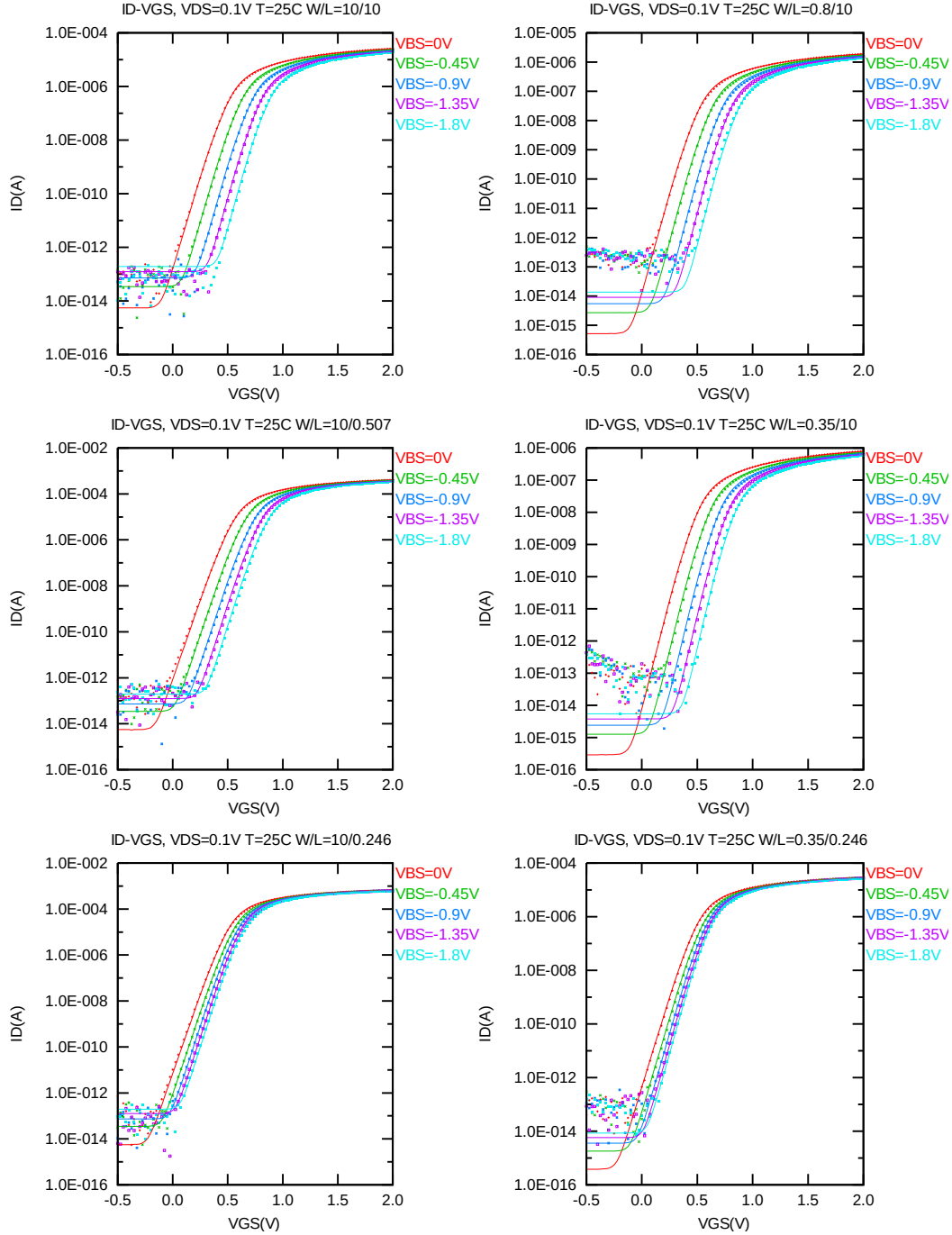


Figure 3-7 Log scaled I_D vs. V_{GS} at $V_{DS} = 0.1V$ for bpw NMOS

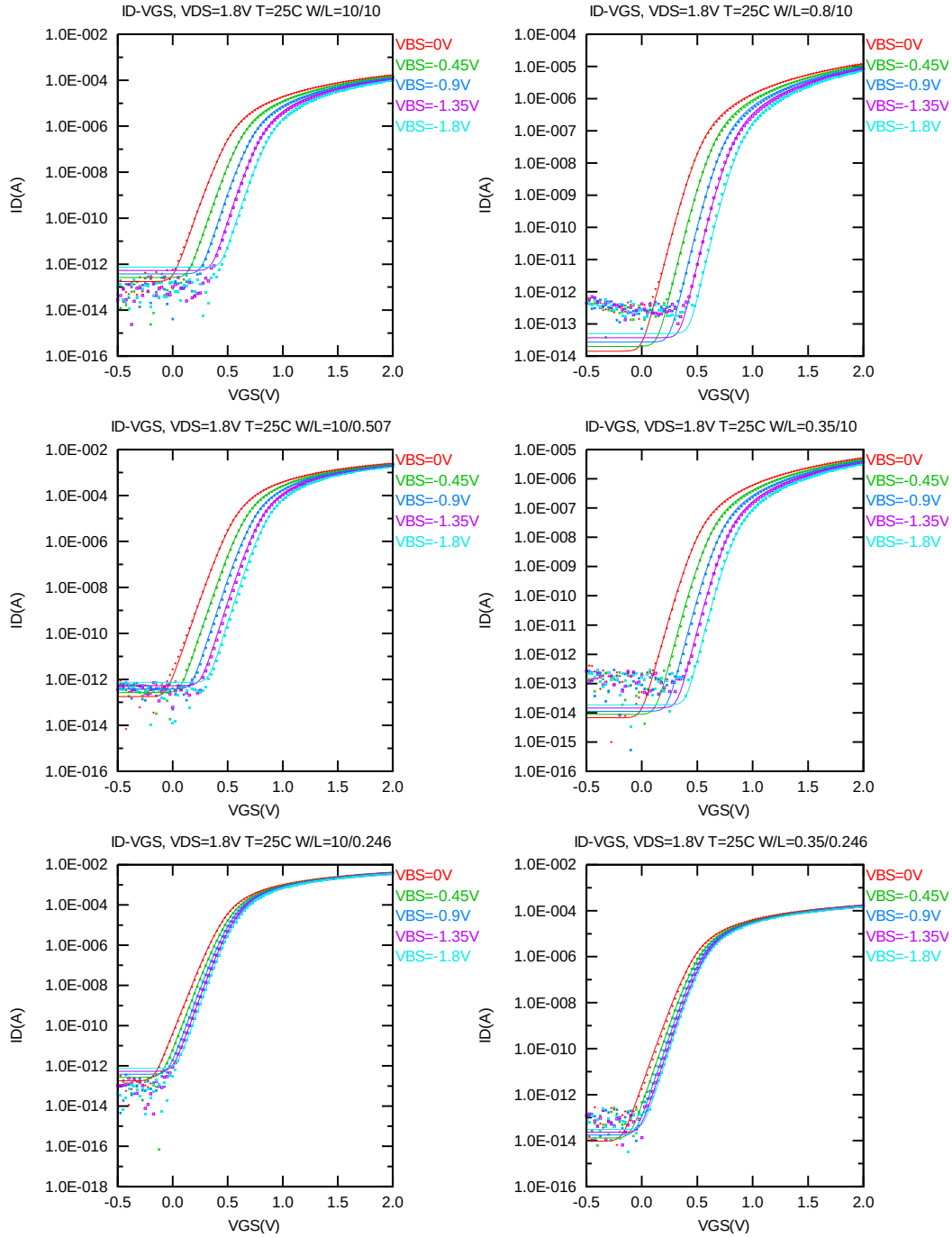


Figure 3-8 Log scaled I_D vs. V_{GS} at $V_{DS} = 1.8V$ for bpw NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

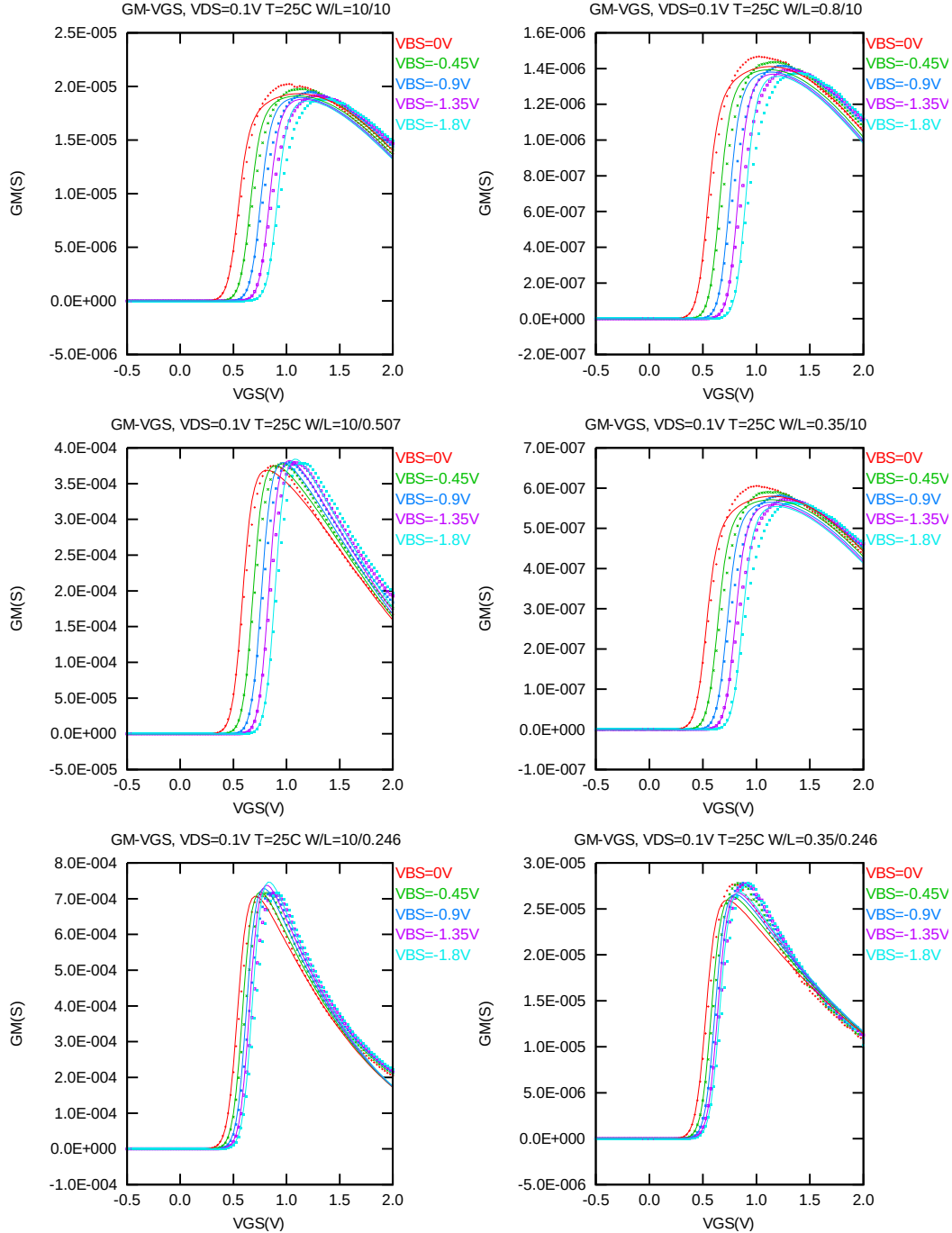


Figure 3-9 G_M vs. V_{GS} at $V_{DS} = 0.1V$ for bpw NMOS

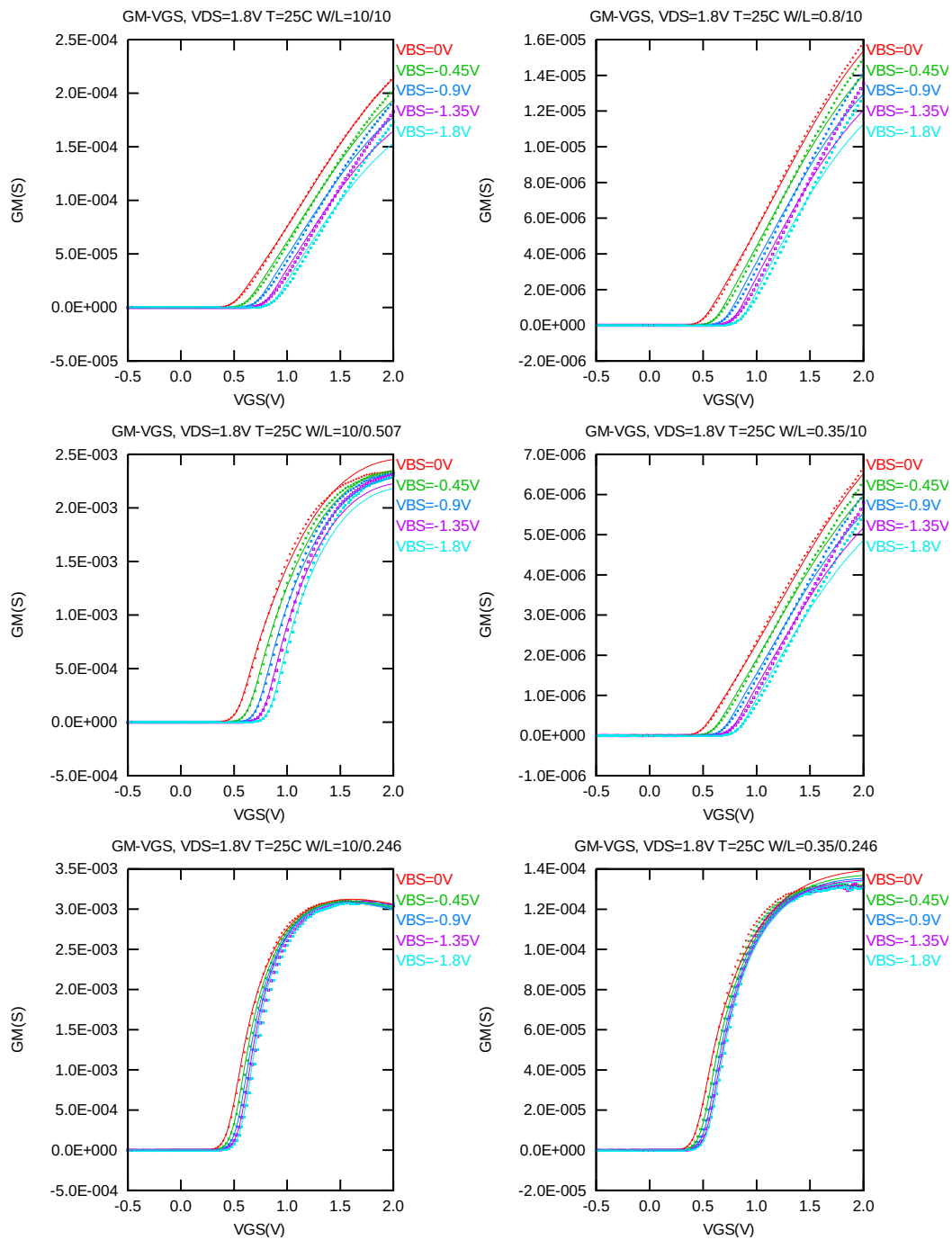


Figure 3-10 G_M vs. V_{GS} at $V_{DS} = 1.8V$ for bpw NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu\text{m})$ at 25°C . The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

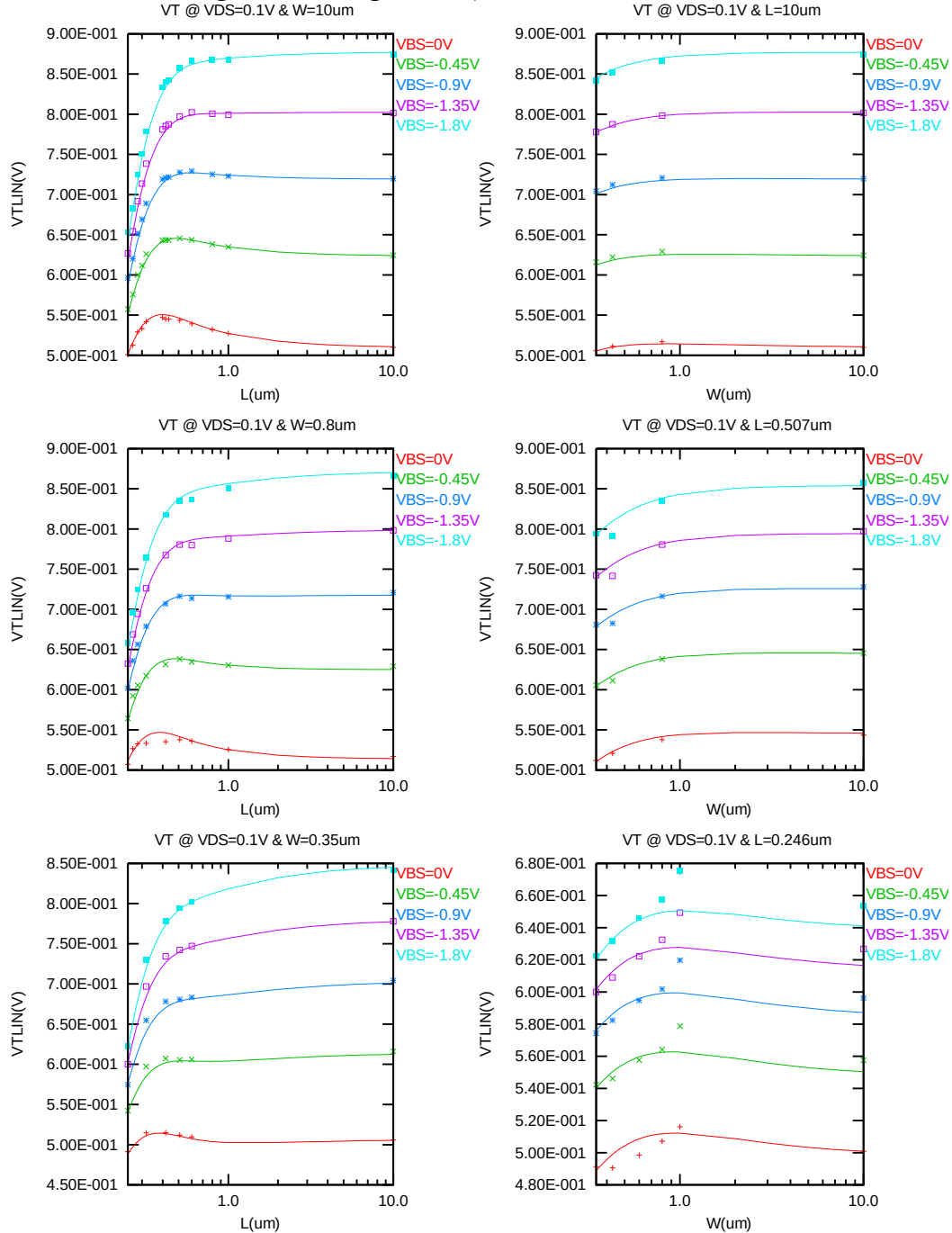


Figure 3-11 V_{TLIN} vs. channel length & channel width for low drain bias ($V_{DS} = 0.1\text{V}$) for bpw NMOS

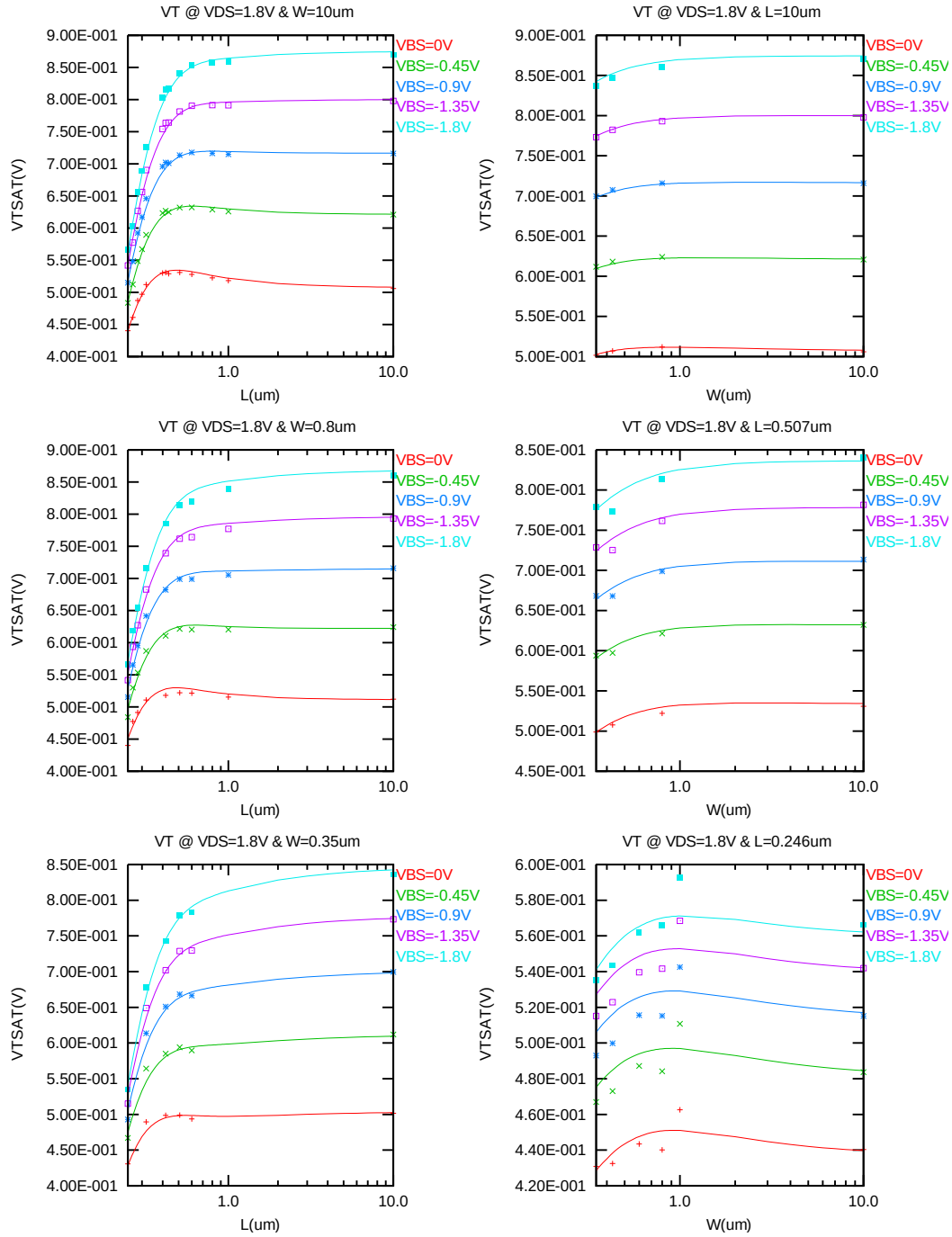


Figure 3-12 V_T vs. channel length & channel width for high drain bias ($V_{DS} = 1.8V$) for bpw NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.8V$ for $V_{BS} = 0V$ to $-1.8V$ at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

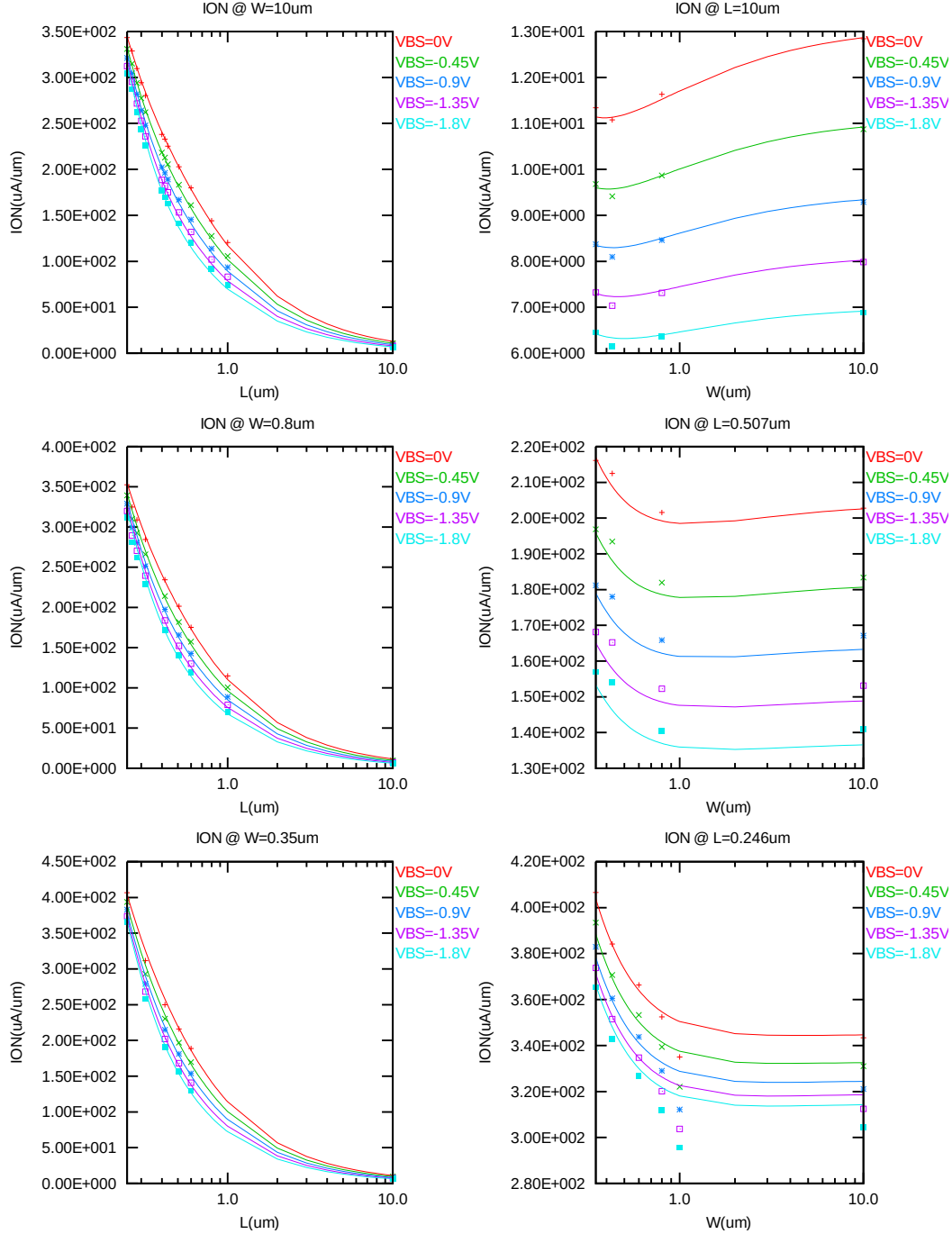


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for bpw NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0V$ to $-1.8V$ at 25 C. A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

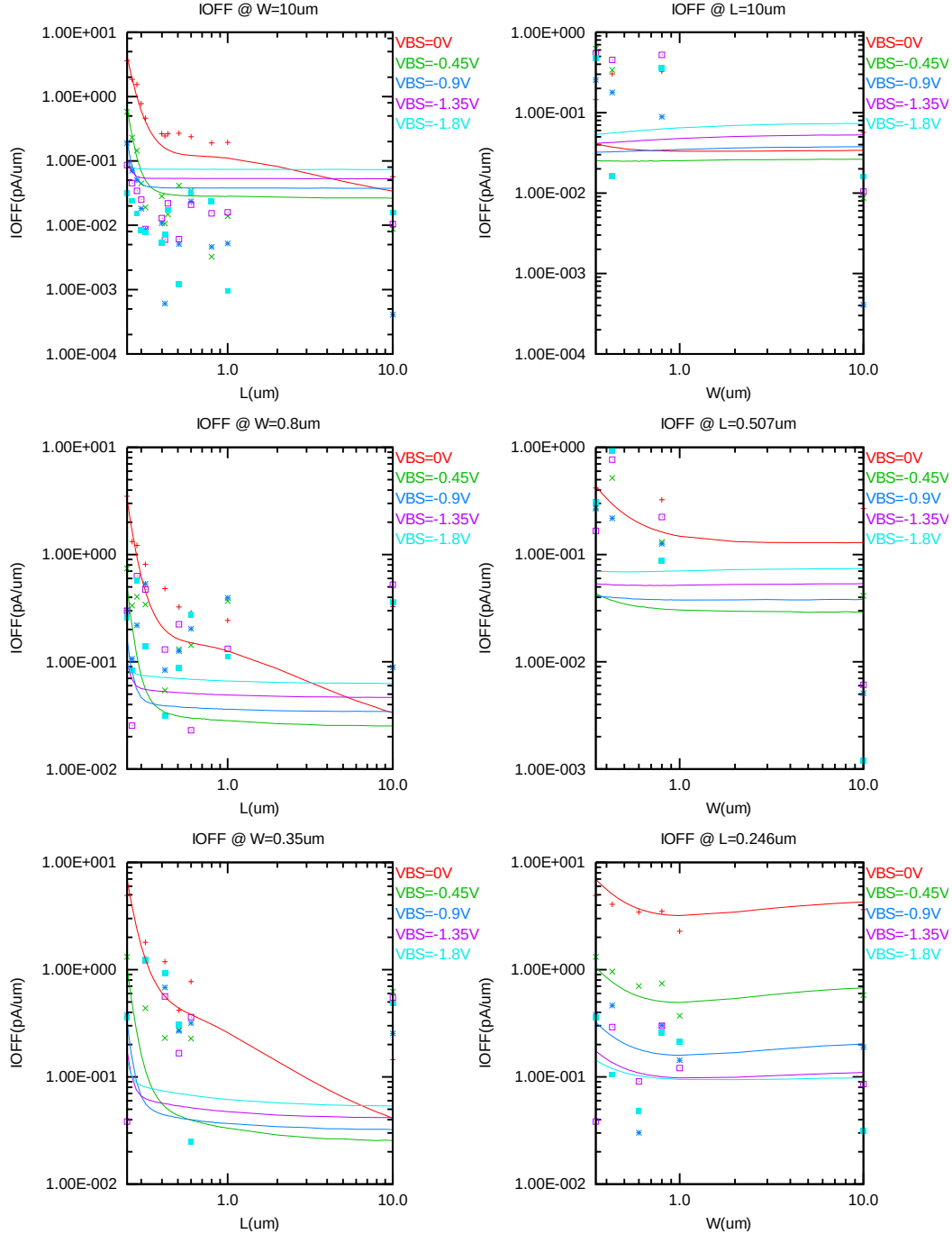


Figure 3-14 I_{OFF} vs. channel length and channel width for bpw NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu m)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

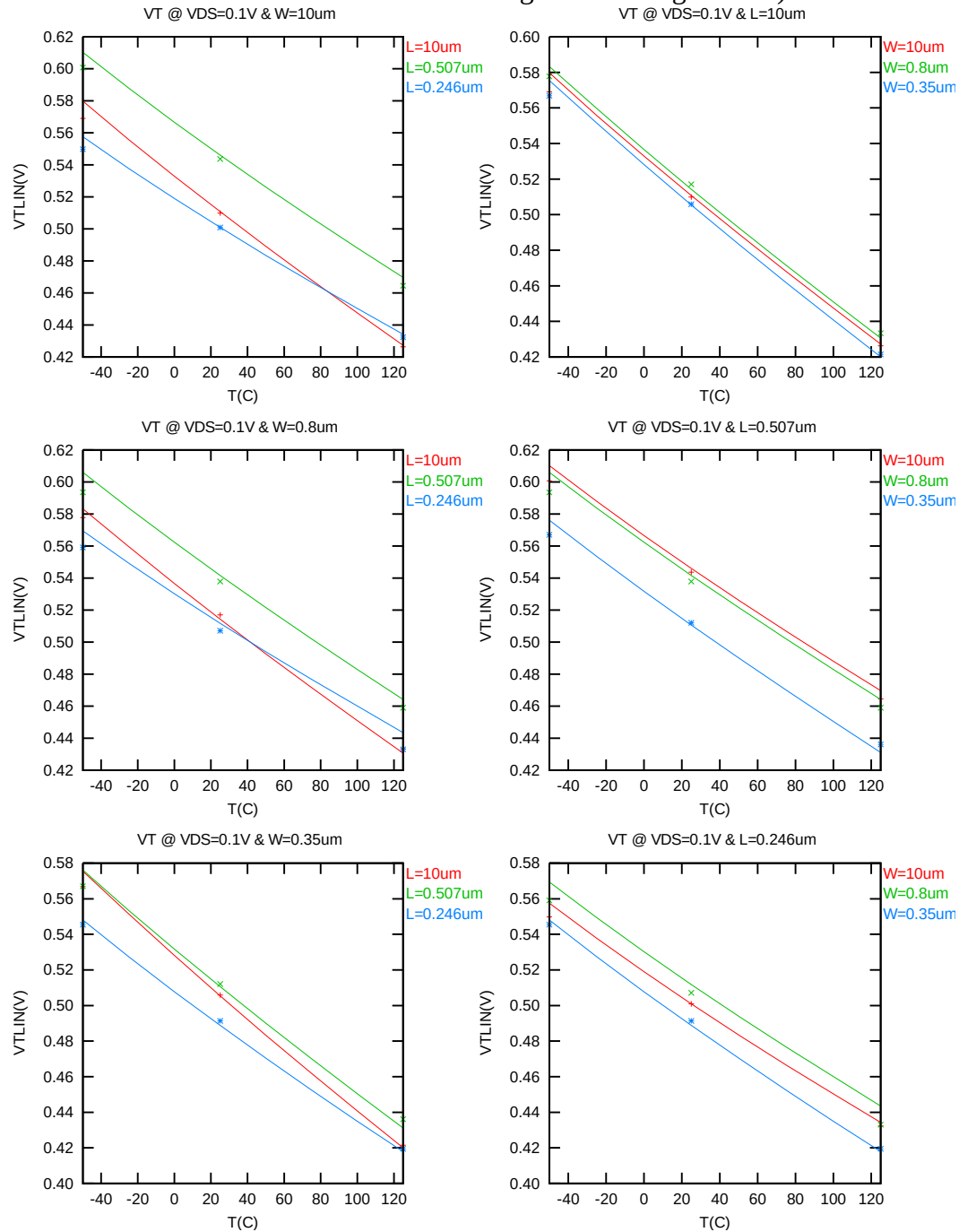


Figure 3-15 V_{TLIN} vs. temperature for bpw NMOS

V_{TSAT} vs. temperature

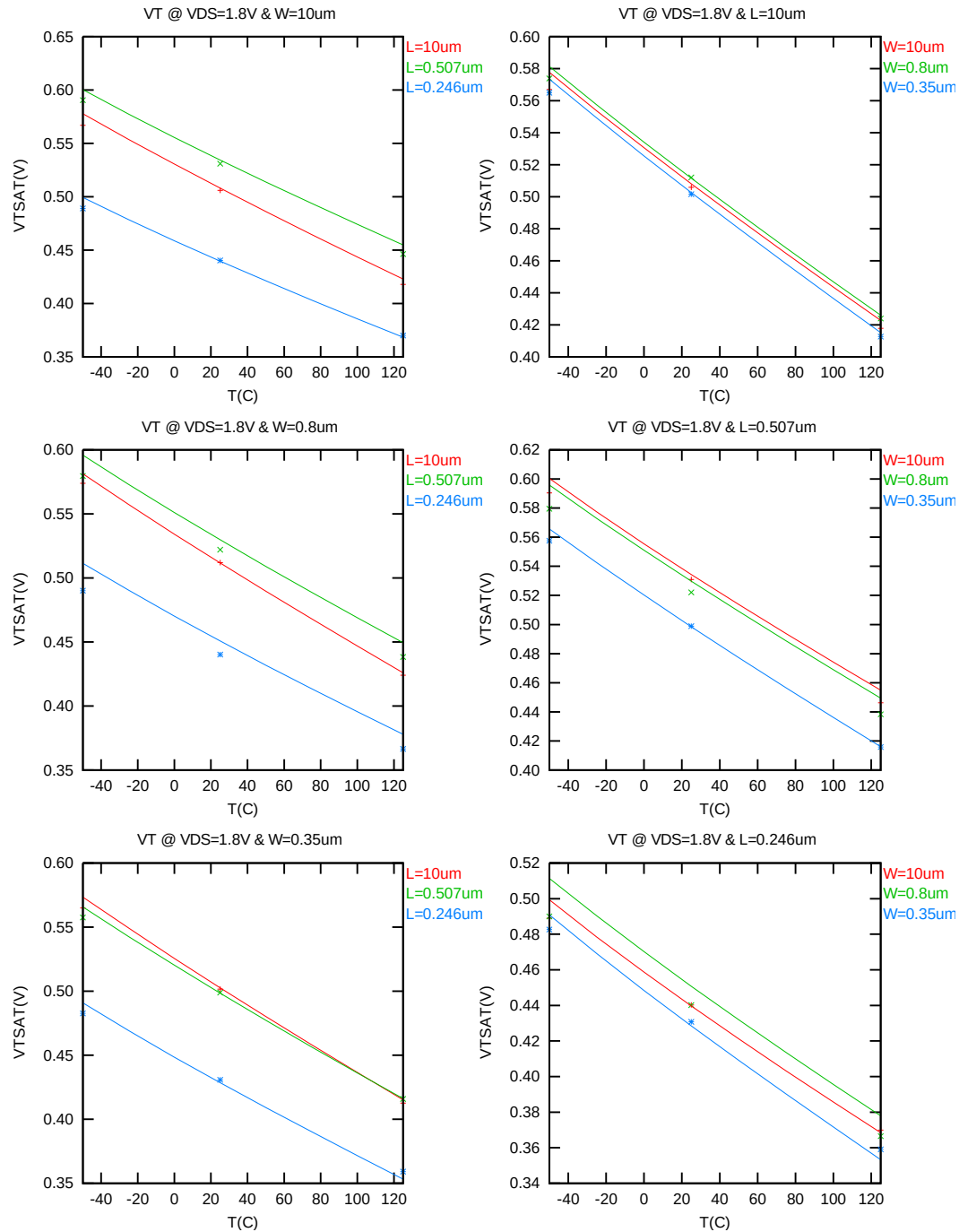


Figure 3-16 V_{TSAT} vs. temperature for bpw NMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.8V$, $V_{BS} = 0V$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

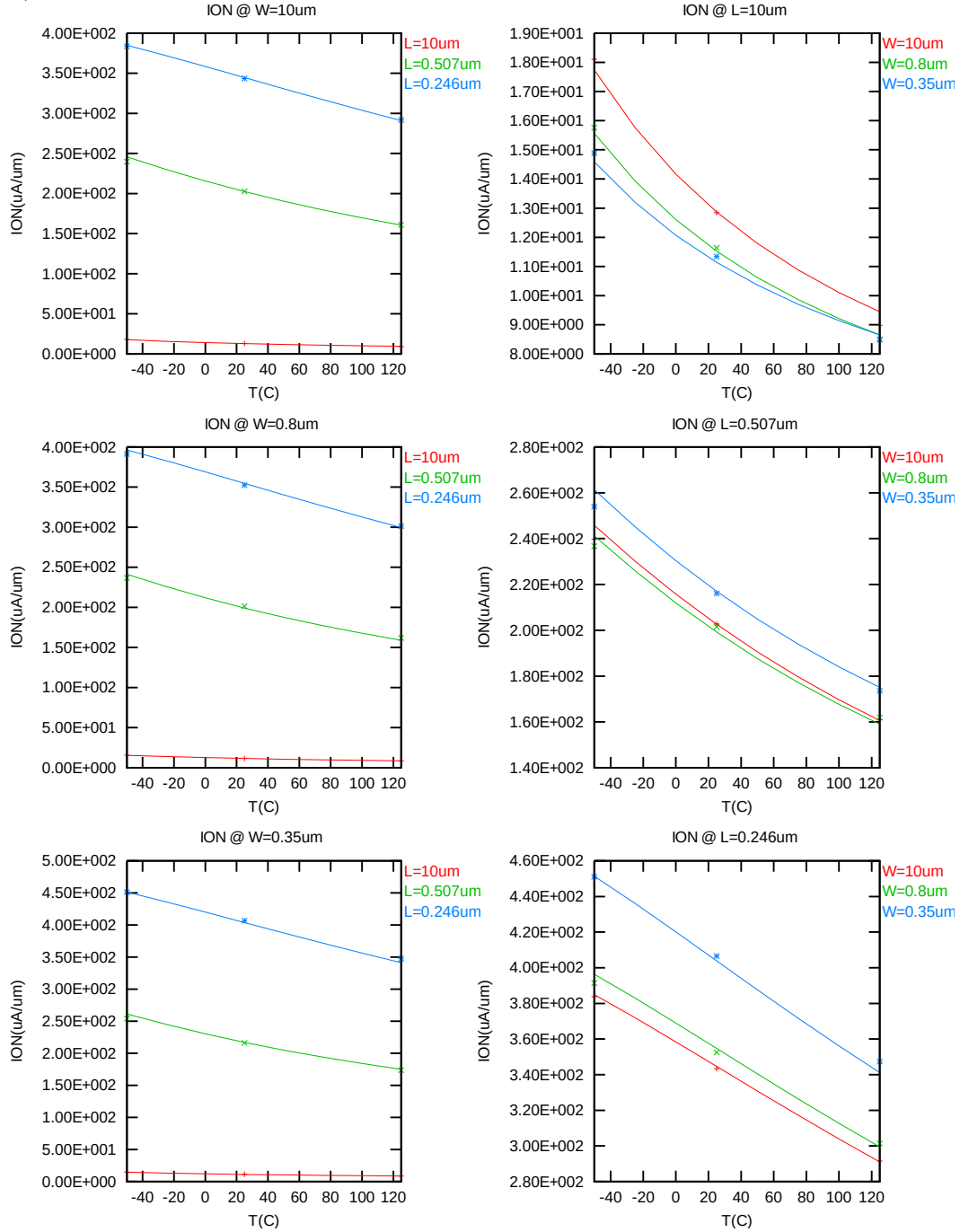


Figure 3-17 Saturation current vs. temperature for bpw NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{DS} = 1.8V$, $V_{GS} = V_{BS} = 0V$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

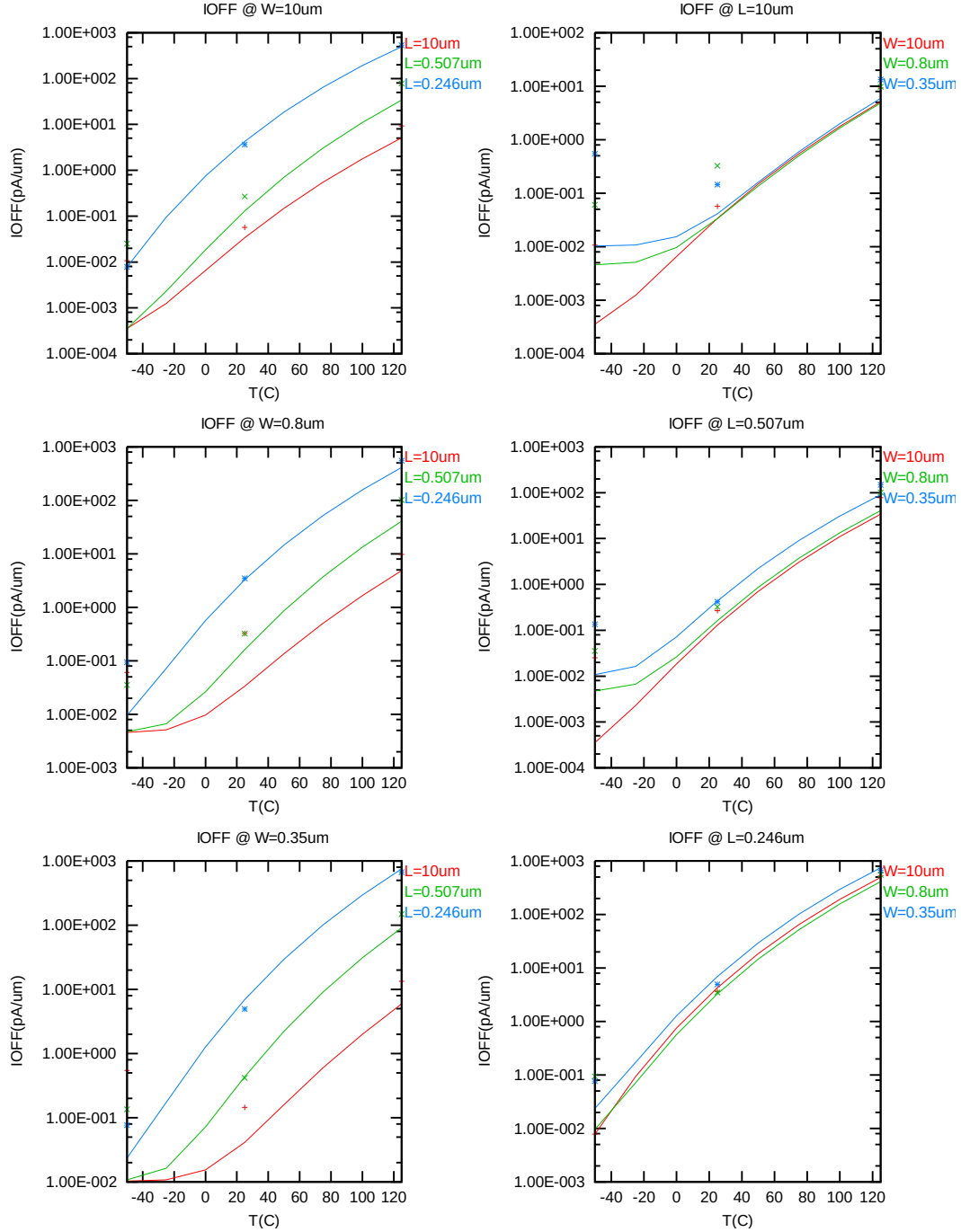


Figure 3-18 I_{OFF} vs. temperature for bpw NMOS

3.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- f. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- g. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- h. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- i. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- j. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 3-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

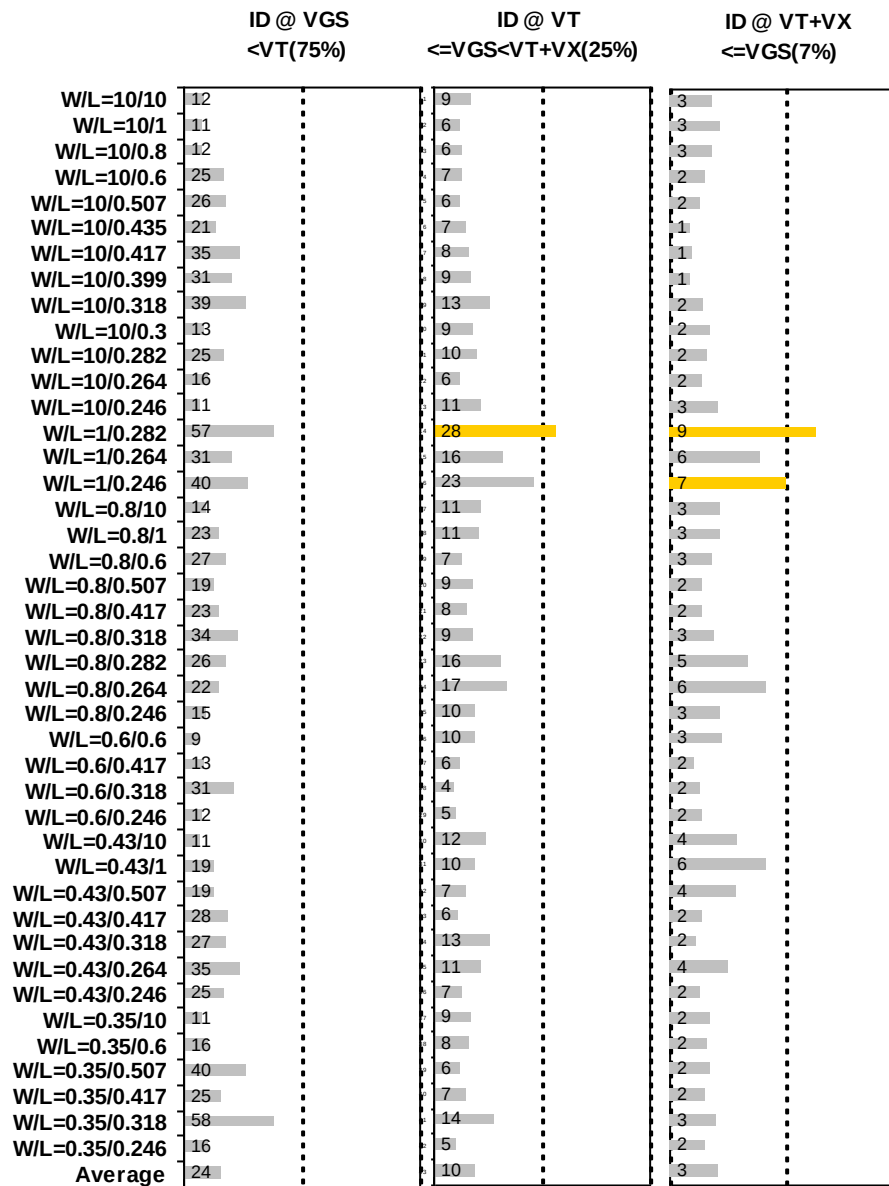


Figure 3-19 I_D accuracy at 25 C for bpw NMOS

G_M and G_{DS} accuracy

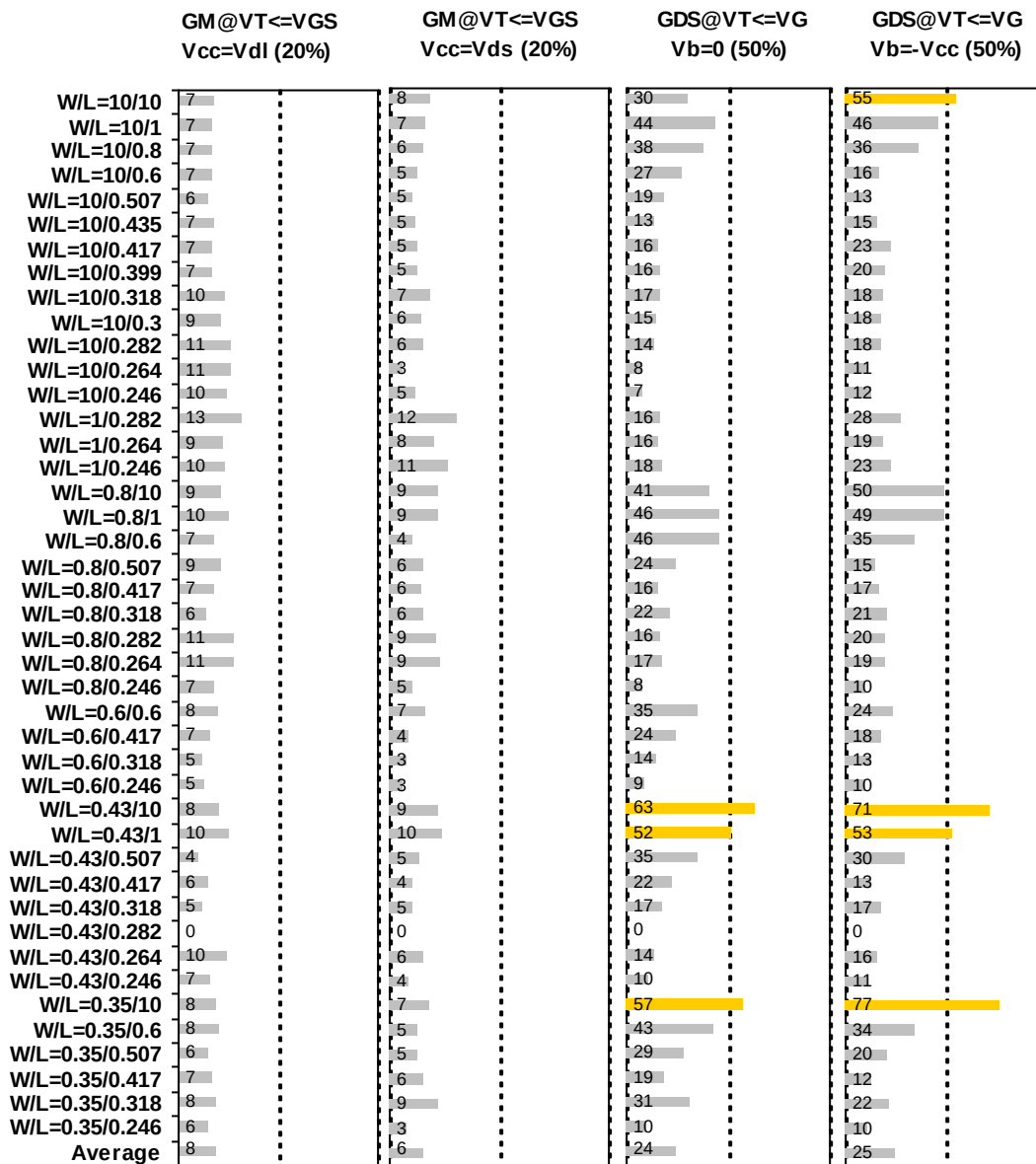


Figure 3-20 G_M and G_{DS} accuracy at 25 C for bpw NMOS

4 1.8V P-ch MOSFET

4.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 4-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0QA	BSW0QA	BSW0QA	BSW0QA
Lot ID	D5FHR.11	D5FHR.11	D5FHR.11	D5FHR.11
Wafer Number	11	11	11	11

4.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
 $I_D = 300\text{nA} * W_{DES} * 0.9 / (L_{DES} * 0.9 + 12\text{nm})$ for NMOS and
 $I_D = -70\text{nA} * W_{DES} * 0.9 / (L_{DES} * 0.9 + 12\text{nm})$ for PMOS, respectively.
The dimension is after 90% shrink and 12nm sizing channel length back.

All parameters are given for $T=25\text{ C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Table 4-2 Electrical parameters for PMOS

All parameters are given for T=25 °C , $V_{BS}=0$ V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W/L= 10 μ m / 10 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.498	-0.496	-0.498	-0.498
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	μ A/ μ m	-2.89	-2.88	-2.89	-2.89

Wide and short channel device (W/L= 10 μ m / 0.246 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.438	-0.438	-0.438	-0.438
V_{TSAT}	$V_{DS} = -1.8$ V	V	-0.410	-0.409	-0.410	-0.410
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	μ A/ μ m	-134.9	-135.28	-134.90	-134.90
I_{OFF}	$V_{DS} = -1.8$ V, $V_{GS}=0$ V	A/ μ m	--	7.2E-13	-6.2E-12	-6.2E-12*

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.494	-0.488	-0.491	-0.494
V_{TSAT}	$V_{DS} = -1.8$ V	V	-0.492	-0.484	-0.490	-0.493
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	μ A/ μ m	-2.81	-2.80	-2.82	-2.81

Narrow and short channel device (W/L= 0.35 μ m / 0.246 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.440	-0.439	-0.438	-0.440
V_{TSAT}	$V_{DS} = -1.8$ V	V	-0.415	-0.414	-0.414	-0.414
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	μ A/ μ m	-123.06	-123.24	-121.00	-123.05
I_{OFF}	$V_{DS} = -1.8$ V, $V_{GS}=0$ V	A/ μ m	--	-3.4E-11	-5.8E-12	-5.7E-12*

Capacitance

CJ	$V_J = 0$ V	F/m ²	--	9.63E-04	9.63E-04	9.63E-04
CJSW	$V_J = 0$ V	F/m	--	1.05E-10	1.05E-10	1.05E-10
CJSWG	$V_J = 0$ V	F/m	--	3.37E-10	3.37E-10	3.37E-10
C _{OV} L	$V_{GS} = 0.5$ V	F/m	--	3.21E-10	2.93E-10	2.93E-10

*gmindc=1e-15

4.3 MOSFET DC Model

4.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 4-3 Geometry of PMOS devices measured for DC parameter extraction

WL	10	1	0.8	0.6	0.507	0.435	0.417	0.399	0.318	0.3	0.282	0.264	0.246
10	X	X	X	X	X	X	X	X	X	X	X	X	X
1											X	X	X
0.8	X	X		X	X		X		X		X	X	X
0.6				X			X		X				X
0.43	X	X			X		X		X		X	X	X
0.35	X			X	X		X		X				X

4.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 4-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.1	-1.8	-1.7
V_{GS} (V)	0.5	-2	-0.025
V_{BS} (V)	0	1.8	0.45

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-2	-0.025
V_{GS} (V)	-0.6	-1.8	-0.3
V_{BS} (V)	0	1.8	1.8

4.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

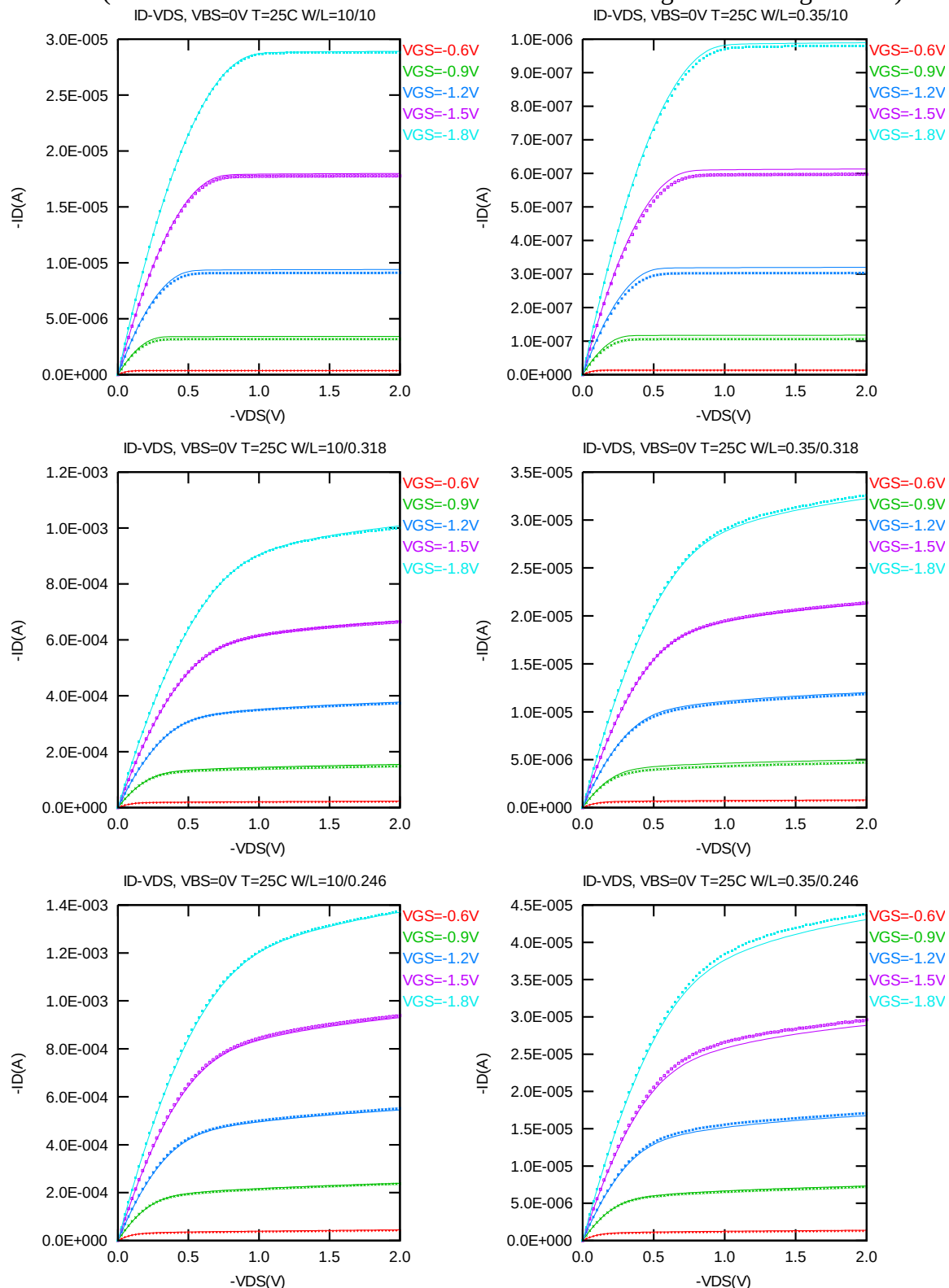


Figure 4-1 I_D vs. V_{DS} at $V_{BS} = 0V$ for PMOS

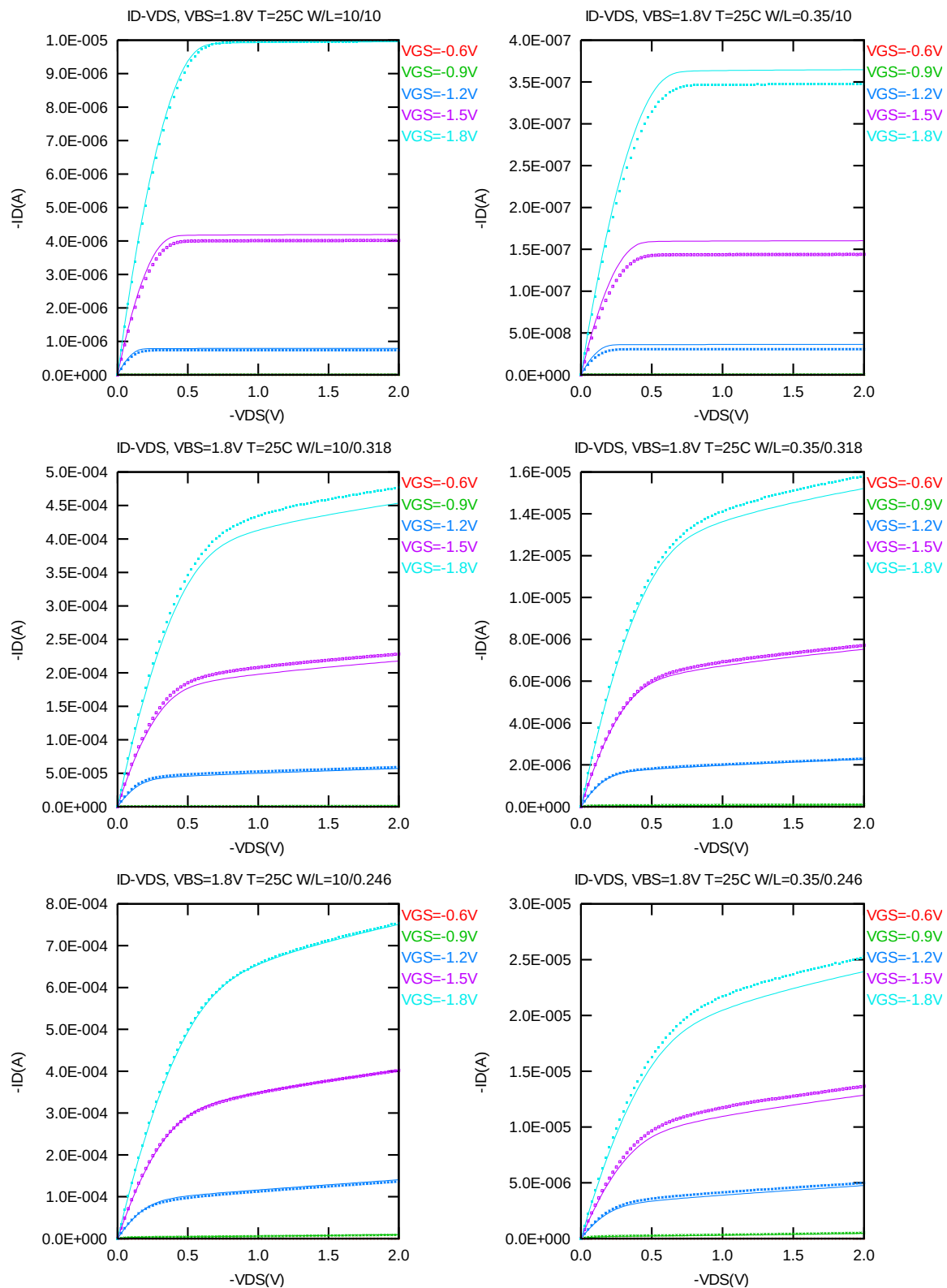


Figure 4-2 I_D vs. V_{DS} at $V_{BS} = 1.8V$ for PMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

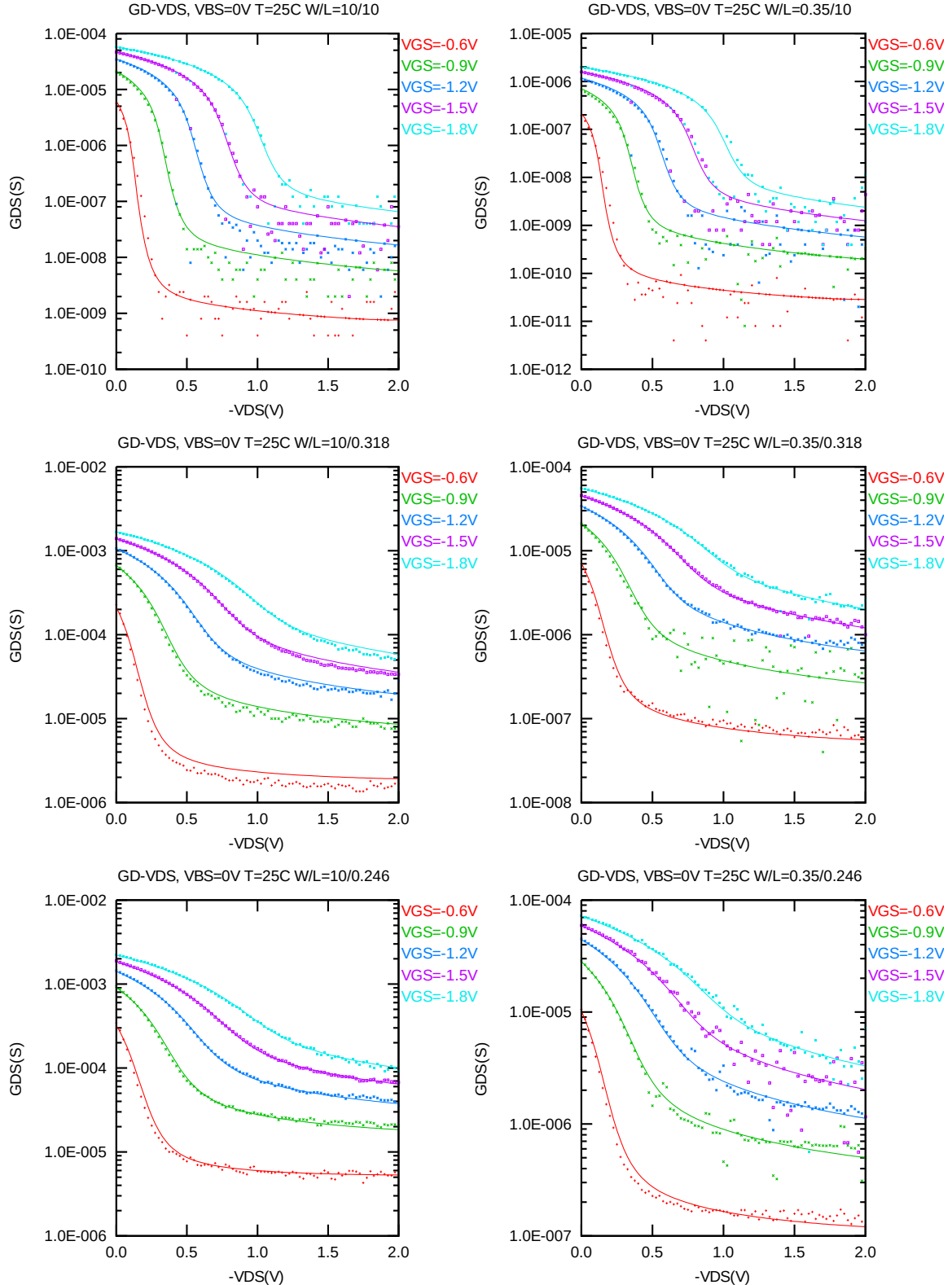
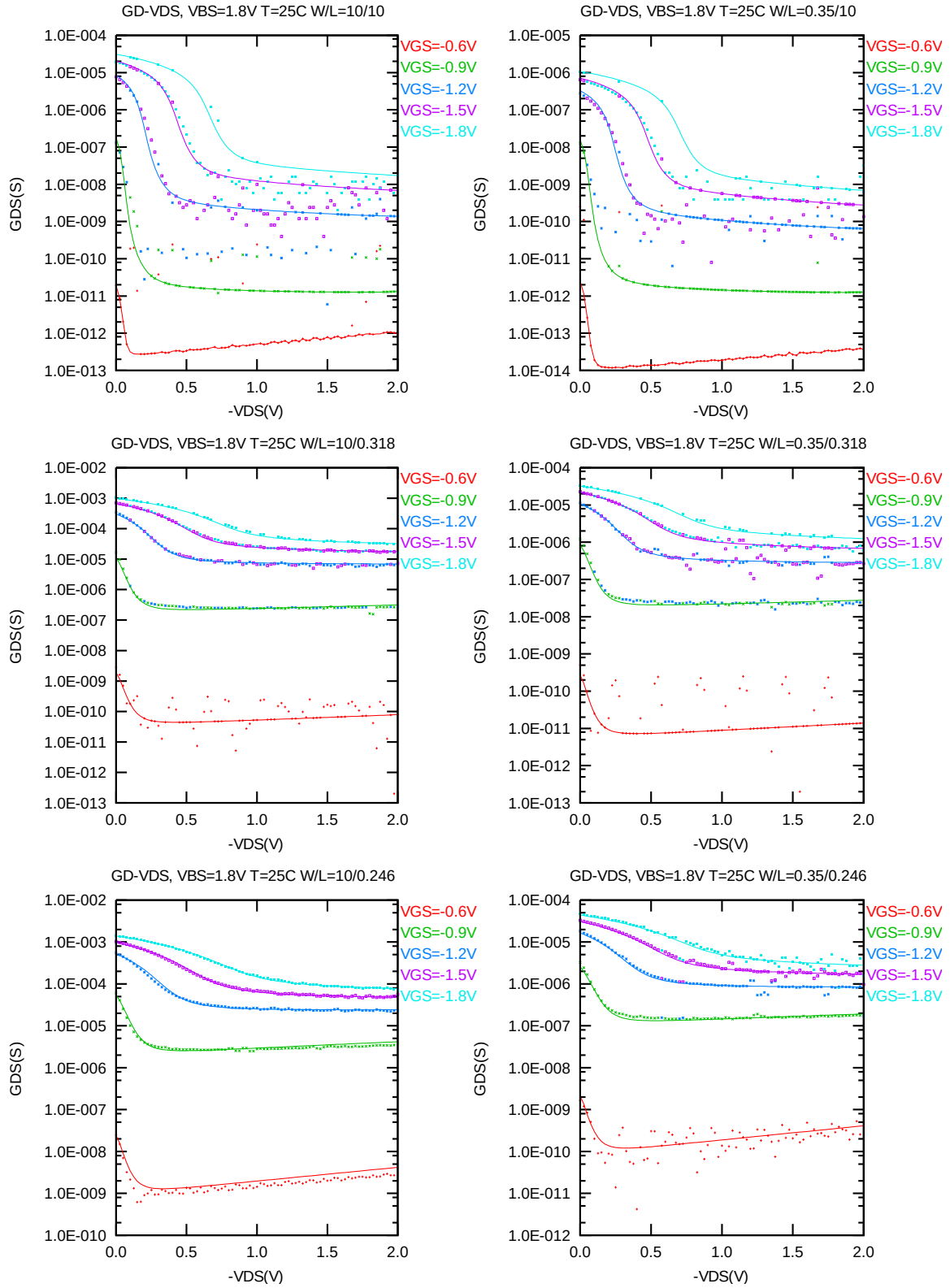


Figure 4-3 G_{DS} vs. V_{DS} at $V_{BS} = 0$ V for PMOS



I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

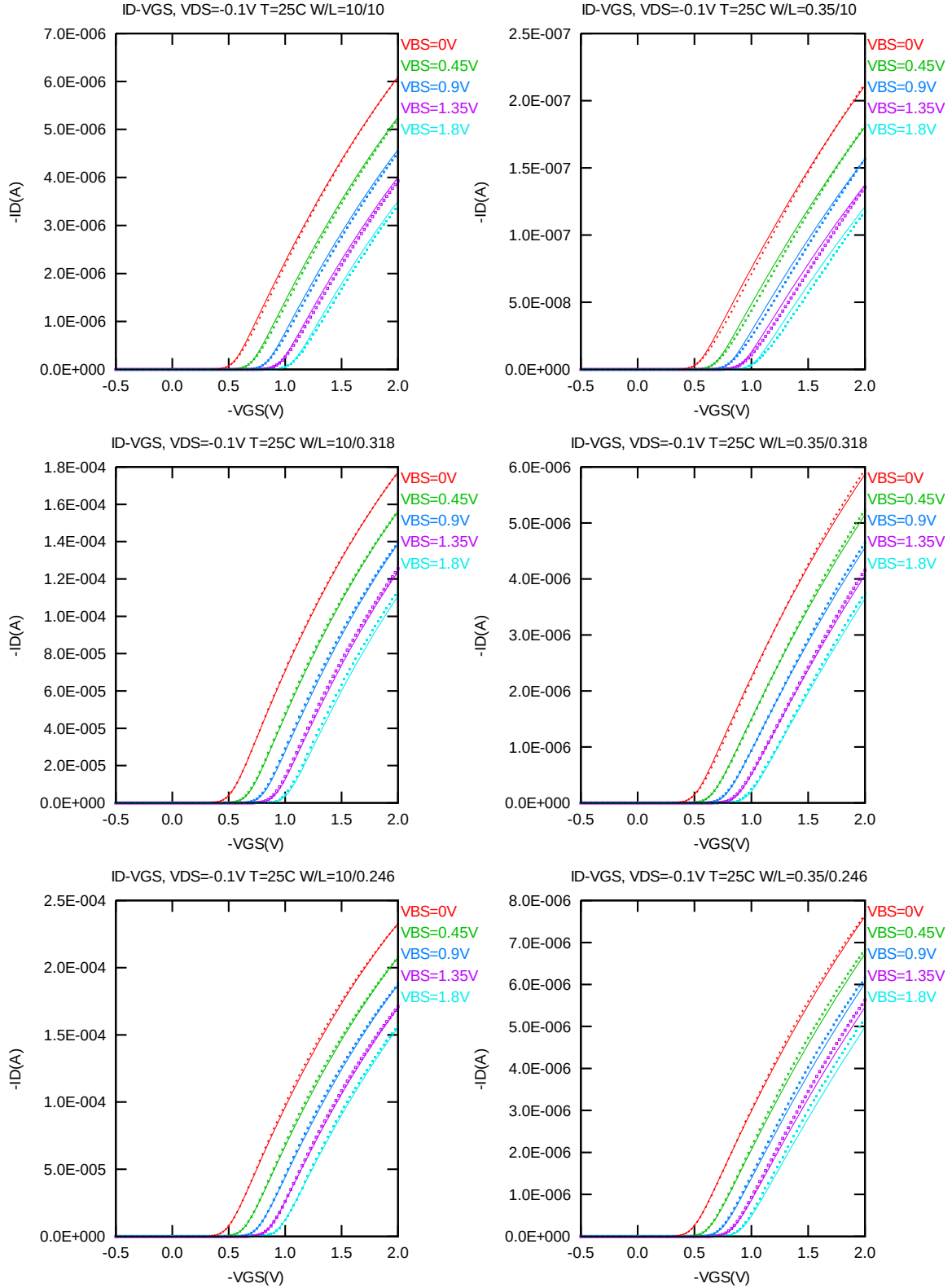


Figure 4-5 I_D vs. V_{GS} at $V_{DS} = -0.1V$ for PMOS

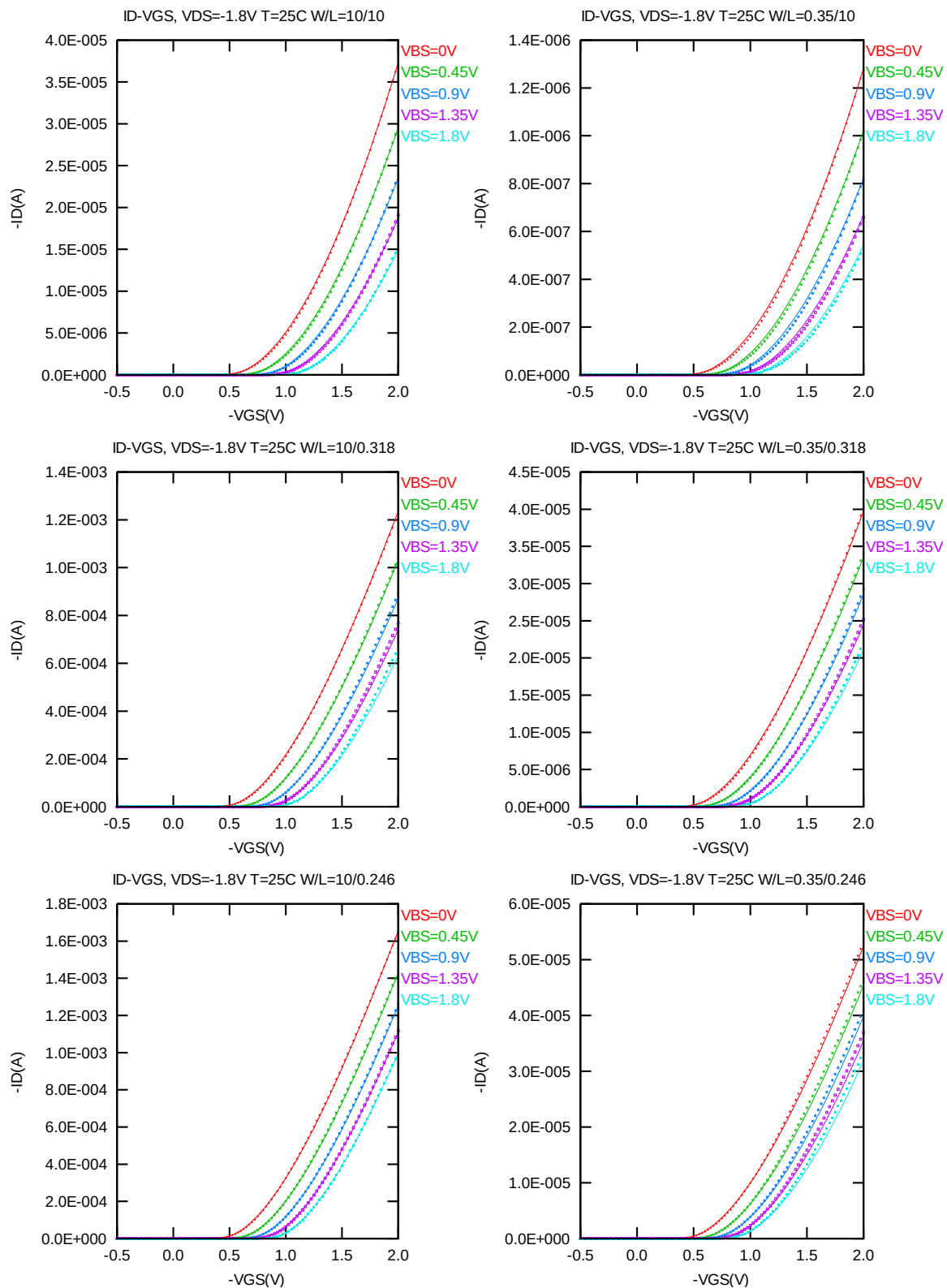


Figure 4-6 I_D vs. V_{GS} at $V_{DS} = -1.8V$ for PMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

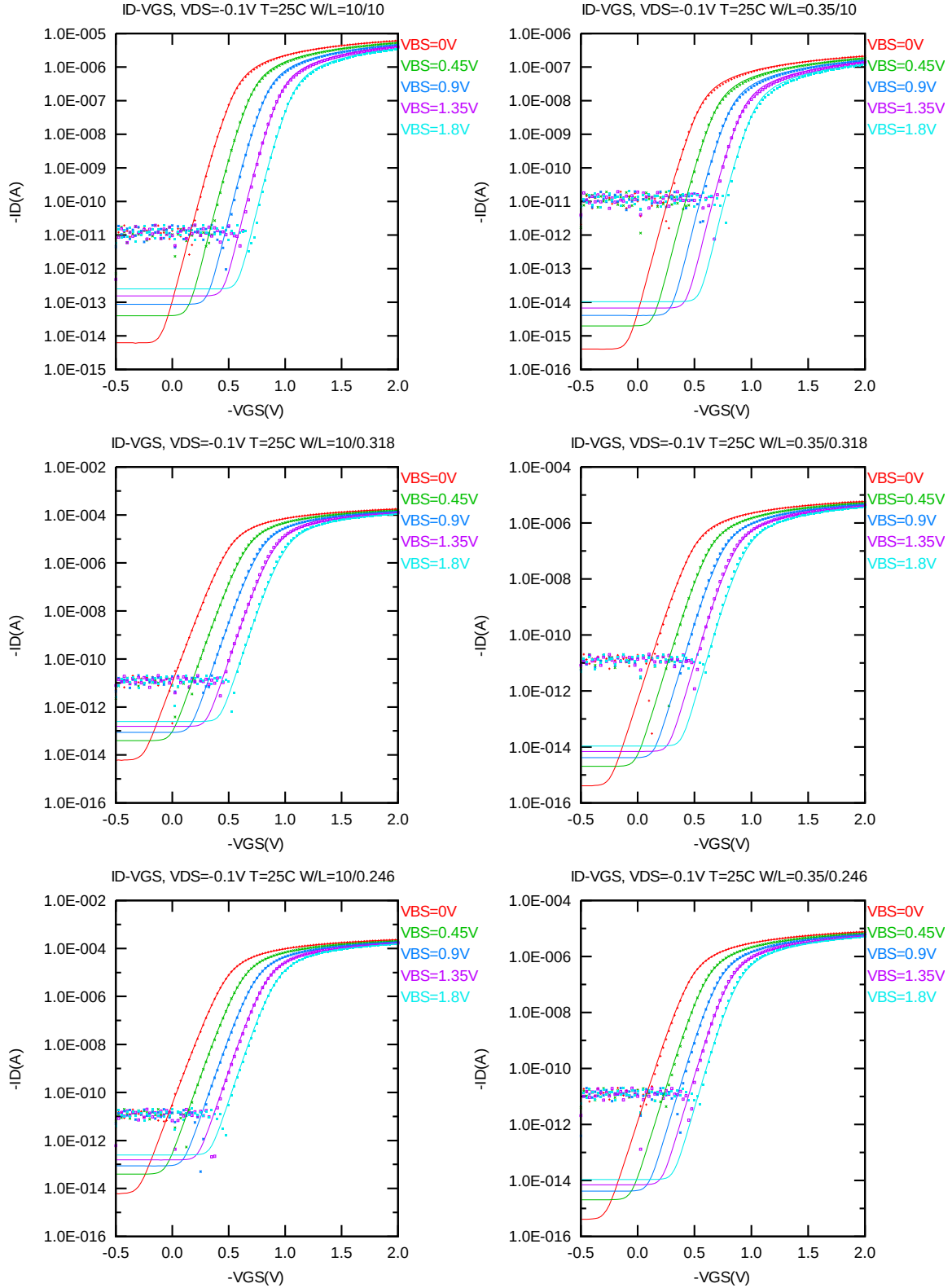


Figure 4-7 Log scaled I_D vs. V_{GS} at $V_{DS} = -0.1$ V for PMOS

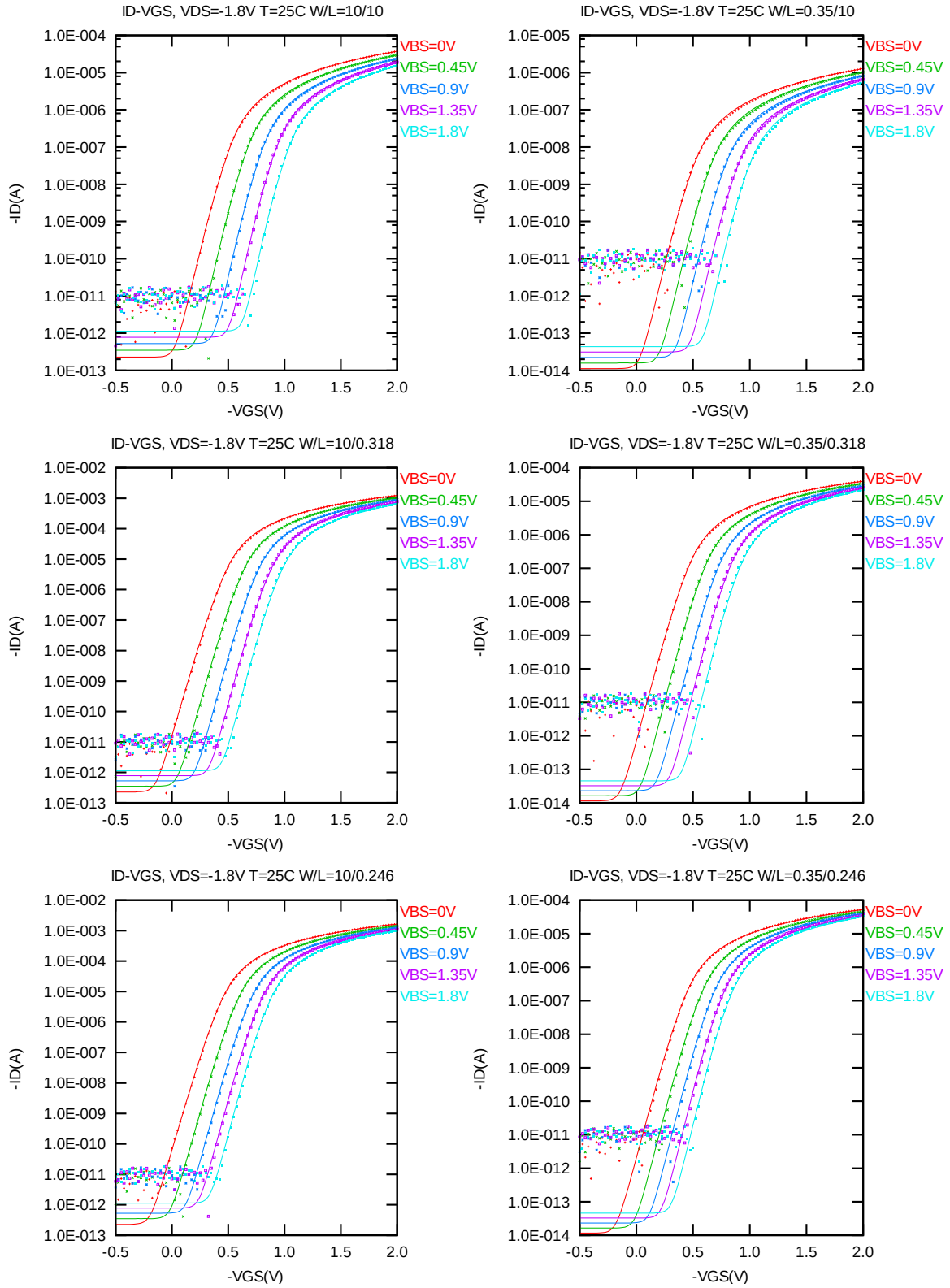


Figure 4-8 Log scaled I_D vs. V_{GS} at $V_{DS} = -1.8V$ for PMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

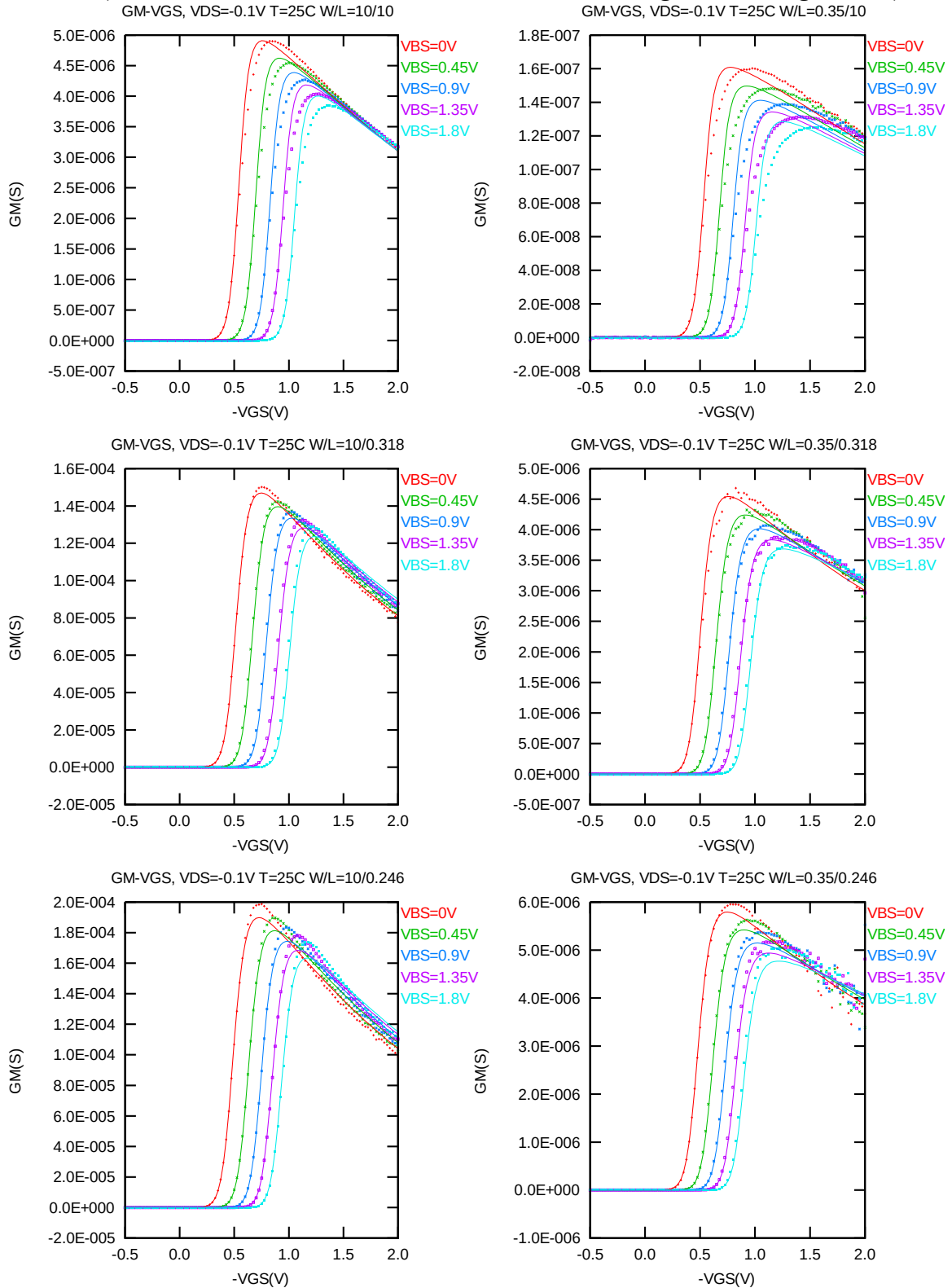


Figure 4-9 G_M vs. V_{GS} at $V_{DS} = -0.1 V$ for PMOS

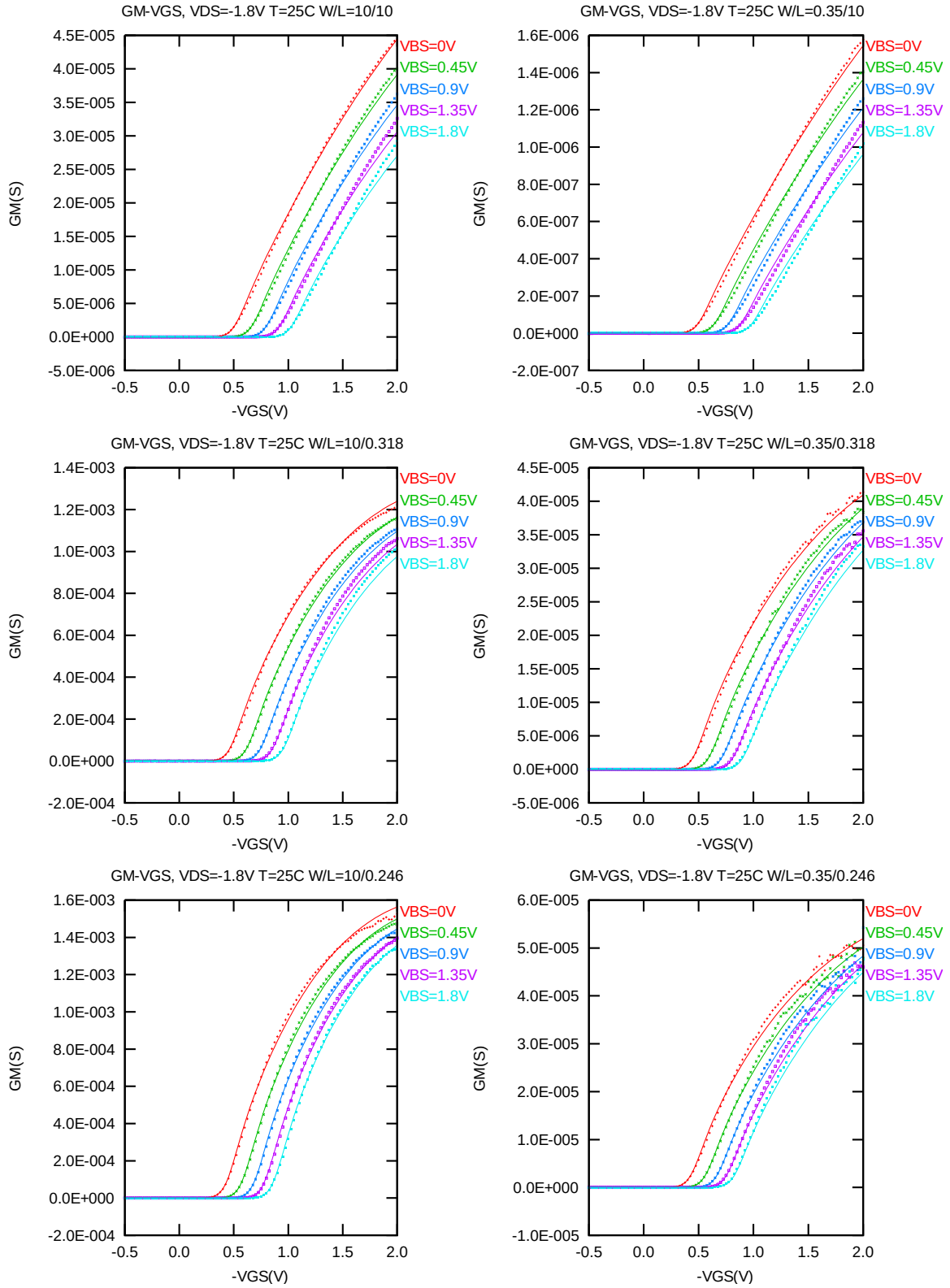


Figure 4-10 G_M vs. V_{GS} at $V_{DS} = -1.8V$ for PMOS

The L and W in the plots refer to LDES and WDES respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu\text{m})$ at 25°C . The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

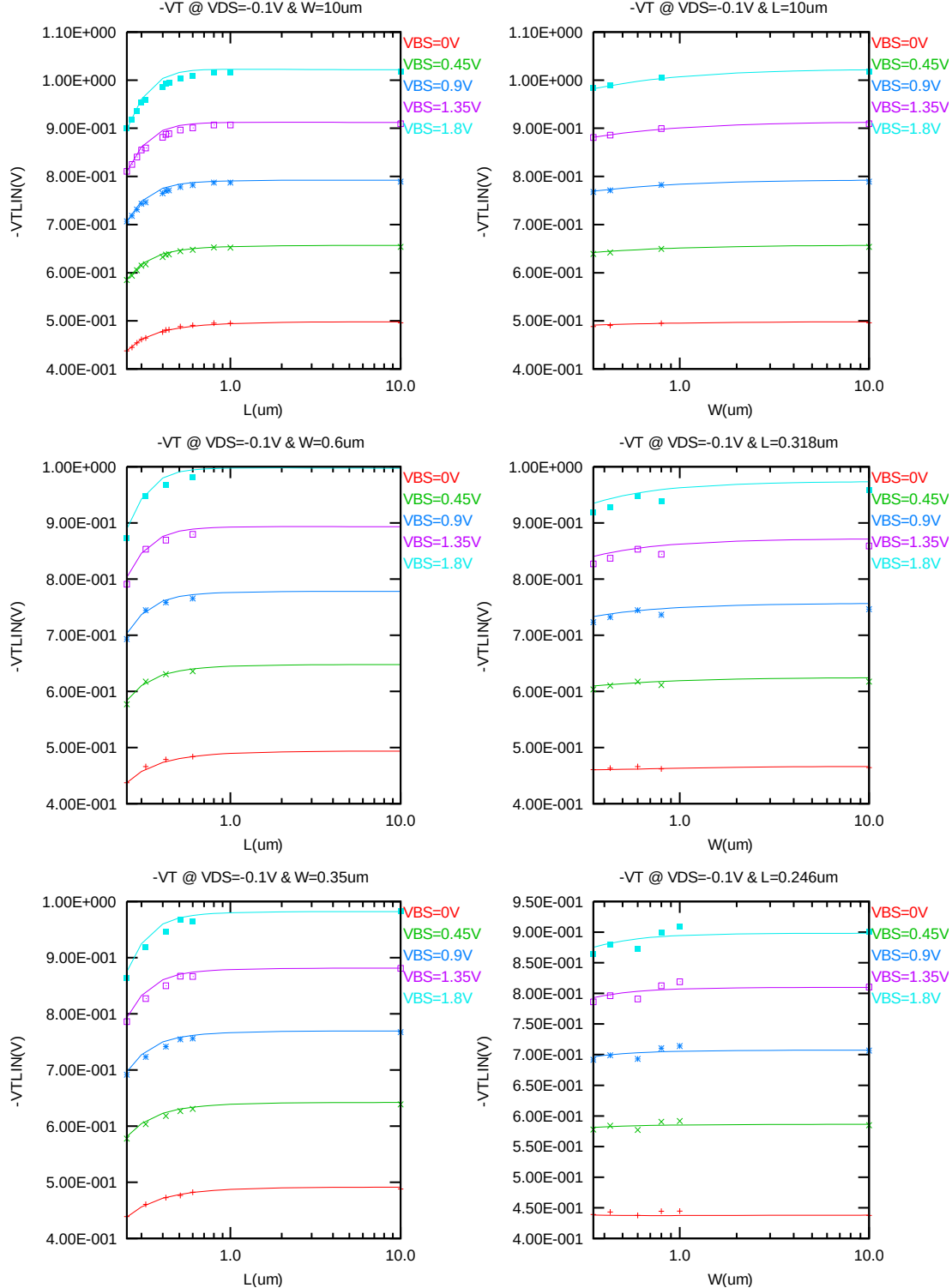


Figure 4-11 V_{TLIN} vs. channel length & channel width for low drain bias ($V_{DS} = -0.1\text{V}$) for PMOS

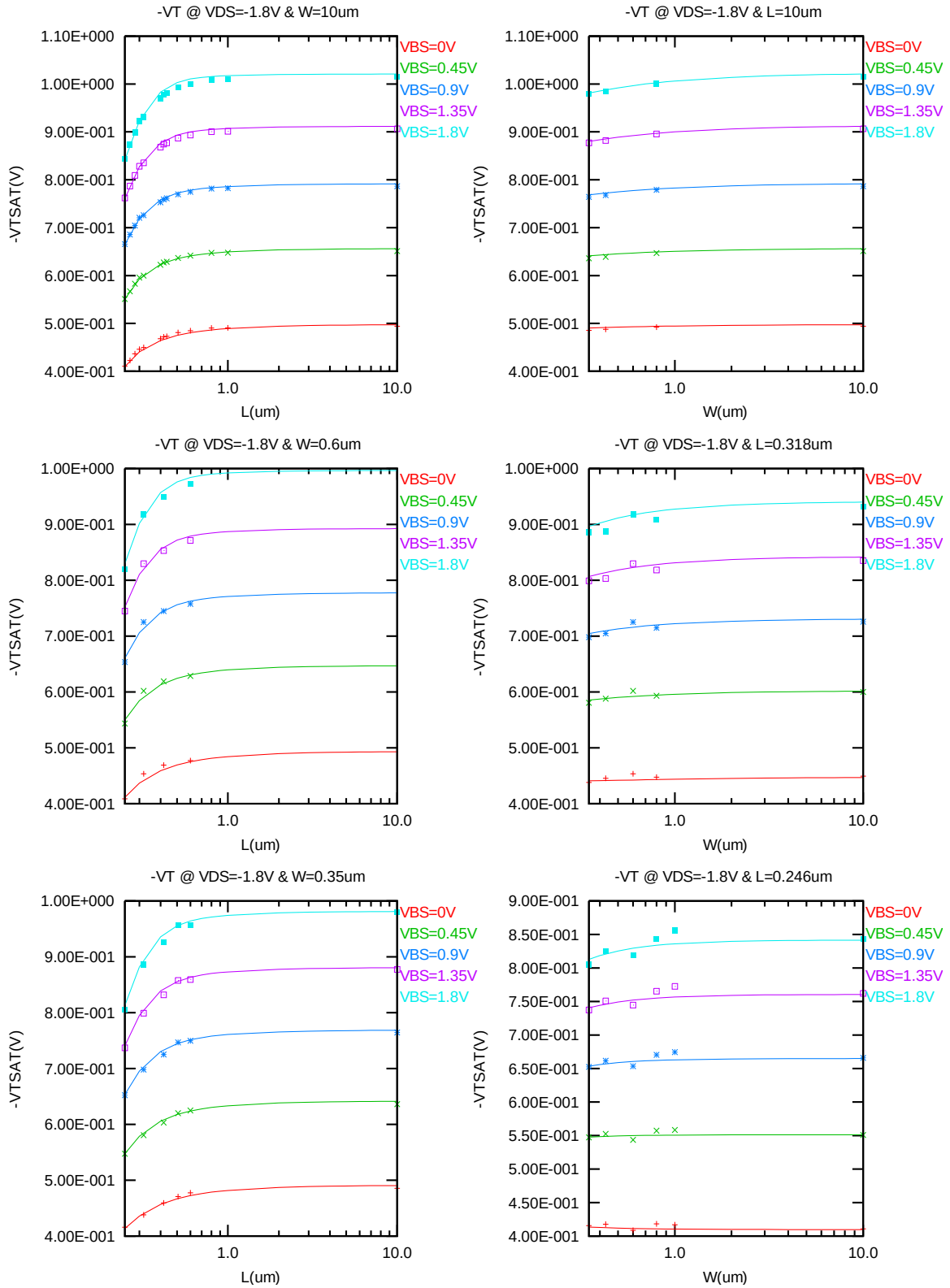


Figure 4-12 V_T vs. channel length & channel width for high drain bias ($V_{DS} = -1.8V$) for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -1.8V$ for $V_{BS} = 0V$ and $V_{BS} = 1.8V$ at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

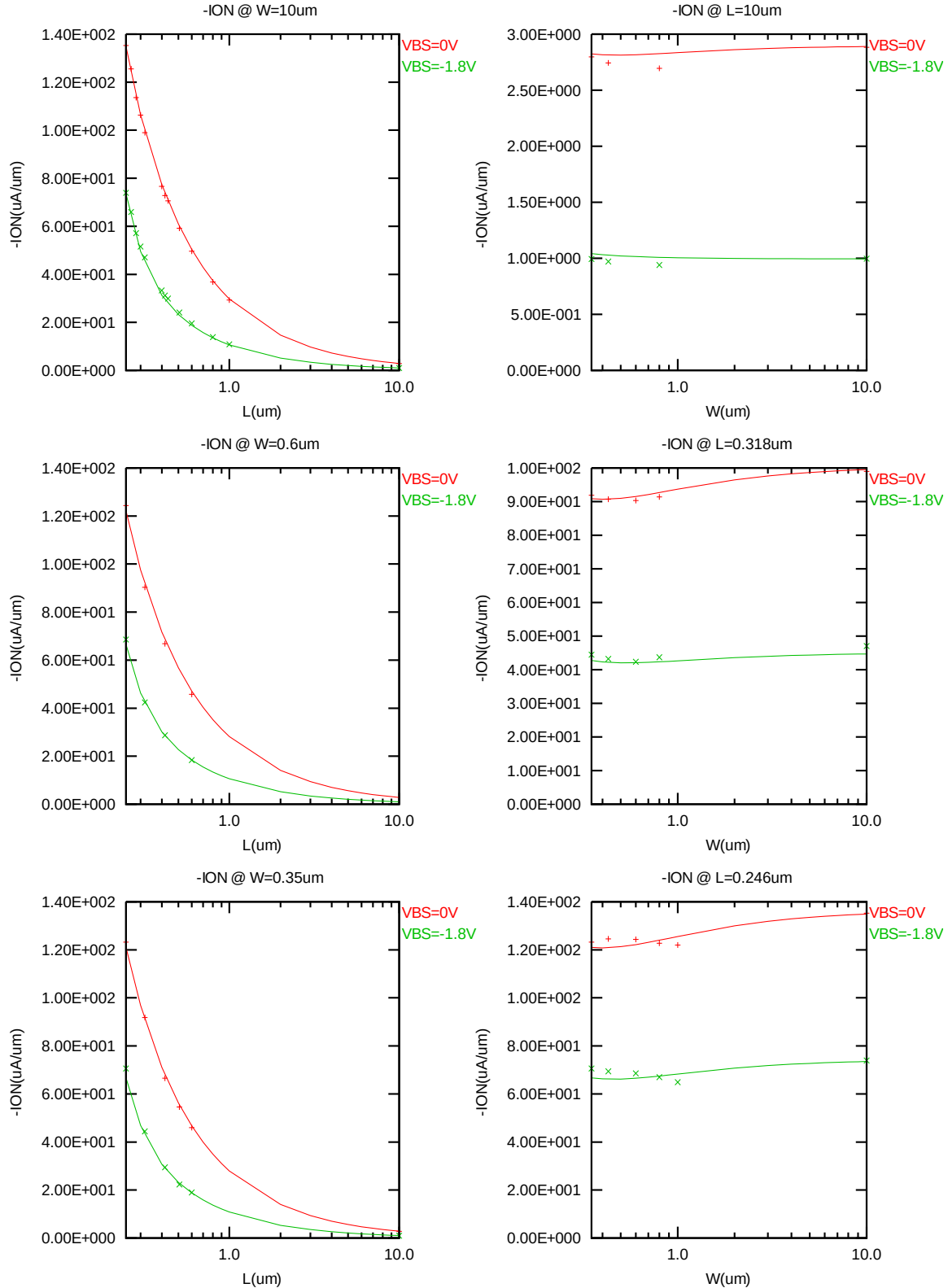


Figure 4-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = -V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0V$ to $1.8V$ at $25^\circ C$. A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

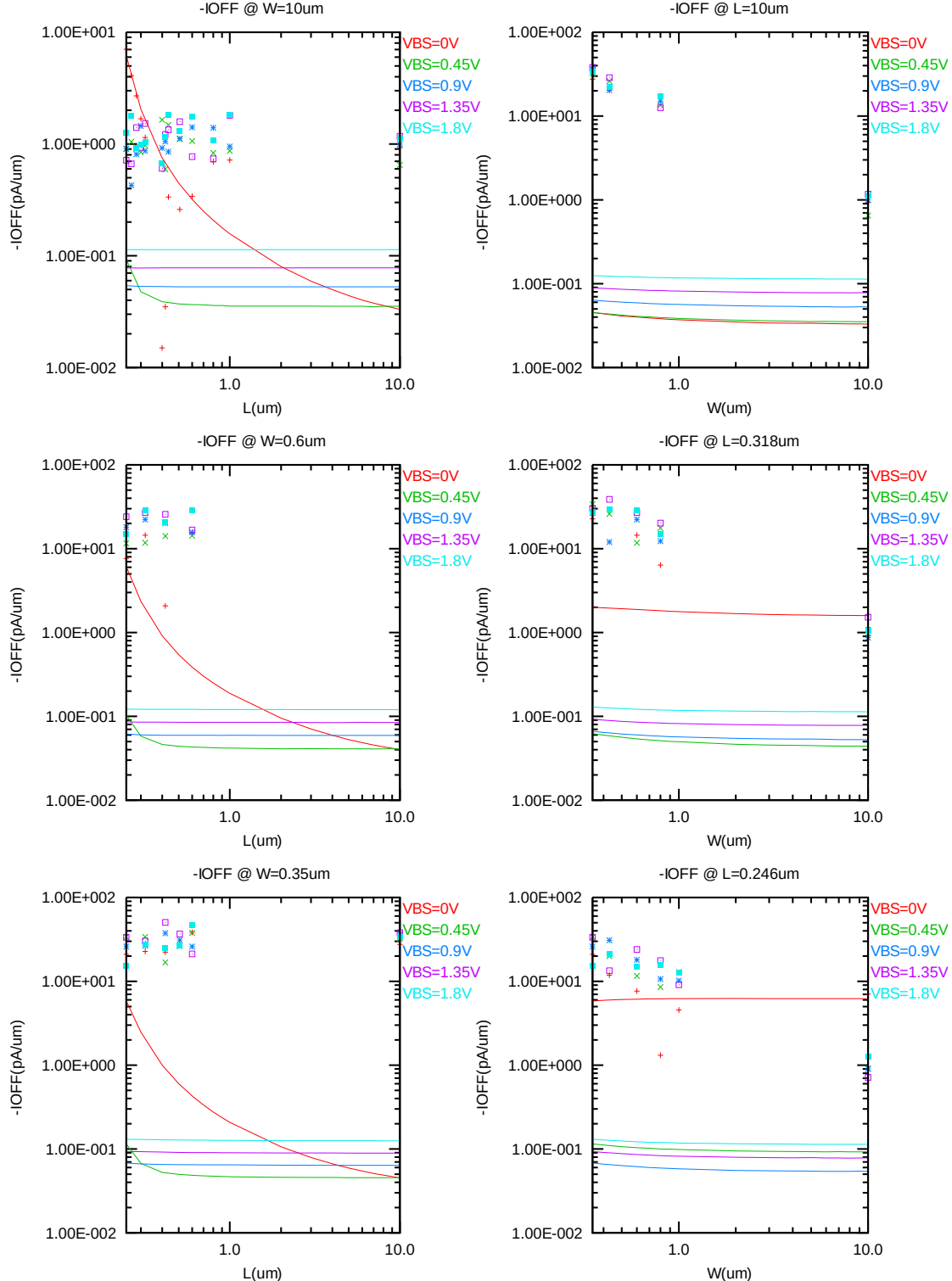


Figure 4-14 I_{OFF} vs. channel length and channel width for PMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = -70nA \cdot (WDES \cdot 0.9) / (LDES \cdot 0.9 + 0.012 \mu m)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

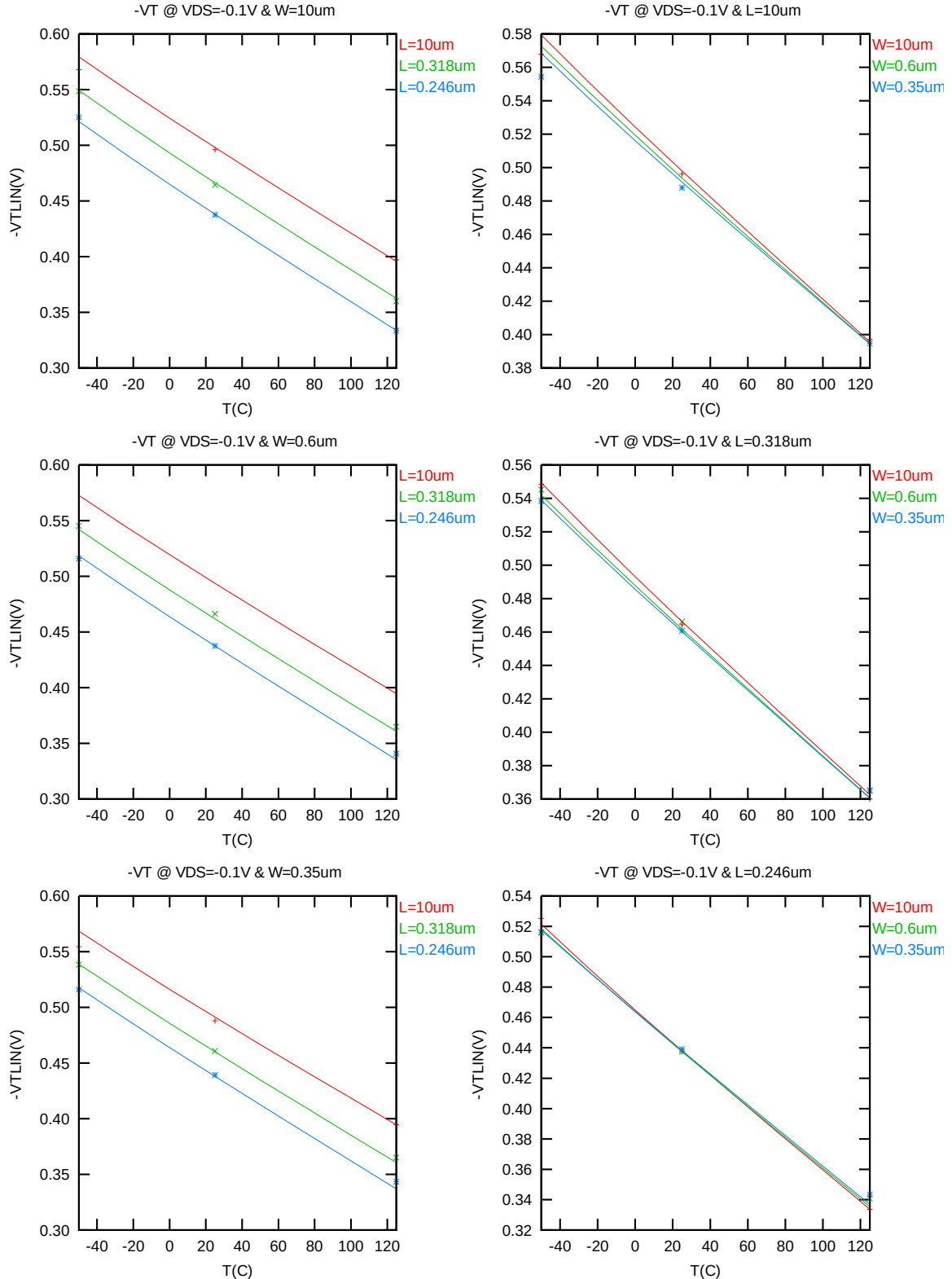


Figure 4-15 V_{TLIN} vs. temperature for PMOS

V_{TSAT} vs. temperature

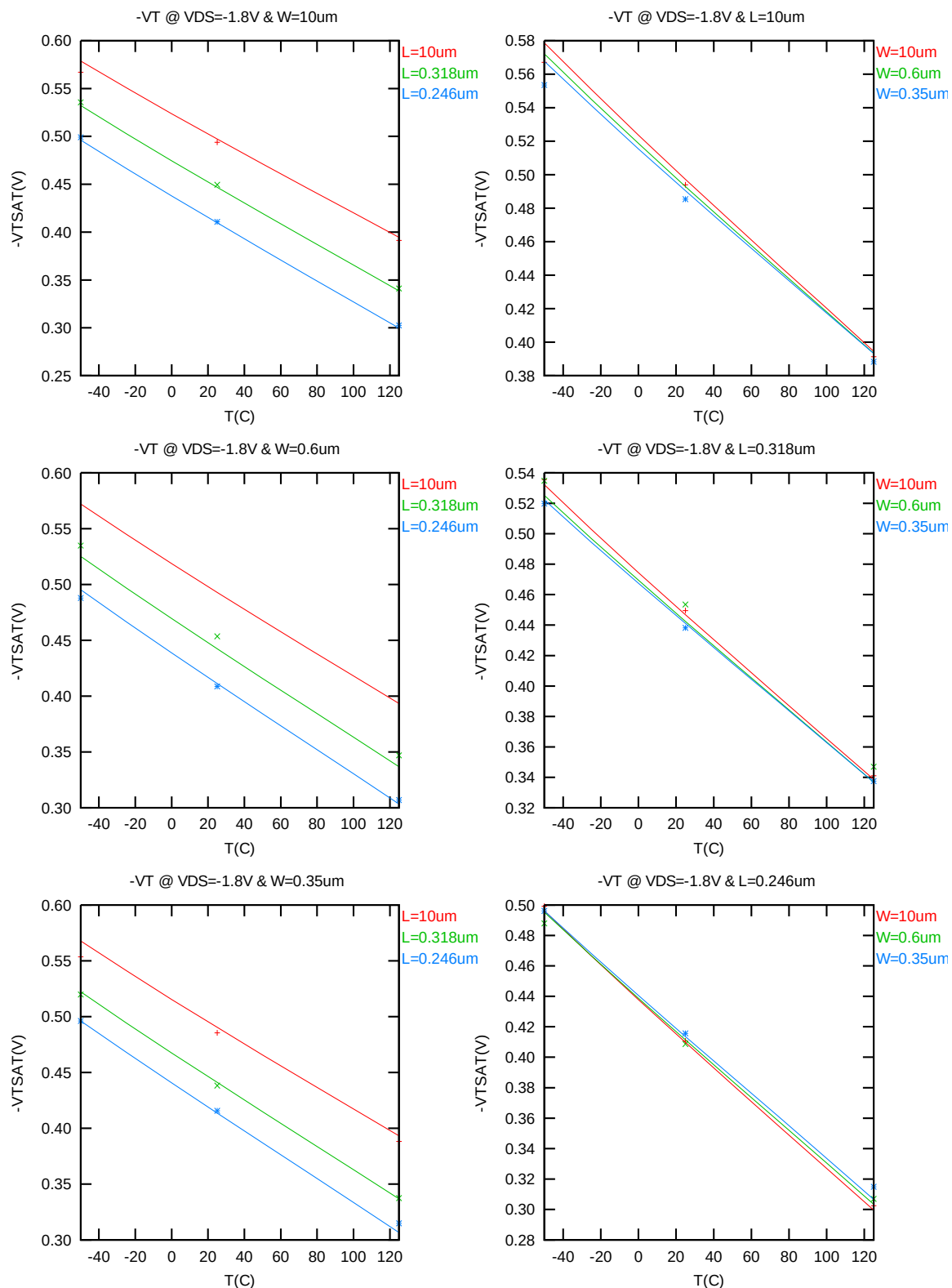


Figure 4-16 V_{TSAT} vs. temperature for PMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = -1.8V$, $V_{BS} = 0V$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

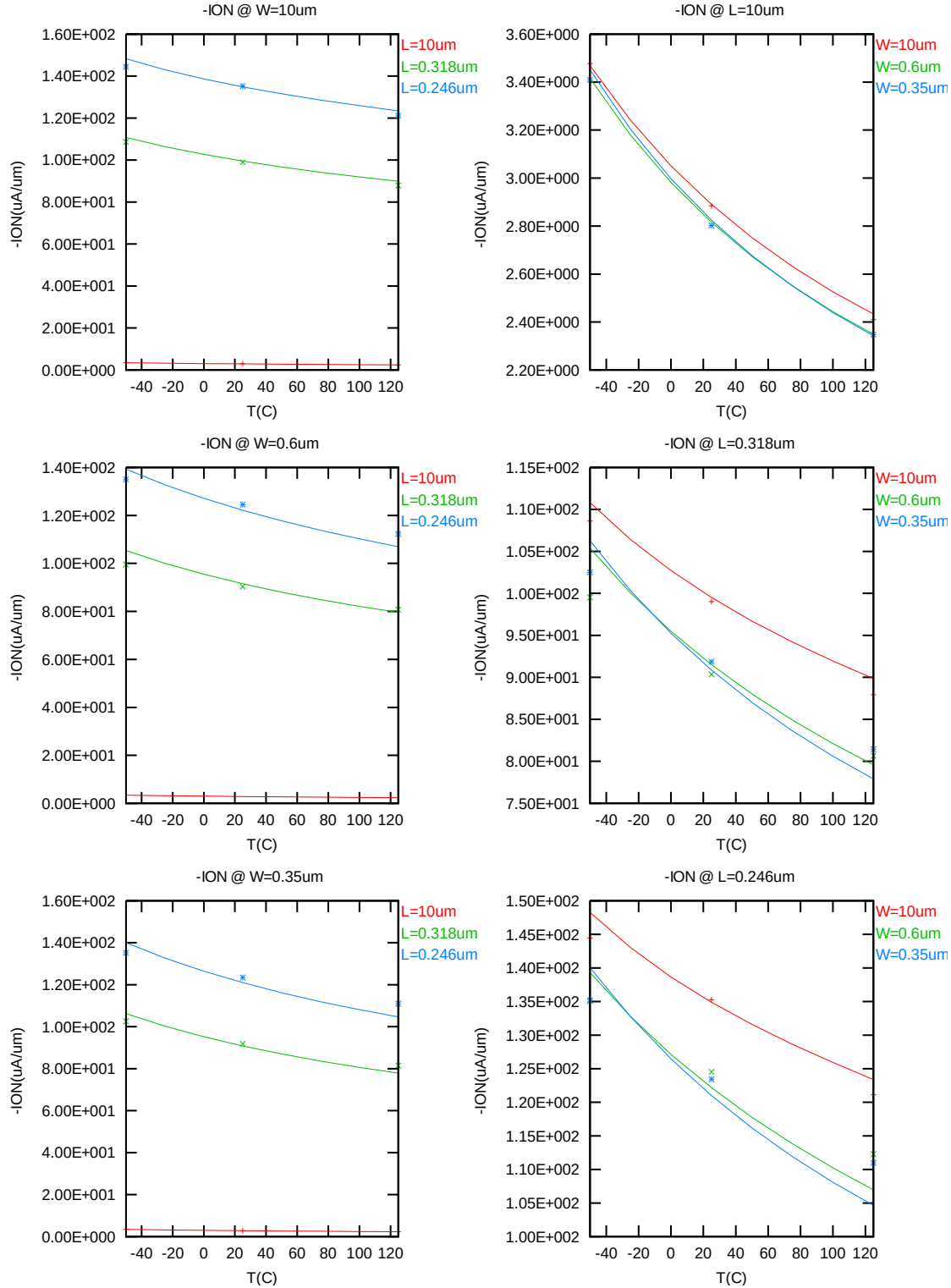


Figure 4-17 Saturation current vs. temperature for PMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{DS} = -1.8V$, $V_{GS} = V_{BS} = 0V$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

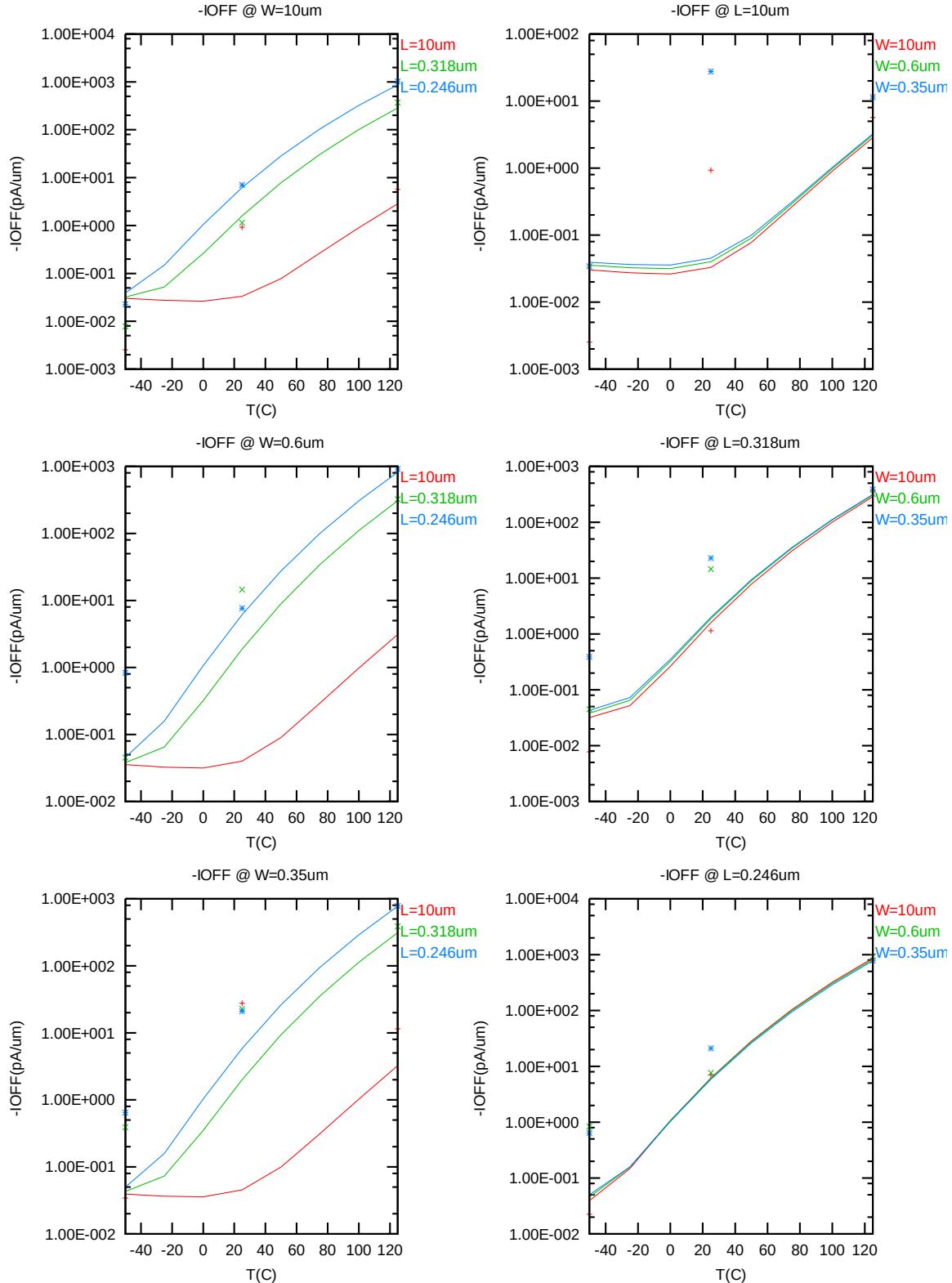


Figure 4-18 I_{OFF} vs. temperature for PMOS

4.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- k. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- l. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- m. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- n. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- o. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 4-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

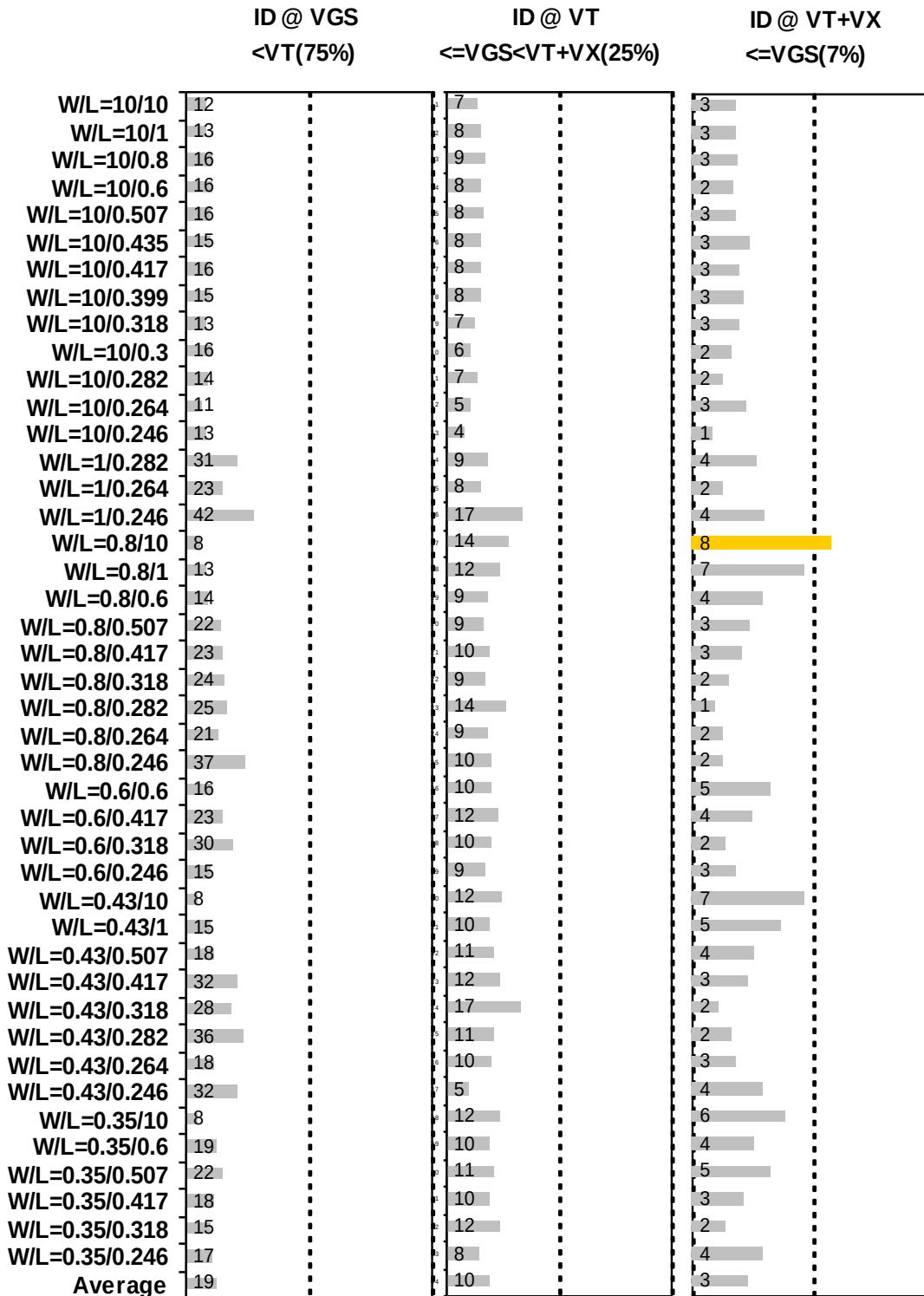


Figure 4-19 I_D accuracy at 25 C for PMOS

G_M and G_{DS} accuracy

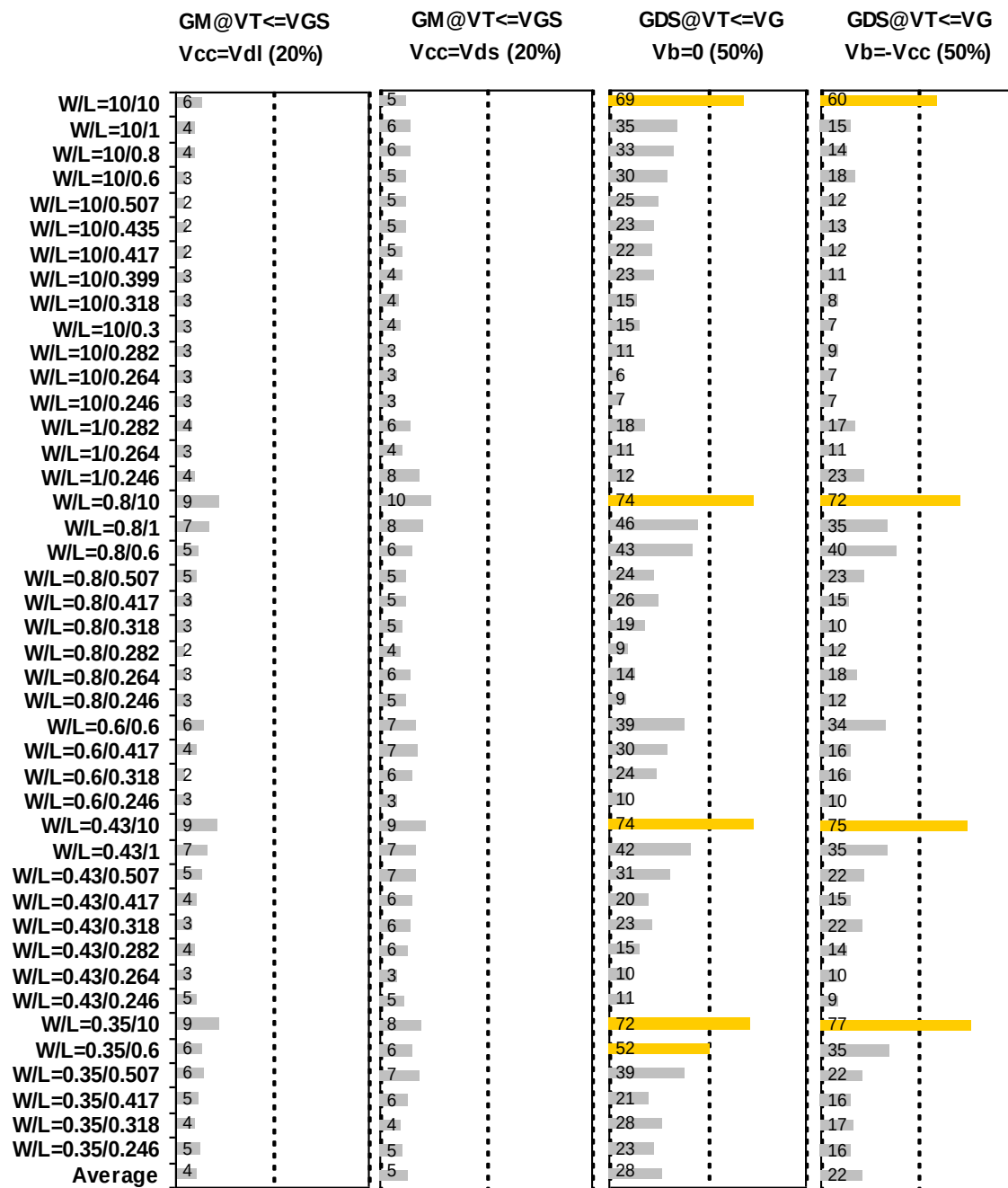


Figure 4-20 G_M and G_{DS} accuracy at 25 C for PMOS

4.4 MOSFET Capacitance Model

4.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 4-6 Measurement structure for PMOS C_{gg} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C _{GDS}	80	80

Table 4-7 Large area measurement structure for PMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	10	10
	1	10
	.5	10
	.246	10
	10	1
	.417	10
	.417	1
	10	.5
	1	1
	.246	.5
	.5	.5
	.417	.25
	10	.16
	10	.25
	.417	.16
	.246	.16
	.5	.16

4.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 4-8 Measurement conditions for PMOS capacitance model

V_{BIAS} [V]	From -3.63 to 3.63 V in 0.066 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

4.4.3 T_{OX} extraction

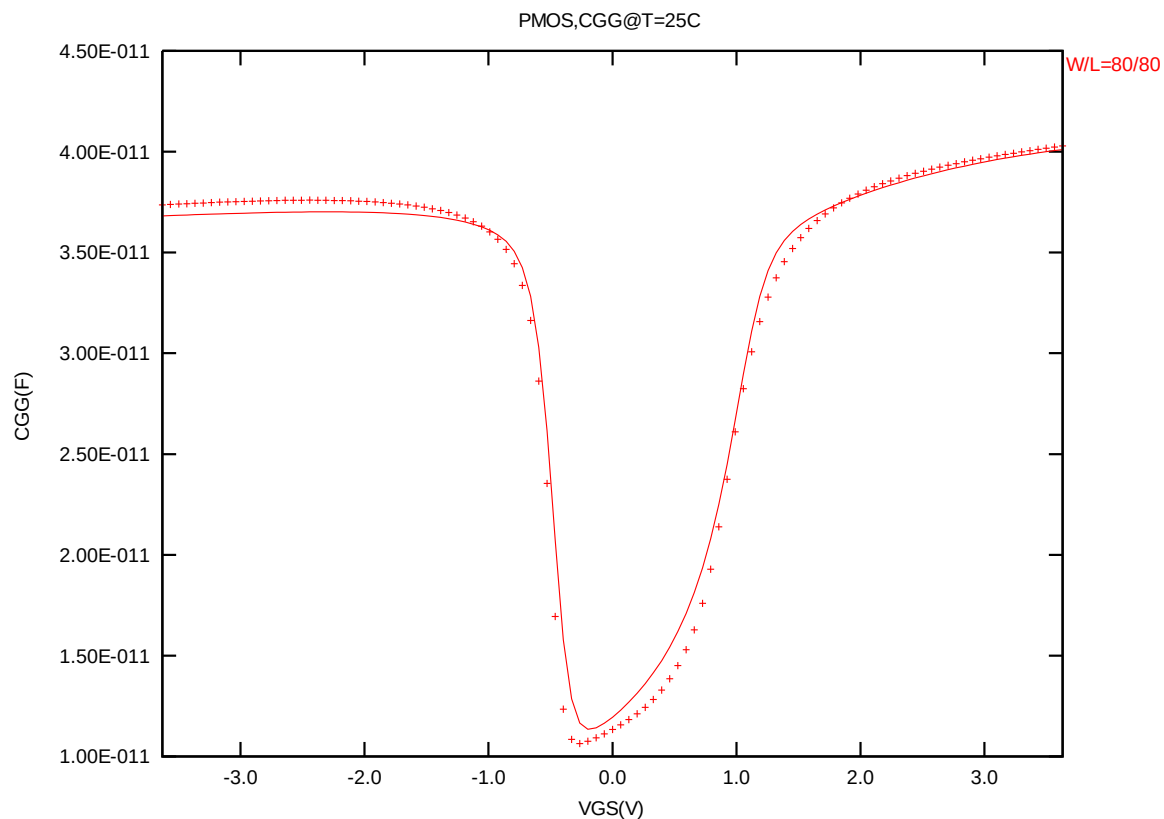


Figure 4-21 Measured and simulated C_{GG} curves

4.4.4 Width and length AC model accuracy

Width and length AC model accuracy

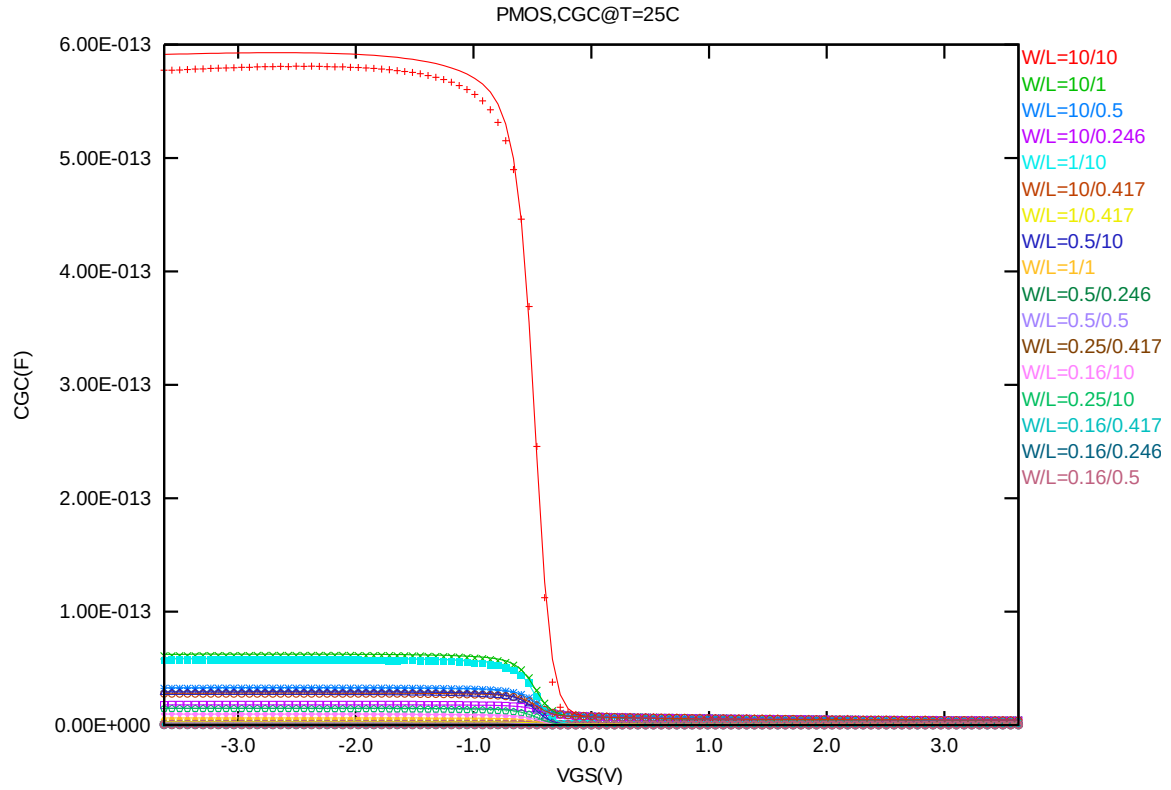


Figure 4-22 Measured and simulated C_{GC} curves for MOSFET measurement structure for PMOS

4.5 Parasitic Source / Drain Junction Capacitance Model

4.5.1 Test devices

The following test structures were used for parameter extraction.

Table 4-9 Junction capacitor structures for PMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.830E-02	Poly finger structure

4.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 4-10 Junction capacitor measurement conditions for PMOS

V _{BIAS} [V]	VH: nwell & VL: p+ From 0 to 3.63 V in 0.033 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,125

4.5.3 Extracted model parameters and verification plots

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

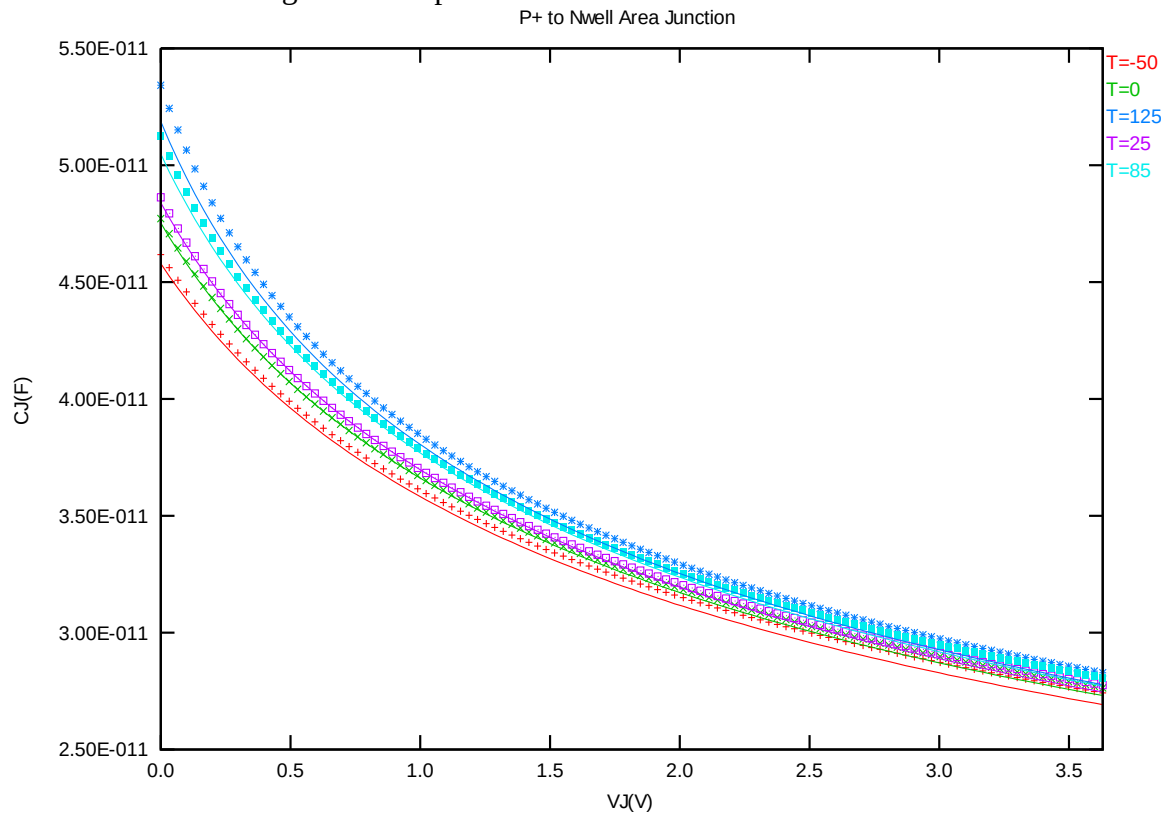


Figure 4-23 Bottom area junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for PMOS

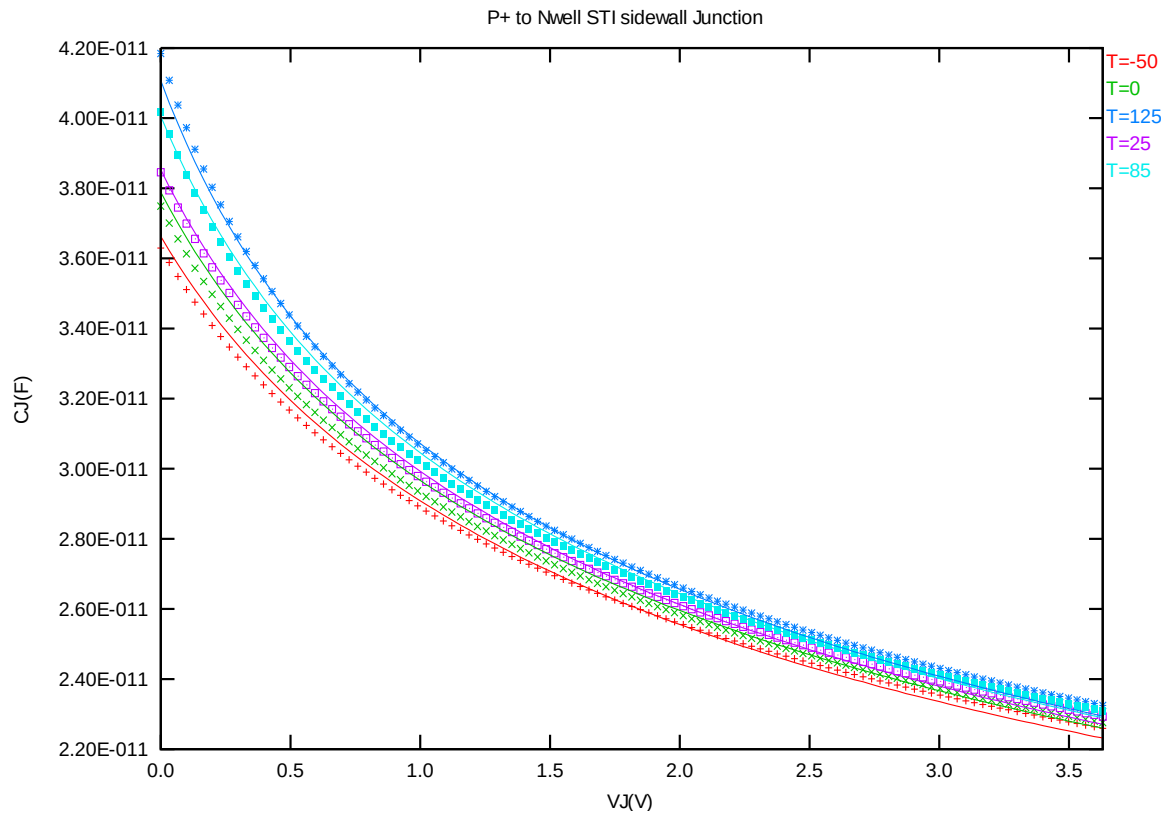


Figure 4-24 STI bounded sidewall junction capacitance at -50 C, 0 C, 25 C, 85 C and 125 C for PMOS

5 Worst Case Model

5.1 Wafer Data vs. Corner Plots

Threshold voltage in linear region V_{TLIN}

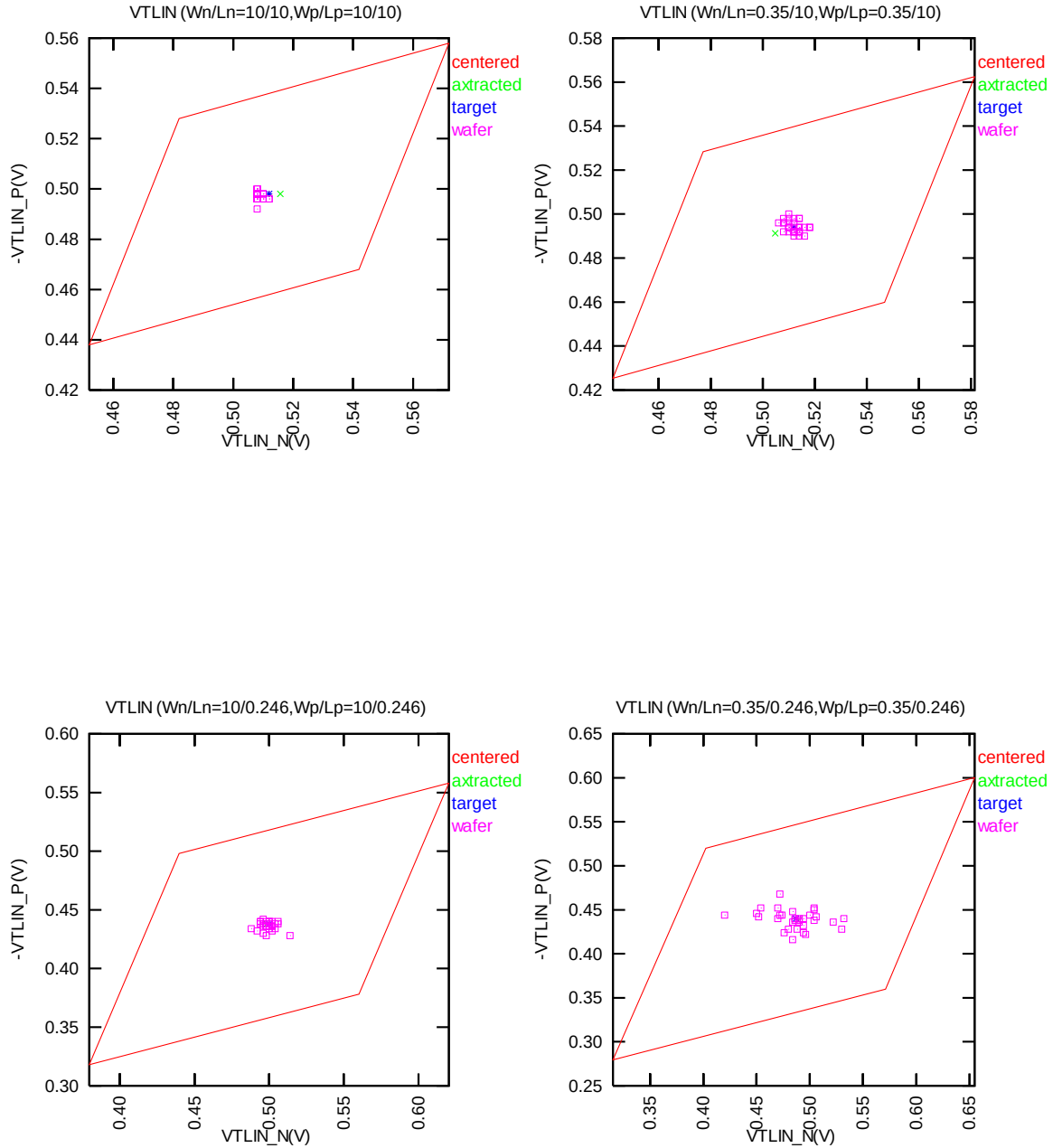


Figure 5-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in saturation region V_{TSAT}

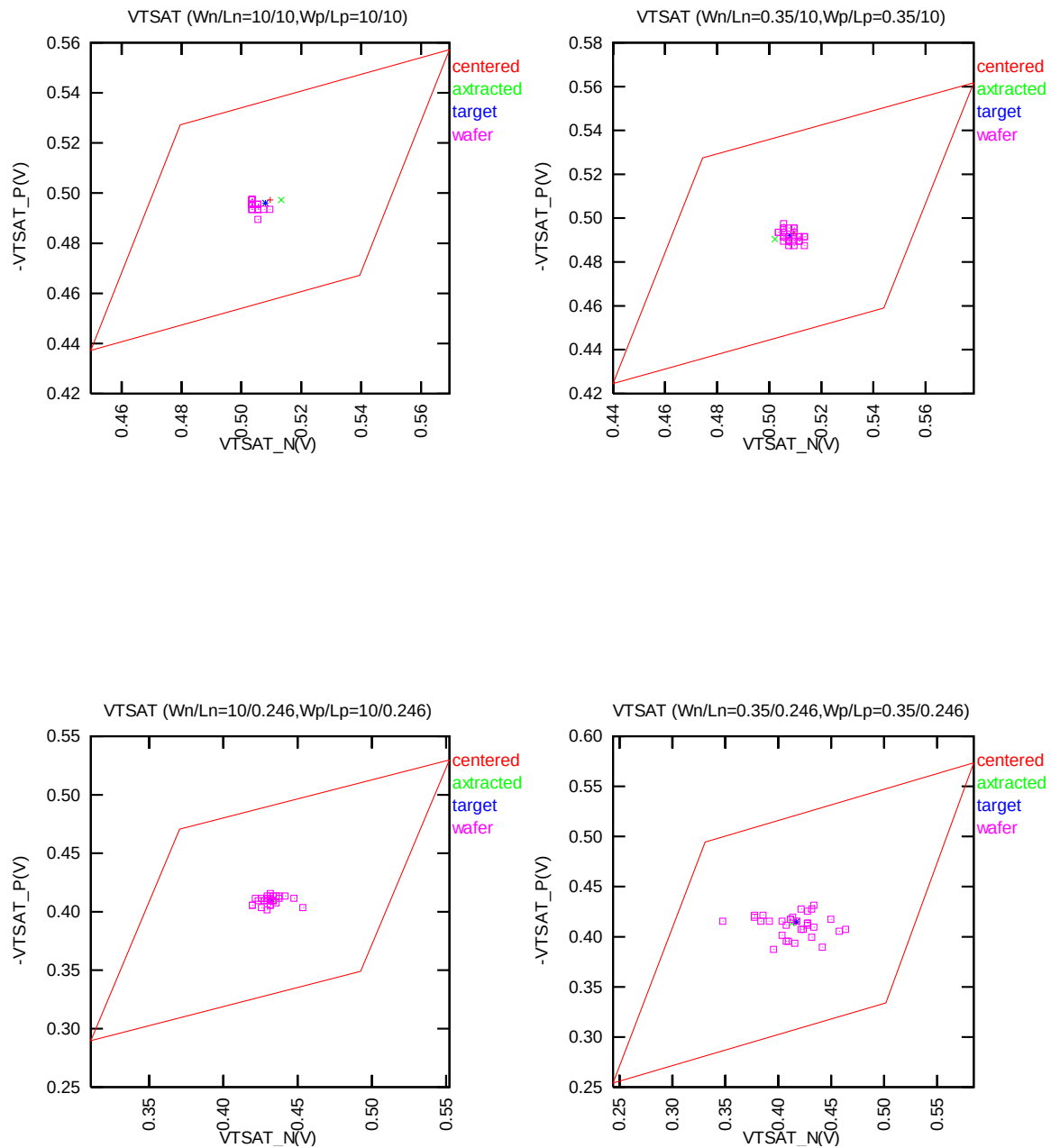


Figure 5-2 Corner plots of threshold voltage at high drain bias

Saturation current I_{ON}

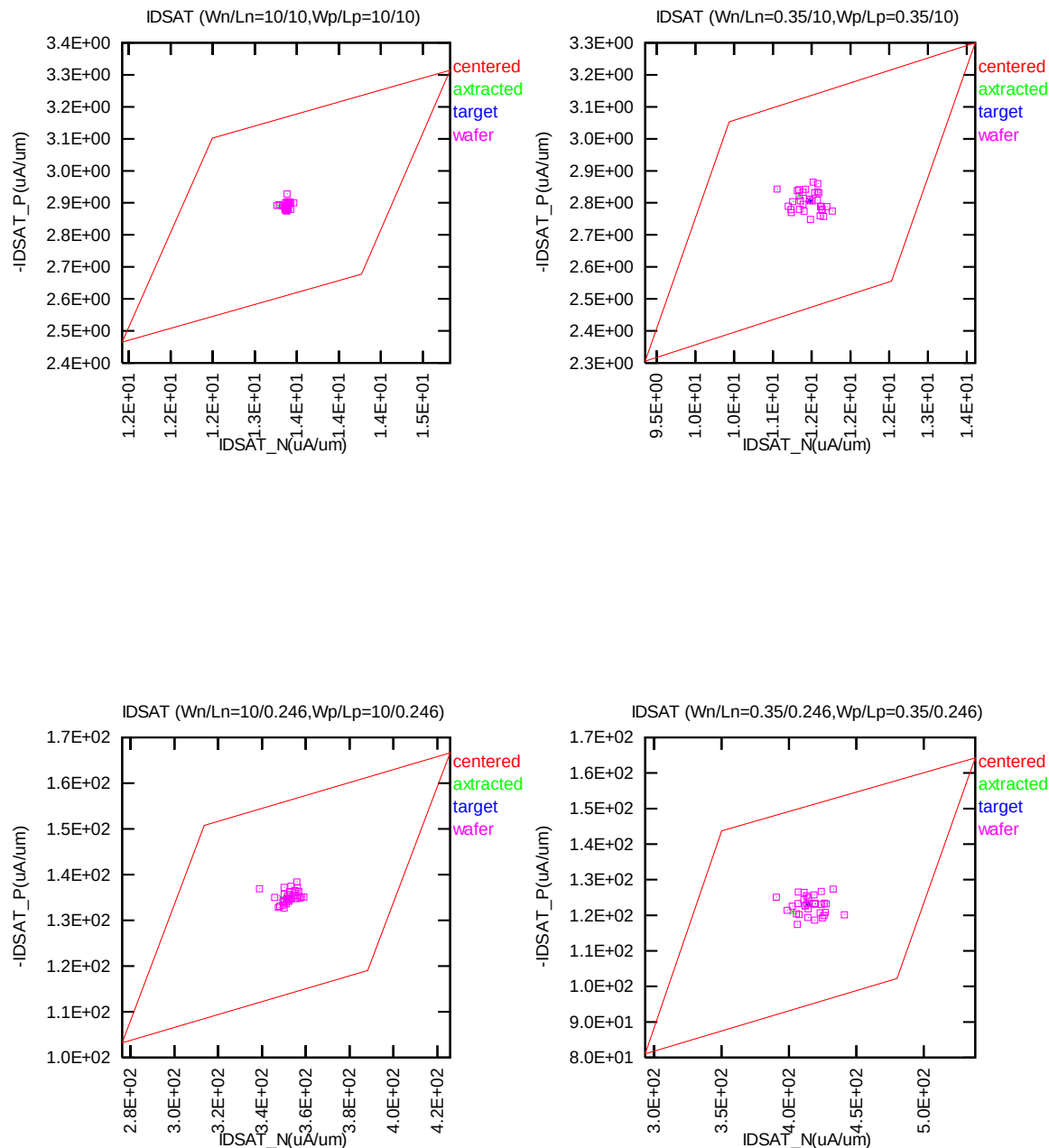


Figure 5-3 Corner plots of on-current

Off state current I_{OFF}

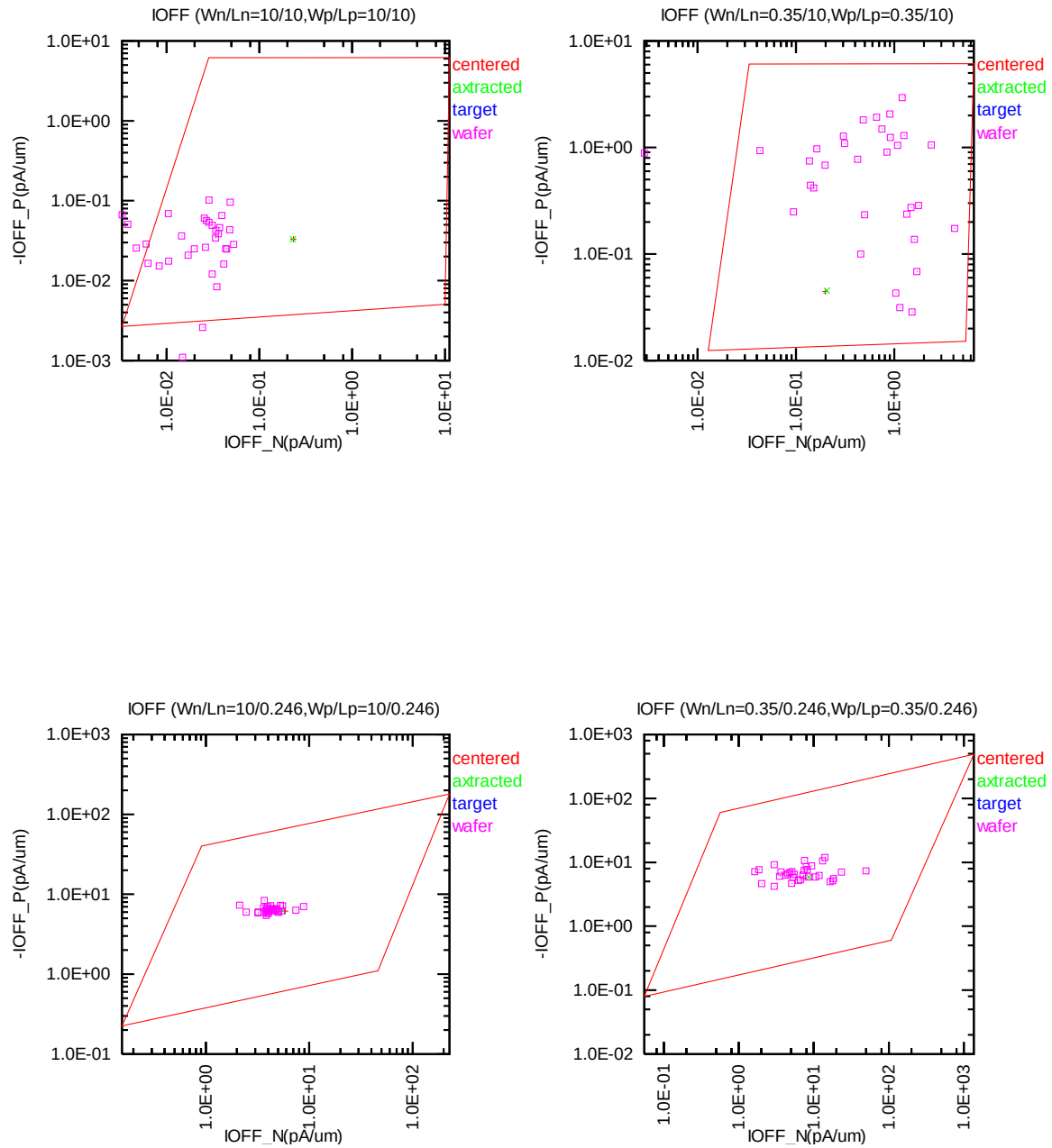


Figure 5-4 Corner plots of off-current

Saturation current I_{ON} v.s. Threshold voltage in linear region V_{TLIN} For bpw NMOS

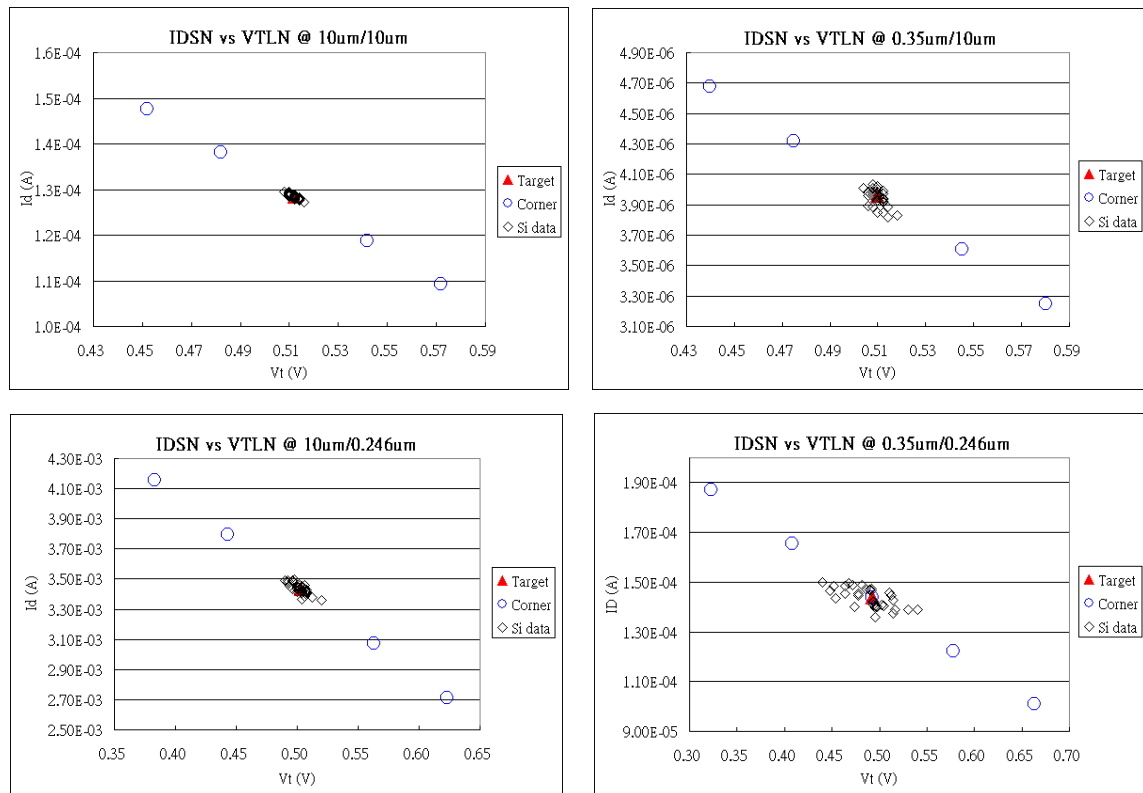


Figure 5-5 Corner plots of on-current v.s. threshold voltage at low drain bias

5.2 Device Performance Reference Tables

** The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The EDR value of size W/L: 10um/0.26um would be the same as the model card simulation result in size of W/L: 10um/0.234um (After 90% shrink without 12nm sizing channel length back).

Table 5-1 NMOS ID current at $V_{GS}=1.8V$, $V_{DS}=0.9V$ and $V_{GS}=1.8V$, $V_{DS}=1.8V$.

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IDSAT (μA)	VDS	0.9V	1.8V	0.9V	1.8V	0.9V	1.8V
TT	9/9	14.30	184.87	13.45	133.19	13.01	98.20
SS	9/9	9.34	156.90	9.41	113.67	9.63	84.38
FF	9/9	20.58	213.04	18.38	152.65	16.97	111.81
SNFP	9/9	11.73	172.04	11.39	124.38	11.32	92.09
FNFP	9/9	17.19	197.94	15.72	142.13	14.84	104.37
TT	9/0.234	662.74	3552.60	629.79	3155.40	602.29	2731.50
SS	9/0.234	325.76	2802.80	337.65	2482.40	350.30	2147.20
FF	9/0.234	1097.90	4290.20	1006.00	3829.40	921.35	3324.80
SNFP	9/0.234	476.94	3187.90	468.43	2818.90	463.32	2432.90
FNFP	9/0.234	874.48	3911.90	814.29	3491.20	760.39	3032.80
TT	0.35/9	0.48	6.03	0.46	4.46	0.46	3.37
SS	0.35/9	0.29	4.87	0.30	3.63	0.32	2.77
FF	0.35/9	0.73	7.20	0.67	5.29	0.63	3.96
SNFP	0.35/9	0.38	5.45	0.38	4.06	0.39	3.07
FNFP	0.35/9	0.60	6.61	0.56	4.87	0.54	3.66
TT	0.35/0.234	31.27	162.46	29.32	145.02	27.70	125.80
SS	0.35/0.234	11.11	115.23	11.86	102.68	12.61	89.14
FF	0.35/0.234	60.43	210.18	54.38	188.38	48.77	163.69
SNFP	0.35/0.234	19.63	137.69	19.27	122.52	19.08	106.09
FNFP	0.35/0.234	45.44	187.61	41.53	168.06	38.06	146.13

Table 5-2 NMOS ID current density at $V_{GS}=1.8V$, $V_{DS}=0.9V$ and $V_{GS}=1.8V$, $V_{DS}=1.8V$.

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IDSAT ($\mu A/\mu m$)	VDS	0.9V	1.8V	0.9V	1.8V	0.9V	1.8V
TT	9/9	1.59	20.54	1.49	14.80	1.45	10.91
SS	9/9	1.04	17.43	1.05	12.63	1.07	9.38
FF	9/9	2.29	23.67	2.04	16.96	1.89	12.42
SNFP	9/9	1.30	19.12	1.27	13.82	1.26	10.23
FNSP	9/9	1.91	21.99	1.75	15.79	1.65	11.60
TT	9/0.234	73.64	394.73	69.98	350.60	66.92	303.50
SS	9/0.234	36.20	311.42	37.52	275.82	38.92	238.58
FF	9/0.234	121.99	476.69	111.78	425.49	102.37	369.42
SNFP	9/0.234	52.99	354.21	52.05	313.21	51.48	270.32
FNSP	9/0.234	97.16	434.66	90.48	387.91	84.49	336.98
TT	0.35/9	1.37	17.22	1.33	12.75	1.32	9.62
SS	0.35/9	0.83	13.91	0.87	10.38	0.92	7.90
FF	0.35/9	2.09	20.58	1.91	15.13	1.80	11.32
SNFP	0.35/9	1.08	15.58	1.08	11.59	1.11	8.78
FNSP	0.35/9	1.71	18.87	1.60	13.92	1.55	10.45
TT	0.35/0.234	89.35	464.17	83.76	414.34	79.13	359.43
SS	0.35/0.234	31.75	329.23	33.87	293.37	36.03	254.70
FF	0.35/0.234	172.65	600.51	155.37	538.23	139.34	467.69
SNFP	0.35/0.234	56.09	393.40	55.05	350.06	54.50	303.11
FNSP	0.35/0.234	129.81	536.03	118.66	480.17	108.73	417.51

Table 5-3 NMOS V_{TLIN} .

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
V _{TH} (V)	V _{DS}	0.1V	1.8V	0.1V	1.8V	0.1V	1.8V
TT	9/9	0.577	0.575	0.512	0.510	0.434	0.430
SS	9/9	0.636	0.635	0.572	0.570	0.495	0.491
FF	9/9	0.517	0.515	0.452	0.450	0.373	0.370
SNFP	9/9	0.607	0.605	0.542	0.540	0.465	0.461
FNSP	9/9	0.547	0.545	0.482	0.480	0.404	0.400
TT	9/0.234	0.556	0.489	0.500	0.432	0.435	0.363
SS	9/0.234	0.676	0.609	0.621	0.553	0.556	0.484
FF	9/0.234	0.436	0.368	0.379	0.311	0.313	0.241
SNFP	9/0.234	0.616	0.550	0.560	0.493	0.495	0.424
FNSP	9/0.234	0.496	0.427	0.440	0.371	0.374	0.301
TT	0.35/9	0.577	0.575	0.512	0.509	0.434	0.429
SS	0.35/9	0.646	0.644	0.582	0.579	0.505	0.500
FF	0.35/9	0.508	0.506	0.442	0.439	0.363	0.358
SNFP	0.35/9	0.611	0.610	0.547	0.544	0.469	0.464
FNSP	0.35/9	0.542	0.540	0.477	0.474	0.398	0.393
TT	0.35/0.234	0.542	0.471	0.487	0.416	0.421	0.349
SS	0.35/0.234	0.709	0.638	0.655	0.585	0.591	0.518
FF	0.35/0.234	0.371	0.299	0.315	0.244	0.248	0.176
SNFP	0.35/0.234	0.626	0.556	0.571	0.502	0.506	0.434
FNSP	0.35/0.234	0.458	0.386	0.402	0.331	0.336	0.263

Table 5-4 NMOS ID current at $V_{GS}=0$ and $V_{DS}=0.1V$ or $V_{DS}=1.8V$ (off current).

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IOFF (A)	VDS	0.1V	1.8V	0.1V	1.8V	0.1V	1.8V
TT	9/9	5.93E-14	2.45E-12	2.18E-13	2.10E-12	3.61E-11	3.89E-11
SS	9/9	1.67E-16	3.55E-15	2.81E-14	3.38E-14	1.03E-11	1.07E-11
FF	9/9	3.04E-12	1.26E-10	3.94E-12	1.01E-10	1.37E-10	2.17E-10
SNFP	9/9	6.11E-15	2.49E-13	6.92E-14	2.63E-13	1.85E-11	1.93E-11
FNSP	9/9	2.74E-12	1.14E-10	3.11E-12	9.01E-11	7.37E-11	1.44E-10
TT	9/0.234	6.80E-14	2.61E-12	6.90E-12	5.27E-11	1.13E-09	5.24E-09
SS	9/0.234	2.22E-16	3.55E-15	1.91E-13	1.38E-12	7.74E-11	3.50E-10
FF	9/0.234	4.06E-12	1.43E-10	2.62E-10	2.05E-09	1.70E-08	7.85E-08
SNFP	9/0.234	6.88E-15	2.65E-13	1.09E-12	8.13E-12	2.87E-10	1.31E-09
FNSP	9/0.234	2.91E-12	1.18E-10	4.58E-11	4.17E-10	4.46E-09	2.09E-08
TT	0.35/9	2.03E-15	7.93E-14	8.89E-15	7.05E-14	1.55E-12	1.66E-12
SS	0.35/9	1.98E-16	3.61E-15	1.12E-15	4.55E-15	3.95E-13	4.15E-13
FF	0.35/9	6.77E-14	2.81E-12	1.21E-13	2.27E-12	7.08E-12	9.02E-12
SNFP	0.35/9	3.75E-16	1.09E-14	2.75E-15	1.19E-14	7.37E-13	7.75E-13
FNSP	0.35/9	5.67E-14	2.35E-12	7.62E-14	1.87E-12	3.42E-12	4.95E-12
TT	0.35/0.234	3.53E-15	1.32E-13	3.30E-13	2.58E-12	5.31E-11	2.49E-10
SS	0.35/0.234	1.98E-16	3.61E-15	2.31E-15	1.92E-14	1.26E-12	5.56E-12
FF	0.35/0.234	5.39E-13	1.18E-11	6.17E-11	4.77E-10	2.60E-09	1.19E-08
SNFP	0.35/0.234	4.93E-16	1.55E-14	2.54E-14	1.98E-13	7.73E-12	3.56E-11
FNSP	0.35/0.234	1.37E-13	5.37E-12	4.52E-12	3.79E-11	3.69E-10	1.74E-09

*gmindc=1e-15

Table 5-5 bpw NMOS ID current at $V_{GS}=1.8V$, $V_{DS}=0.9V$ and $V_{GS}=1.8V$, $V_{DS}=1.8V$.

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IDSAT (μA)	VDS	0.9V	1.8V	0.9V	1.8V	0.9V	1.8V
TT	9/9	13.02	176.27	12.62	127.97	12.45	93.97
SS	9/9	8.31	148.60	8.63	108.67	9.03	80.59
FF	9/9	19.10	204.24	17.53	147.28	16.49	107.18
SNFP	9/9	10.56	162.22	10.56	118.32	10.71	87.38
FNSP	9/9	15.79	190.40	14.88	137.62	14.33	100.52
TT	9/0.234	630.02	3439.70	597.19	3091.70	558.30	2622.70
SS	9/0.234	303.47	2795.40	305.88	2440.80	303.93	2008.80
FF	9/0.234	1044.80	4087.50	973.58	3742.50	887.28	3243.70
SNFP	9/0.234	455.74	3120.50	441.29	2766.20	422.23	2311.50
FNSP	9/0.234	825.18	3759.00	773.66	3417.00	712.79	2936.90
TT	0.35/9	0.43	5.75	0.44	4.40	0.46	3.43
SS	0.35/9	0.25	4.66	0.28	3.60	0.31	2.84
FF	0.35/9	0.66	6.86	0.64	5.20	0.63	4.00
SNFP	0.35/9	0.34	5.20	0.36	4.00	0.38	3.14
FNSP	0.35/9	0.54	6.30	0.53	4.80	0.54	3.71
TT	0.35/0.234	26.95	159.82	26.01	144.03	25.08	122.71
SS	0.35/0.234	8.65	117.38	9.36	100.82	10.03	81.83
FF	0.35/0.234	53.99	202.58	50.83	187.24	47.22	164.30
SNFP	0.35/0.234	16.54	138.96	16.55	122.43	16.56	101.73
FNSP	0.35/0.234	39.56	180.77	37.57	165.64	35.46	143.92

**Table 5-6 bpw NMOS ID current density at $V_{GS}=1.8V$, $V_{DS}=0.9V$ and $V_{GS}=1.8V$,
 $V_{DS}=1.8V$.**

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IDSAT ($\mu A/\mu m$)	VDS	0.9V	1.8V	0.9V	1.8V	0.9V	1.8V
TT	9/9	1.45	19.59	1.40	14.22	1.38	10.44
SS	9/9	0.92	16.51	0.96	12.07	1.00	8.95
FF	9/9	2.12	22.69	1.95	16.36	1.83	11.91
SNFP	9/9	1.17	18.02	1.17	13.15	1.19	9.71
FNSP	9/9	1.75	21.16	1.65	15.29	1.59	11.17
TT	9/0.234	70.00	382.19	66.35	343.52	62.03	291.41
SS	9/0.234	33.72	310.60	33.99	271.20	33.77	223.20
FF	9/0.234	116.09	454.17	108.18	415.83	98.59	360.41
SNFP	9/0.234	50.64	346.72	49.03	307.36	46.91	256.83
FNSP	9/0.234	91.69	417.67	85.96	379.67	79.20	326.32
TT	0.35/9	1.23	16.42	1.25	12.57	1.31	9.79
SS	0.35/9	0.73	13.31	0.80	10.30	0.90	8.12
FF	0.35/9	1.90	19.60	1.82	14.85	1.80	11.42
SNFP	0.35/9	0.96	14.85	1.02	11.43	1.10	8.96
FNSP	0.35/9	1.53	18.01	1.51	13.71	1.54	10.60
TT	0.35/0.234	77.01	456.63	74.32	411.51	71.64	350.60
SS	0.35/0.234	24.72	335.37	26.75	288.06	28.67	233.79
FF	0.35/0.234	154.26	578.80	145.22	534.97	134.92	469.43
SNFP	0.35/0.234	47.27	397.03	47.27	349.80	47.30	290.66
FNSP	0.35/0.234	113.02	516.49	107.35	473.26	101.30	411.20

Table 5-7 bpw NMOS V_{TLIN} and V_{TSAT} .

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
V_{TH} (V)	V_{DS}	0.1V	1.8V	0.1V	1.8V	0.1V	1.8V
TT	9/9	0.582	0.580	0.512	0.510	0.428	0.424
SS	9/9	0.641	0.639	0.572	0.570	0.490	0.485
FF	9/9	0.522	0.520	0.452	0.450	0.367	0.362
SNFP	9/9	0.611	0.609	0.542	0.540	0.459	0.454
FNSP	9/9	0.552	0.550	0.482	0.480	0.398	0.393
TT	9/0.234	0.561	0.502	0.503	0.442	0.436	0.370
SS	9/0.234	0.679	0.621	0.624	0.562	0.559	0.492
FF	9/0.234	0.442	0.383	0.383	0.321	0.314	0.248
SNFP	9/0.234	0.620	0.561	0.563	0.502	0.497	0.431
FNSP	9/0.234	0.501	0.443	0.443	0.382	0.374	0.309
TT	0.35/9	0.580	0.578	0.510	0.507	0.424	0.419
SS	0.35/9	0.649	0.647	0.580	0.577	0.496	0.490
FF	0.35/9	0.511	0.509	0.440	0.437	0.352	0.347
SNFP	0.35/9	0.615	0.613	0.545	0.542	0.460	0.455
FNSP	0.35/9	0.546	0.544	0.475	0.472	0.388	0.383
TT	0.35/0.234	0.554	0.489	0.493	0.425	0.420	0.347
SS	0.35/0.234	0.721	0.656	0.663	0.595	0.594	0.521
FF	0.35/0.234	0.386	0.321	0.323	0.254	0.247	0.174
SNFP	0.35/0.234	0.638	0.572	0.578	0.510	0.507	0.434
FNSP	0.35/0.234	0.471	0.405	0.408	0.339	0.333	0.260

Table 5-8 bpw NMOS ID current at $V_{GS}=0$ and $V_{DS}=0.1V$ or $V_{DS}= 1.8V$ (off current).

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IOFF (A)	VDS	0.1V	1.8V	0.1V	1.8V	0.1V	1.8V
TT	9/9	2.78E-16	5.33E-15	1.34E-13	2.95E-13	3.70E-11	4.58E-11
SS	9/9	2.22E-16	3.55E-15	2.24E-14	2.66E-14	1.02E-11	1.06E-11
FF	9/9	1.27E-14	4.90E-13	3.74E-12	9.86E-11	3.13E-10	4.98E-09
SNFP	9/9	1.67E-16	3.55E-15	5.14E-14	7.28E-14	1.86E-11	2.01E-11
FNSP	9/9	8.99E-15	3.62E-13	2.51E-12	7.24E-11	2.00E-10	3.64E-09
TT	9/0.234	4.61E-15	5.15E-14	4.90E-12	3.09E-11	9.40E-10	3.92E-09
SS	9/0.234	2.22E-16	5.33E-15	1.33E-13	8.10E-13	6.09E-11	2.46E-10
FF	9/0.234	5.91E-13	6.91E-12	2.00E-10	1.36E-09	1.53E-08	6.72E-08
SNFP	9/0.234	5.00E-16	7.11E-15	7.57E-13	4.73E-12	2.28E-10	9.49E-10
FNSP	9/0.234	6.11E-14	9.33E-13	3.43E-11	2.75E-10	4.00E-09	1.97E-08
TT	0.35/9	2.01E-16	3.61E-15	6.47E-15	1.32E-14	1.73E-12	1.95E-12
SS	0.35/9	2.01E-16	3.61E-15	9.82E-16	4.33E-15	4.18E-13	4.38E-13
FF	0.35/9	5.38E-16	1.54E-14	1.23E-13	2.41E-12	1.23E-11	1.25E-10
SNFP	0.35/9	1.98E-16	3.61E-15	2.28E-15	6.05E-15	7.99E-13	8.48E-13
FNSP	0.35/9	3.85E-16	1.07E-14	6.17E-14	1.45E-12	6.32E-12	7.44E-11
TT	0.35/0.234	3.61E-16	5.94E-15	2.04E-13	1.63E-12	4.38E-11	2.19E-10
SS	0.35/0.234	2.01E-16	3.61E-15	1.30E-15	1.19E-14	8.77E-13	3.97E-12
FF	0.35/0.234	1.95E-13	2.92E-12	4.37E-11	3.50E-10	2.42E-09	1.17E-08
SNFP	0.35/0.234	2.05E-16	3.66E-15	1.43E-14	1.15E-13	5.75E-12	2.84E-11
FNSP	0.35/0.234	6.18E-15	1.01E-13	3.13E-12	2.71E-11	3.40E-10	1.80E-09

*gmindc=1e-15

Table 5-9 PMOS ID current at $V_{GS} = -1.8V$, $V_{DS} = -0.9V$ and $V_{GS} = -1.8V$, $V_{DS} = -1.8V$.

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IDSAT (μA)	-VDS	0.9V	1.8V	0.9V	1.8V	0.9V	1.8V
TT	9/9	3.04	34.64	3.41	28.86	3.89	24.31
SS	9/9	1.97	29.31	2.39	24.61	2.89	20.90
FF	9/9	4.40	40.03	4.66	33.11	5.07	27.65
SNFP	9/9	3.67	37.37	4.00	30.98	4.45	25.95
FNSP	9/9	2.47	31.92	2.88	26.74	3.37	22.64
TT	9/0.234	169.41	1333.50	192.41	1212.80	220.72	1109.00
SS	9/0.234	81.04	1000.80	104.86	927.14	133.97	866.24
FF	9/0.234	296.99	1676.60	310.50	1498.30	329.96	1342.30
SNFP	9/0.234	229.35	1504.40	248.84	1355.50	273.84	1225.90
FNSP	9/0.234	119.74	1165.00	144.03	1070.00	173.59	989.80
TT	0.35/9	0.12	1.33	0.13	1.09	0.14	0.90
SS	0.35/9	0.07	1.08	0.08	0.89	0.10	0.75
FF	0.35/9	0.18	1.58	0.18	1.28	0.19	1.05
SNFP	0.35/9	0.15	1.46	0.15	1.19	0.17	0.98
FNSP	0.35/9	0.09	1.21	0.11	0.99	0.12	0.83
TT	0.35/0.234	6.21	49.80	6.61	43.07	7.23	37.28
SS	0.35/0.234	2.23	31.94	2.83	28.35	3.55	25.19
FF	0.35/0.234	12.67	67.98	12.31	57.48	12.37	48.60
SNFP	0.35/0.234	9.15	58.87	9.25	50.32	9.66	43.03
FNSP	0.35/0.234	3.90	40.84	4.46	35.79	5.18	31.40

Table 5-10 PMOS ID current density at $V_{GS} = -1.8V$, $V_{DS} = -0.9V$ and $V_{GS} = -1.8V$, $V_{DS} = -1.8V$.

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IDSAT ($\mu A/\mu m$)	-VDS	0.9V	1.8V	0.9V	1.8V	0.9V	1.8V
TT	9/9	0.34	3.85	0.38	3.21	0.43	2.70
SS	9/9	0.22	3.26	0.27	2.73	0.32	2.32
FF	9/9	0.49	4.45	0.52	3.68	0.56	3.07
SNFP	9/9	0.41	4.15	0.44	3.44	0.49	2.88
FNSP	9/9	0.27	3.55	0.32	2.97	0.37	2.52
TT	9/0.234	18.82	148.17	21.38	134.76	24.52	123.22
SS	9/0.234	9.00	111.20	11.65	103.02	14.89	96.25
FF	9/0.234	33.00	186.29	34.50	166.48	36.66	149.14
SNFP	9/0.234	25.48	167.16	27.65	150.61	30.43	136.21
FNSP	9/0.234	13.30	129.44	16.00	118.89	19.29	109.98
TT	0.35/9	0.34	3.80	0.37	3.11	0.41	2.58
SS	0.35/9	0.21	3.10	0.24	2.56	0.29	2.14
FF	0.35/9	0.51	4.51	0.53	3.66	0.55	3.01
SNFP	0.35/9	0.42	4.16	0.44	3.39	0.48	2.79
FNSP	0.35/9	0.27	3.44	0.30	2.84	0.34	2.36
TT	0.35/0.234	17.74	142.28	18.88	123.05	20.65	106.53
SS	0.35/0.234	6.38	91.24	8.08	81.01	10.13	71.98
FF	0.35/0.234	36.19	194.24	35.17	164.22	35.35	138.87
SNFP	0.35/0.234	26.15	168.20	26.44	143.76	27.60	122.95
FNSP	0.35/0.234	11.16	116.69	12.76	102.26	14.80	89.73

Table 5-11 PMOS V_{TLIN} .

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-VTH (V)	-VDS	0.1V	1.8V	0.1V	1.8V	0.1V	1.8V
TT	9/9	0.579	0.579	0.498	0.497	0.396	0.394
SS	9/9	0.639	0.638	0.558	0.557	0.457	0.456
FF	9/9	0.520	0.519	0.438	0.437	0.334	0.333
SNFP	9/9	0.550	0.549	0.468	0.467	0.365	0.364
FNSP	9/9	0.609	0.608	0.528	0.527	0.427	0.425
TT	9/0.234	0.521	0.496	0.438	0.410	0.334	0.300
SS	9/0.234	0.641	0.615	0.558	0.530	0.456	0.422
FF	9/0.234	0.402	0.377	0.318	0.290	0.212	0.177
SNFP	9/0.234	0.462	0.436	0.378	0.349	0.273	0.238
FNSP	9/0.234	0.581	0.557	0.498	0.471	0.395	0.362
TT	0.35/9	0.571	0.571	0.494	0.493	0.397	0.396
SS	0.35/9	0.639	0.638	0.563	0.562	0.468	0.466
FF	0.35/9	0.503	0.503	0.425	0.424	0.327	0.326
SNFP	0.35/9	0.537	0.537	0.460	0.459	0.362	0.361
FNSP	0.35/9	0.605	0.604	0.528	0.527	0.433	0.431
TT	0.35/0.234	0.520	0.497	0.440	0.414	0.339	0.307
SS	0.35/0.234	0.678	0.654	0.600	0.573	0.503	0.471
FF	0.35/0.234	0.362	0.340	0.279	0.254	0.173	0.141
SNFP	0.35/0.234	0.441	0.418	0.360	0.334	0.256	0.224
FNSP	0.35/0.234	0.598	0.576	0.520	0.495	0.421	0.390

Table 5-12 PMOS ID current at $V_{GS}=0$ and $V_{DS}=-0.1V$ or $V_{DS}=-1.8V$ (off current).

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IOFF (A)	-VDS	0.1V	1.8V	0.1V	1.8V	0.1V	1.8V
TT	9/9	5.55E-15	2.74E-13	1.08E-13	3.11E-13	2.76E-11	2.82E-11
SS	9/9	2.22E-16	7.11E-15	1.99E-14	2.66E-14	9.01E-12	9.23E-12
FF	9/9	1.48E-12	7.42E-11	2.01E-12	5.58E-11	9.58E-11	1.37E-10
SNFP	9/9	1.48E-12	7.42E-11	1.71E-12	5.55E-11	5.29E-11	9.40E-11
FNSP	9/9	3.33E-16	7.11E-15	4.36E-14	4.97E-14	1.52E-11	1.55E-11
TT	9/0.234	4.54E-14	3.52E-13	3.02E-11	5.54E-11	4.43E-09	7.75E-09
SS	9/0.234	6.66E-16	7.11E-15	1.11E-12	2.01E-12	3.70E-10	6.36E-10
FF	9/0.234	4.81E-12	8.09E-11	8.42E-10	1.62E-09	5.15E-08	9.15E-08
SNFP	9/0.234	1.85E-12	7.50E-11	1.65E-10	3.62E-10	1.55E-08	2.77E-08
FNSP	9/0.234	4.55E-15	1.60E-14	5.57E-12	9.95E-12	1.25E-09	2.14E-09
TT	0.35/9	4.06E-16	1.39E-14	5.08E-15	1.60E-14	1.15E-12	1.18E-12
SS	0.35/9	2.05E-16	3.77E-15	9.12E-16	4.39E-15	3.78E-13	3.95E-13
FF	0.35/9	5.66E-14	2.83E-12	8.79E-14	2.14E-12	4.52E-12	6.13E-12
SNFP	0.35/9	5.66E-14	2.83E-12	6.85E-14	2.12E-12	2.32E-12	3.89E-12
FNSP	0.35/9	2.05E-16	3.77E-15	1.96E-15	5.50E-15	6.18E-13	6.38E-13
TT	0.35/0.234	2.22E-15	1.74E-14	1.21E-12	1.98E-12	1.66E-10	2.66E-10
SS	0.35/0.234	2.05E-16	3.77E-15	1.50E-14	2.79E-14	5.95E-12	9.42E-12
FF	0.35/0.234	7.07E-13	4.09E-12	1.02E-10	1.71E-10	4.46E-09	7.37E-09
SNFP	0.35/0.234	9.43E-14	3.03E-12	1.14E-11	2.11E-11	8.87E-10	1.46E-09
FNSP	0.35/0.234	2.95E-16	3.83E-15	1.29E-13	2.11E-13	3.05E-11	4.81E-11

*gmindc=1e-15

6 Shallow Trench Isolation (STI) or LOD Stress Effect

6.1 Simulation Results for NMOS at SA=SB=0.318 μm (SAREF=2.2 μm)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot (W_{\text{DES}}^{*0.9}) / (L_{\text{DES}}^{*0.9} + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

The simulation results for the NMOS characterization are summarized in the table below.

Table 6-1 Simulation results for NMOS

All parameters are given for $T=25^\circ\text{C}$, $V_{\text{BS}}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device ($W/L = 10\ \mu\text{m} / 10\ \mu\text{m}$)

V_{TLIN}	$V_{\text{DS}} = 0.1\text{ V}$	V	0.513
I_{ON}	$V_{\text{DS}}=V_{\text{GS}} = 1.8\text{ V}$	$\mu\text{A} / \mu\text{m}$	13.26

Wide and short channel device ($W/L = 10\ \mu\text{m} / 0.246\ \mu\text{m}$)

V_{TLIN}	$V_{\text{DS}} = 0.1\text{ V}$	V	0.521
V_{TSAT}	$V_{\text{DS}} = 1.8\text{ V}$	V	0.457
I_{ON}	$V_{\text{DS}}=V_{\text{GS}} = 1.8\text{ V}$	$\mu\text{A}/\mu\text{m}$	341.77
I_{OFF}	$V_{\text{DS}} = 1.8\text{ V}, V_{\text{GS}}=0\text{ V}$	$\text{A}/\mu\text{m}$	$2.8\text{E}-12^*$

Narrow and long channel device ($W/L = 0.35\ \mu\text{m} / 10\ \mu\text{m}$)

V_{TLIN}	$V_{\text{DS}} = 0.1\text{ V}$	V	0.513
V_{TSAT}	$V_{\text{DS}} = 1.8\text{ V}$	V	0.510
I_{ON}	$V_{\text{DS}}=V_{\text{GS}} = 1.8\text{ V}$	$\mu\text{A}/\mu\text{m}$	11.38

Narrow and short channel device ($W/L = 0.35\ \mu\text{m} / 0.246\ \mu\text{m}$)

V_{TLIN}	$V_{\text{DS}} = 0.1\text{ V}$	V	0.508
V_{TSAT}	$V_{\text{DS}} = 1.8\text{ V}$	V	0.442
I_{ON}	$V_{\text{DS}}=V_{\text{GS}} = 1.8\text{ V}$	$\mu\text{A}/\mu\text{m}$	404.89
I_{OFF}	$V_{\text{DS}} = 1.8\text{ V}, V_{\text{GS}}=0\text{ V}$	$\text{A}/\mu\text{m}$	$3.5\text{E}-12^*$

*gmindc=1e-15

6.1.1 Model verification plots for NMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D=300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C . The dimension is after 90% shrink and 12nm sizing channel length back.

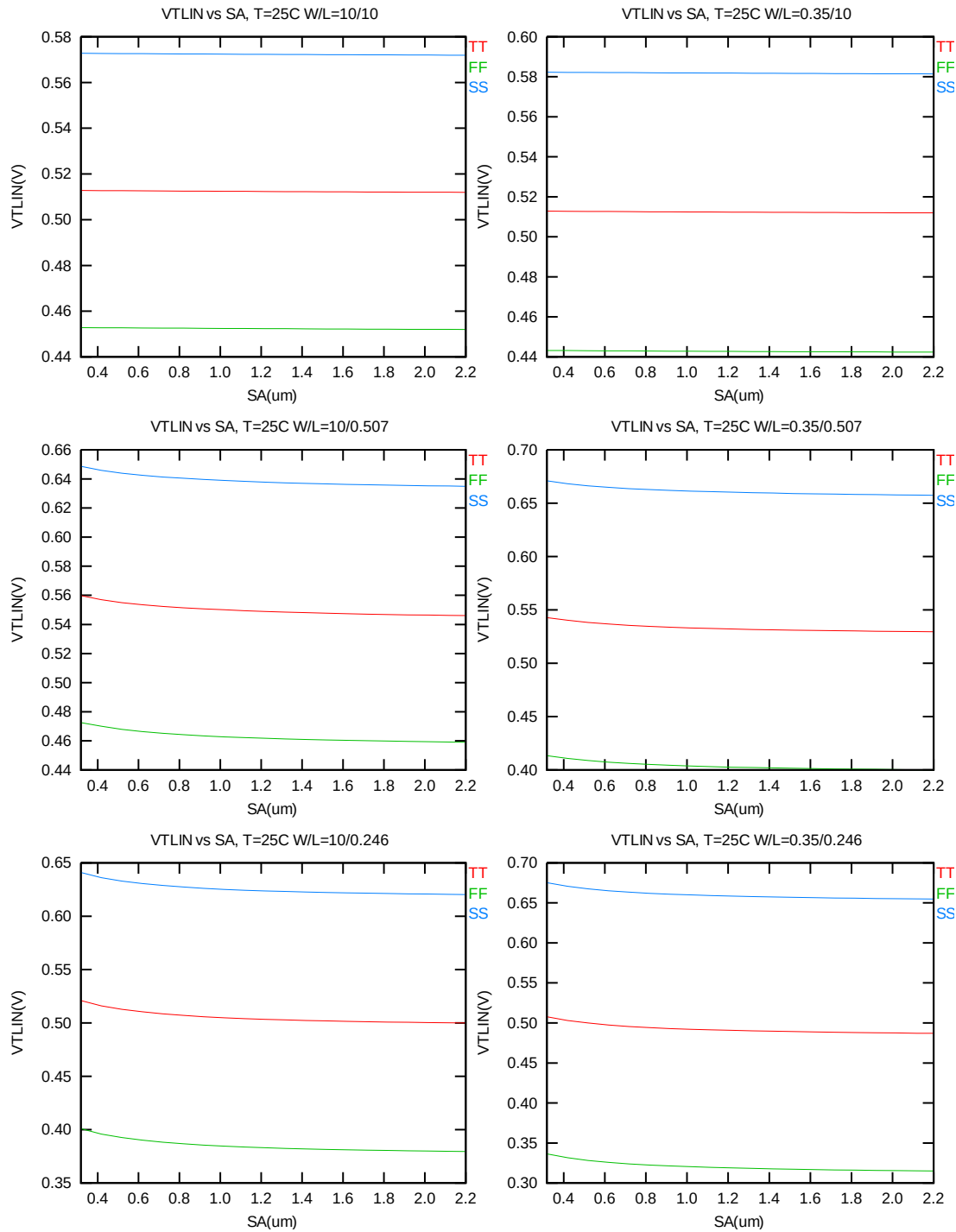


Figure 6-1 V_T vs. S_A for low drain bias ($V_{DS} = 0.1\text{V}$) for NMOS

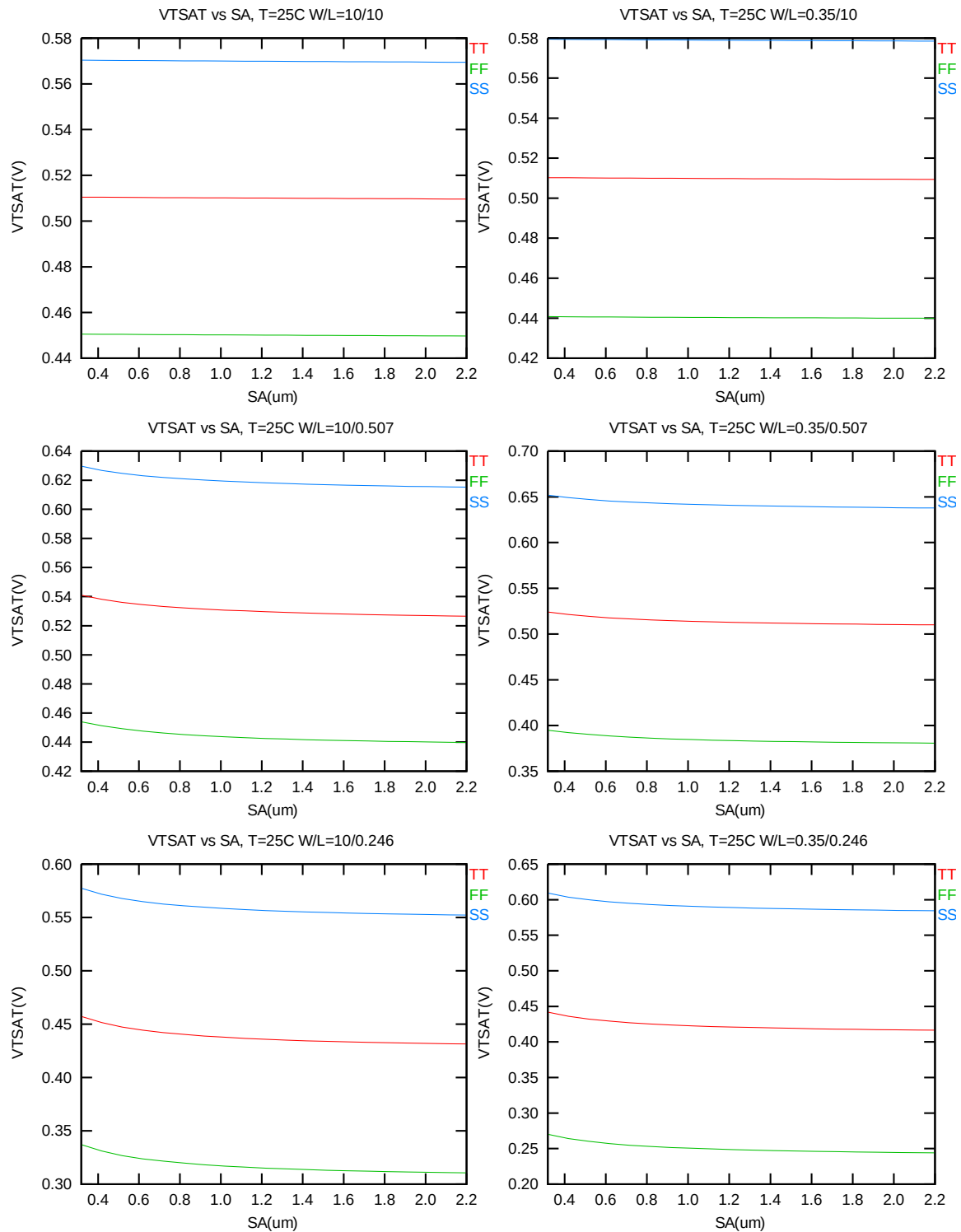


Figure 6-2 V_T vs. SA for high drain bias ($V_{DS} = 1.8V$) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}=1.8V$ for $V_{BS}=0$ at 25C.

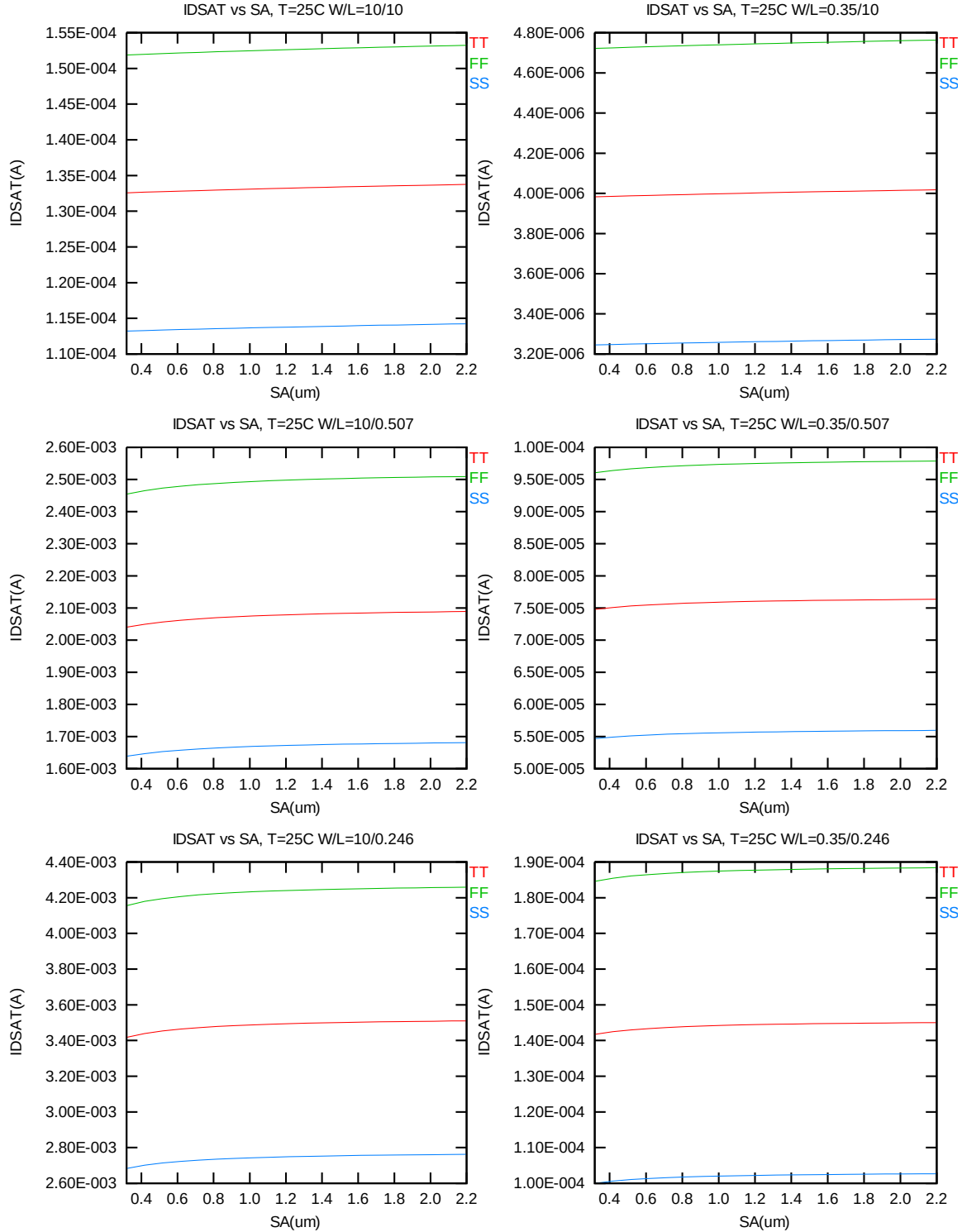


Figure 6-3 Saturation current I_{ON} vs. SA for NMOS

6.2 Simulation Results for PMOS at SA=0.318 um (SAREF=2.2um)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70nA * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu m)$. The dimension is after 90% shrink and 12nm sizing channel length back.

The simulation results for the PMOS characterization are summarized in the table below.

Table 6-2 Simulation results for PMOS

All parameters are given for T=25 °C , $V_{BS}=0$ V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device (W/L= 10 μm / 10 μm)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.498
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	$\mu A / \mu m$	-2.92

Wide and short channel device (W/L= 10 μm / 0.246 μm)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.439
V_{TSAT}	$V_{DS} = -1.8$ V	V	-0.411
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	$\mu A / \mu m$	-143.57
I_{OFF}	$V_{DS} = -1.8$ V, $V_{GS}=0$ V	A/ μm	-5.8E-12*

Narrow and long channel device (W/L= 0.35 μm / 10 μm)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.494
V_{TSAT}	$V_{DS} = -1.8$ V	V	-0.493
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	$\mu A / \mu m$	-2.83

Narrow and short channel device (W/L= 0.35 μm / 0.246 μm)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.441
V_{TSAT}	$V_{DS} = -1.8$ V	V	-0.415
I_{ON}	$V_{DS}=V_{GS} = -1.8$ V	$\mu A / \mu m$	-130.40
I_{OFF}	$V_{DS} = -1.8$ V, $V_{GS}=0$ V	A/ μm	-5.3E-12 *

*gmindc=1e-15

6.2.1 Model verification plots for PMOS

The L and W in the plots refer to LDES and WDES respectively

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C . The dimension is after 90% shrink and 12nm sizing channel length back.

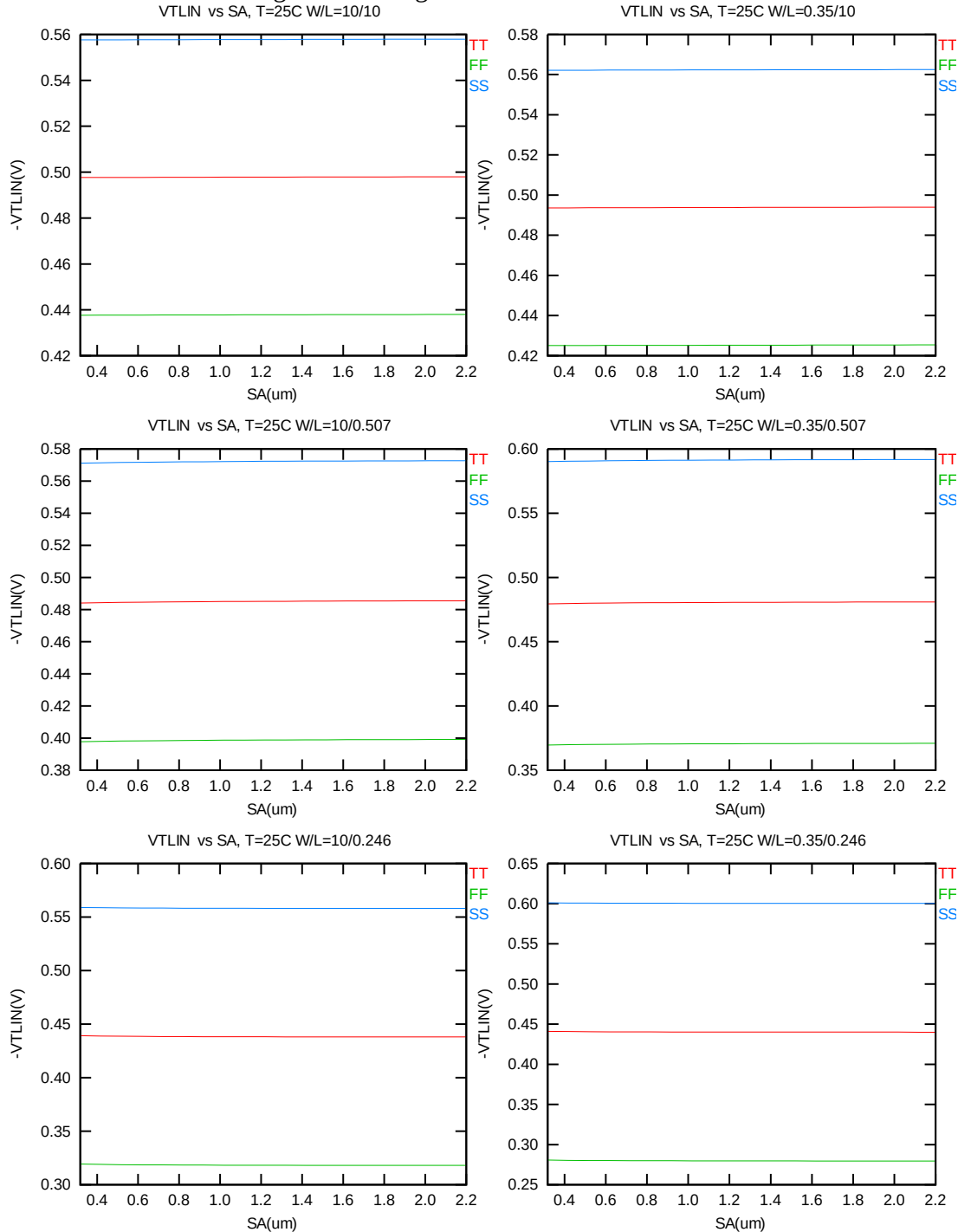


Figure 6-4 V_T vs. SA for low drain bias ($V_{DS} = -0.1\text{V}$) for PMOS

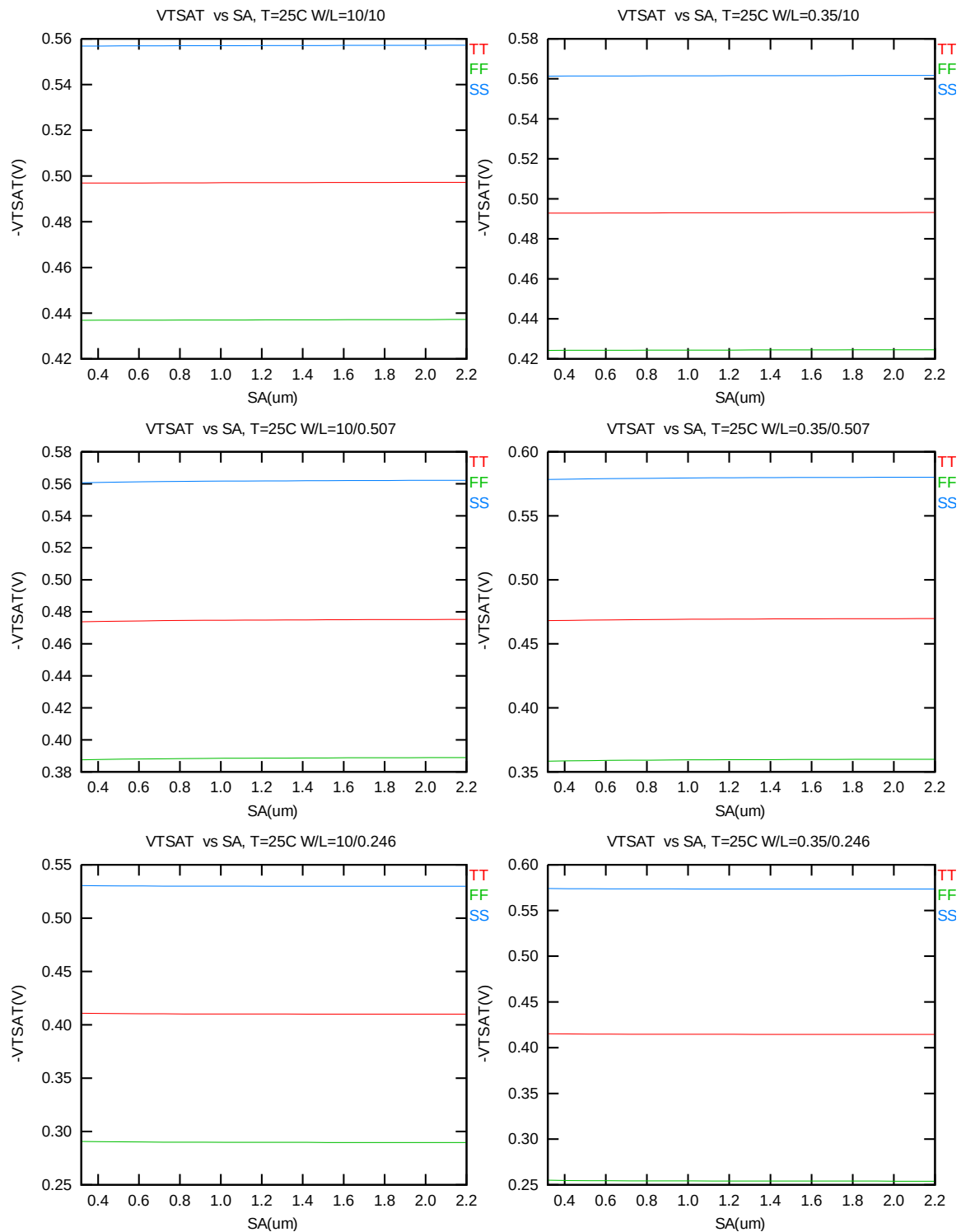


Figure 6-5 V_T vs. S_A for high drain bias ($V_{DS} = -1.8V$) for PMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}= -1.8V$ for $V_{BS}=0$ at 25C.

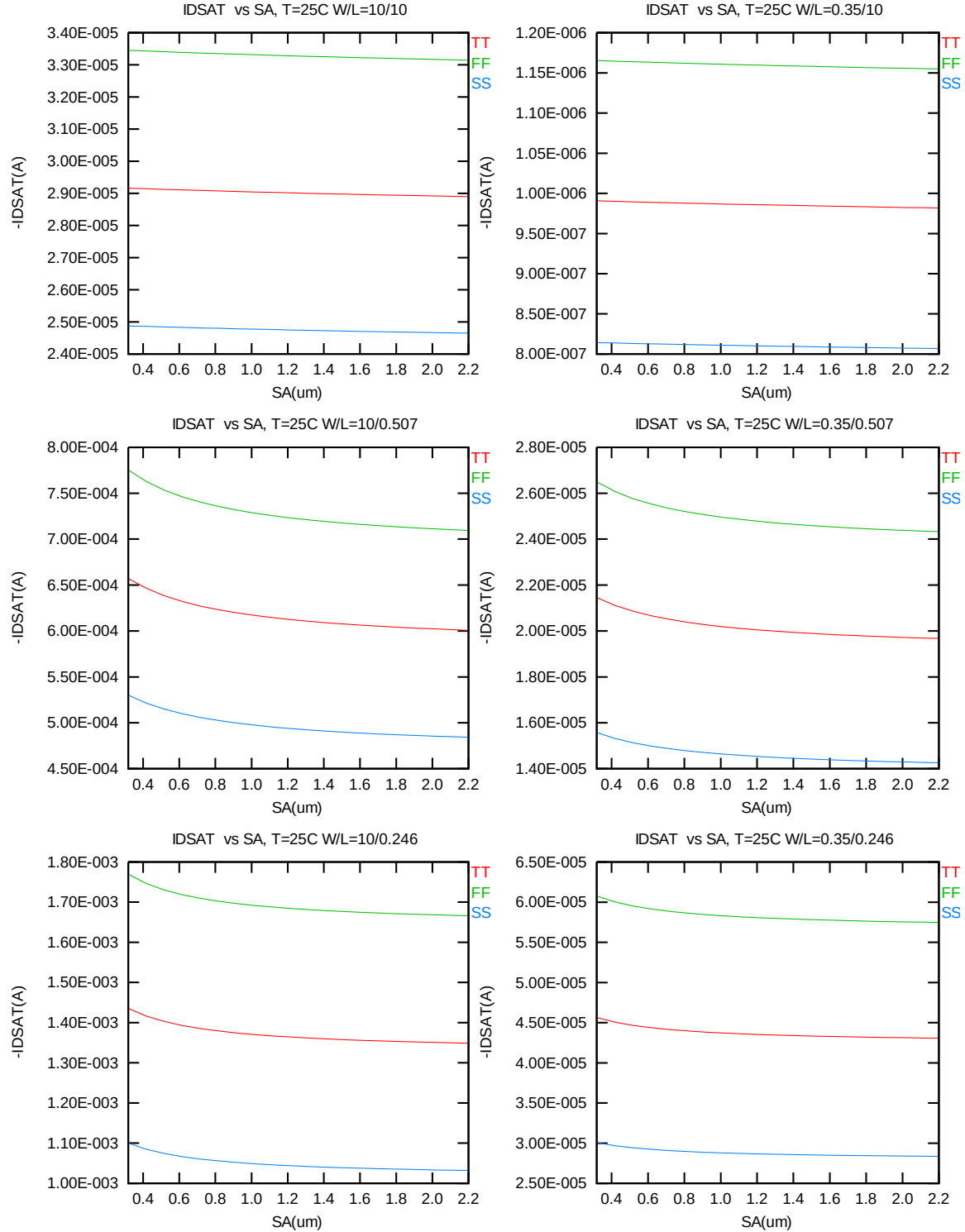


Figure 6-6 Saturation current I_{ON} vs. SA for PMOS

6.3 V_T and I_{ON} Variation v. SA for both N/PMOS

Delta V_T is defined as $V_T(SA) - V_T(2.2\mu m)$ for NMOS and $-[V_T(SA) - V_T(2.2\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SA) - I_{ON}(2.2\mu m)]/I_{ON}(2.2\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SA

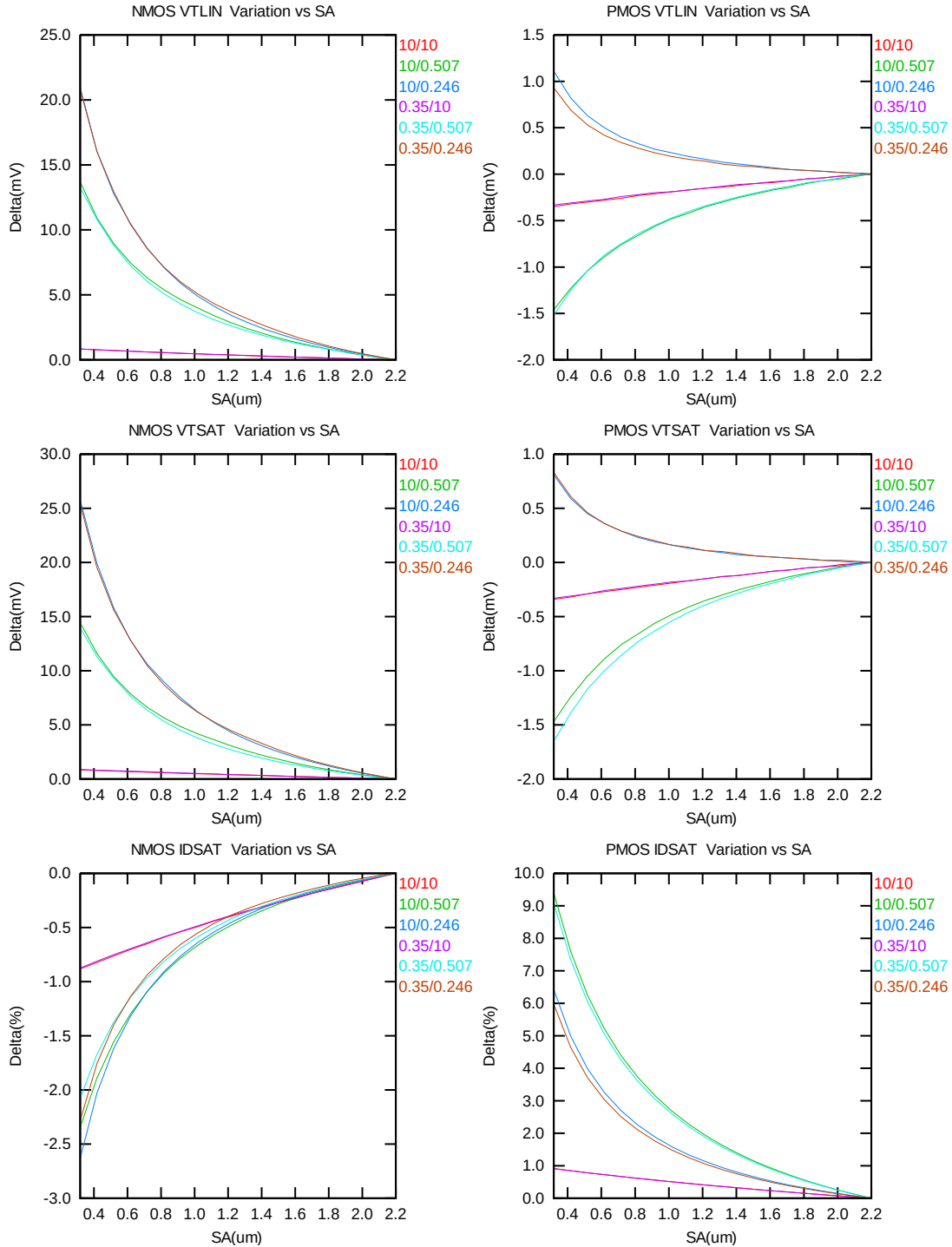


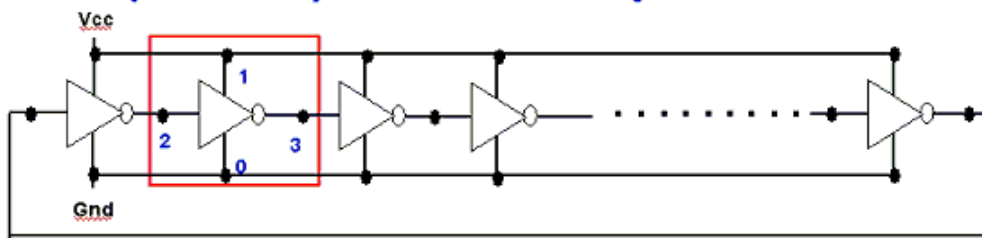
Figure 6-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

7 Dynamic Performance

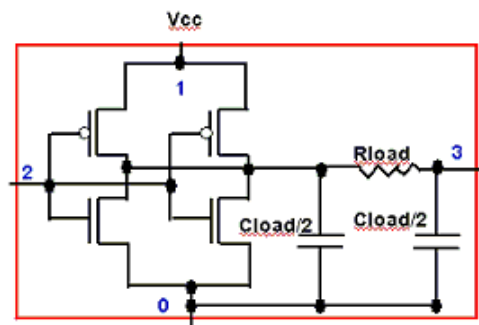
7.1 Test Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted.

R.O. (Inverter) circuit description



Ring Oscillator circuit



single stage R.O. circuit

Figure 7-1 Ring oscillator netlist description

7.2 Dynamic Performance for SA=0.318um

Table 7-1 Ring oscillator (inverter) structure description for SA=SB=0.318um

R.O. (Inverter) Structure description		
Contact size	0.144 um X 0.144 um	
Polysilicon gate to contact spacing	0.12 um	
Contact to diffusion edge spacing	0.054 um	
Without RC loading	R = 0 ohm	C = 0 fF
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	5 um X 1	
Channel length for n-ch MOSFET	0.246 um	
Channel width for p-ch MOSFET	6.5 um X 1	
Channel Length for p-ch MOSFET	0.246 um	

Table 7-2 Electrical parameters for ring oscillator (ps/stage) for SA=SB=0.318um

Parameter	Condition	Unit	Target	Centered Value
T_{pd}	$V_{cc} = 1.8V$	ps/stage	--	43.8

The simulation results at different temperatures and using different worst-case models are listed below.

Table 7-3 Ring oscillator simulation results (ps/stage) for SA=SB=0.318um

Inverter, R=0, C=0f, Fanout=1, Vcc=1.8V				Inverter, R=0, C=0f, Fanout=1, T=25C			
Temp(C)	TT	FF	SS	Vcc(V)	TT	FF	SS
-50	38.327	30.468	50.739	1.20	80.694	57.050	121.702
0	42.067	33.672	55.238	1.40	61.936	46.379	87.223
25	43.832	35.212	57.331	1.60	50.941	39.709	68.690
85	47.864	38.775	61.994	1.80	43.832	35.212	57.331
125	50.404	41.064	64.863	1.98	39.356	32.287	50.400

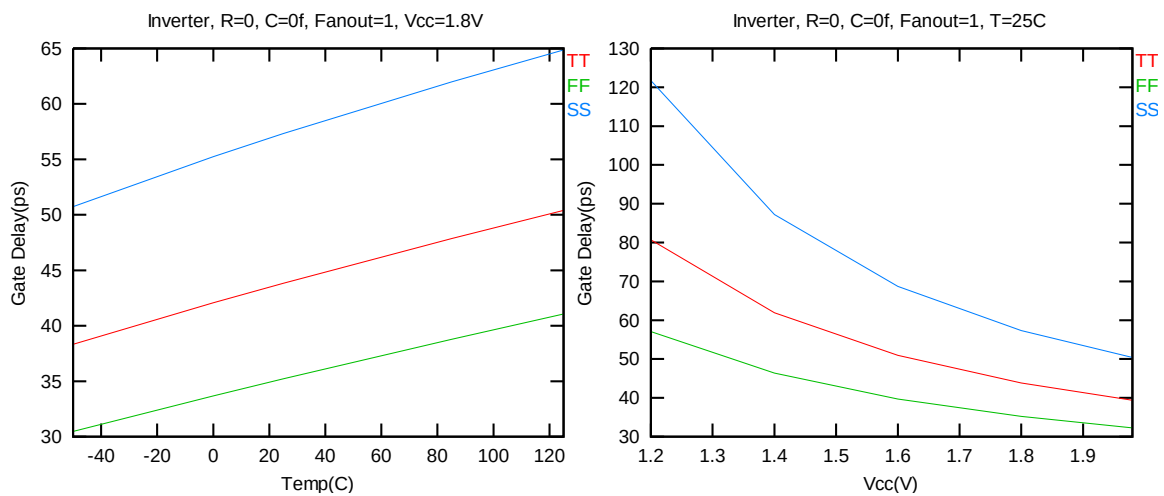
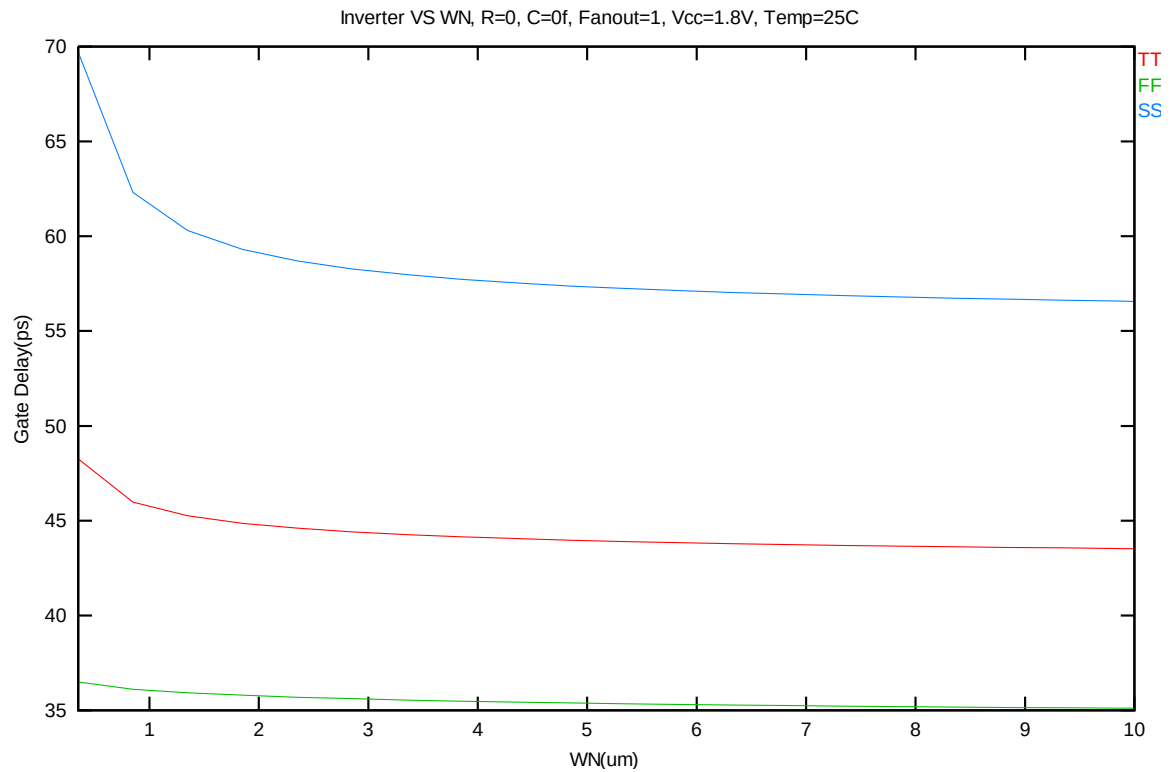


Figure 7-2 Ring oscillator (inverter) simulation for SA=SB=0.318um

The (Inverter) ring oscillator gate delay simulation vs. width for SA=SB=0.318um



**Figure 7-3 Ring oscillator (inverter) gate delay simulation vs. width for
SA=SB=0.318um**

8 Noise Characteristics

8.1 Verification Plots

NMOS, W=10um L=10um

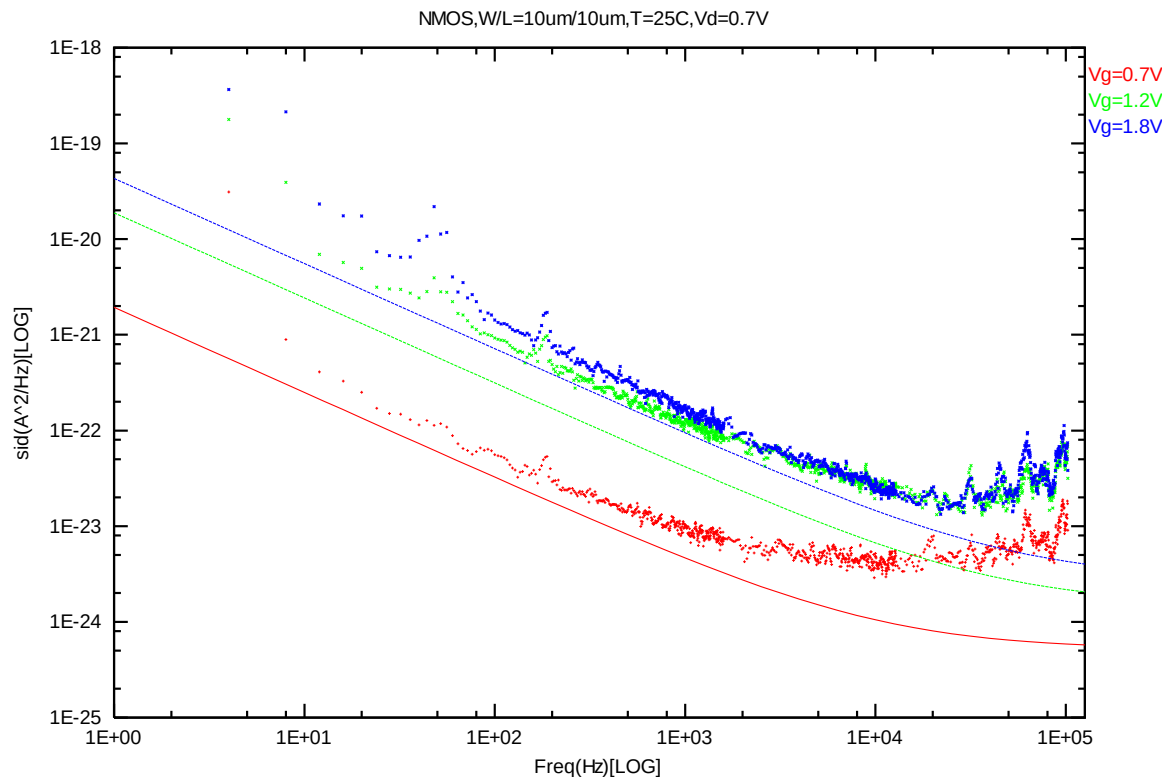


Figure 8-1 NMOS W/L=10um/10um @ Vd=0.7V

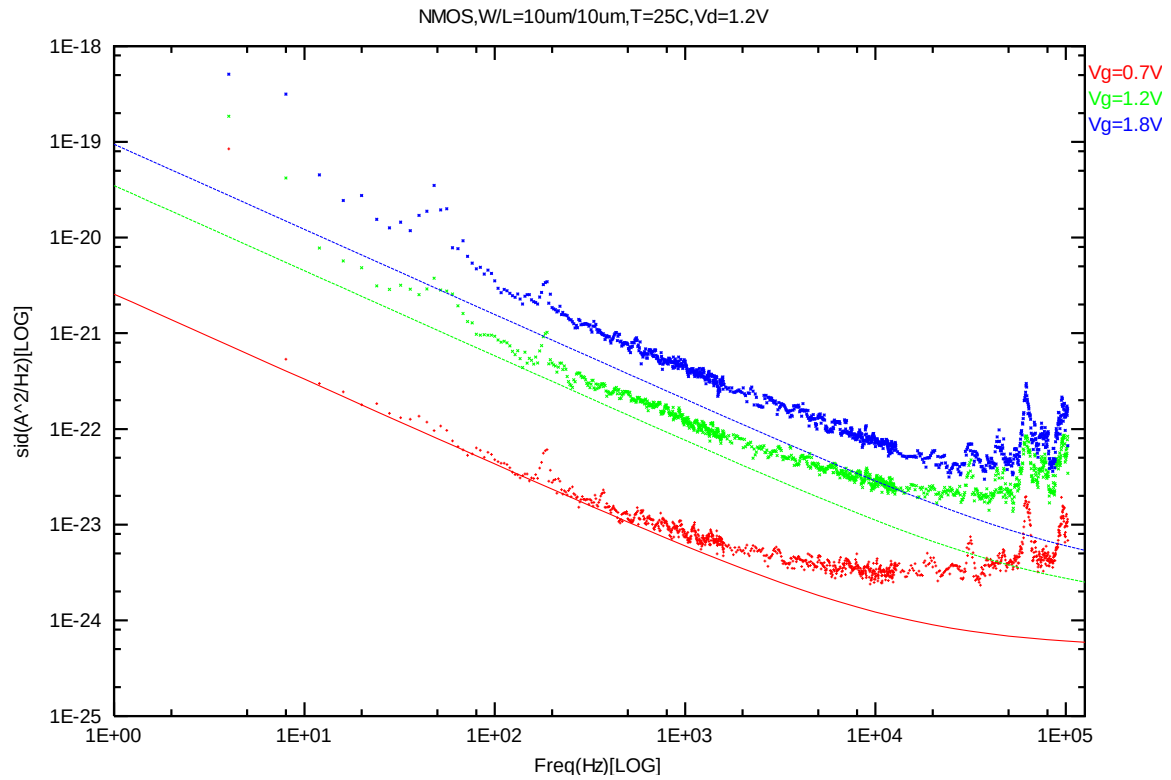


Figure 8-2 NMOS W/L=10um/10um @ Vd=1.2V

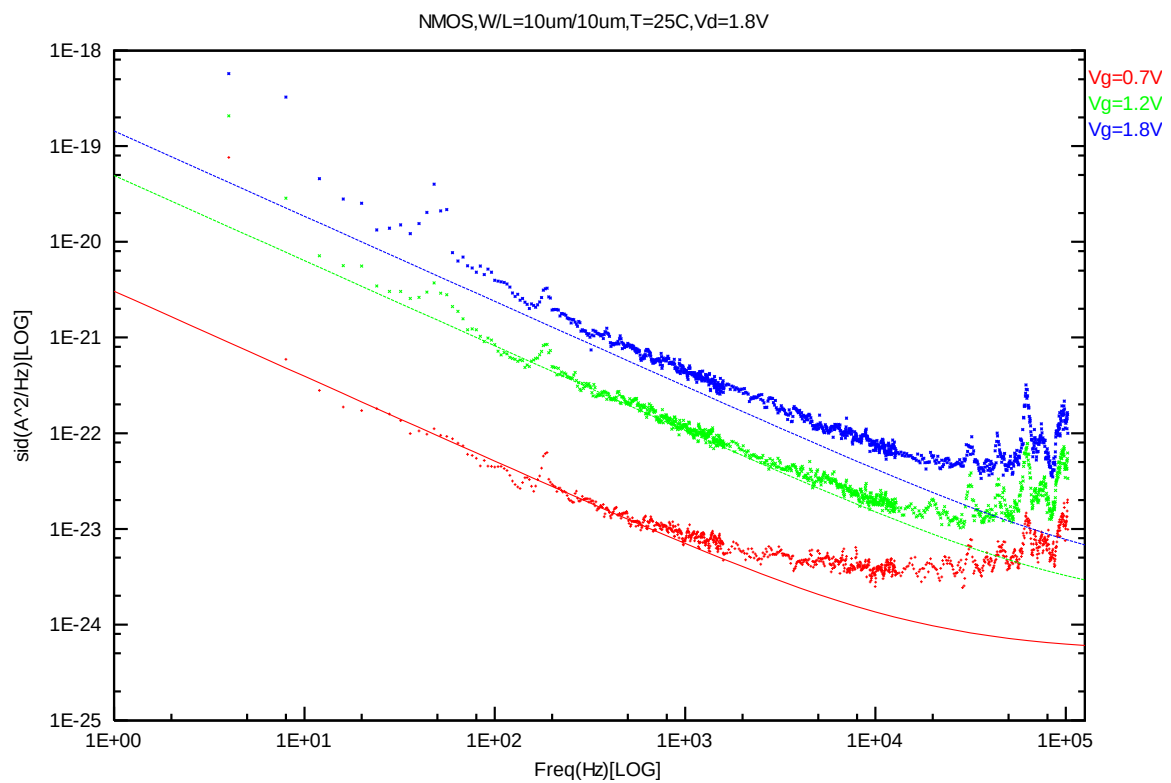


Figure 8-3 NMOS W/L=10um/10um @ Vd=1.8V

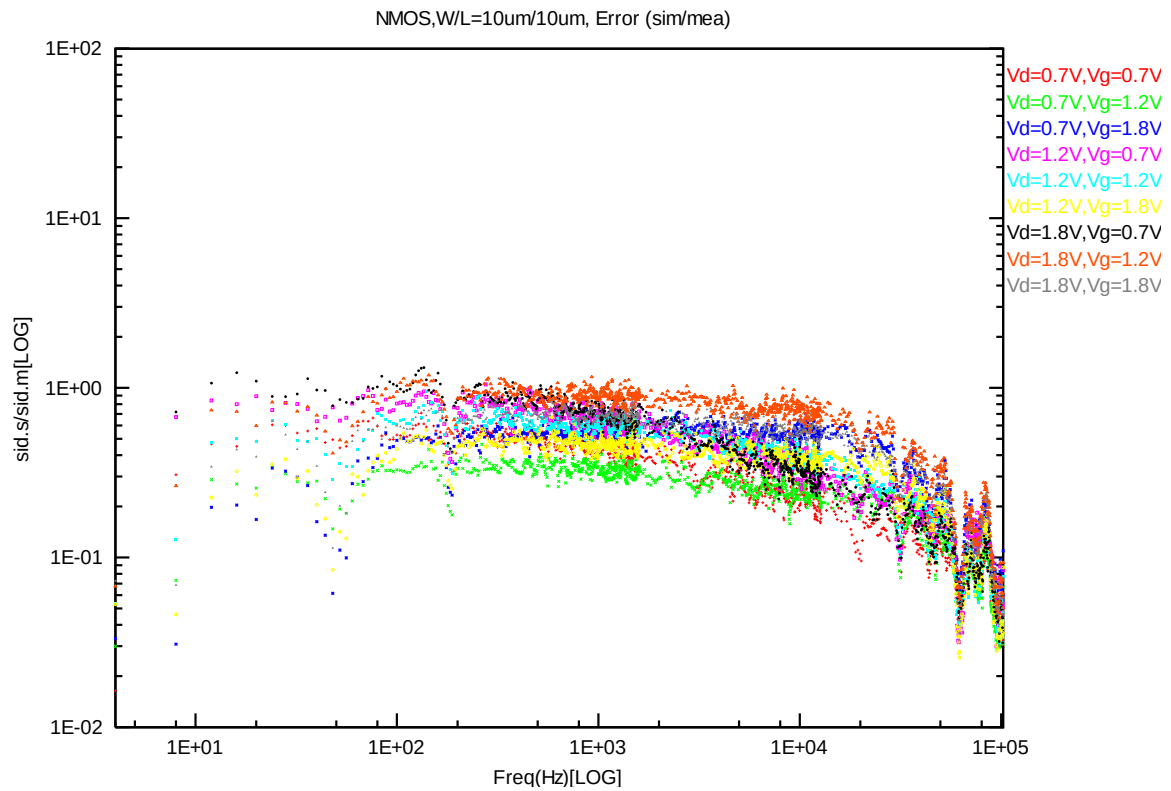


Figure 8-4 NMOS W/L=10um/10um error distribution plot

NMOS, W=10um L=1um

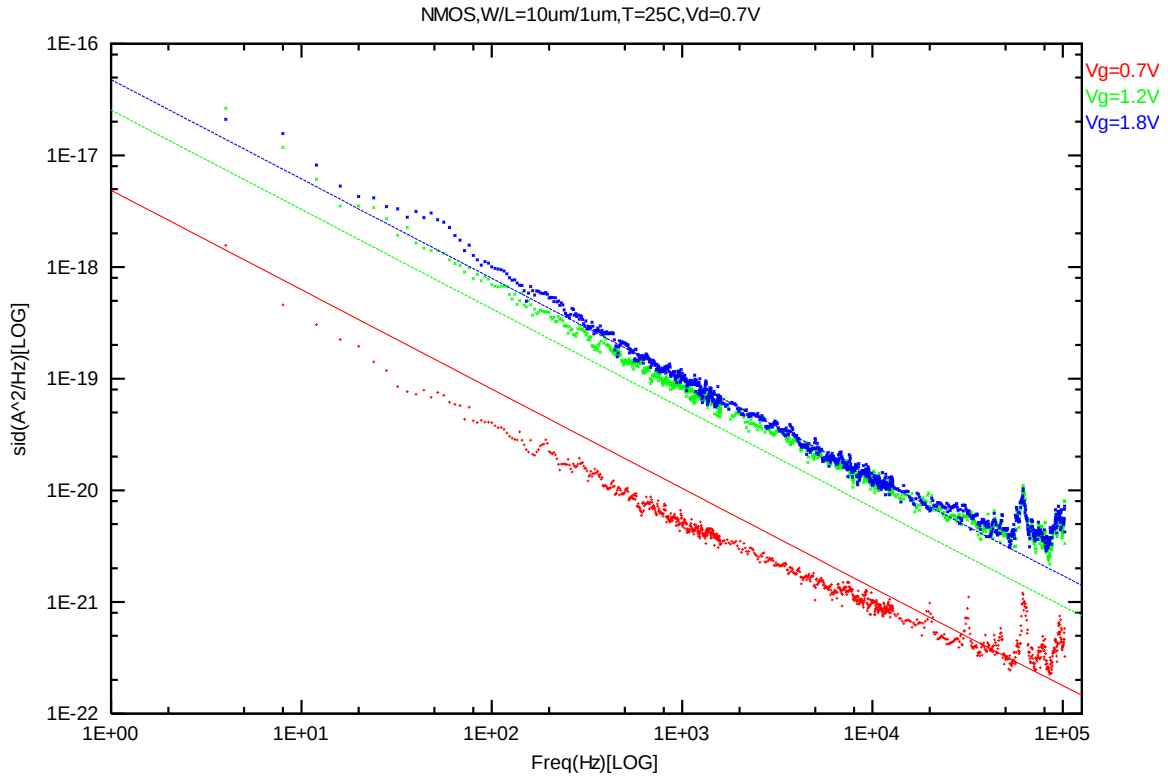


Figure 8-5 NMOS W/L=10um/1um @ Vd=0.7V

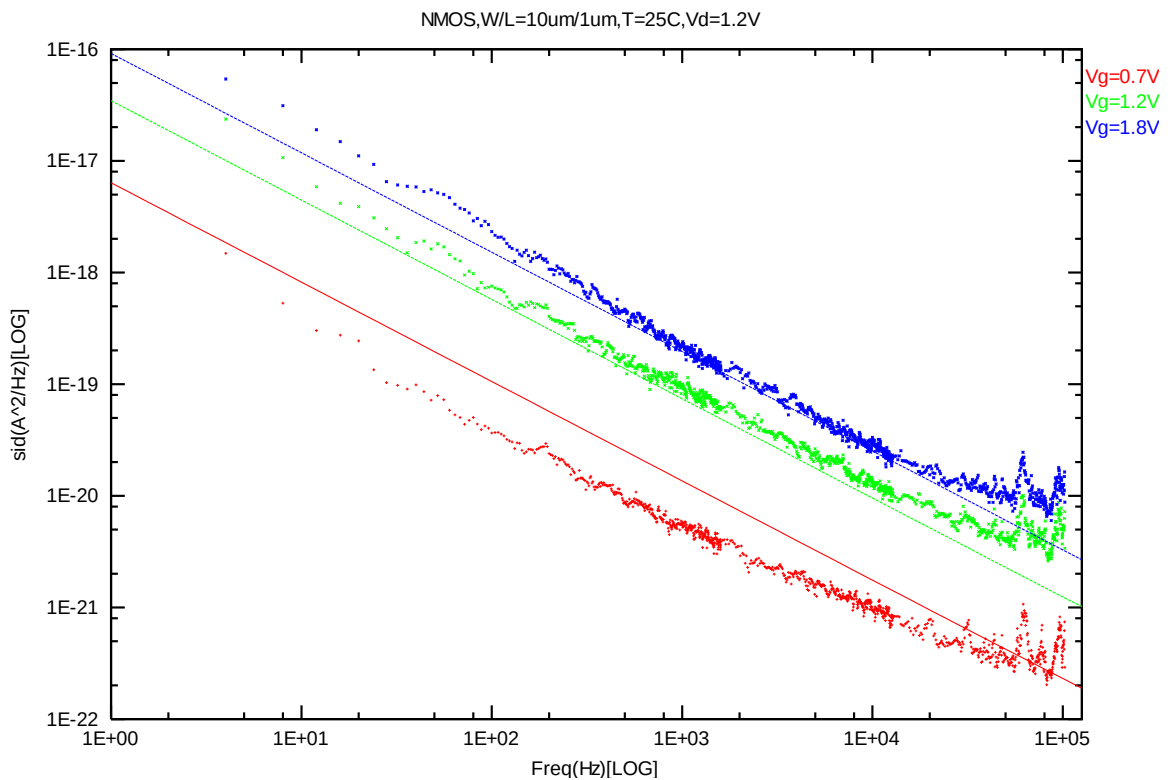
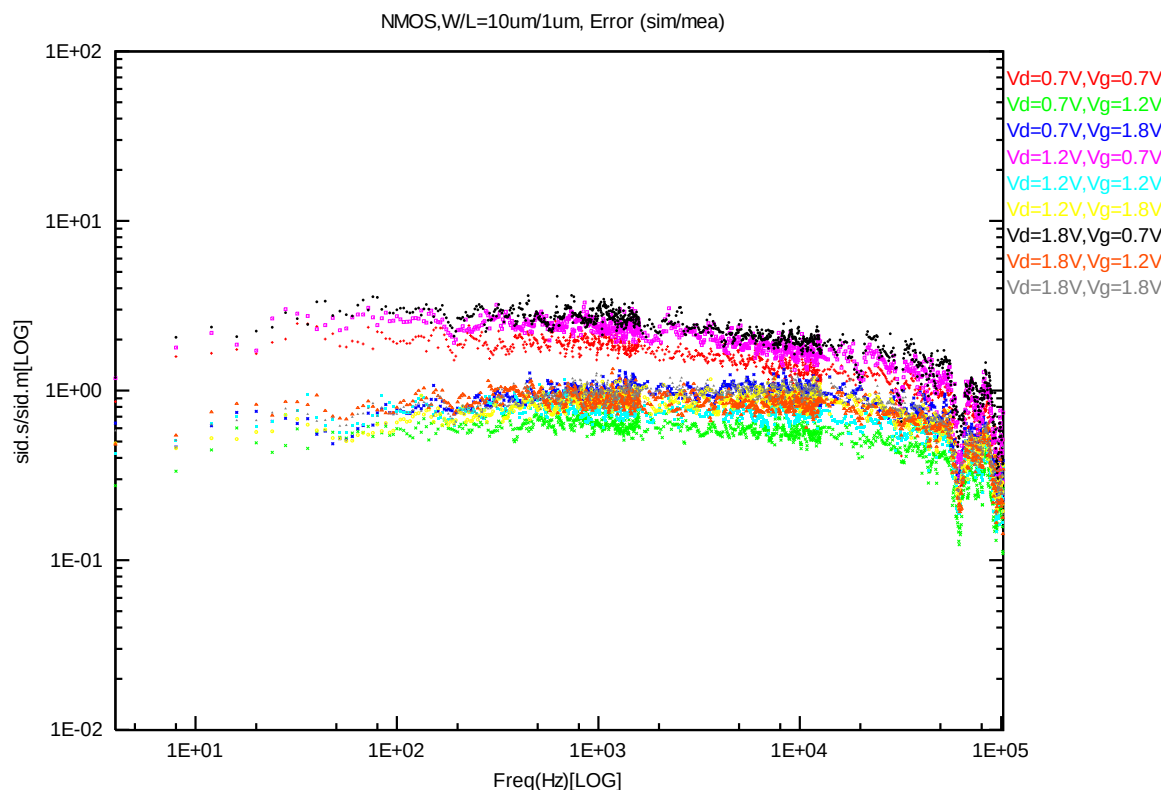
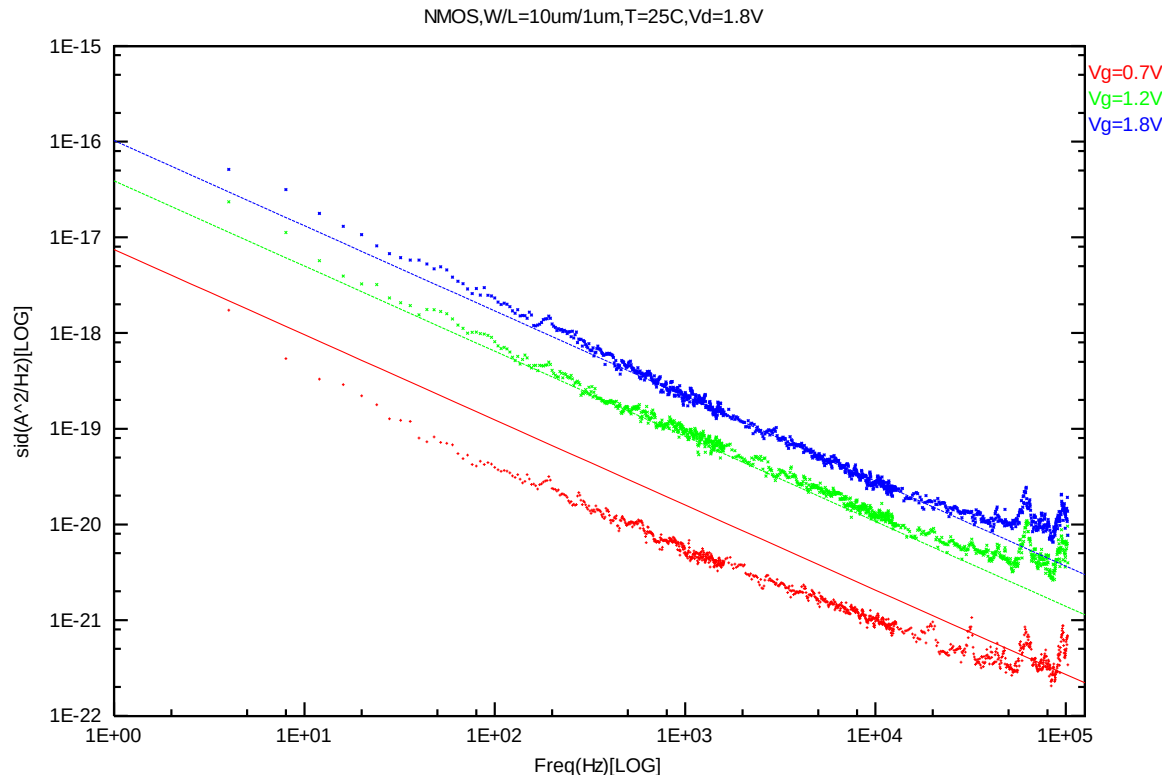


Figure 8-6 NMOS W/L=10um/1um @ Vd=1.2V



NMOS, W=10um L=0.507um

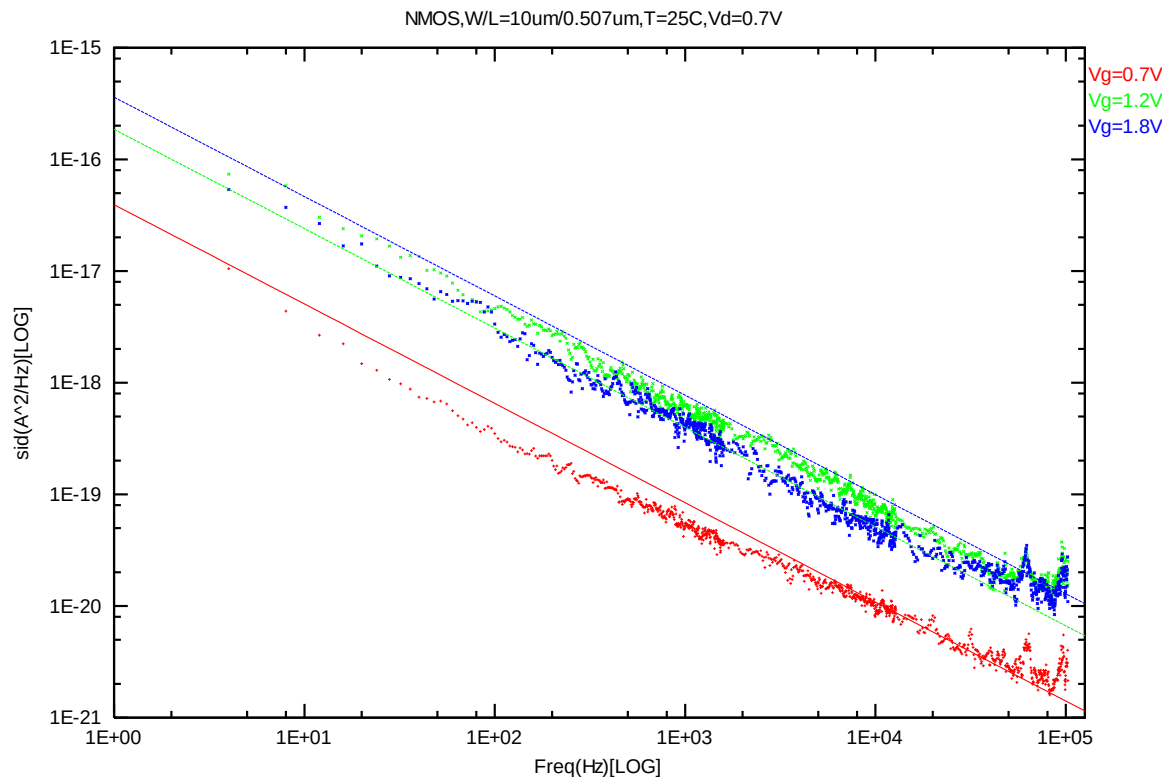


Figure 8-9 NMOS W/L=10um/0.507um @ Vd=0.7V

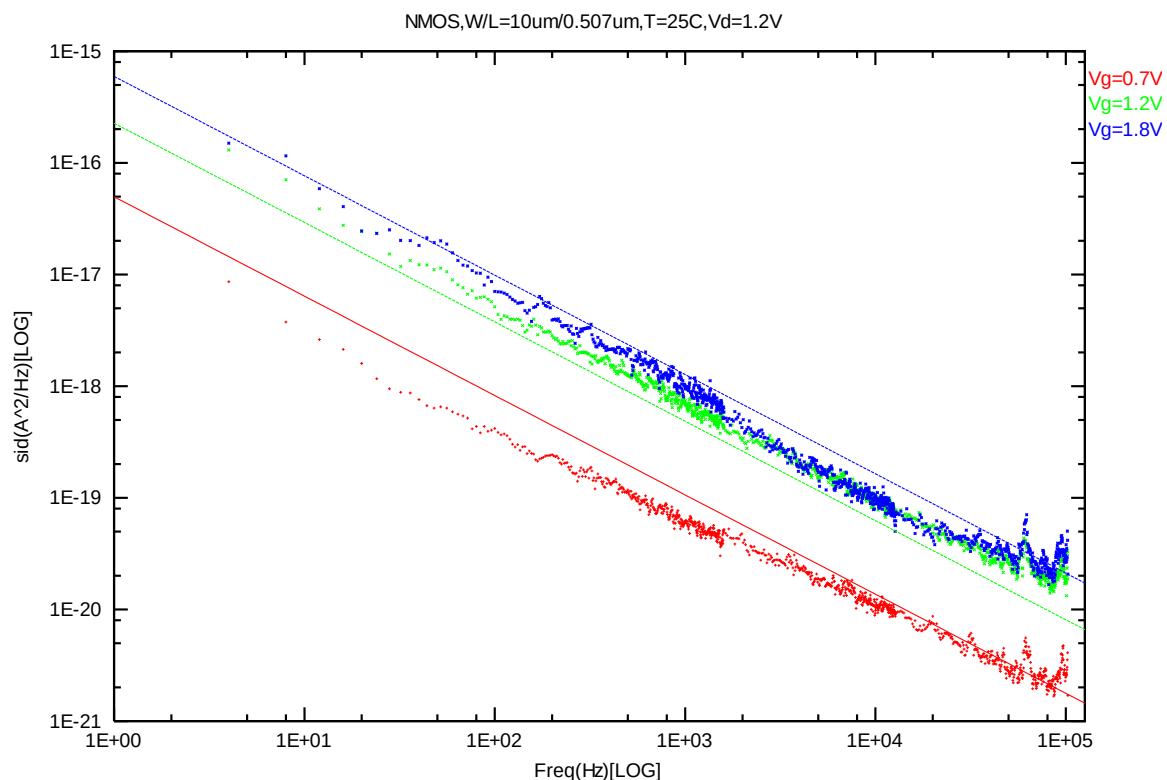


Figure 8-10 NMOS W/L=10um/0.507um @ Vd=1.2V

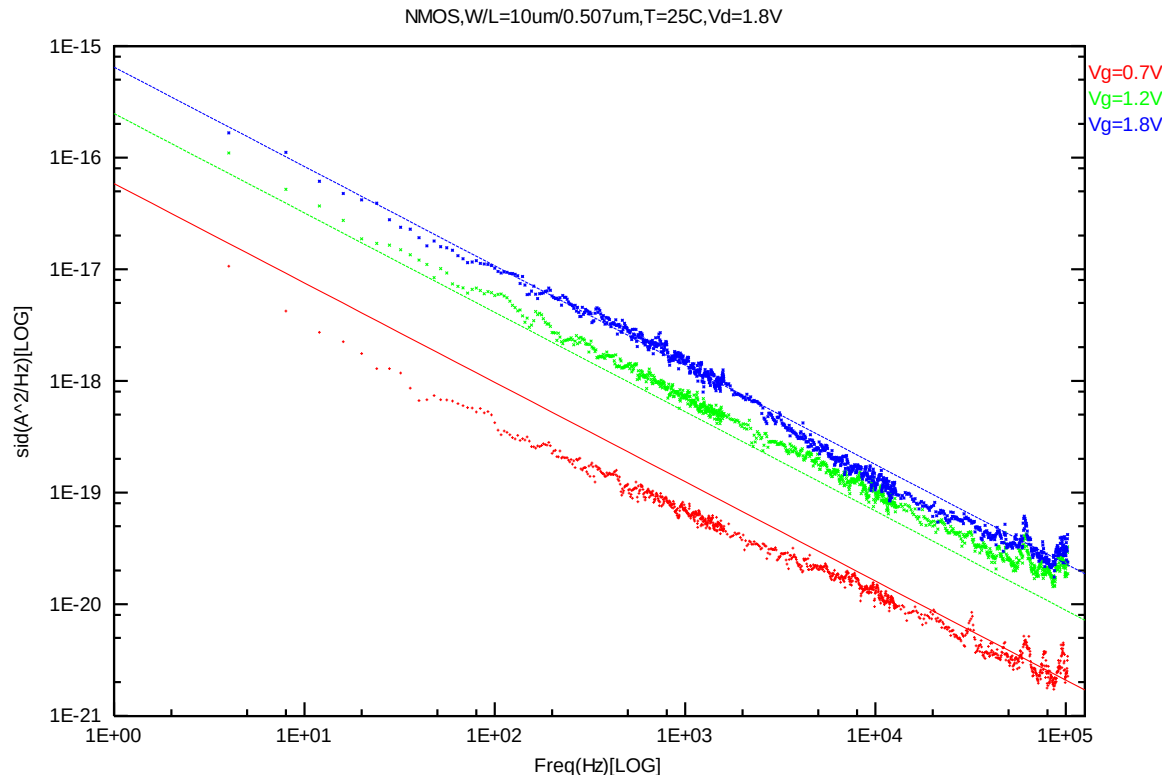


Figure 8-11 NMOS W/L=10um/0.507um @ Vd=1.8V

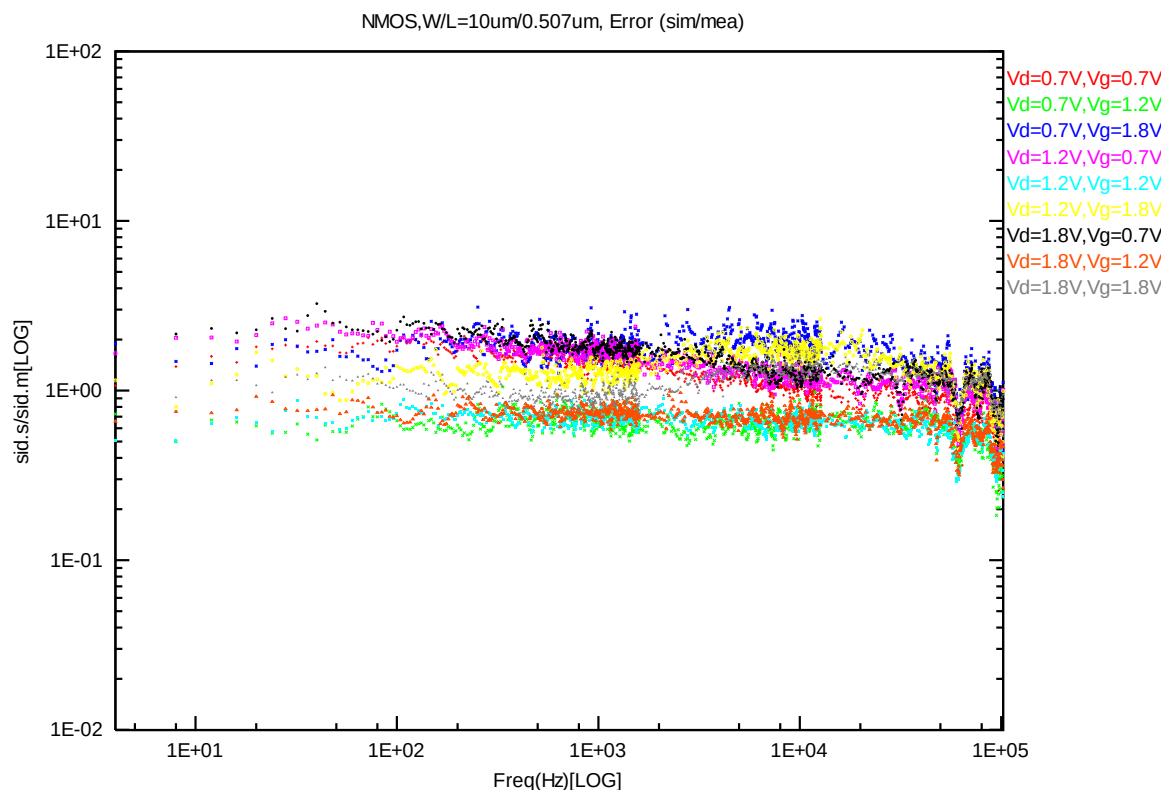


Figure 8-12 NMOS W/L=10um/0.507um error distribution plot

NMOS, W=10um L=0.417um

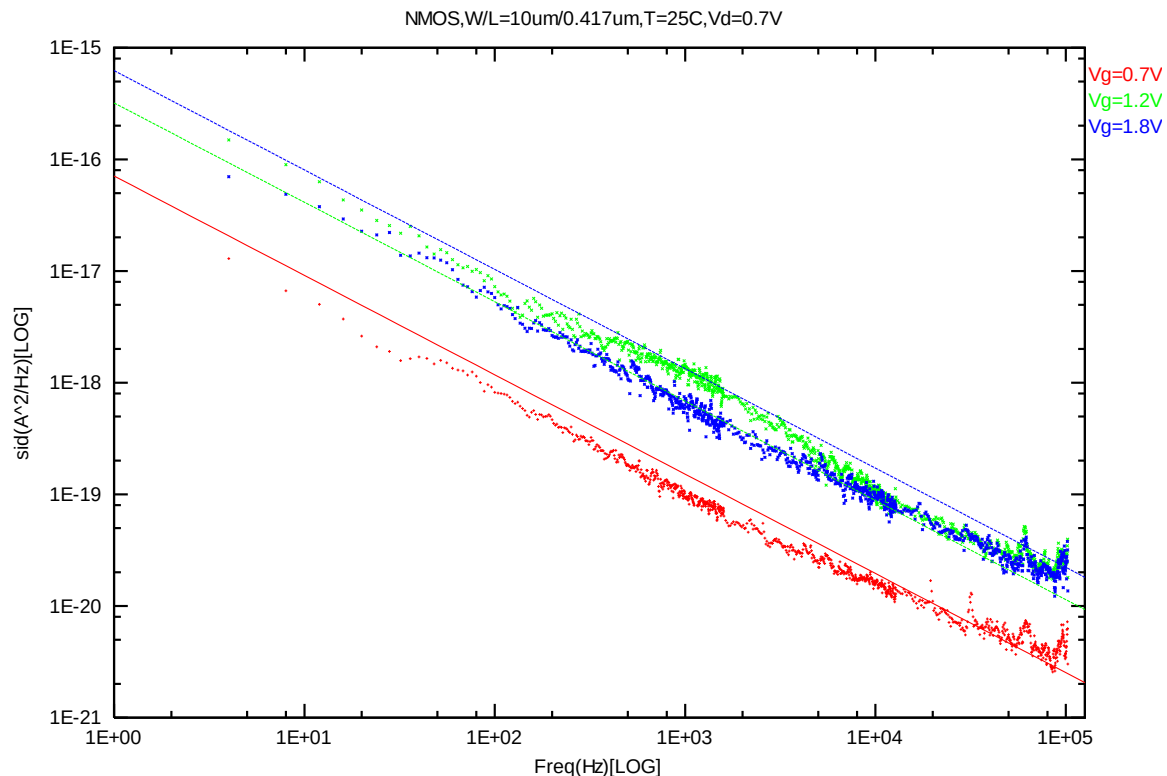


Figure 8-13 NMOS W/L=10um/0.417um @ Vd=0.7V

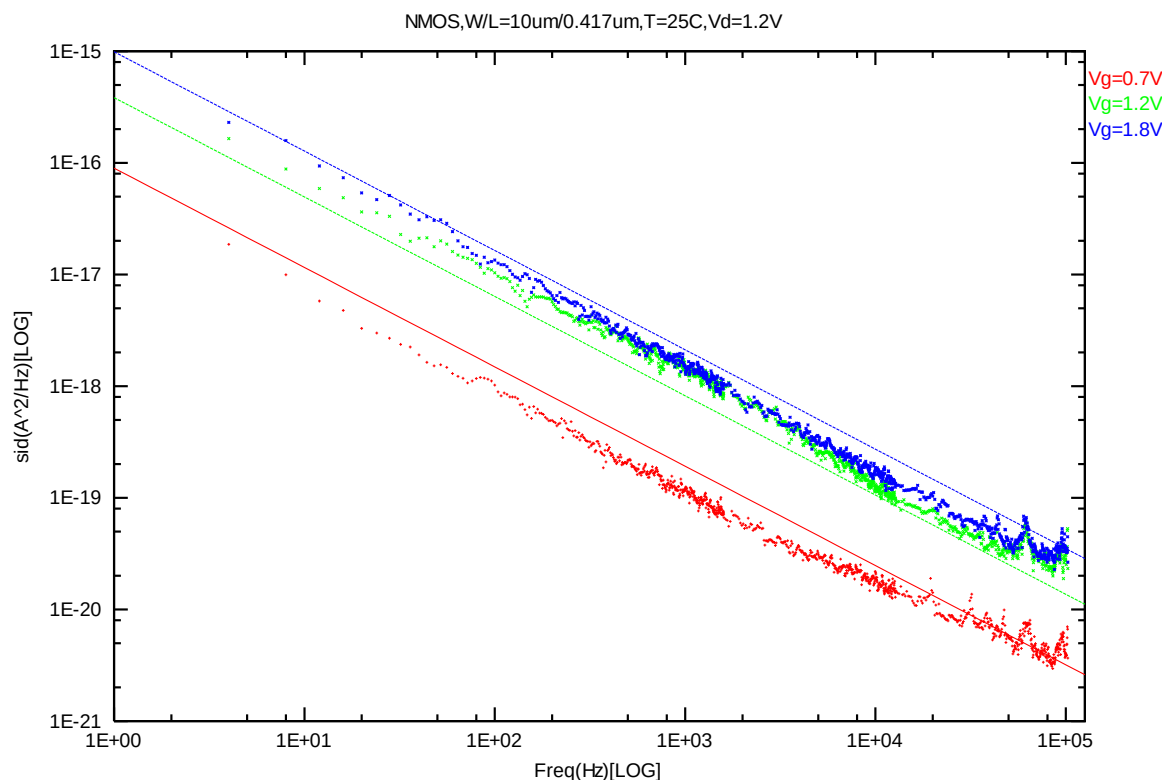


Figure 8-14 NMOS W/L=10um/0.417um @ Vd=1.2V

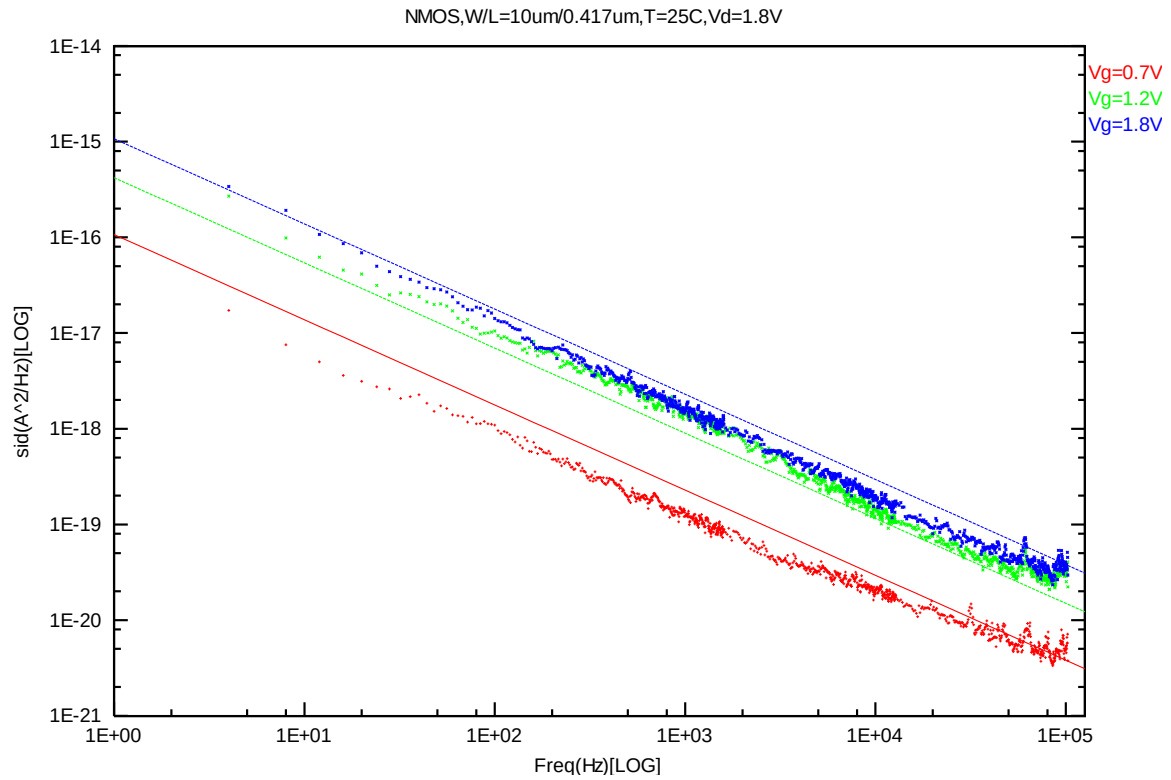


Figure 8-15 NMOS W/L=10um/0.417um @ Vd=1.8V

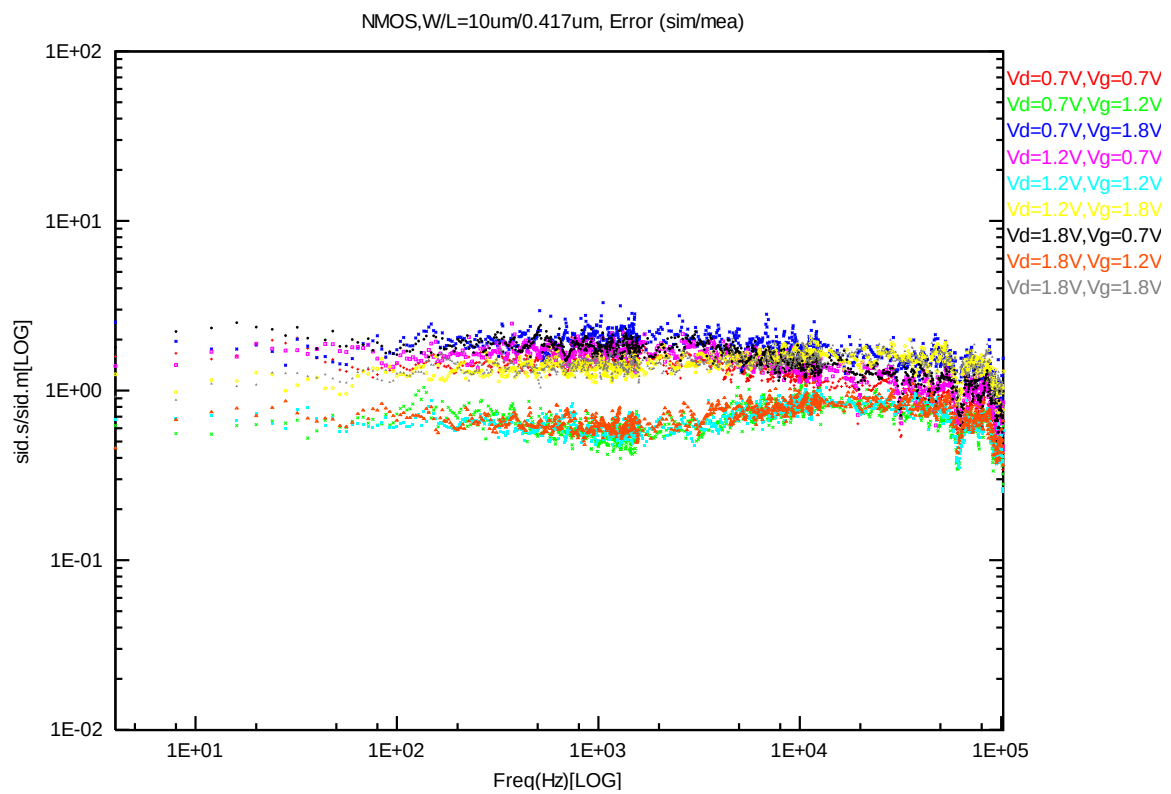


Figure 8-16 NMOS W/L=10um/0.417um error distribution plot

NMOS, W=10um L=0.246um

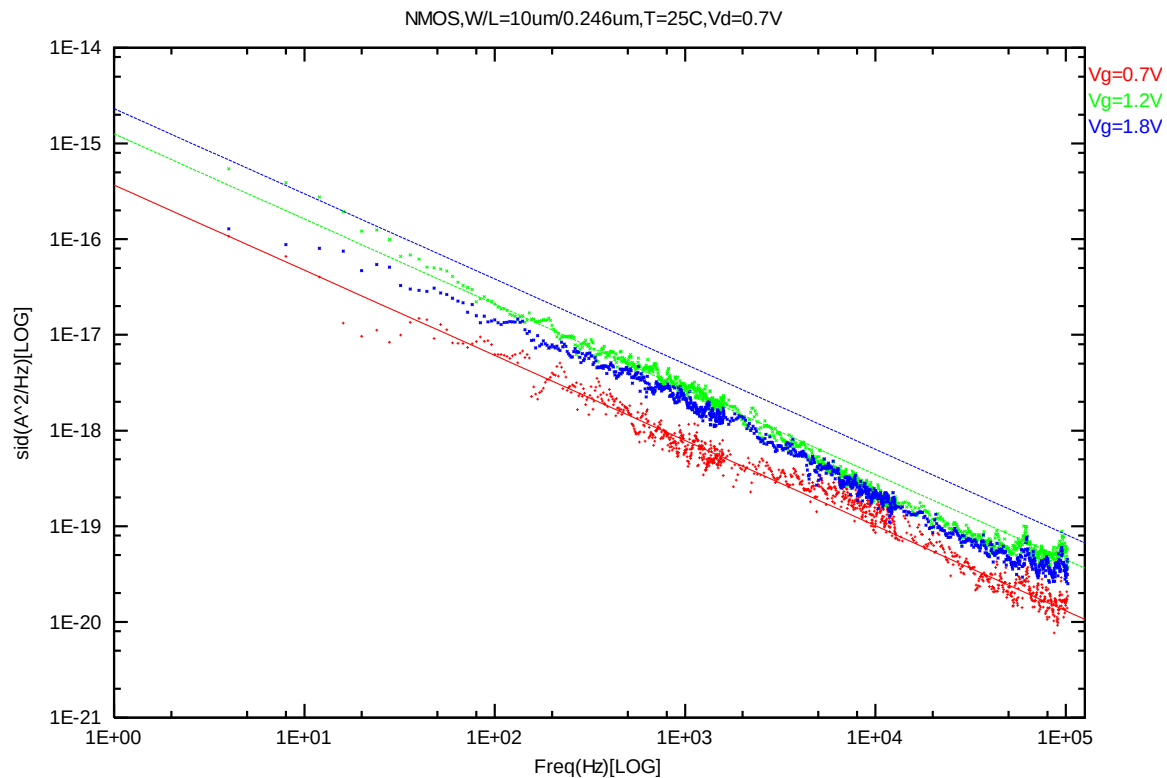


Figure 8-17 NMOS W/L=10um/0.246um @ Vd=0.7V

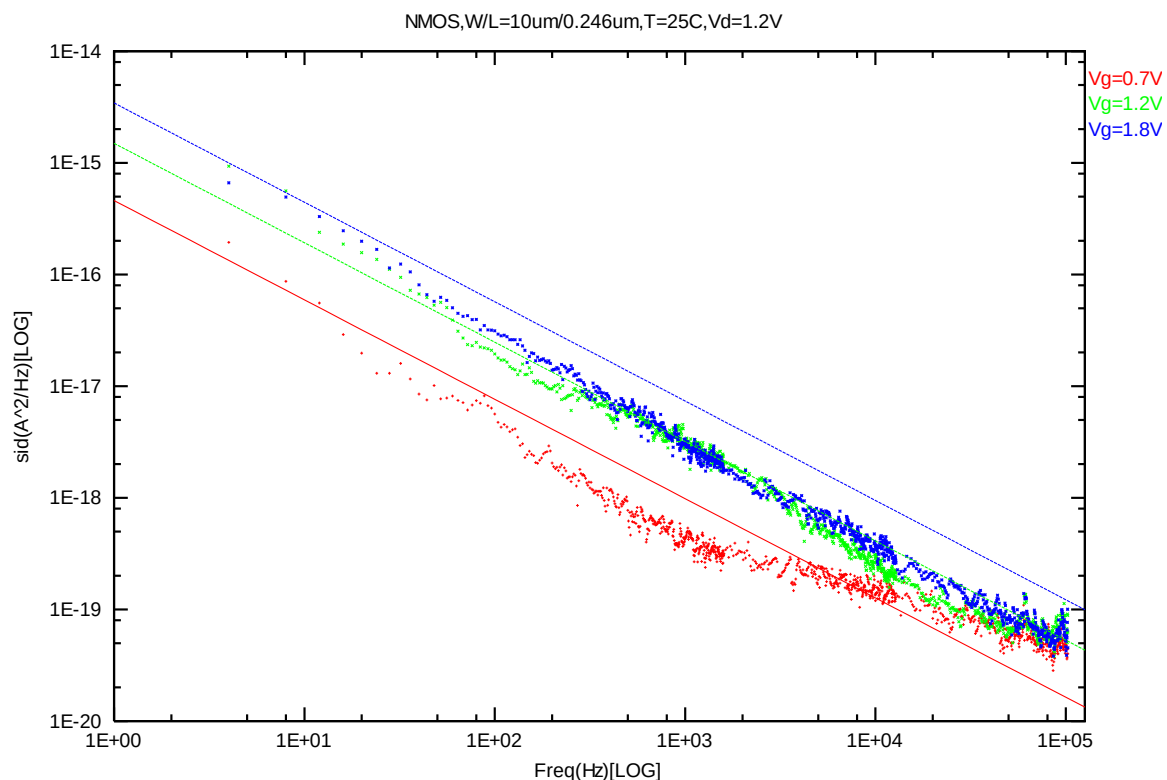


Figure 8-18 NMOS W/L=10um/0.246um @ Vd=1.2V

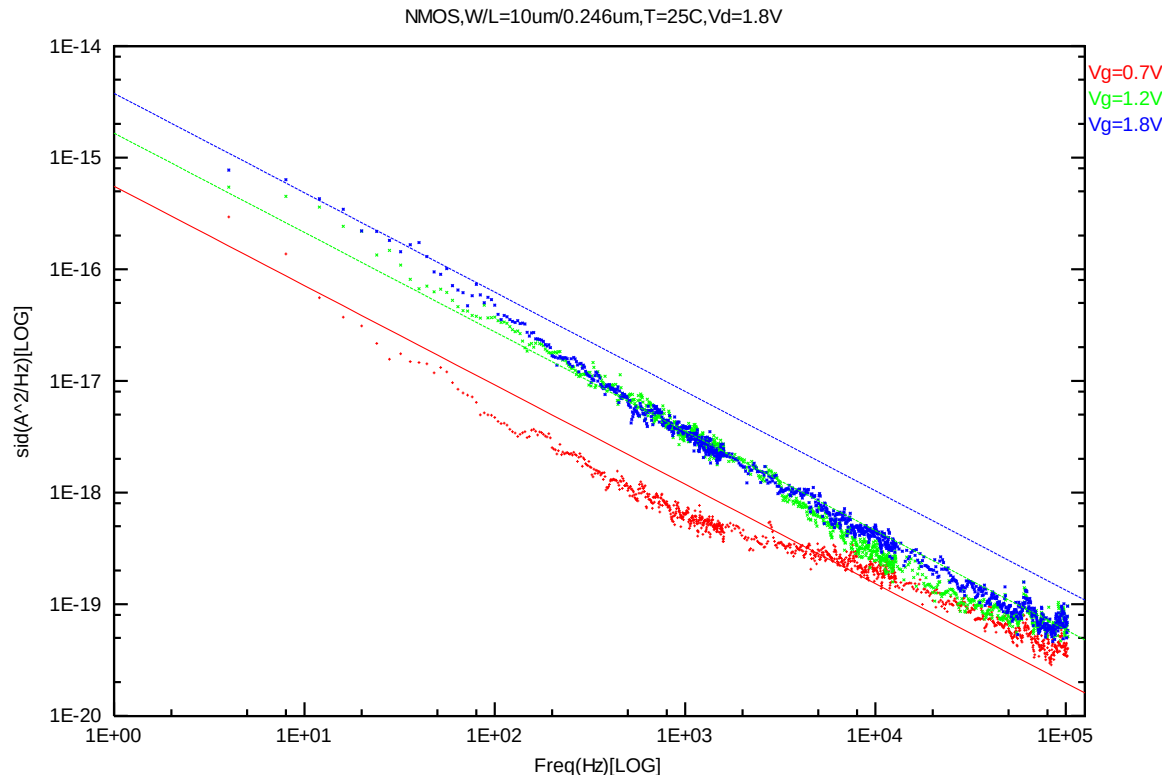


Figure 8-19 NMOS W/L=10um/0.246um @ Vd=1.8V

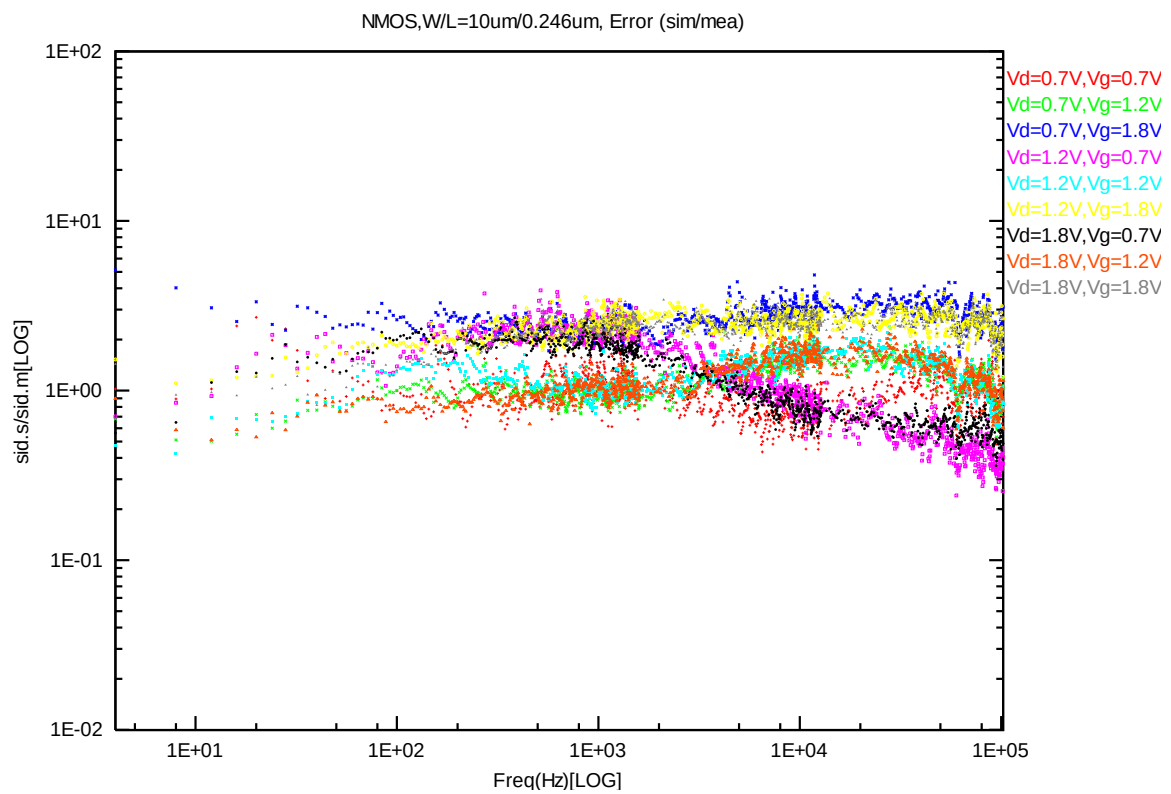


Figure 8-20 NMOS W/L=10um/0.246um error distribution plot

PMOS, W=10um L=10um

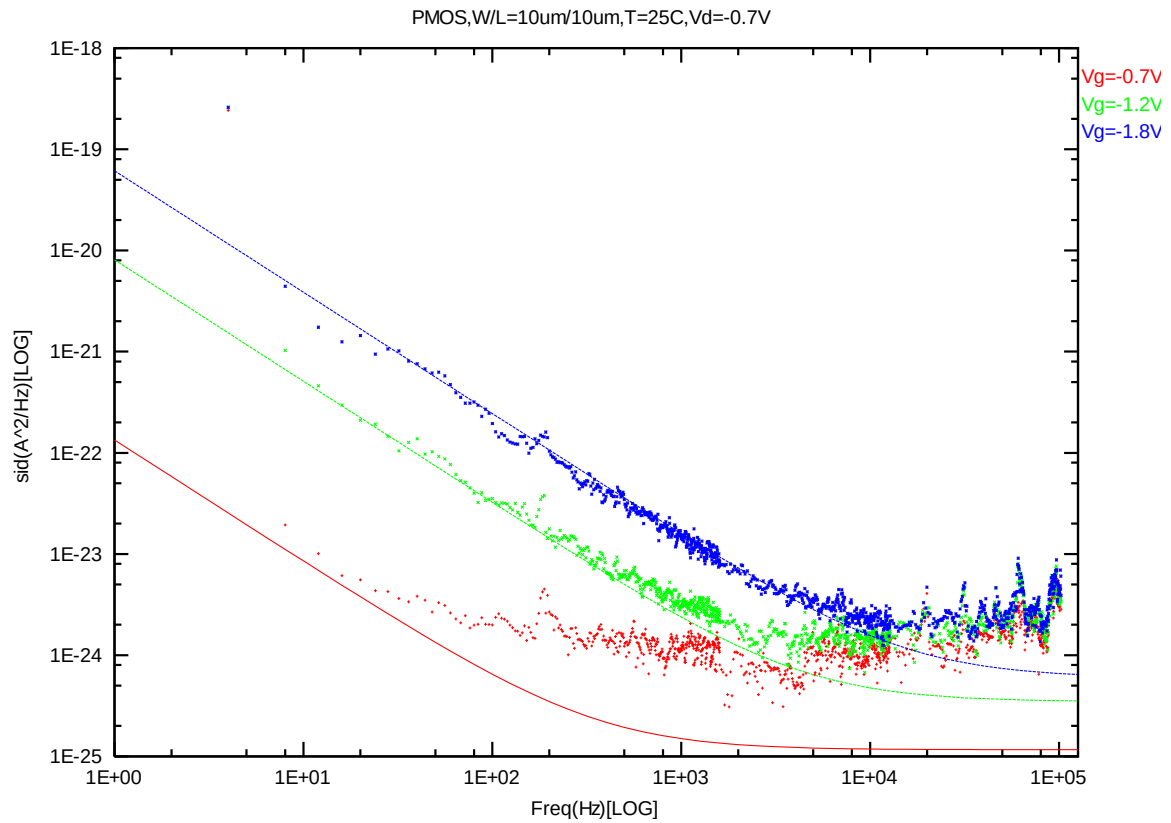


Figure 8-21 PMOS W/L=10um/10um @ Vd=-0.7V

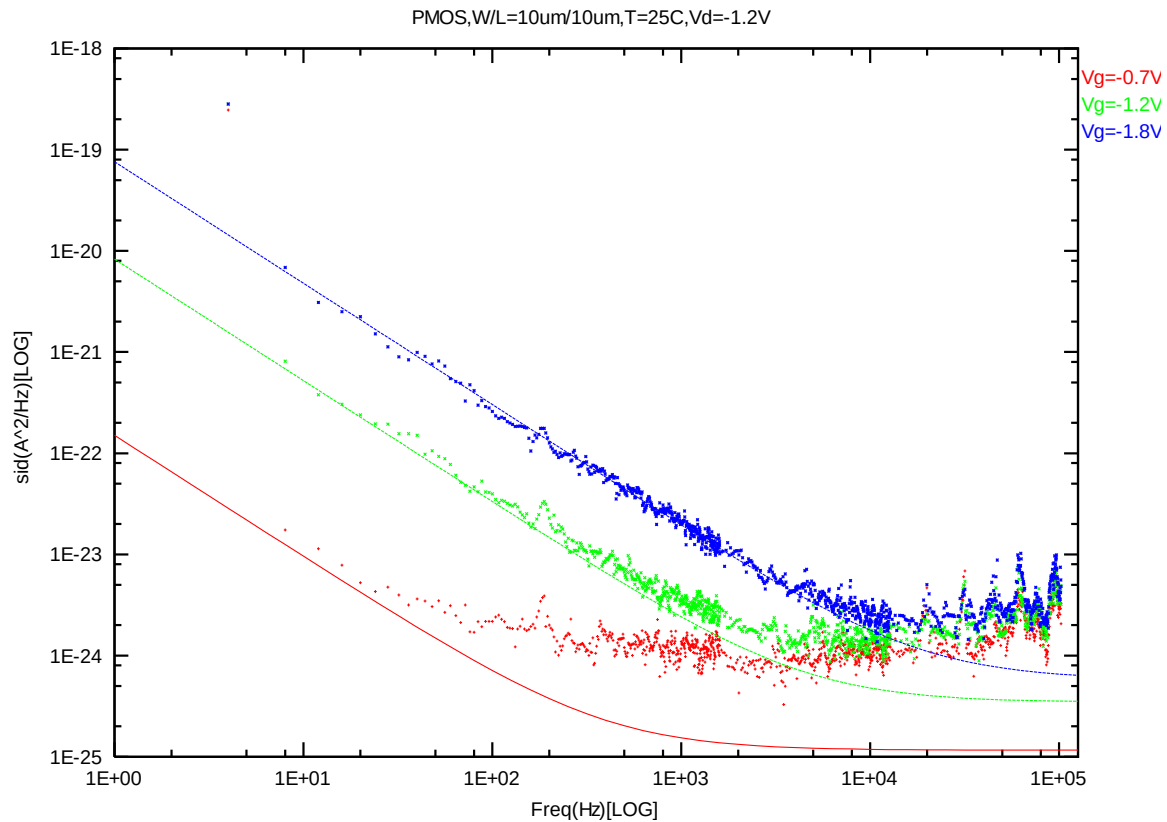


Figure 8-22 PMOS W/L=10um/10um @ Vd=-1.2V

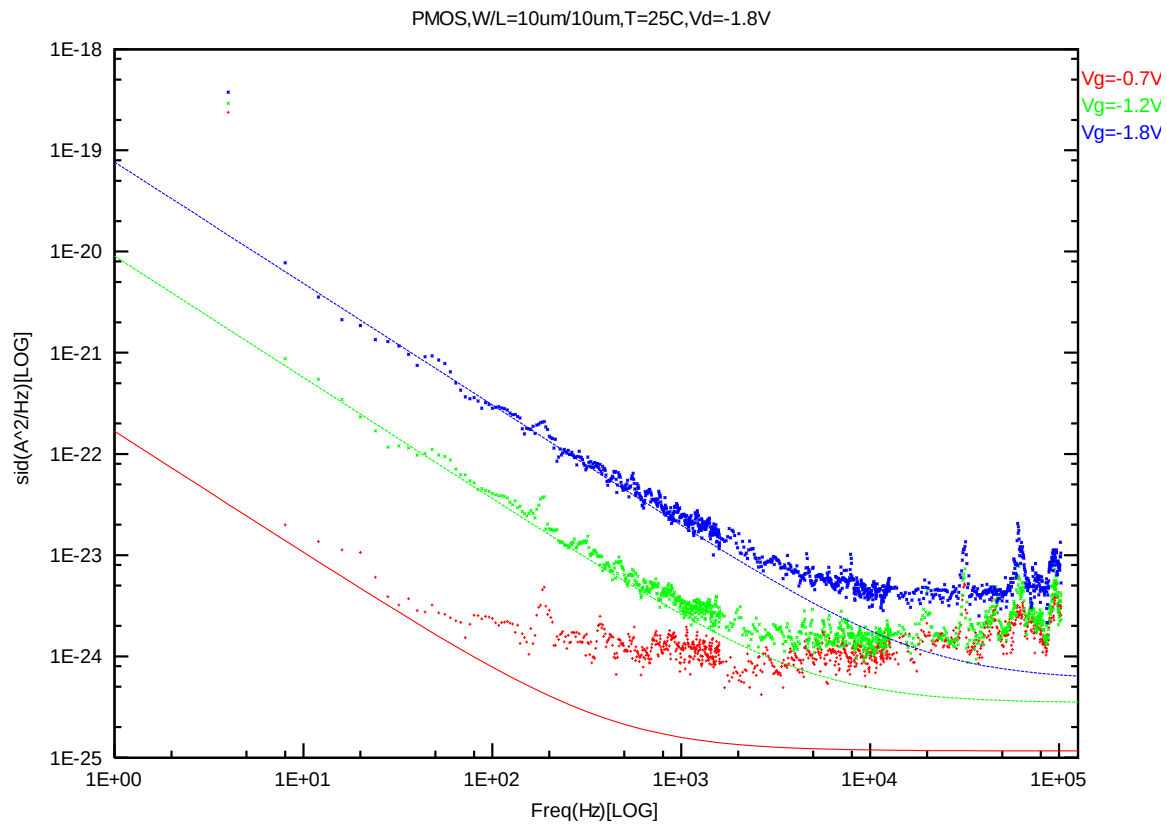


Figure 8-23 PMOS W/L=10um/10um @ Vd=-1.8V

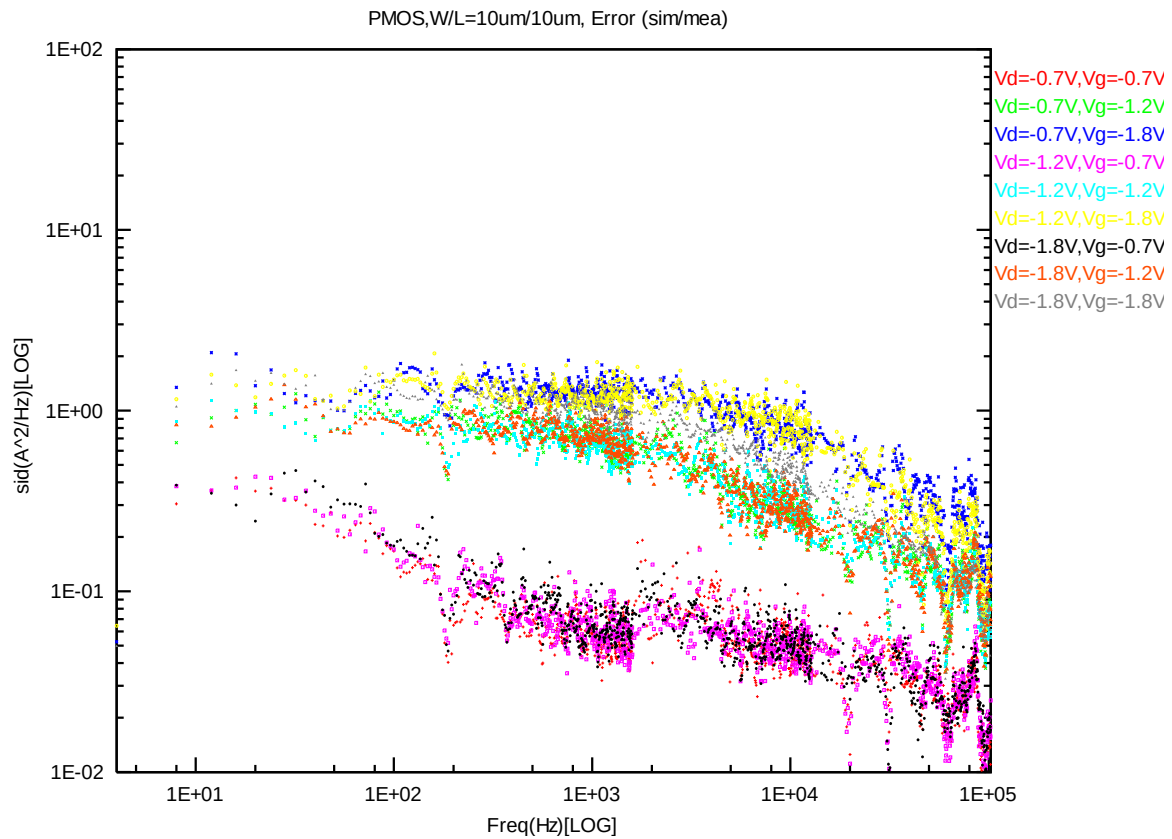


Figure 8-24 PMOS W/L=10um/10um error distribution plot

PMOS, W=10um L=1um

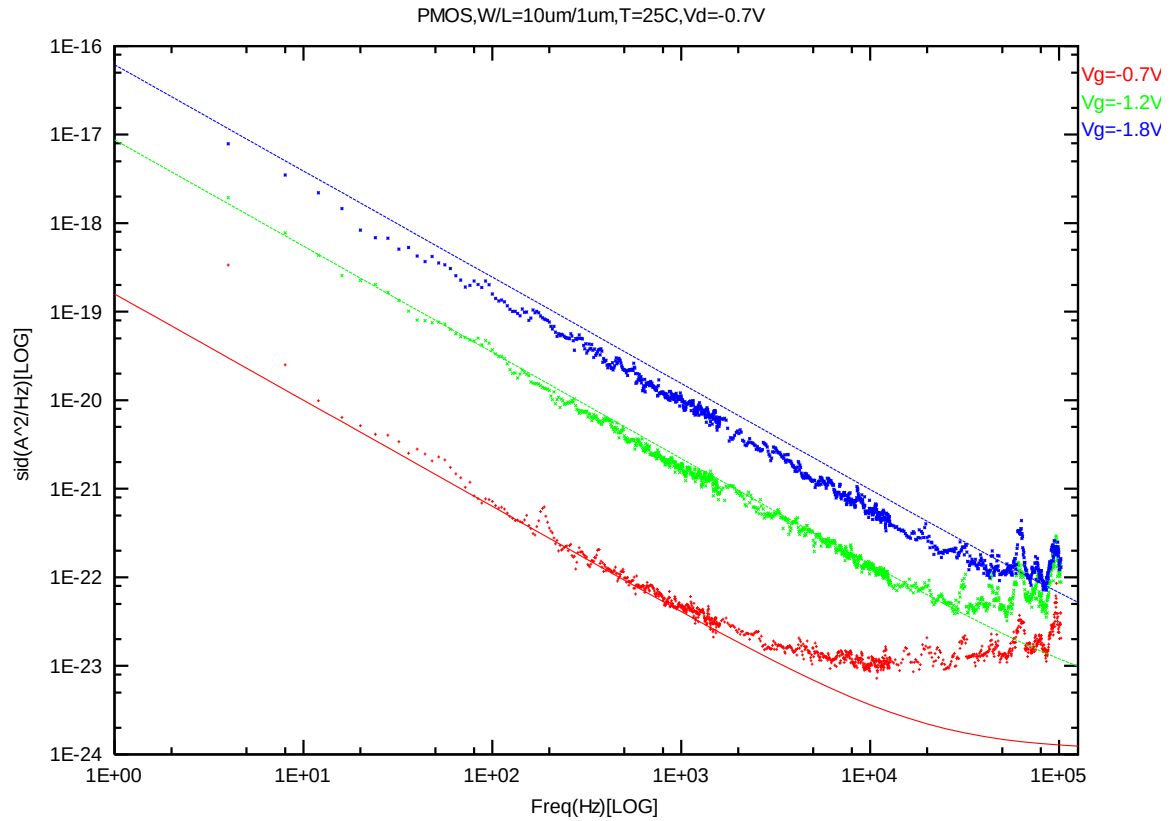


Figure 8-25 PMOS W/L=10um/1um @ Vd=-0.7V

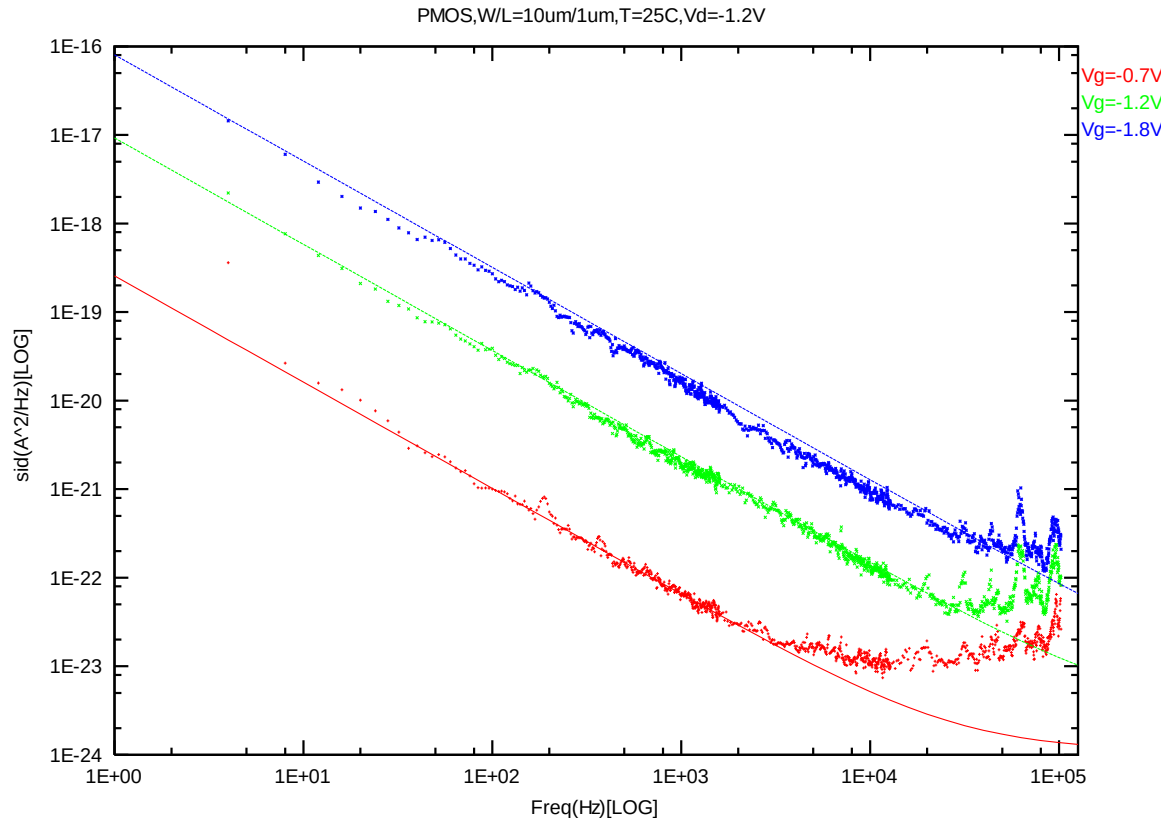


Figure 8-26 PMOS W/L=10um/1um @ Vd=-1.2V

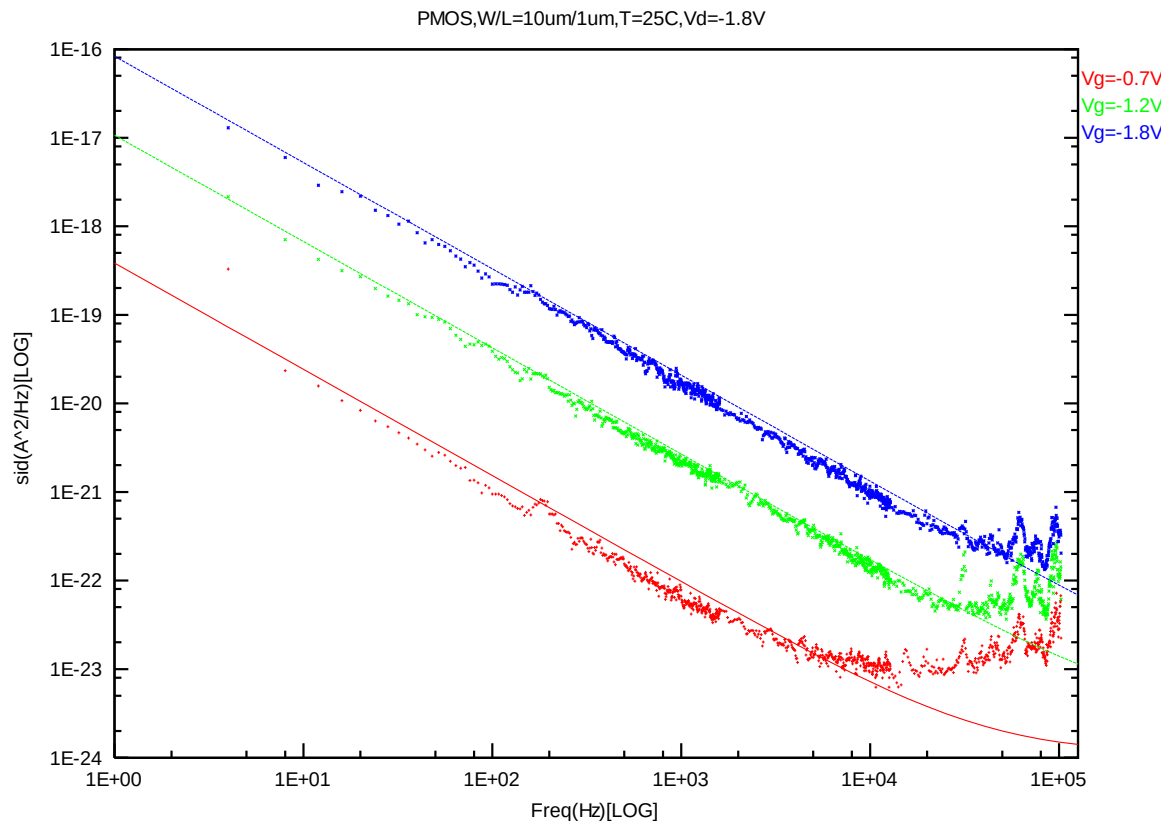


Figure 8-27 PMOS W/L=10um/1um @ Vd=-1.8V

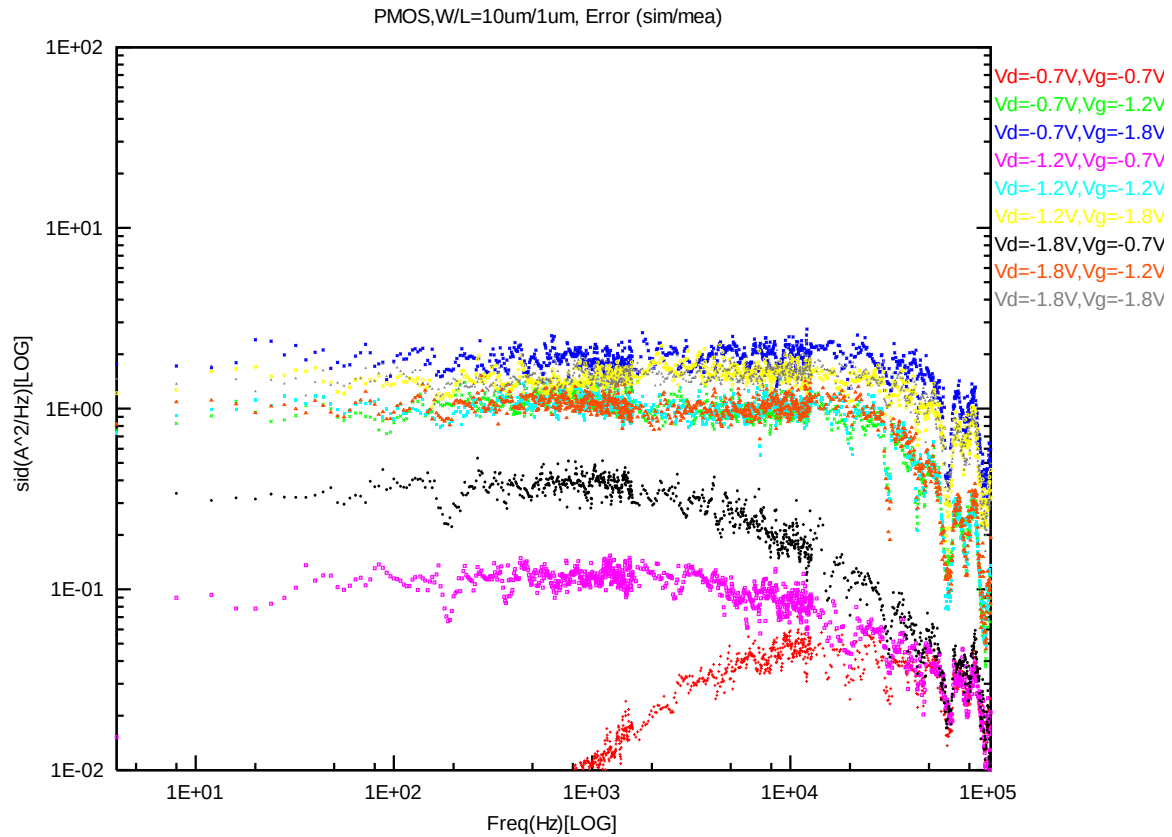


Figure 8-28 PMOS W/L=10um/1um error distribution plot

PMOS, W=10um L=0.507um

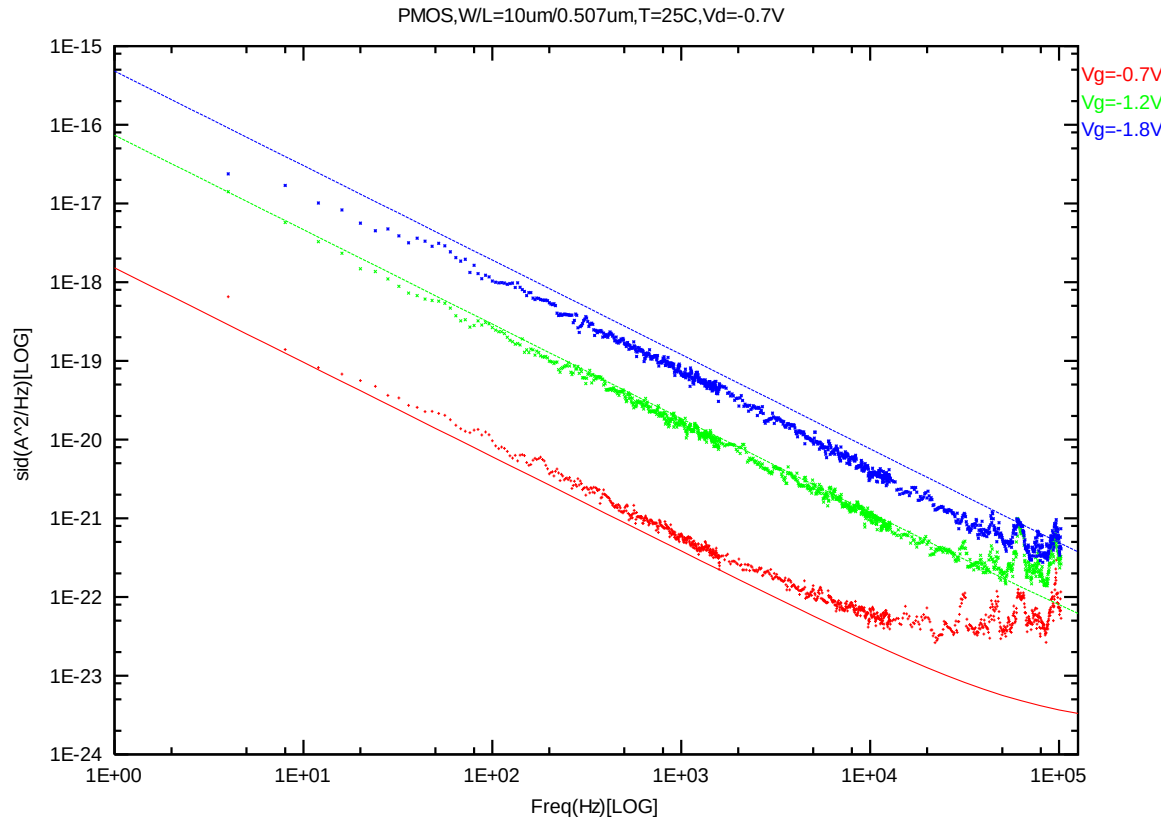


Figure 8-29 PMOS W/L=10um/0.507um @ Vd=-0.7V

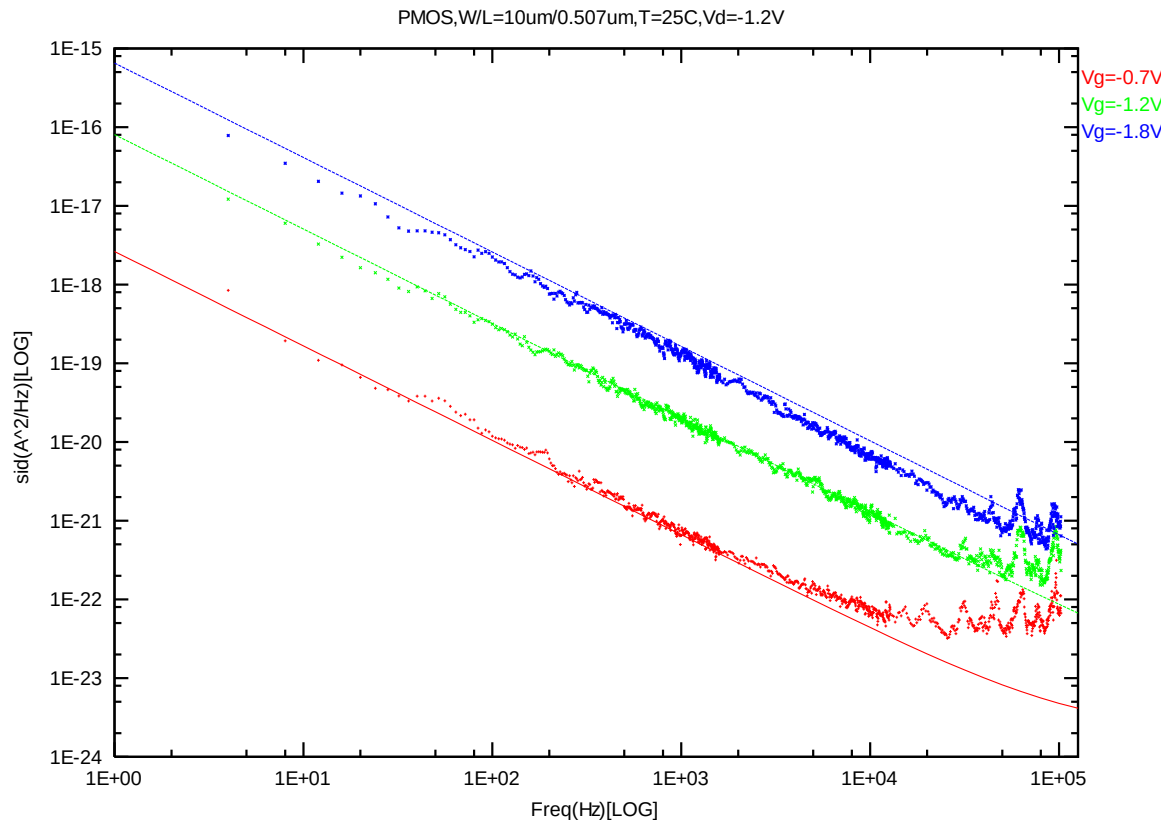


Figure 8-30 PMOS W/L=10um/0.507um @ Vd=-1.2V

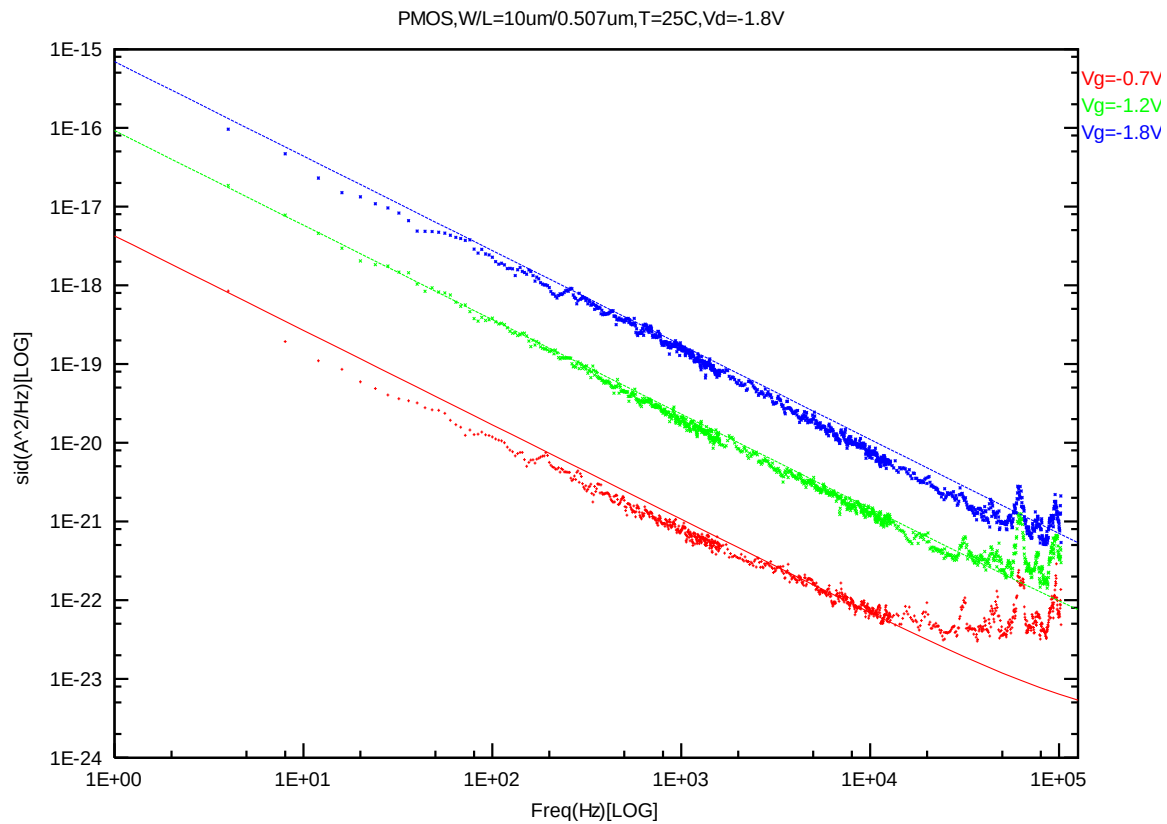


Figure 8-31 PMOS W/L=10um/0.507um @ Vd=-1.8V

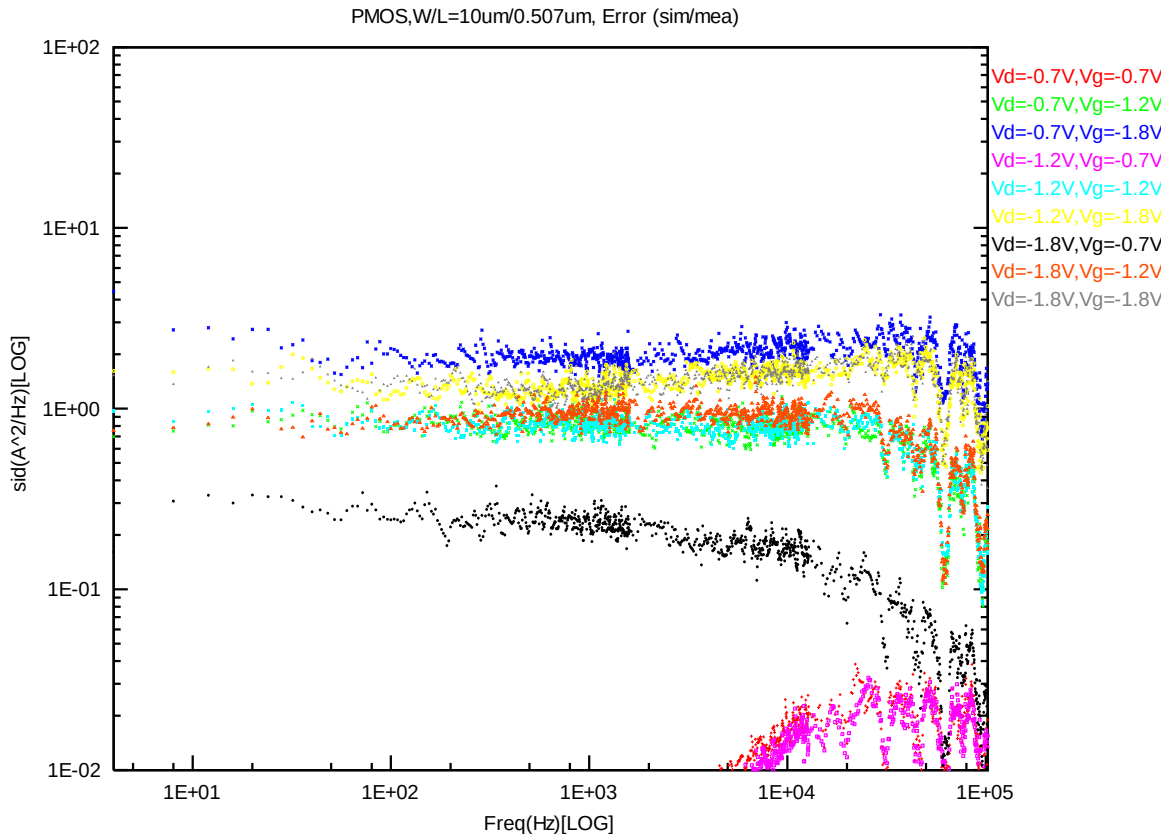


Figure 8-32 PMOS W/L=10um/0.507um error distribution plot

PMOS, W=10um L=0.417um

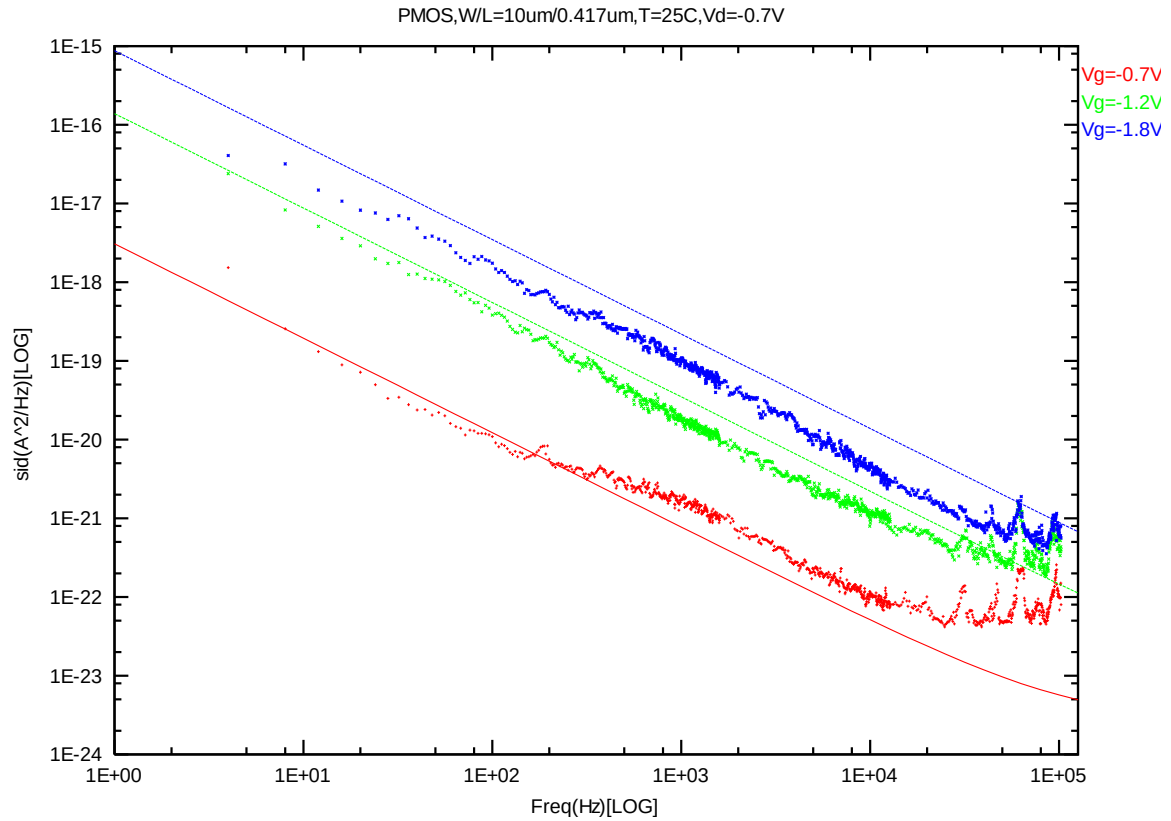


Figure 8-33 PMOS W/L=10um/0.417um @ Vd=-0.7V

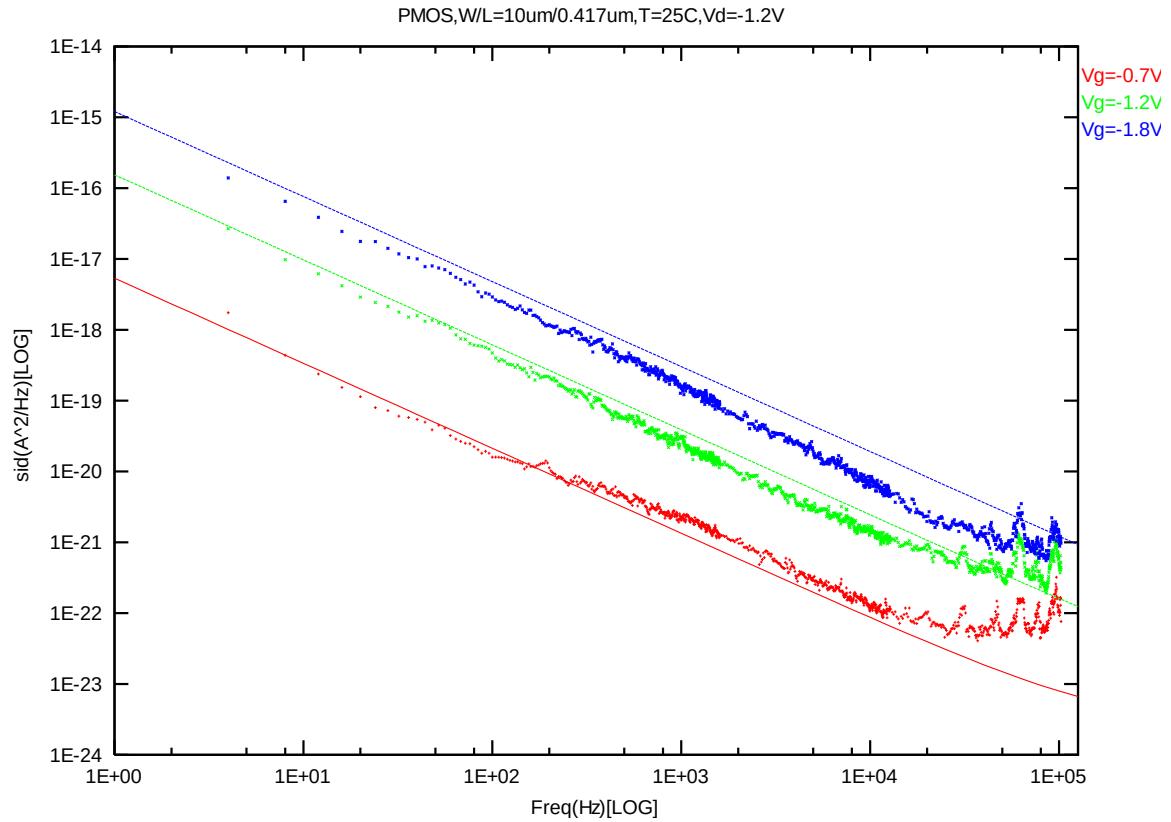


Figure 8-34 PMOS W/L=10um/0.417um @ Vd=-1.2V

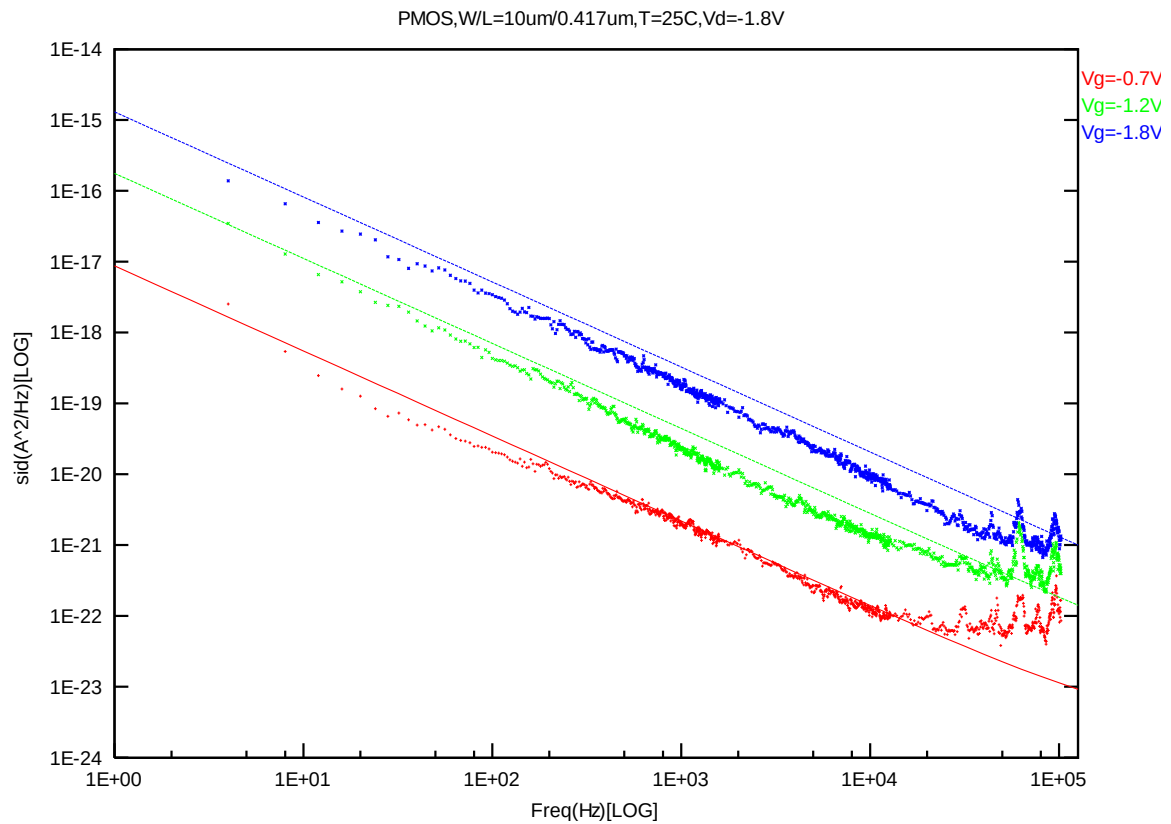


Figure 8-35 PMOS W/L=10um/0.417um @ Vd=-1.8V

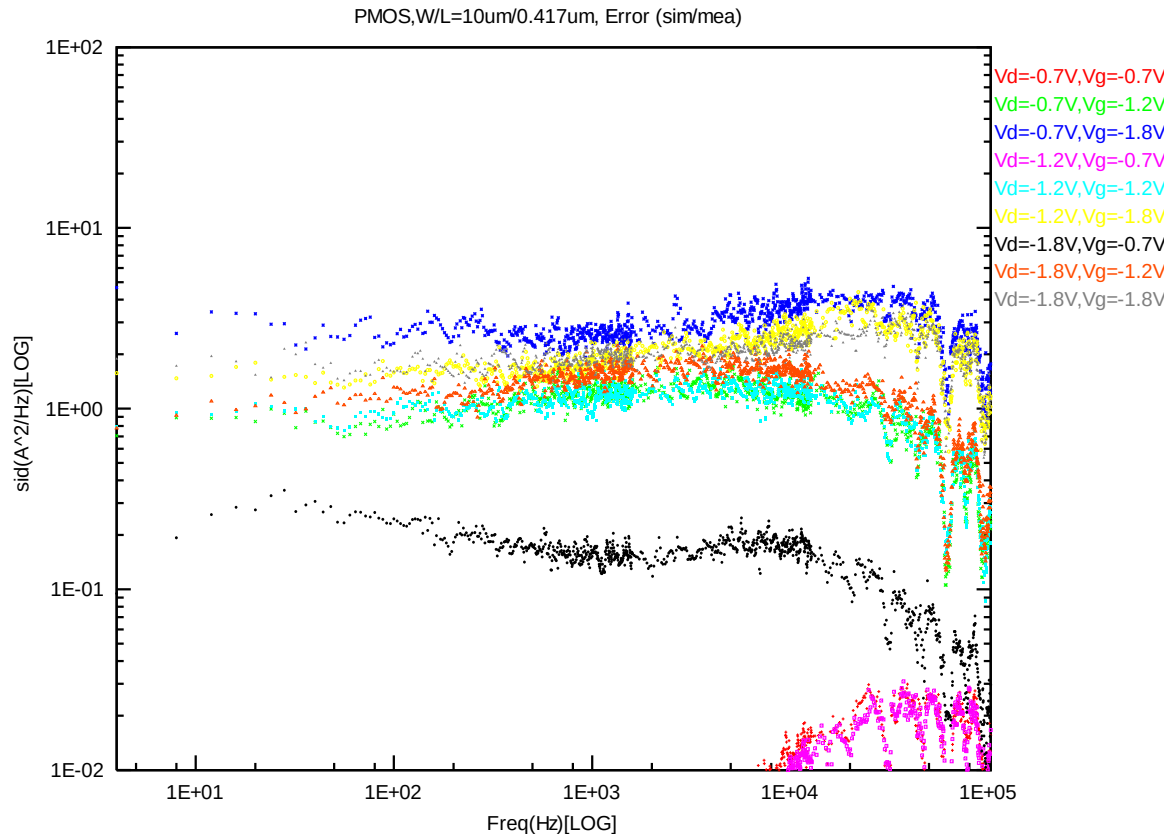


Figure 8-36 PMOS W/L=10um/0.417um error distribution plot

PMOS, W=10um L=0.246um

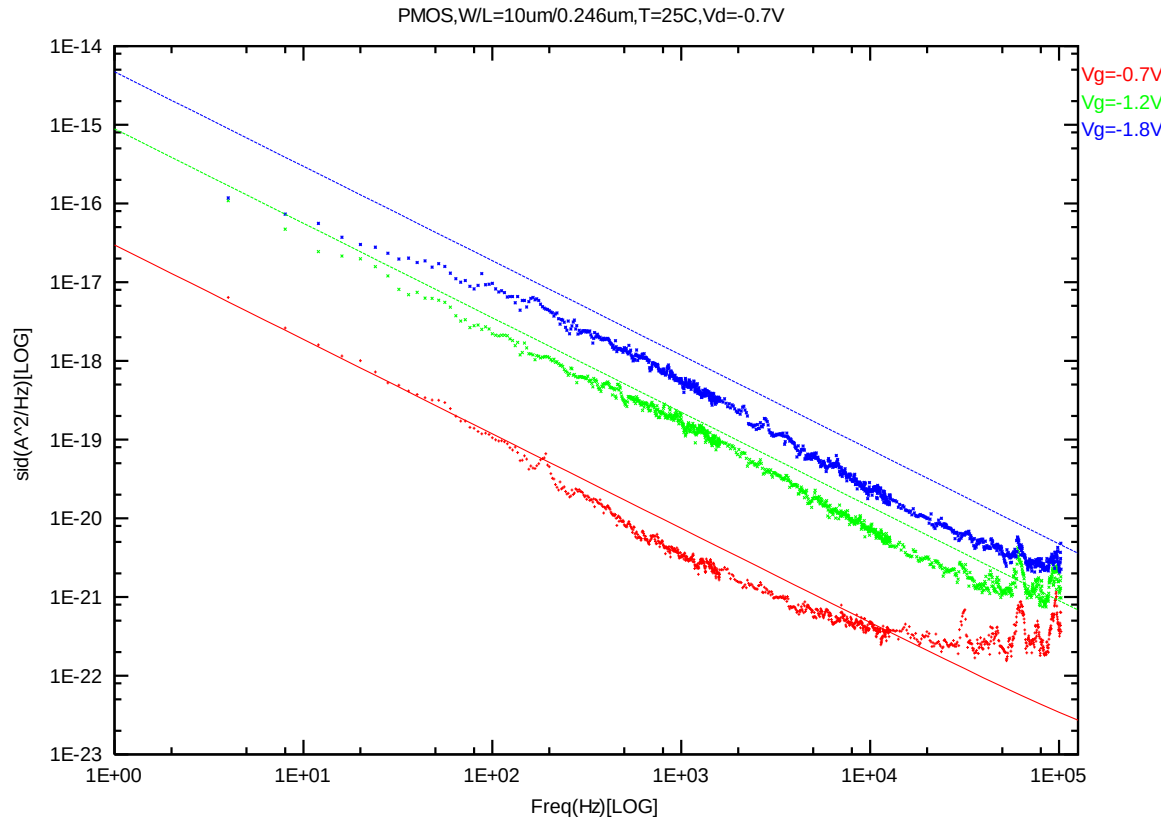


Figure 8-37 PMOS W/L=10um/0.246um @ Vd=-0.7V

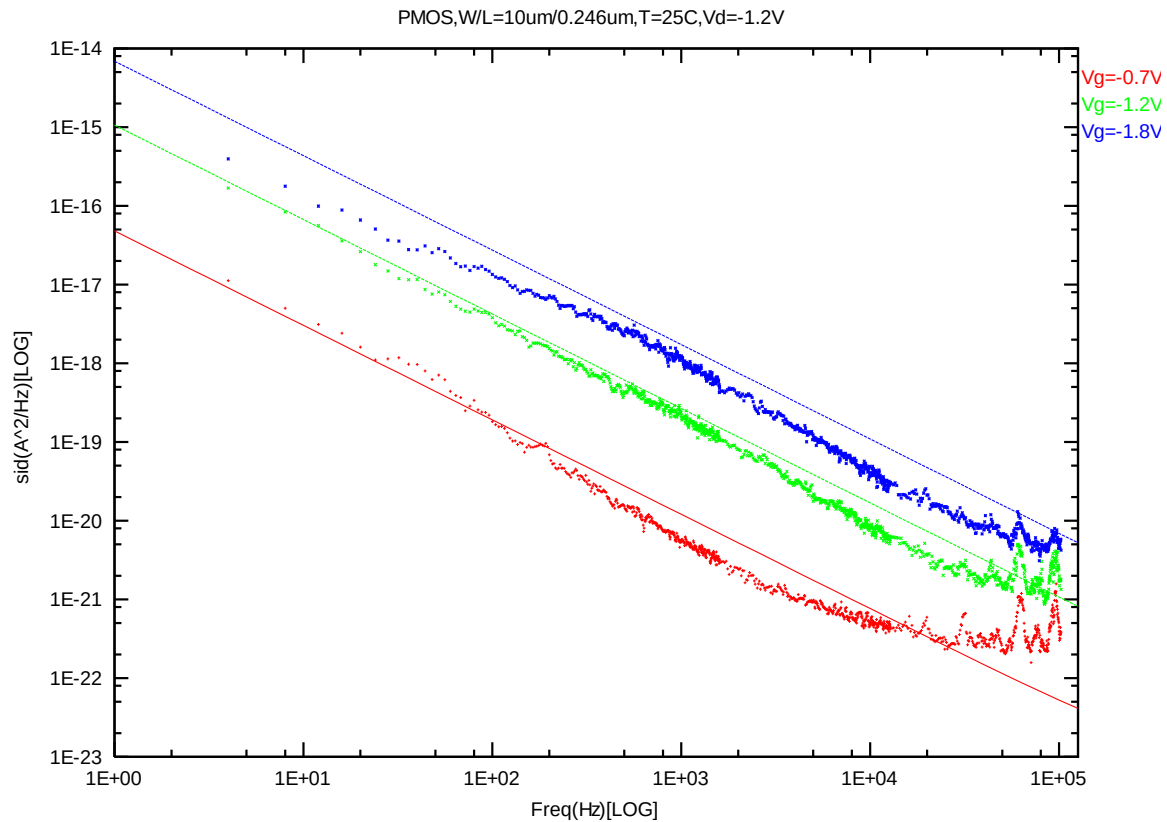


Figure 8-38 PMOS W/L=10um/0.246um @ Vd=-1.2V

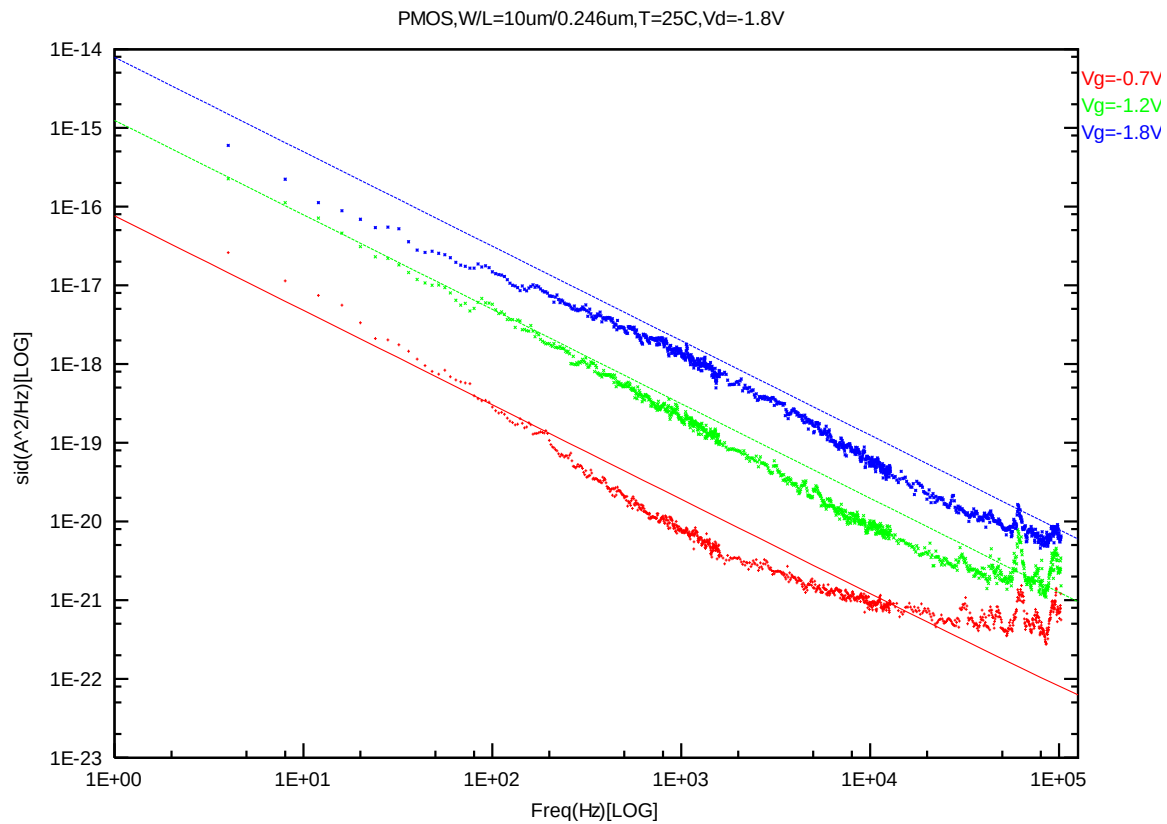


Figure 8-39 PMOS W/L=10um/0.246um @ Vd=-1.8V

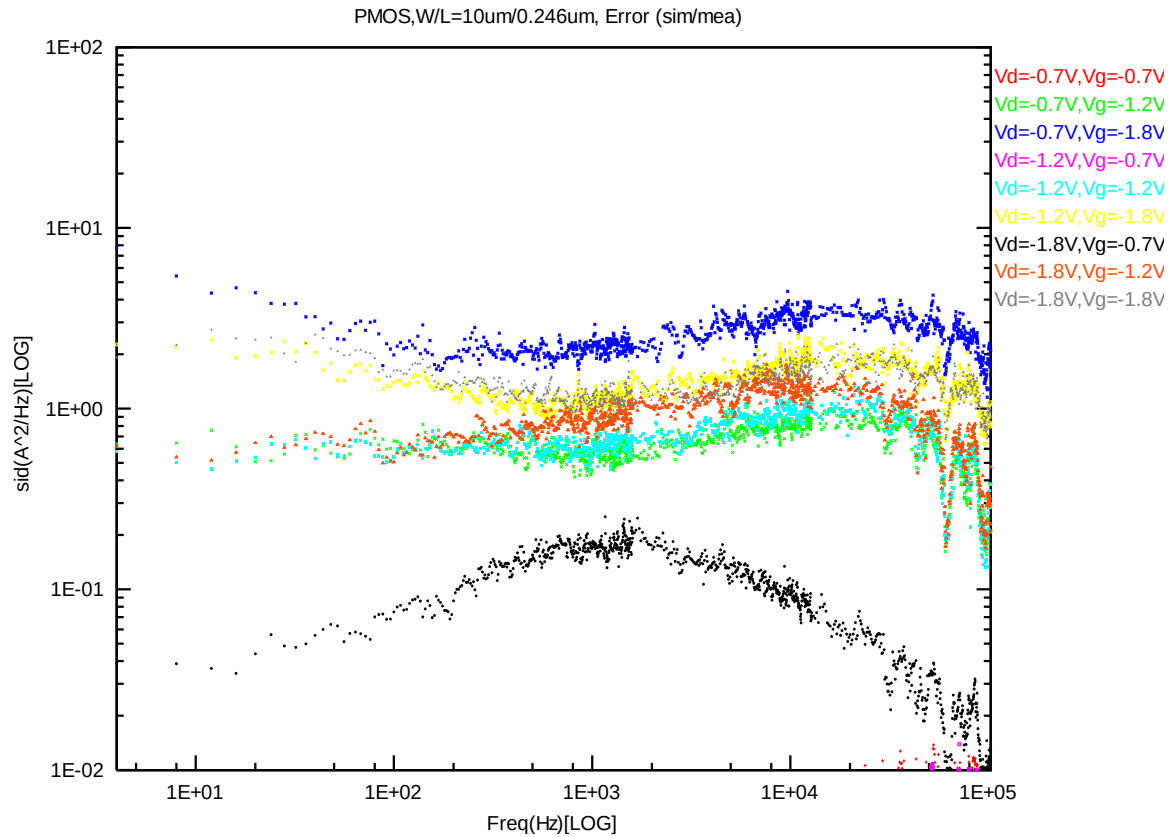


Figure 8-40 PMOS W/L=10um/0.246um error distribution plot

8.2 Trend Plots at Frequency=100Hz

A. NMOS

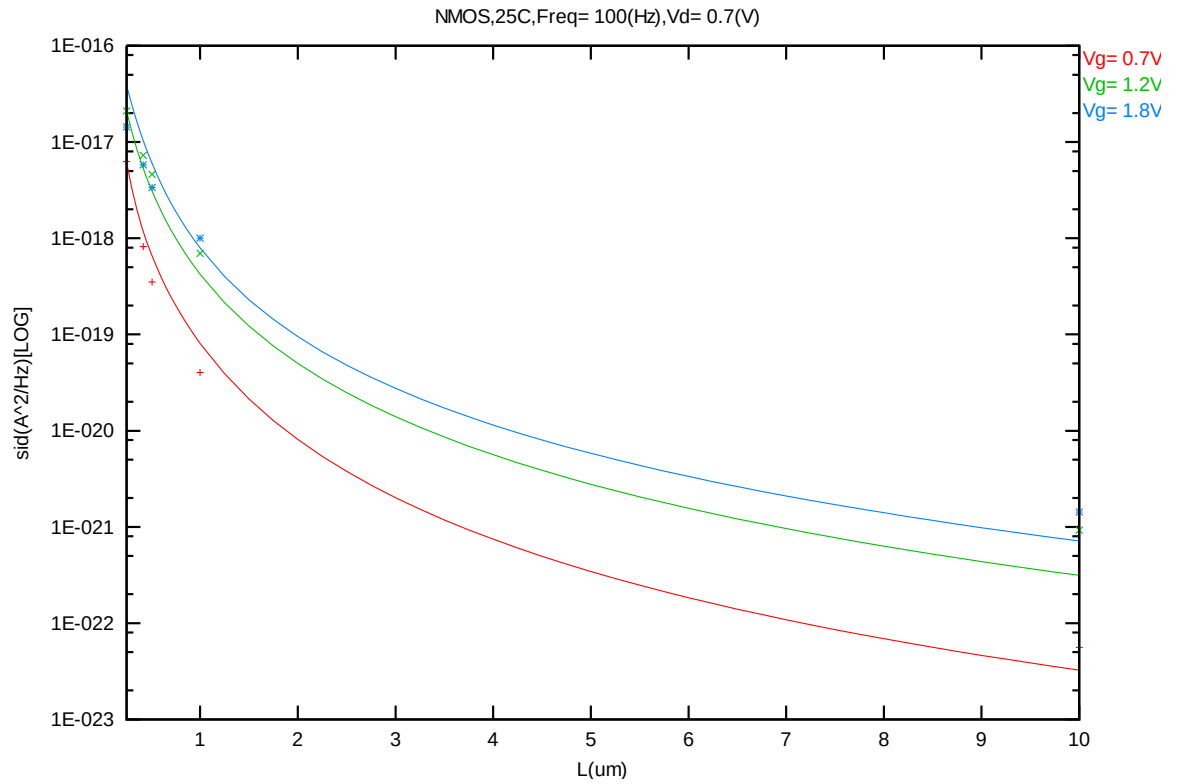


Figure 8-41 Noise vs. Length for NMOS @ Vd=0.7V

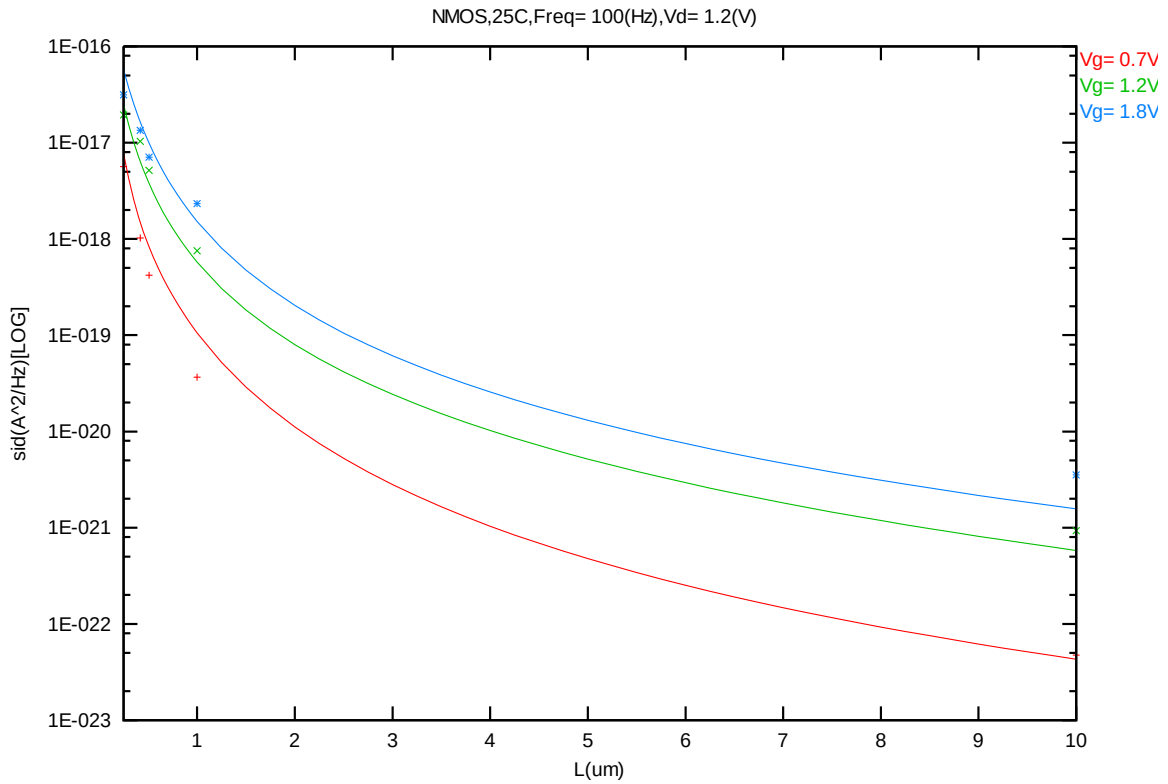


Figure 8-42 Noise vs. Length for NMOS @ Vd=1.2V

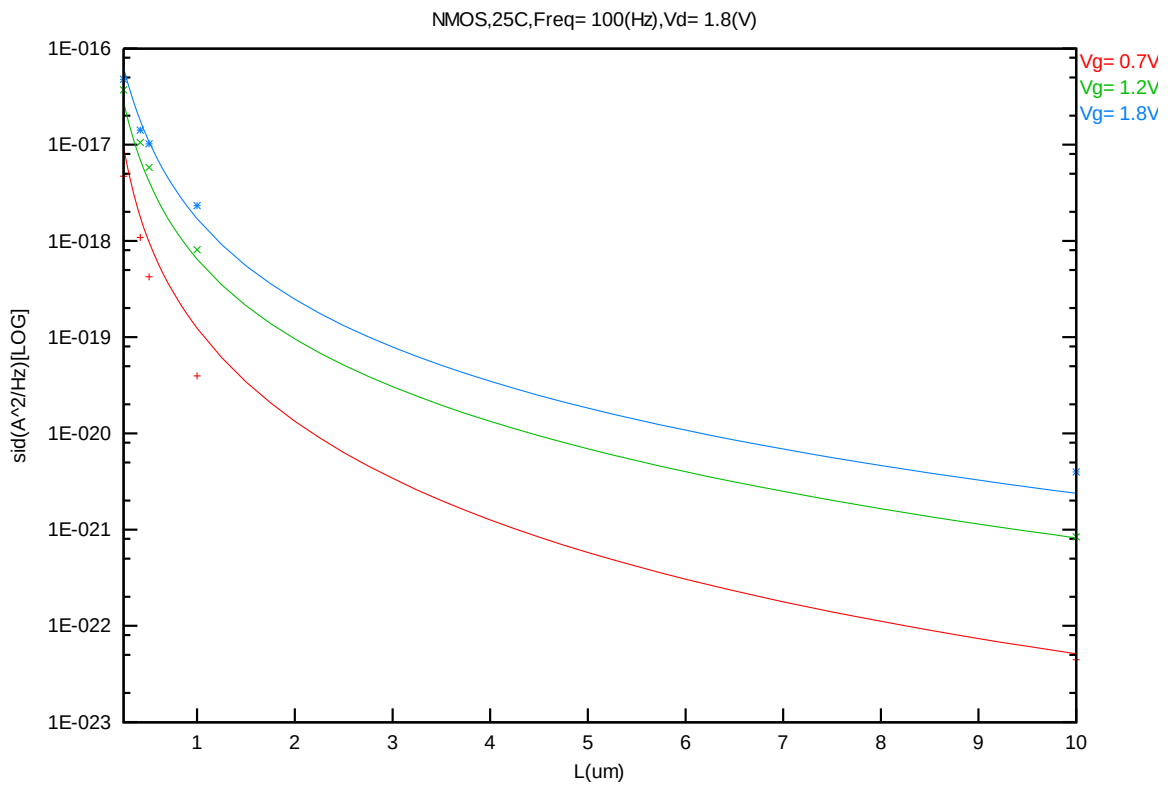


Figure 8-43 Noise vs. Length for NMOS @ Vd=1.8V

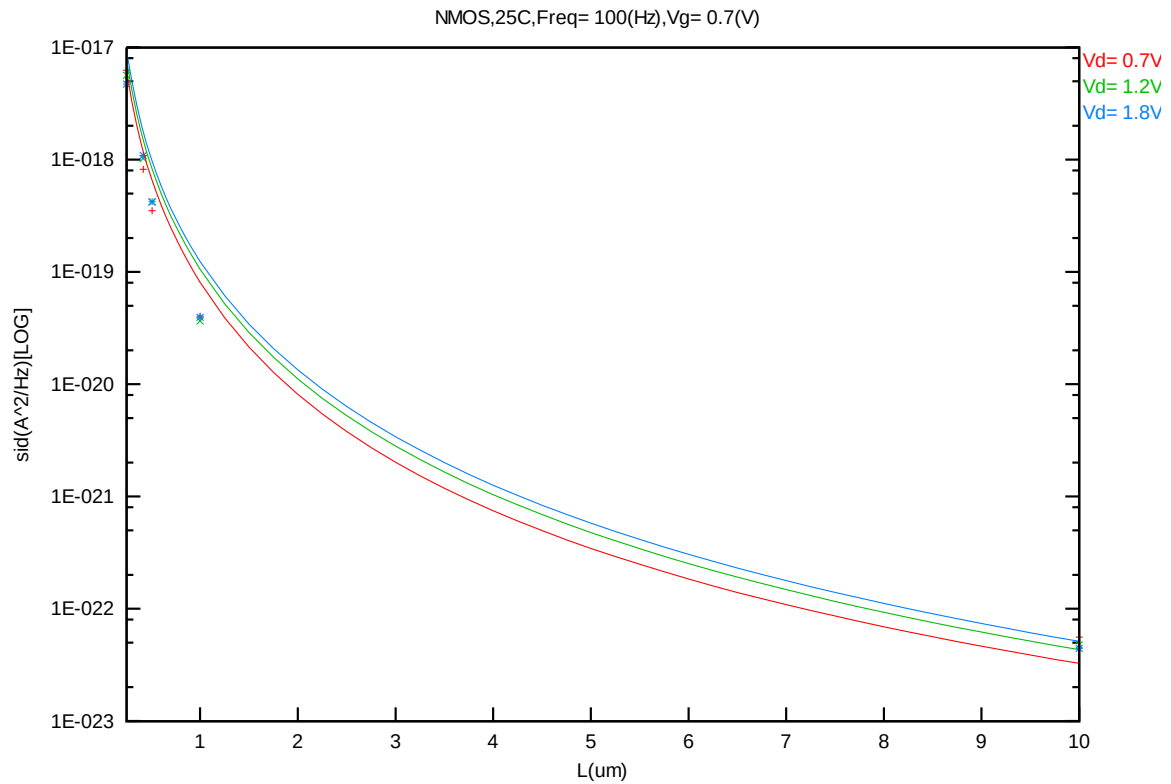


Figure 8-44 Noise vs. Length for NMOS @ Vg=0.7V

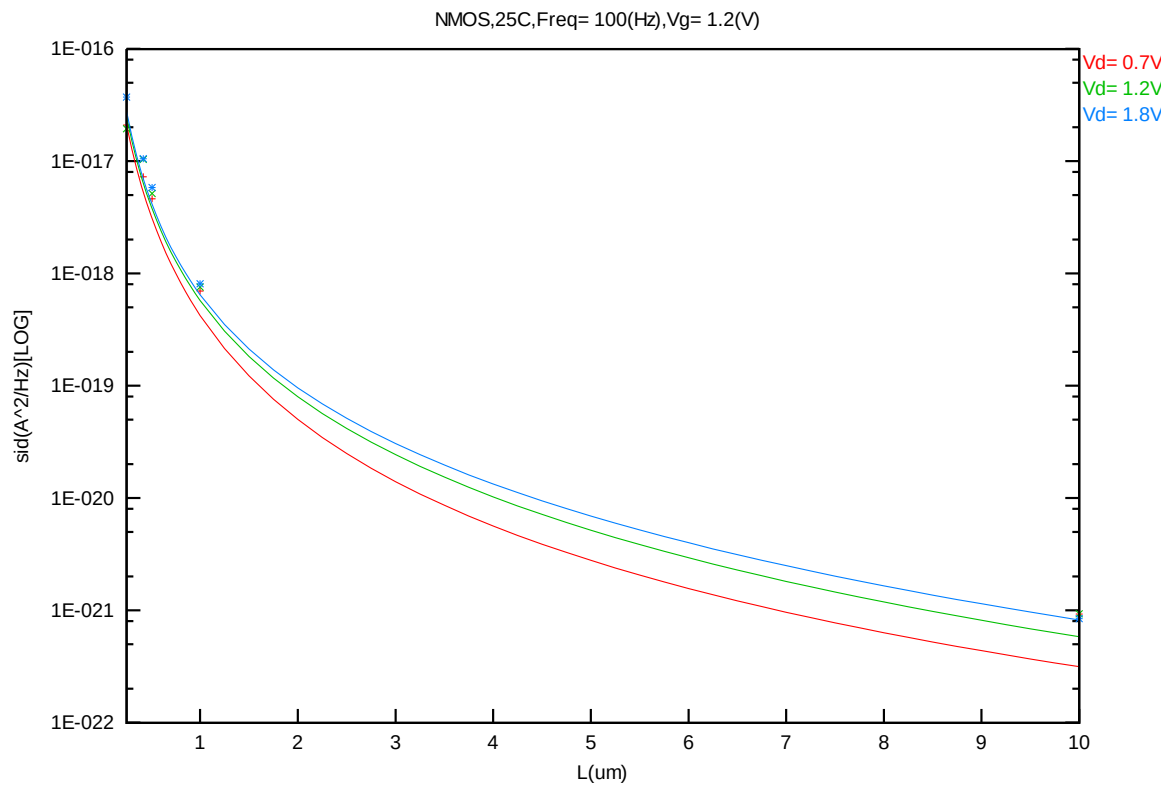


Figure 8-45 Noise vs. Length for NMOS @ Vg=1.2V

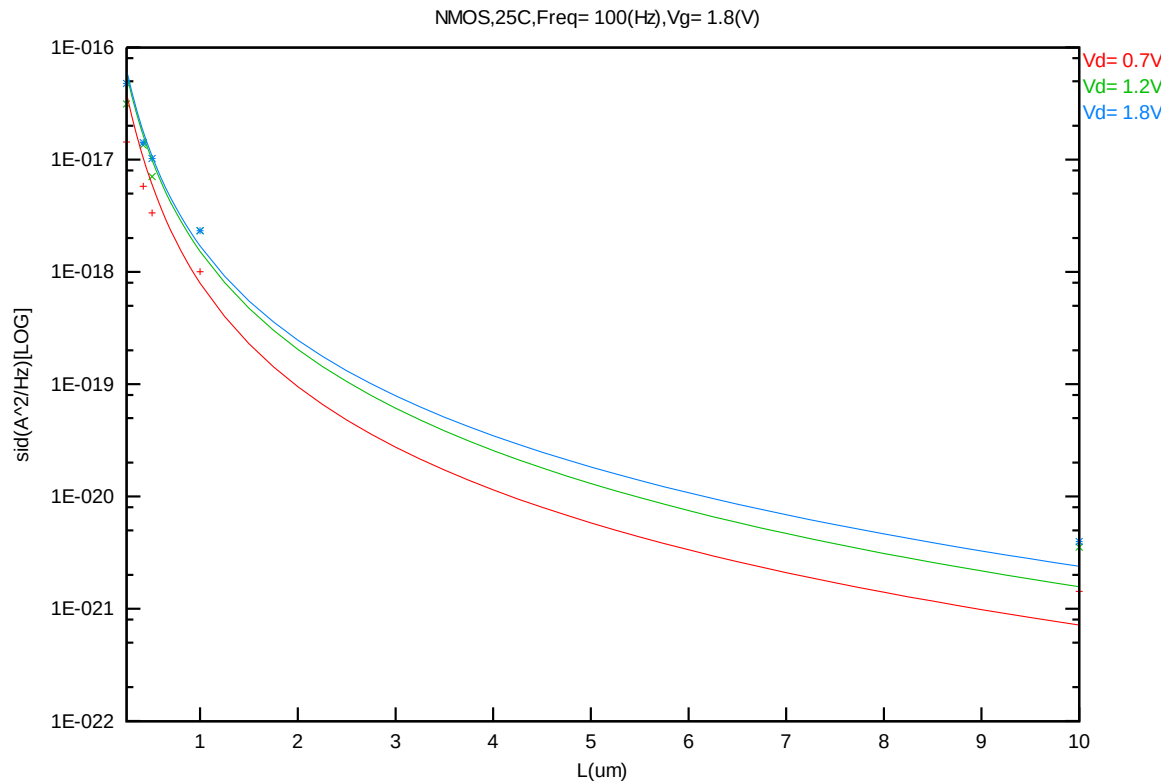


Figure 8-46 Noise vs. Length for NMOS @ Vg=1.8V

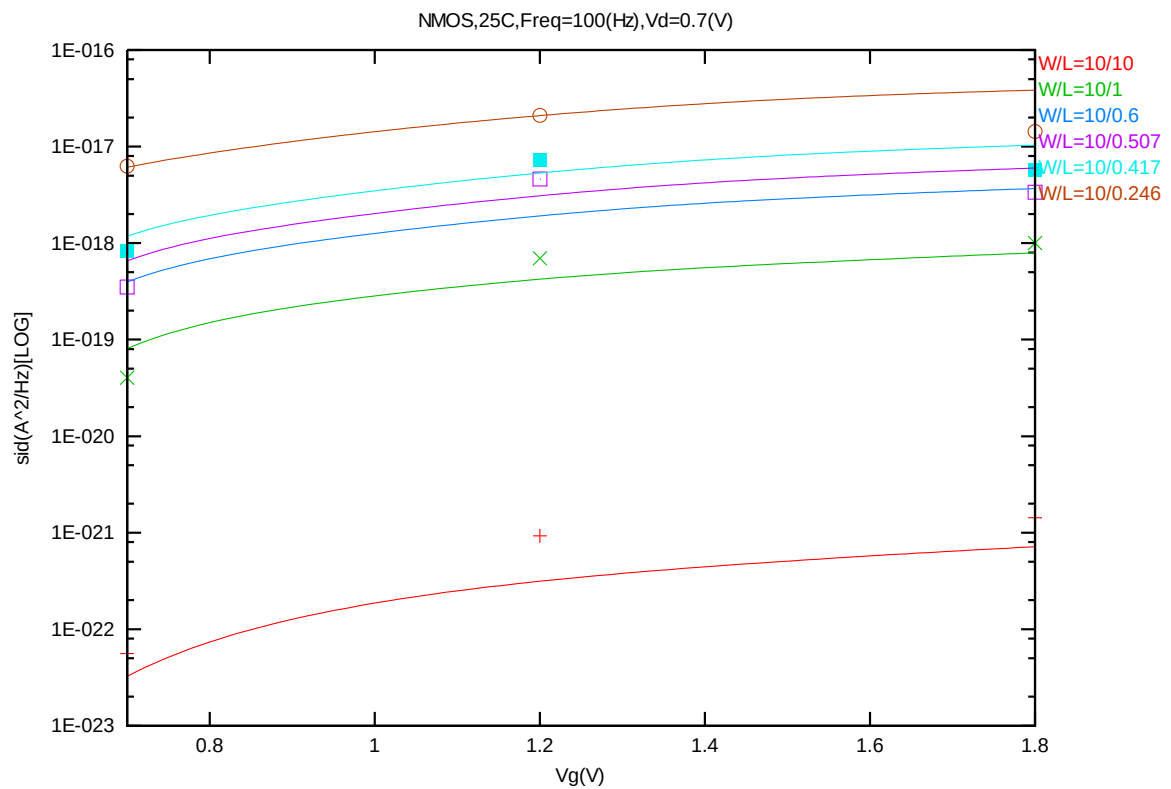


Figure 8-47 Noise vs. Vg for NMOS @ Vd=0.7V

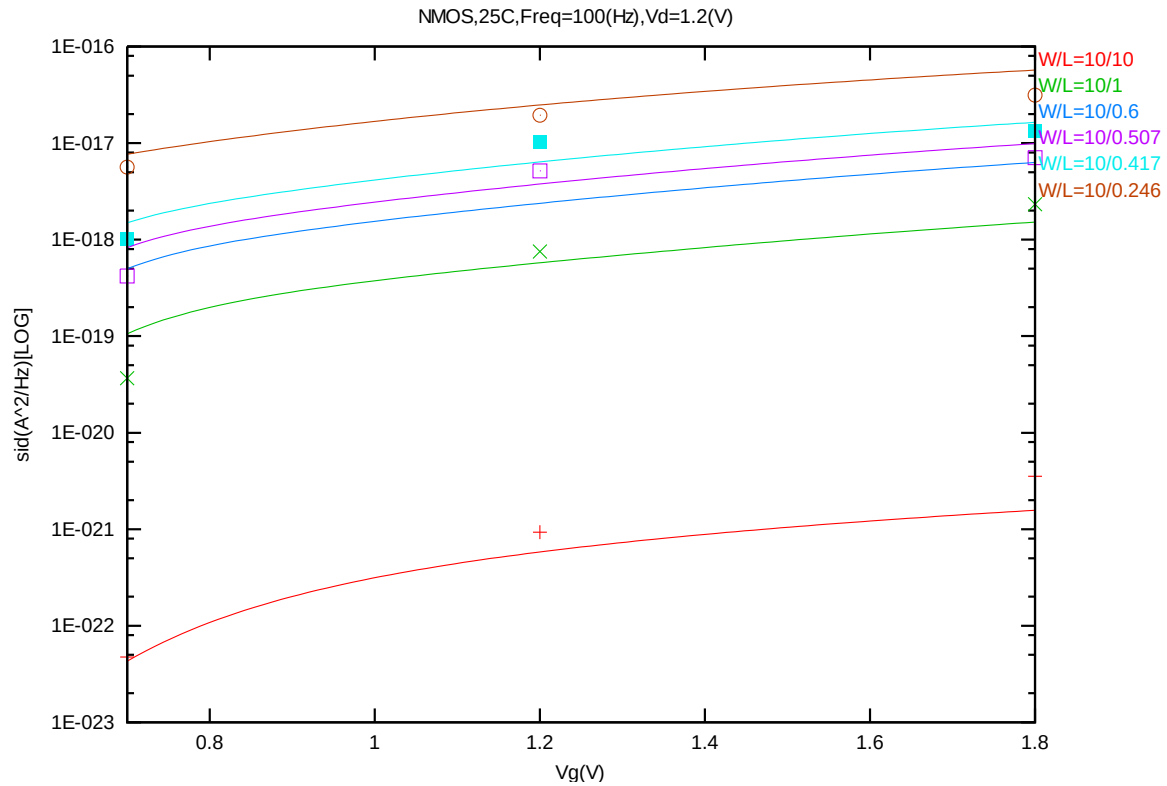


Figure 8-48 Noise vs. Vg for NMOS @ Vd=1.2V

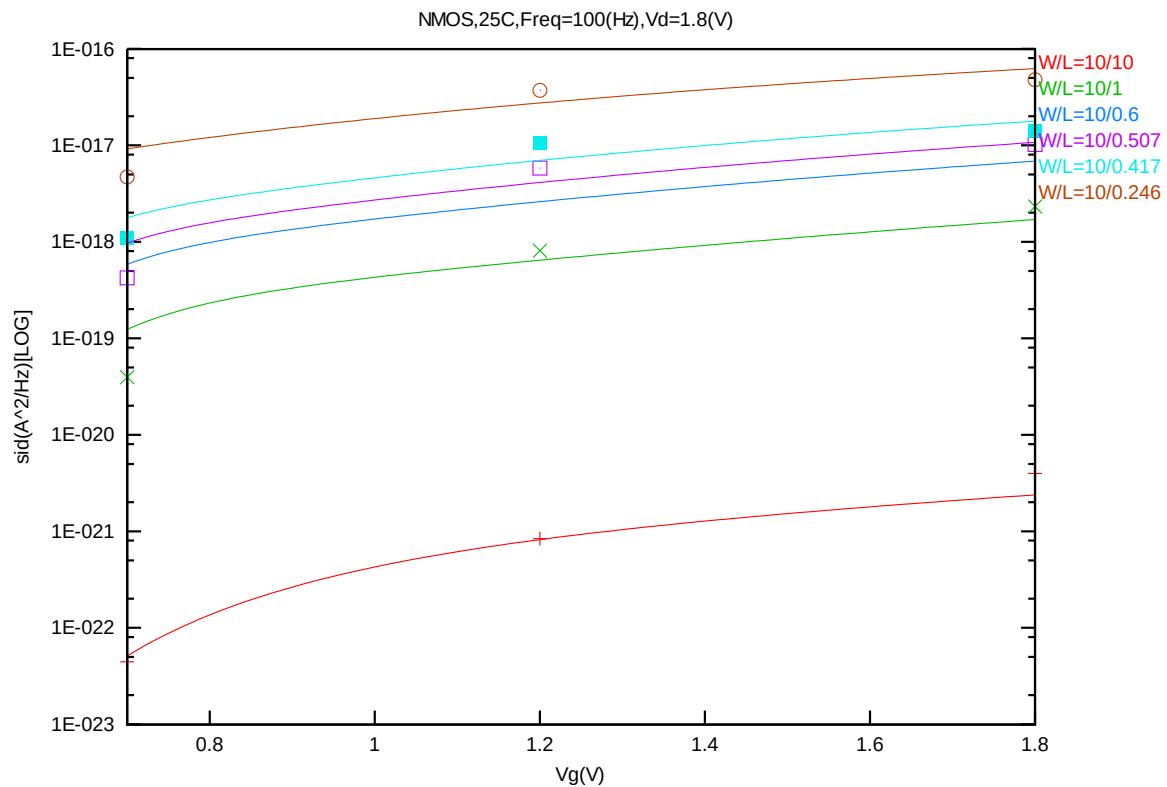
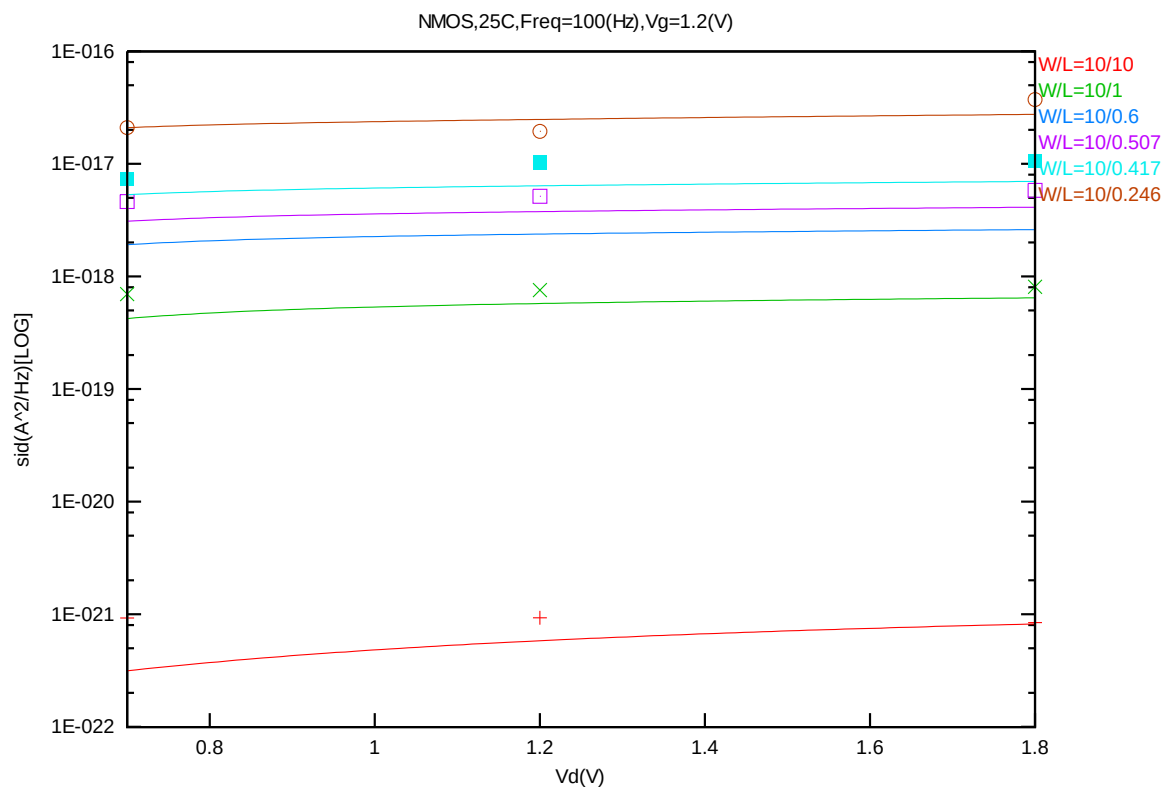
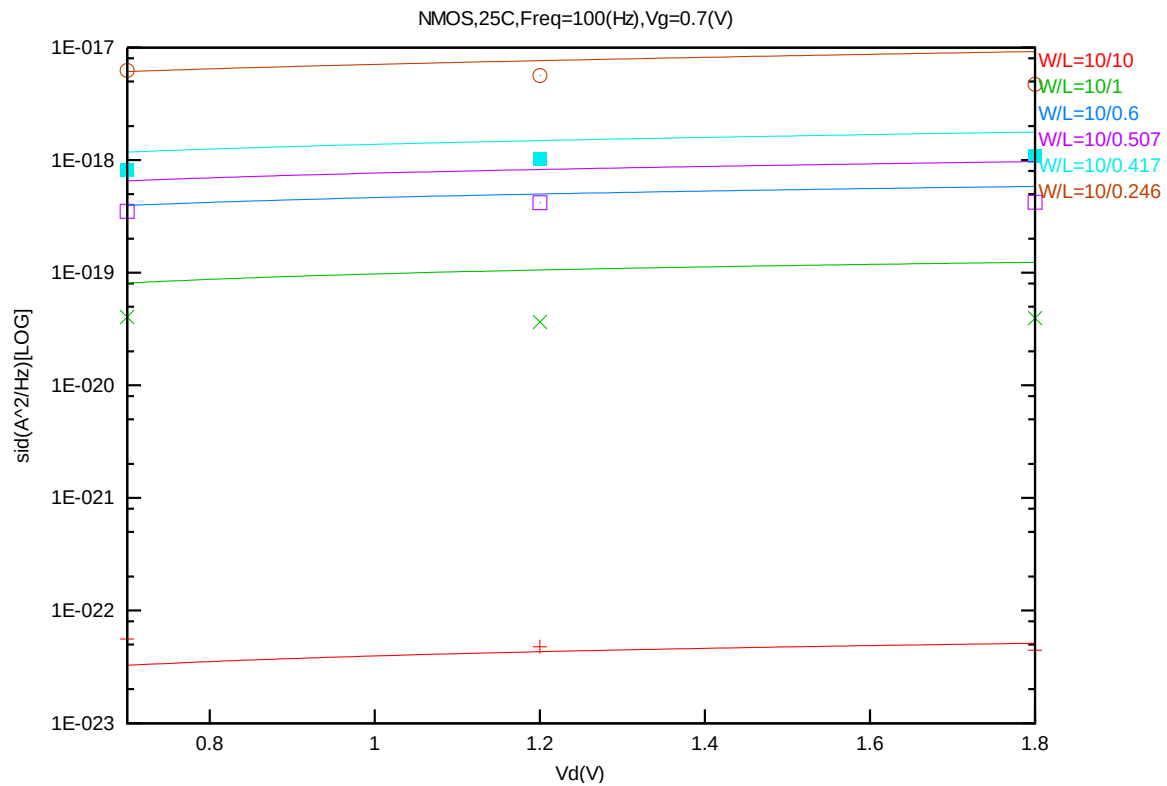


Figure 8-49 Noise vs. Vg for NMOS @ Vd=1.8V



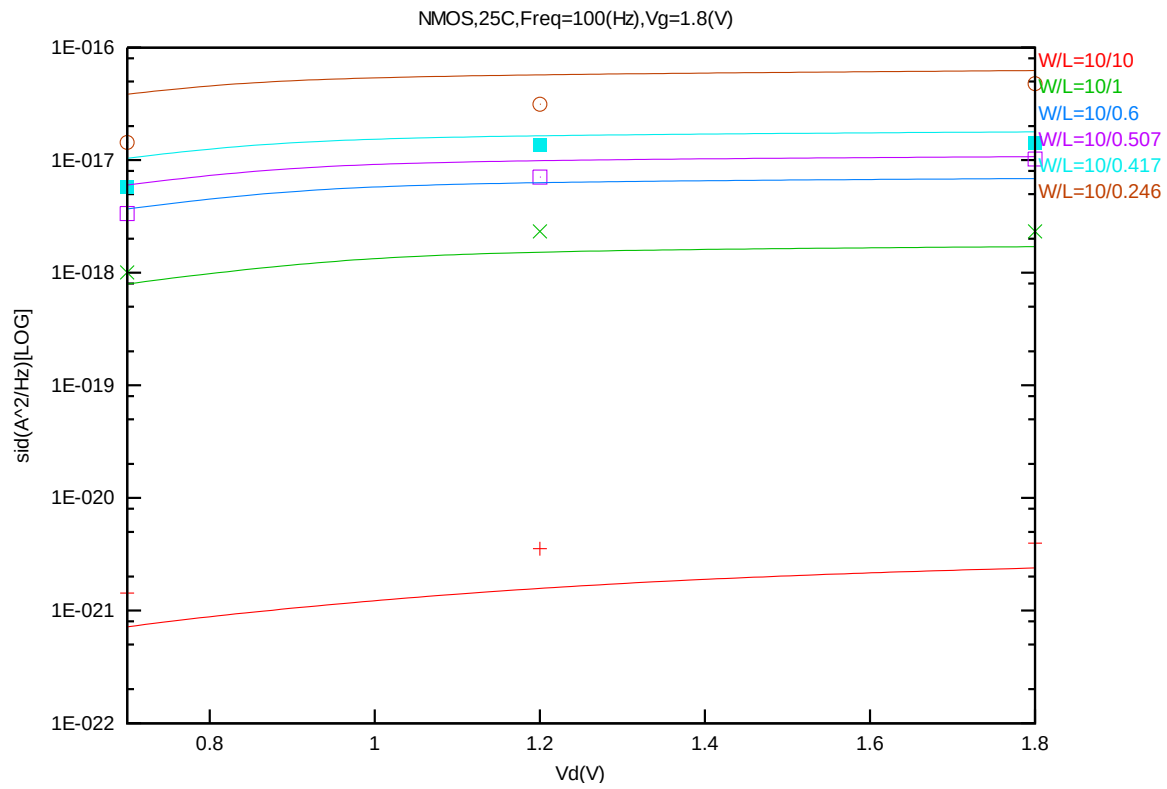


Figure 8-52 Noise vs. V_d for NMOS @ $V_g=1.8V$

B. PMOS

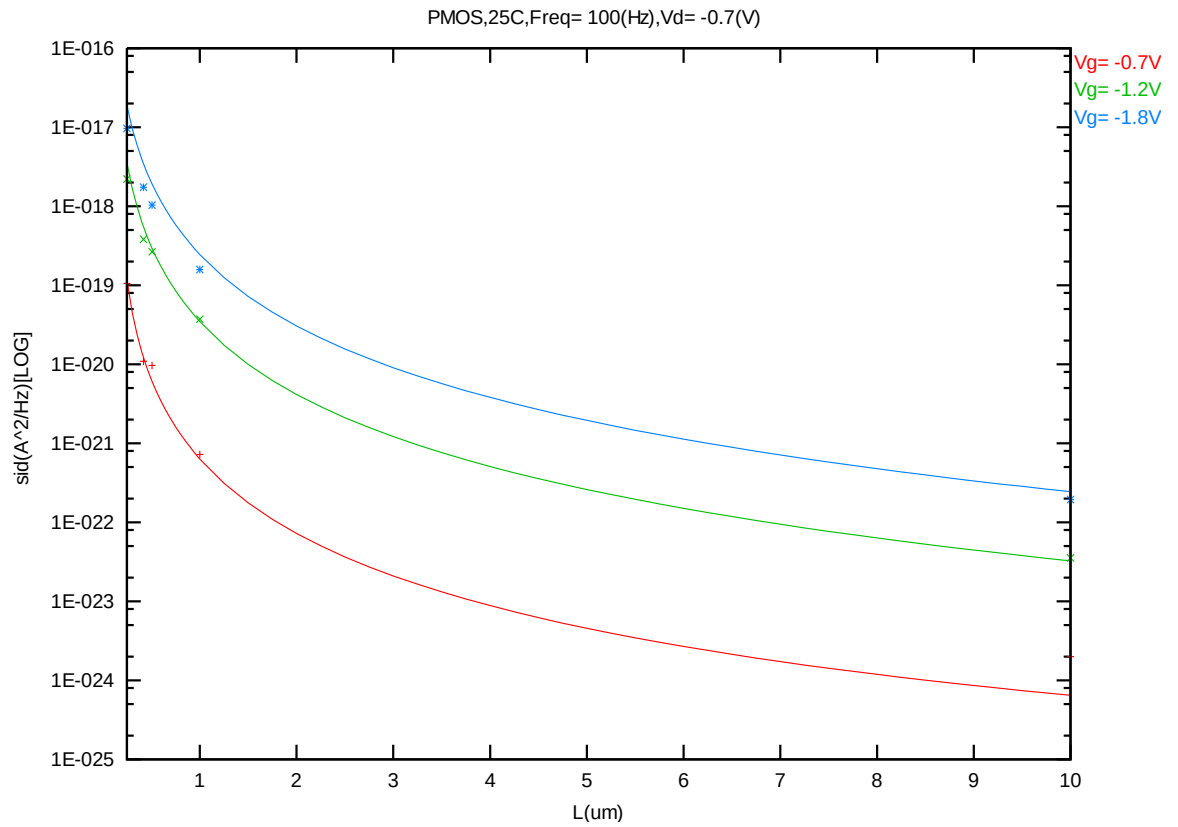


Figure 8-53 Noise vs. Length for PMOS @ $Vd=-0.7V$

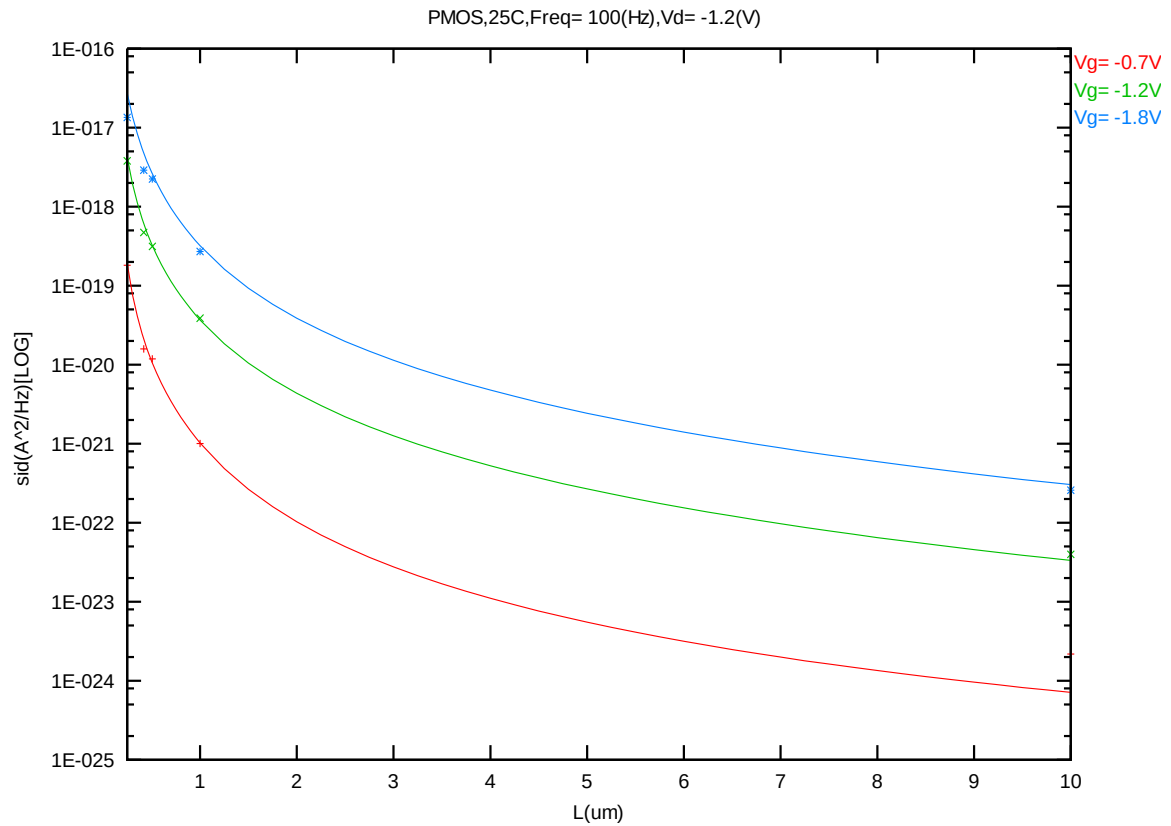


Figure 8-54 Noise vs. Length for PMOS @ Vd=-1.2V

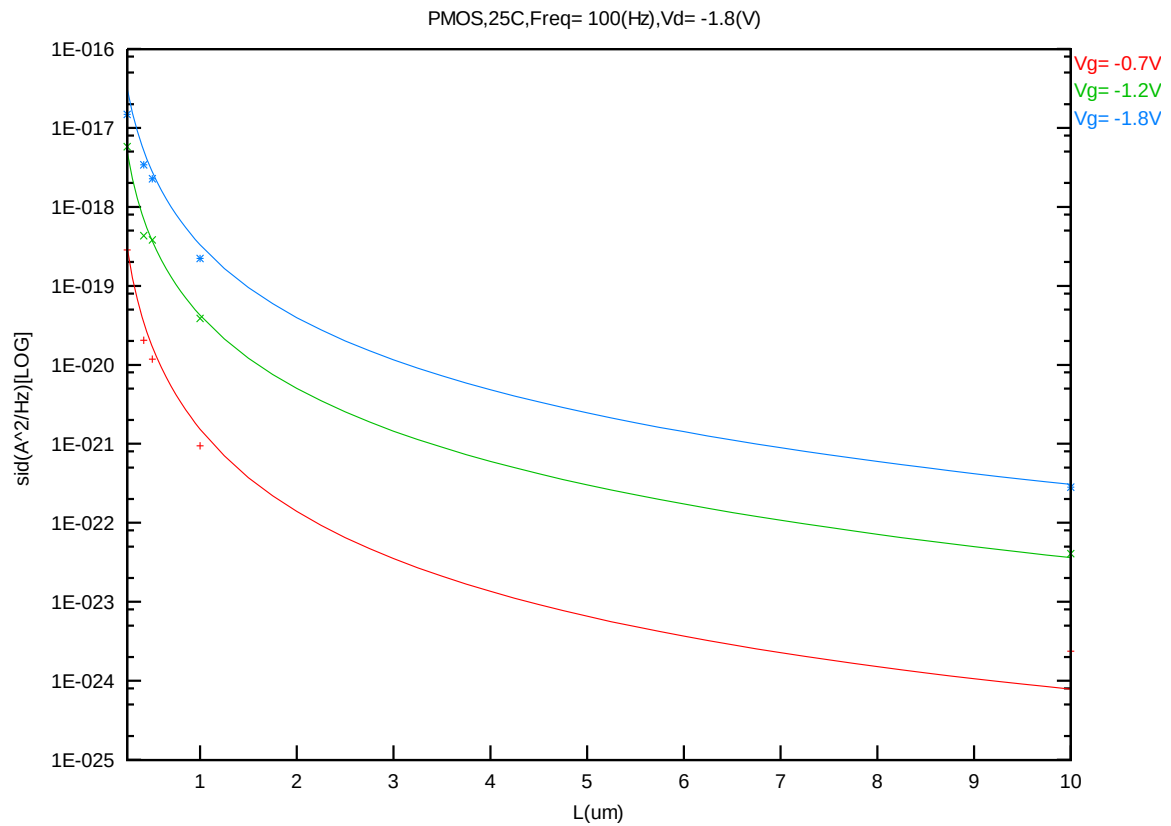


Figure 8-55 Noise vs. Length for PMOS @ Vd=-1.8V

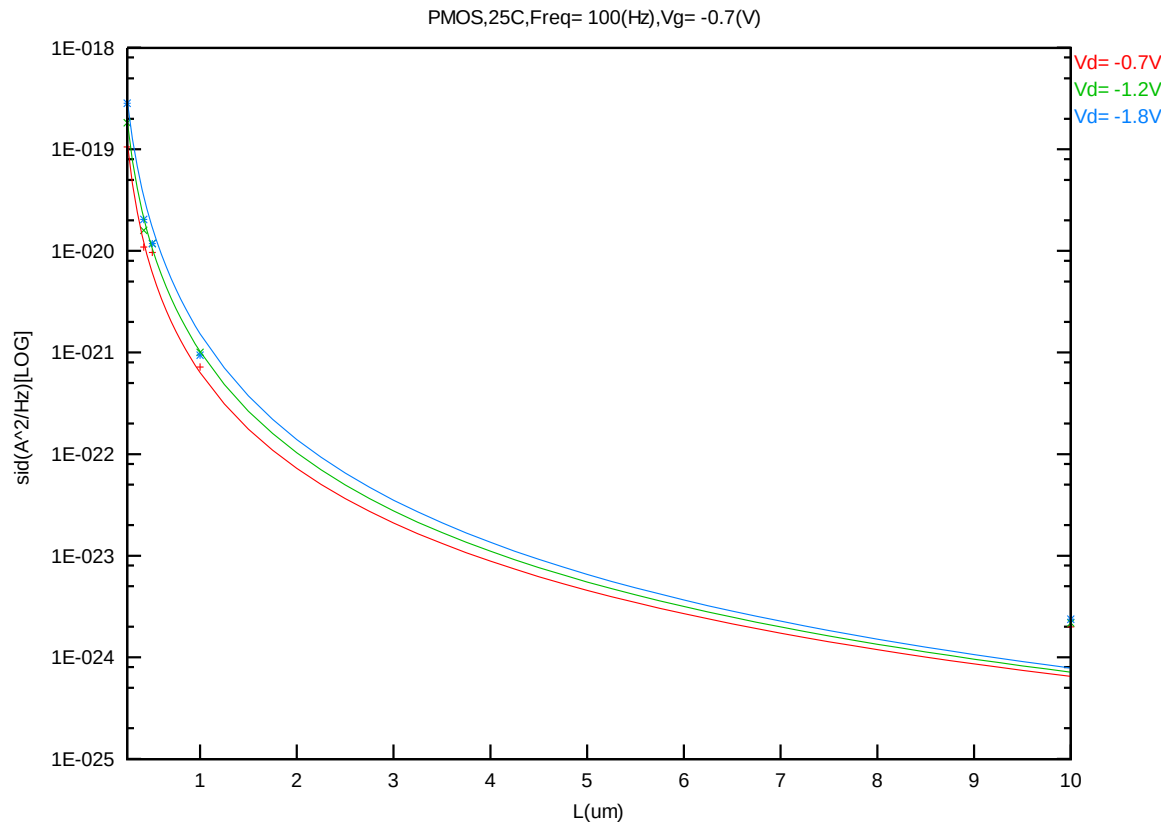


Figure 8-56 Noise vs. Length for PMOS @ Vg=-0.7V

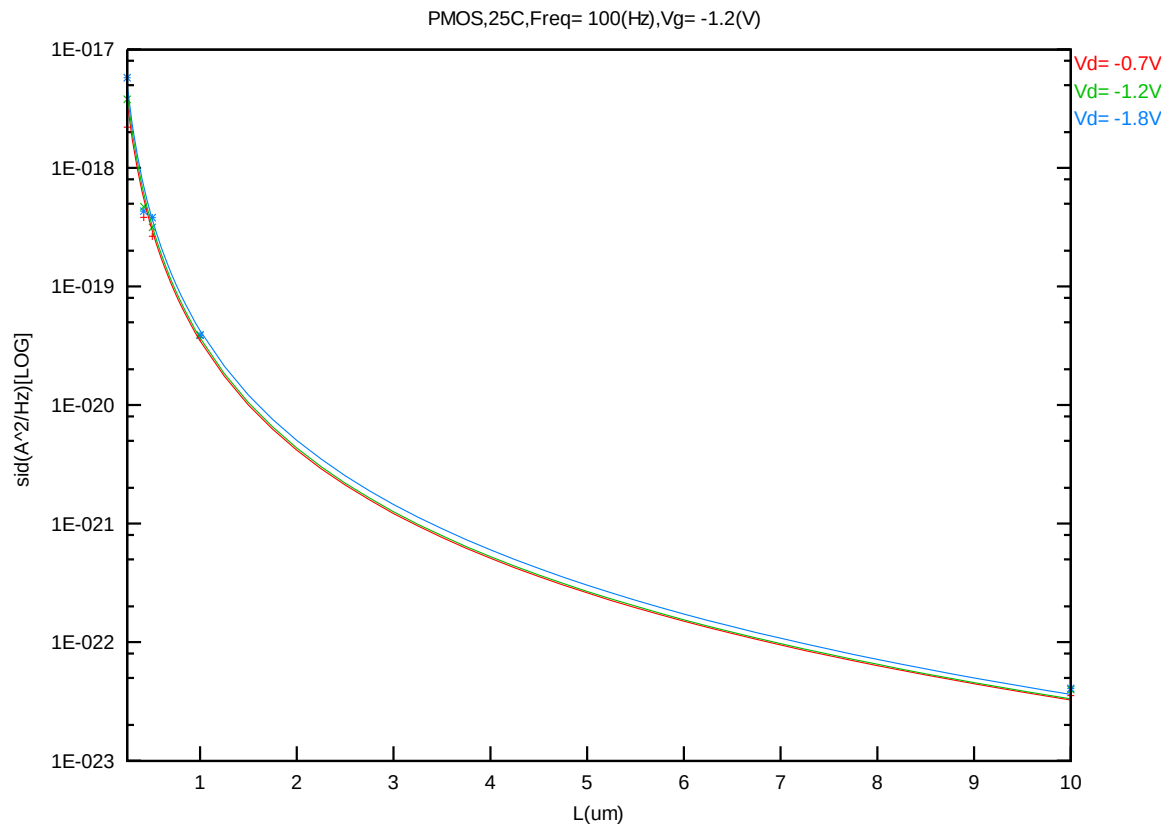


Figure 8-57 Noise vs. Length for PMOS @ Vg=-1.2V

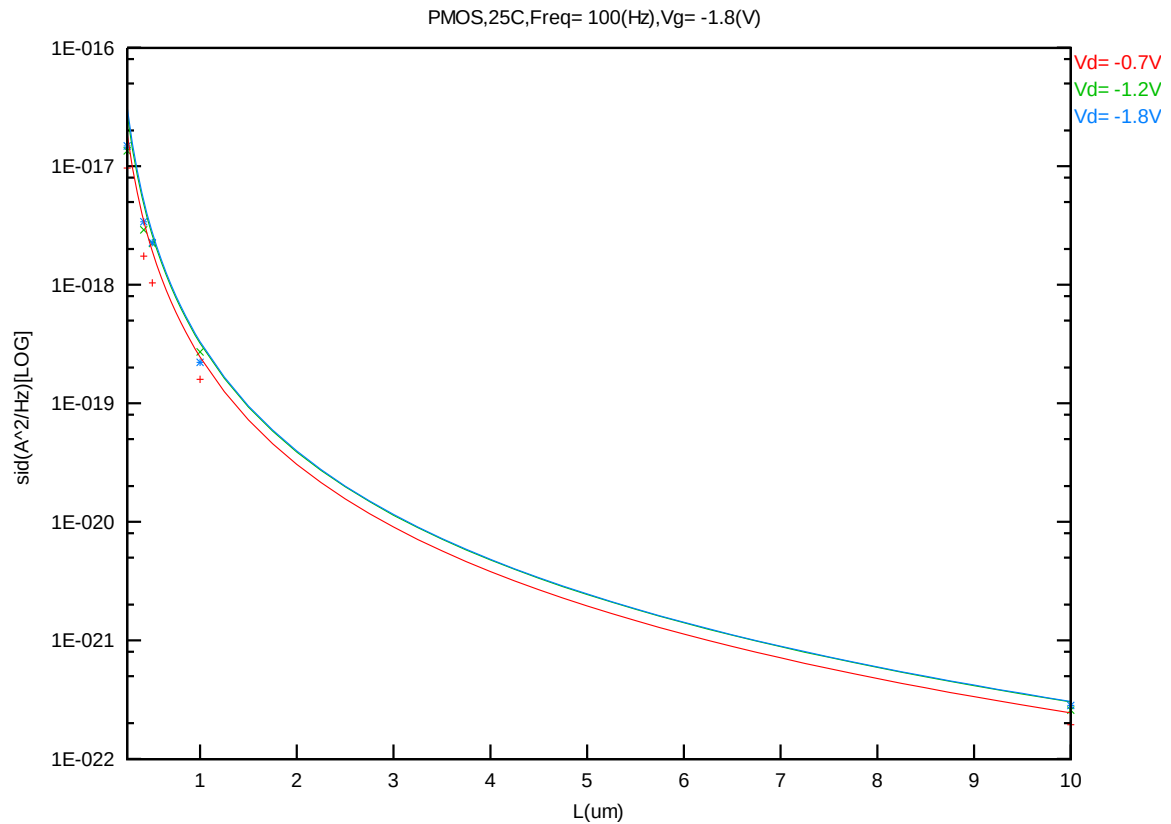


Figure 8-58 Noise vs. Length for PMOS @ Vg=-1.8V

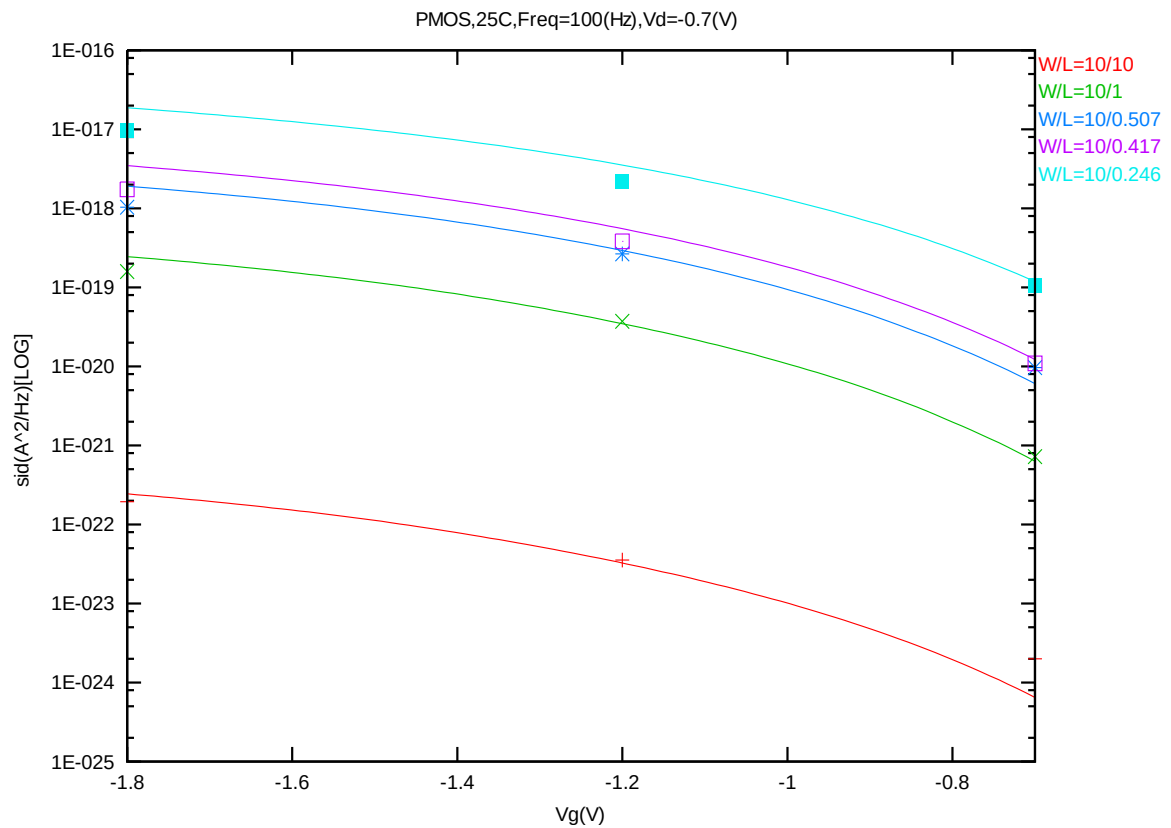


Figure 8-59 Noise vs. Vg for PMOS @ Vd=-0.7V

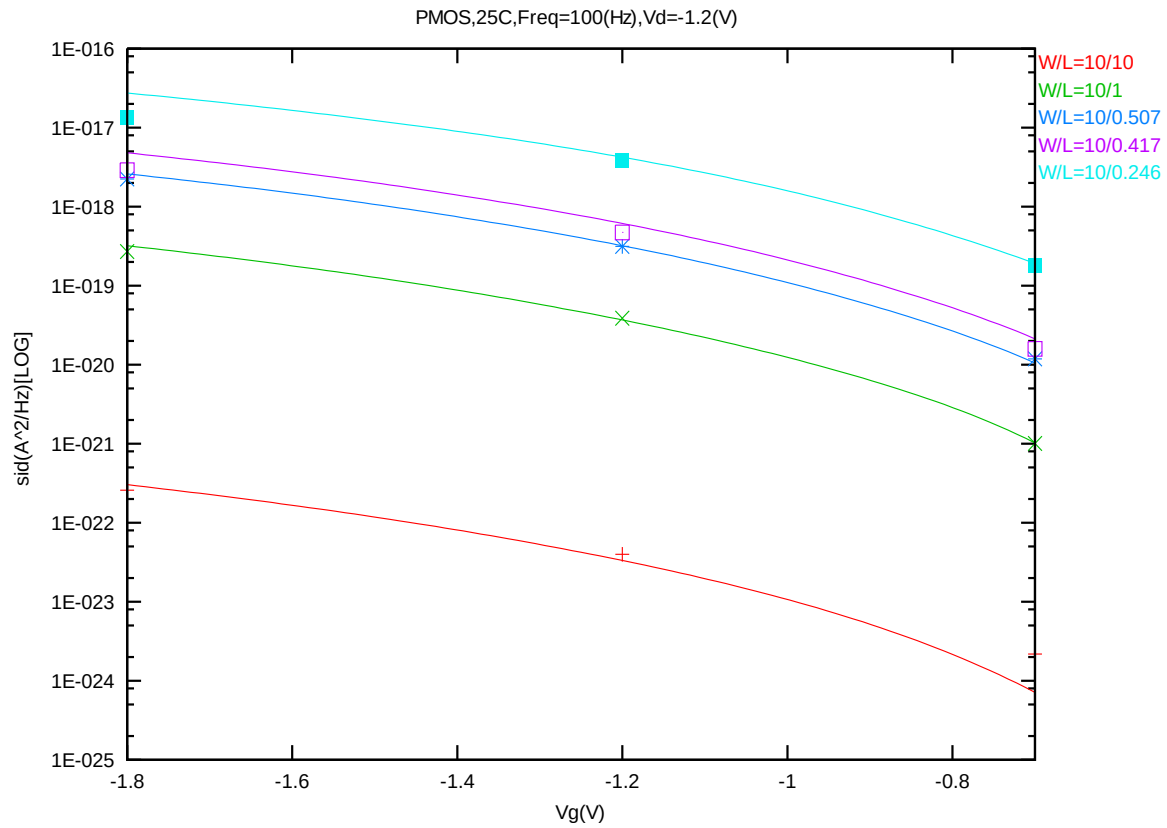


Figure 8-60 Noise vs. Vg for PMOS @ Vd=-1.2V

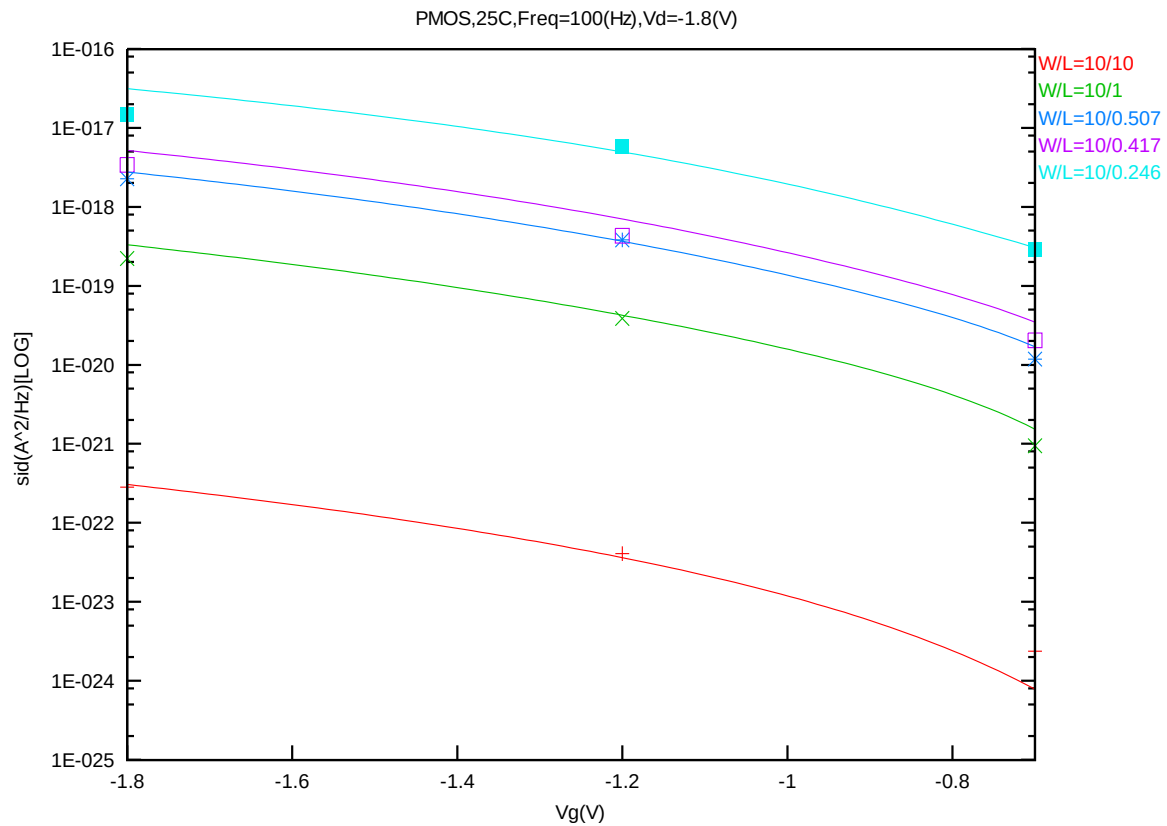


Figure 8-61 Noise vs. Vg for PMOS @ Vd=-1.8V

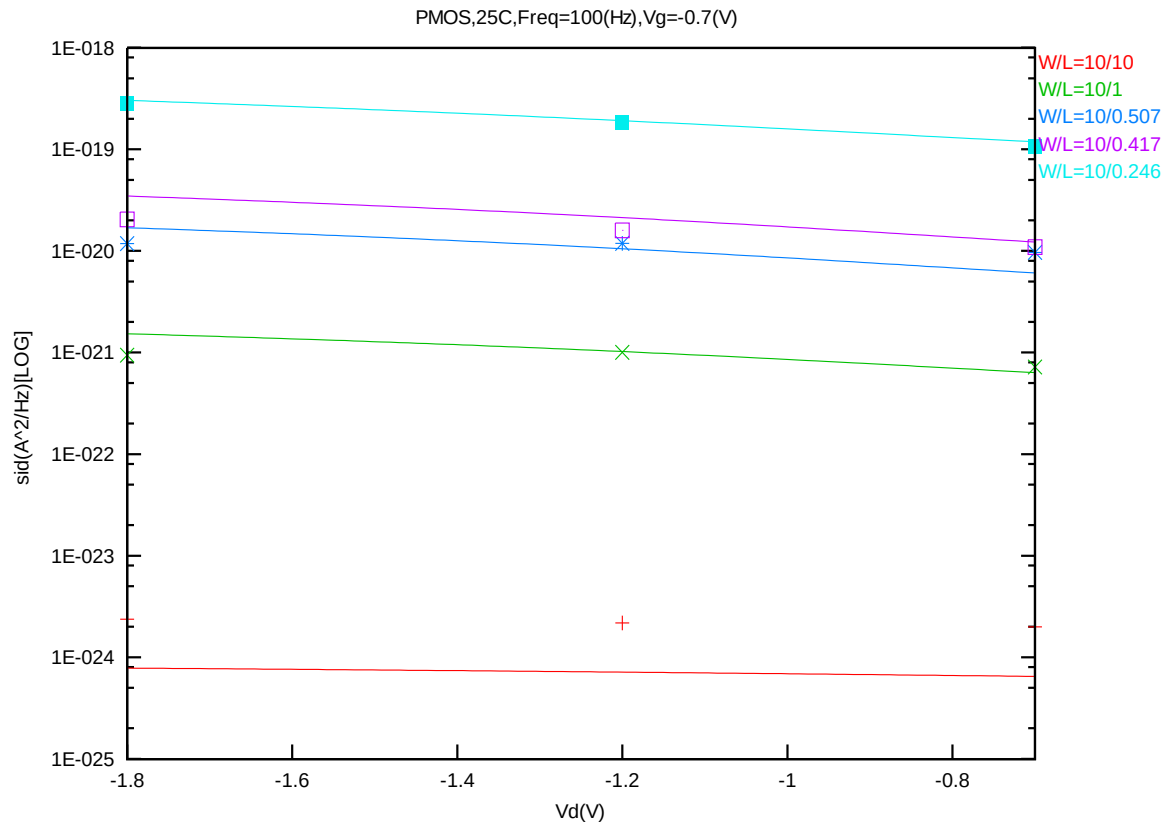


Figure 8-62 Noise vs. Vd for PMOS @ Vg=-0.7V

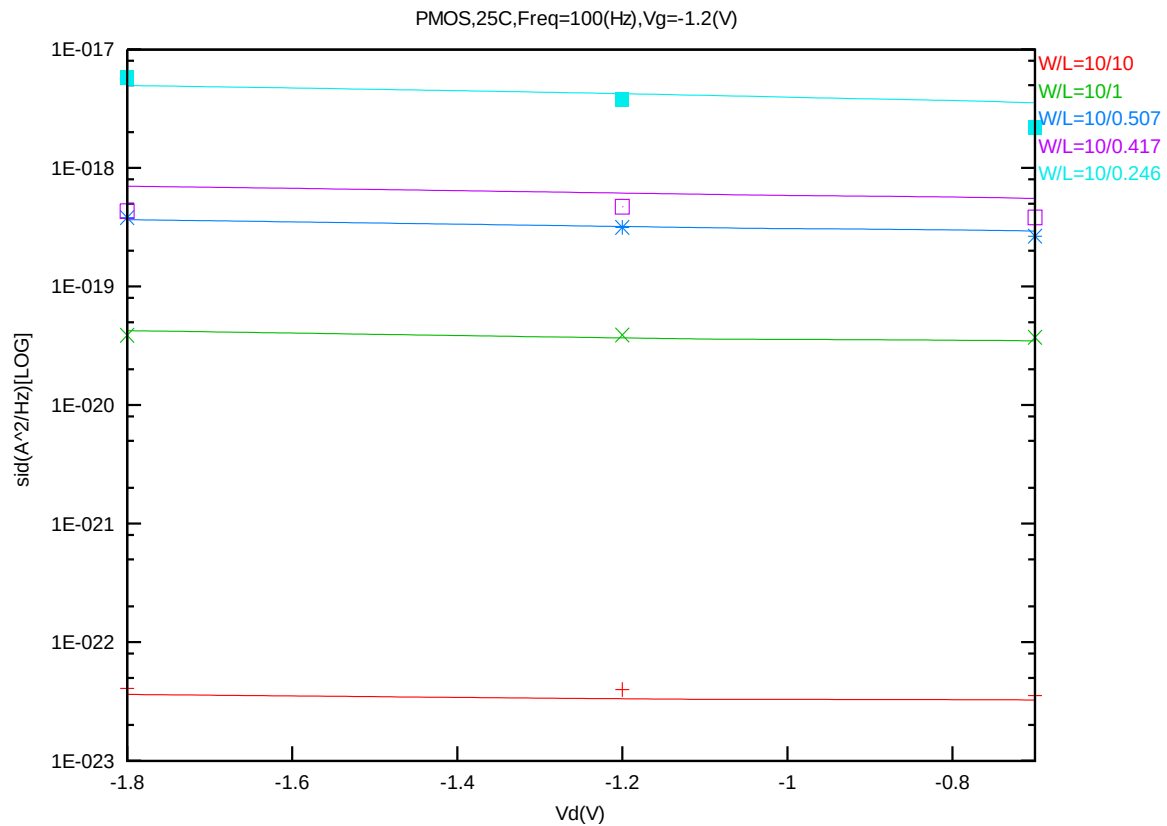


Figure 8-63 Noise vs. Vd for PMOS @ Vg=-1.2V

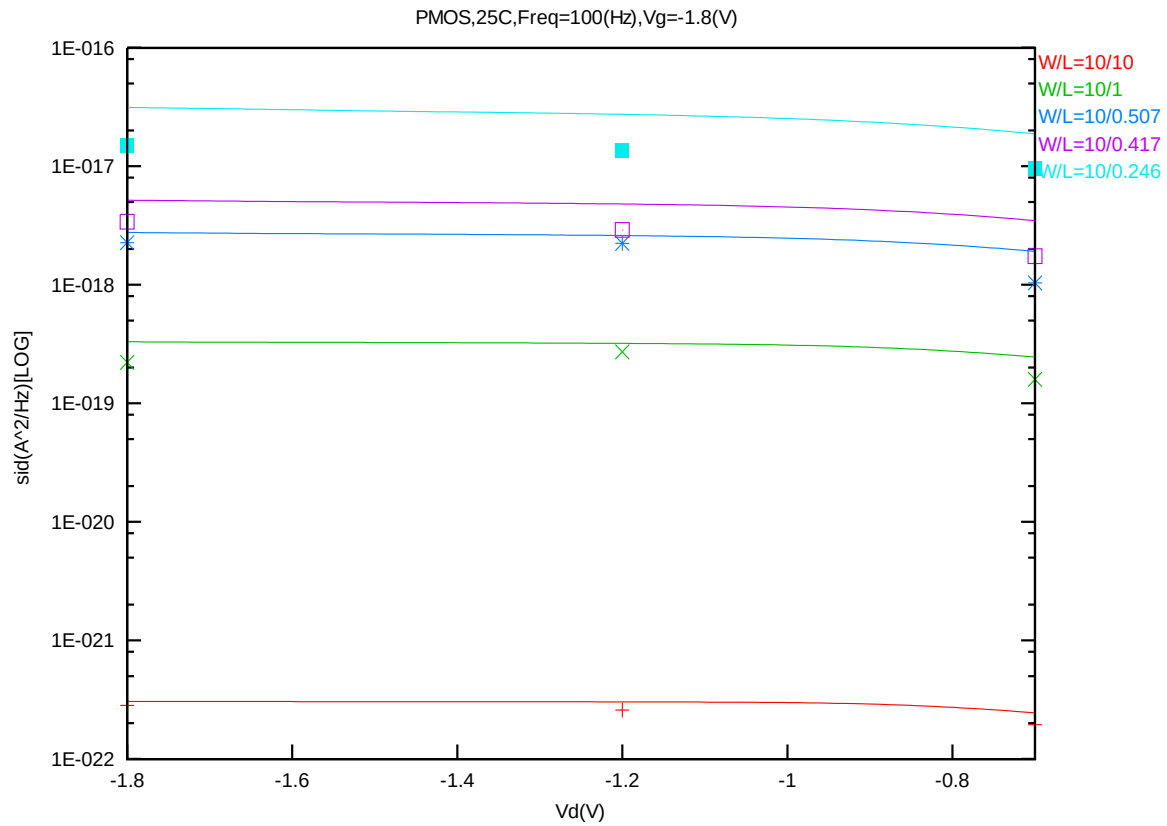


Figure 8-64 Noise vs. Vd for PMOS @ Vg=-1.8V

9 Appendix

9.1 Wafer Data vs. Corner Plots

Threshold voltage in linear region VTLIN

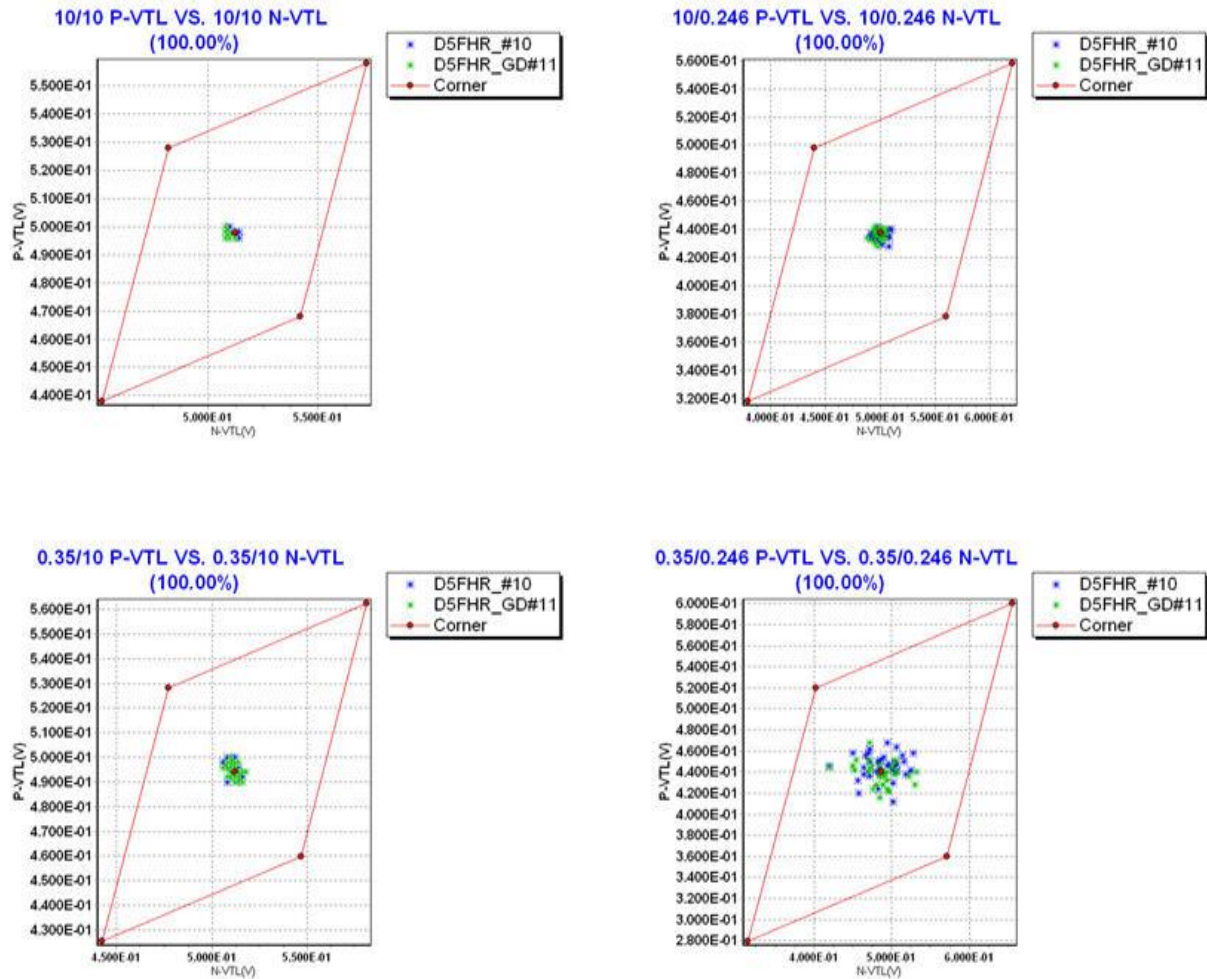


Figure 9-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in linear region VTSAT

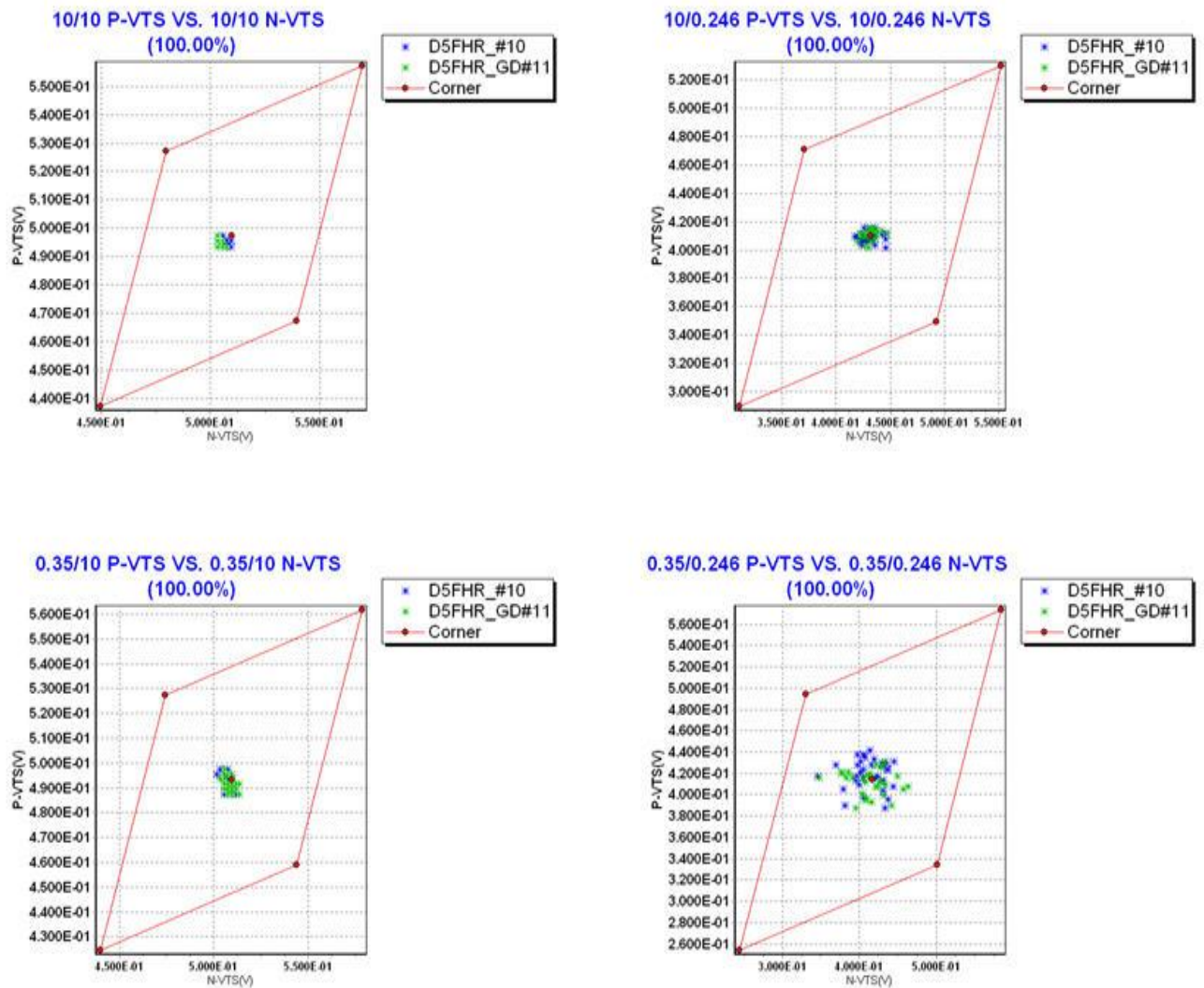


Figure 9-2 Corner plots of threshold voltage at high drain bias

Saturation current I_{ON}

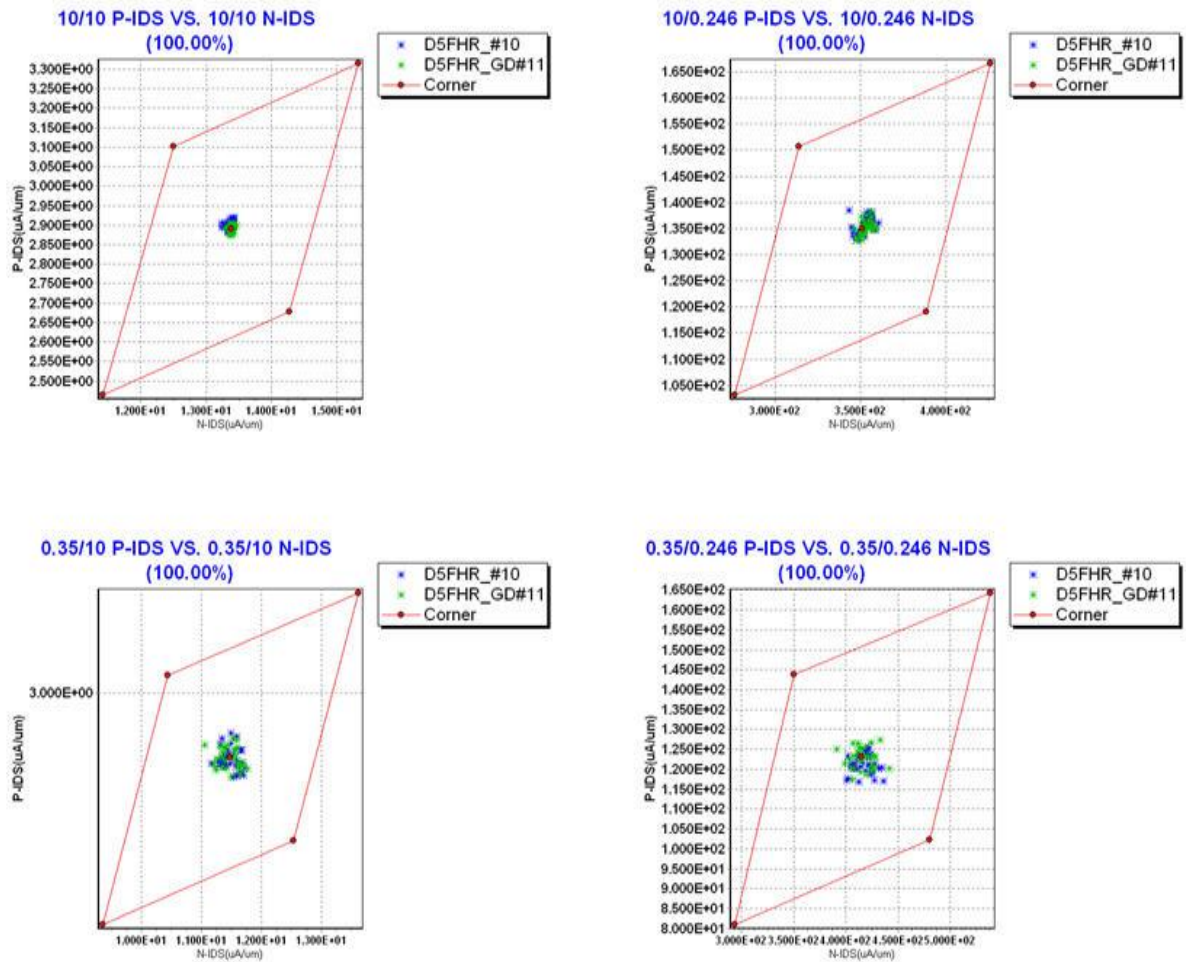


Figure 9-3 Corner plots of on-current

Off state current I_{OFF}

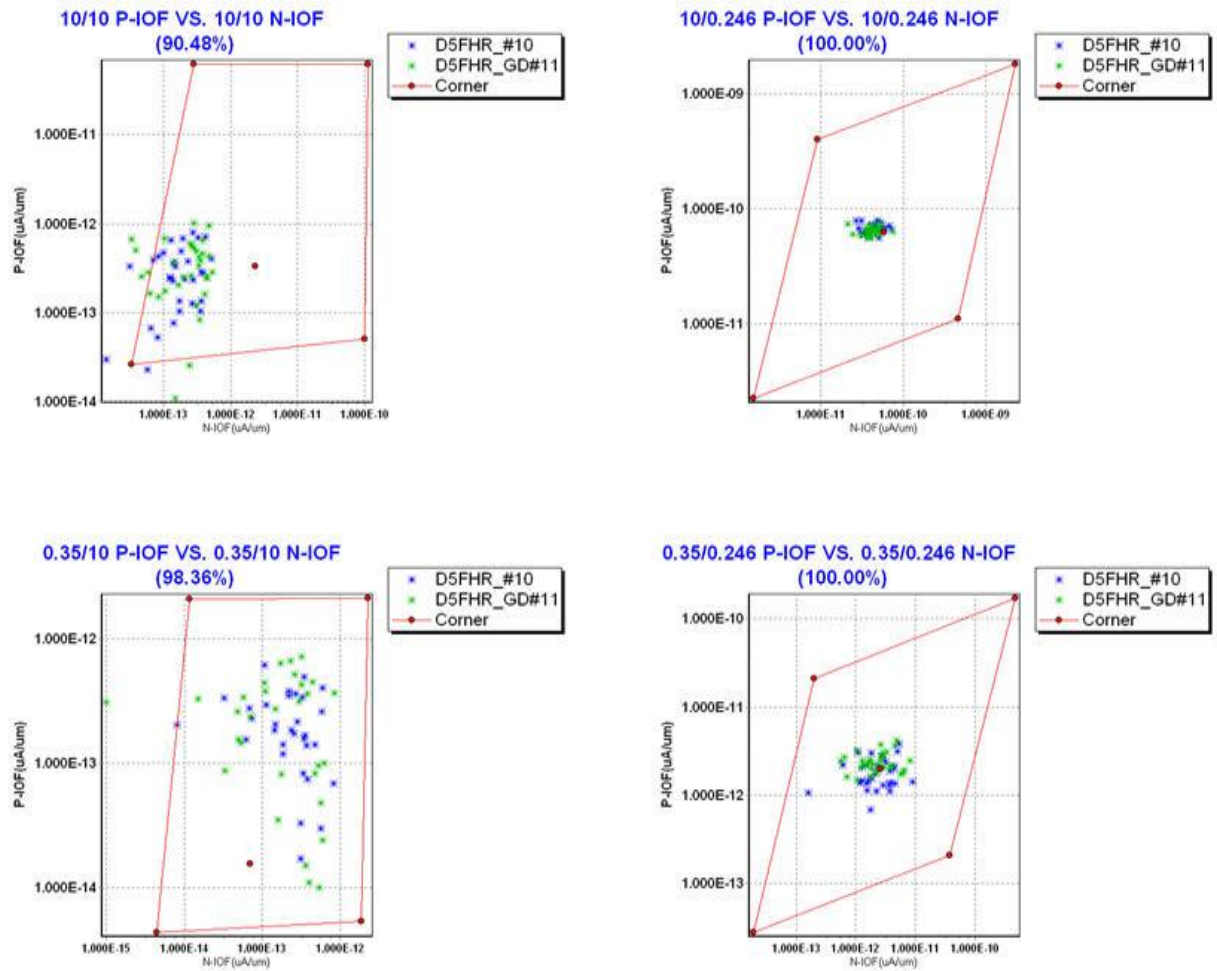


Figure 9-4 Corner plots of off-current

9.2 HSPICE Warning Message

Table 9-1 Hspice warning message

NO.	Warning Message	Explanation
1.	-	-

9.3 ELDO Warning Message

Table 9-2 Eldo warning message

NO.	Warning Message	Explanation
1.	-	-

9.4 SPECTRE Warning Message

Table 9-3 Spectre warning message

NO.	Warning Message	Explanation
1.	-	-